

HPS Gaussian-Process Resonance Search

Analysis Note v5.0.1

Draft for full-2021 unblinding review

Emrys Peets

Stanford University and SLAC National Accelerator Laboratory

Matthew Gignac

SLAC National Accelerator Laboratory

Eden Hsu

Stanford University and SLAC National Accelerator Laboratory

9 September 2026

Abstract

This note consolidates the HPS Gaussian-process prompt-resonance analysis and its subsequent validation studies for review before full-2021 unblinding. The analyzed datasets are full 2015, full 2016 and the released 2021 10% sample. A common coupling combines these datasets through a profiled Poisson likelihood with correlated Gaussian constraints on the interpolated background. The primary displayed limits are pointwise, bounded, asymptotic 90% CL_s limits. Conditional expected-limit bands, empirical tail refinement, background-calibration comparisons, mass-resolution variations and candidate extractions are presented with their respective assumptions. The strongest excess in the combined 19–250 MeV search occurs at 66 MeV, with local $p_0 = 2.845 \times 10^{-3}$ ($Z_{\text{local}} = 2.765$). The resolution-based Šidák approximation gives an equivalent global p -value of 0.0959 ($Z = 1.305$), using $N_{\text{eff}} = 35.38$ effective trials.

Contents

1	Introduction	4
1.1	Published HPS prompt-resonance benchmarks	4
1.2	Complementary prompt-visible constraints	5
1.3	Motivation for the present GPR analysis	6
1.4	Scope of this note	9
2	HPS apparatus and data sets	10
2.1	Detector configurations for the prompt search	10
2.2	Data-taking campaigns and analysis scope	11
3	Event selection, mass resolution, and radiative fraction	15
3.1	Prompt event selection	15
3.2	Kinematic distributions after the 2021 preselection	17
3.3	2015 mass resolution	22
3.4	2016 mass resolution	23
3.5	2021 mass resolution	24
3.6	Radiative fraction and the ε^2 normalization	26
3.6.1	2015 radiative fraction	27
3.6.2	2016 radiative fraction	27
3.6.3	2021 radiative fraction	28
3.7	Systematic penalties from the radiative fraction	29
4	Methodology	32
4.1	Background interpolation around a blinded mass window	32
4.2	Search interval and GP fit support	32
4.3	Observable, scan grid, and blinding	32
4.4	Log-space preprocessing and the choice $\alpha = 1/y$	33
4.5	Kernel choice and distinct regularization roles	35
4.6	Resolution-scaled length-scale parameterization and bounds	36
4.7	Blind-window prediction in count space	40
4.8	Profiled background implementation	41
4.9	Signal model and conversion to ε^2	42
4.10	Statistical inference in a single dataset	44
4.10.1	Local significance	44
4.10.2	CL_s upper limits	45
4.11	Asymptotic and toy calibration	45
4.12	Matched-reference injection-extraction diagnostics	46
4.13	Observed-limit toy diagnostics	47
4.14	Shared- ε^2 combination across datasets	49
4.15	Global significance and the look-elsewhere treatment	50
4.16	Signal absorption and alternative background-profile treatments	51
4.17	Dimuon-threshold interpretation	51
4.18	Why the background uncertainty is profiled	52
4.19	What the signal injections establish	53
4.20	A GP calculation of scan-global significance	55
5	Background model validation with pseudo-experiments	58
5.1	What the validation asks	58
5.2	Optimizing the blind window	59
5.3	Generating spectra and the role of functional forms	60
5.4	2021 validation ensembles at four effective exposures	66
5.5	Targeted validation of the rising threshold	68
5.6	Matched refits and fit-quality checks	69

5.7	How closure is judged	69
5.8	Spurious-signal results from 100 pseudoexperiments	70
5.9	Recovery of injected signals	74
5.10	Choosing the 2021 GP support	74
5.10.1	Source shape and candidate supports	74
5.10.2	Initial scan	76
5.10.3	A practical rule and an independent check	76
5.11	Validation evidence and resulting decisions	78
6	Results for the released data	79
6.1	The complete observed search and conditional expected bands	79
6.2	Conditional full-2021 response compared with BaBar	81
6.3	Signal-yield and coupling limits for each dataset	83
6.4	Combining data sets through a shared coupling	86
6.5	Local asymptotic scans	87
6.6	Conditional expected bands for the individual and overlapping scopes	87
6.7	Empirical limit-tail diagnostics and unresolved tails	91
6.8	Varying the mass resolution in the signal template	94
6.9	Signal extraction from the observed data	96
6.10	Correlated fluctuations and signal-induced echoes	104
6.11	Local probabilities and the conditional GP global studies	109
6.12	Agreement of fitted rates across campaigns	113
7	Status for the full-data unblinding review	114
A	Change log and study history	115
B	Alternative background modeling through calibration	119
C	Candidate removal and traditional background fits	131
D	Joint scan of deficits	141
E	Historical validation studies and their role in v5	142
F	Historical pseudoexperiment regimes and model decisions	143
G	Historical comparisons with the 2015 and 2016 resonance searches	163
H	Exploratory 2015 search on the 15–20 MeV rising edge	168

1 Introduction

The Heavy Photon Search (HPS) prompt-resonance program is formulated in the minimal visible dark-photon model, in which a new broken $U(1)_D$ gauge boson mixes kinetically with the Standard-Model photon [1, 2]. After diagonalizing the gauge fields, the interaction terms relevant for the prompt $A' \rightarrow e^+e^-$ search can be written schematically as

$$\mathcal{L} \supset -\frac{1}{4}F'_{\mu\nu}F'^{\mu\nu} - \frac{\varepsilon}{2}F'_{\mu\nu}F^{\mu\nu} + \frac{1}{2}m_{A'}^2 A'_\mu A'^\mu + \varepsilon e J_{\text{EM}}^\mu A'_\mu. \quad (1)$$

Here ε is the kinetic-mixing parameter, $m_{A'}$ is the dark-photon mass, and J_{EM}^μ is the ordinary electromagnetic current. In the low-mass region considered here, the minimal visible model predicts a narrow prompt dielectron resonance. Its natural width is negligible compared with the HPS detector resolution, so the observed line shape is set by the experimental mass resolution [3, 4]. This narrow-width regime makes a resonance search well suited to the 10 MeV to 300 MeV mass window: fixed-target electroproduction can reveal a visibly decaying vector as a localized excess above the smooth trident spectrum, in a coupling range that includes established dark-sector and lepton-anomaly benchmarks.

In the fixed-target approximation relevant for HPS, prompt dark-photon production is normalized to the radiative trident rate,

$$\sigma(eZ \rightarrow eZA') \simeq \frac{3\pi\varepsilon^2}{2\alpha_{\text{EM}}} m_{A'} \left. \frac{d\sigma_{\gamma^*}}{dm} \right|_{m=m_{A'}}, \quad (2)$$

the analysis extracts a narrow signal yield from the smooth prompt e^+e^- spectrum and then converts that yield into a limit or best fit for ε^2 [2, 3, 4]. The physics interpretation rests on the minimal prompt A' model and the radiative-trident normalization; Gaussian-process regression (GPR) provides the statistical description of the smooth background.

1.1 Published HPS prompt-resonance benchmarks

The published HPS prompt searches provide the immediate experimental context for the current GPR analysis. The 2015 engineering-run publication analyzed 1170 nb^{-1} at 1.056 GeV over 19 MeV to 81 MeV and reported a lowest local p -value of 1.7×10^{-3} at 37.7 MeV; after the look-elsewhere correction the global p -value was 17% [3]. The 2016 prompt search analyzed the full engineering-run sample, $10\,608 \text{ nb}^{-1}$ at 2.3 GeV, over 39 MeV to 179 MeV. Its lowest local p -value was 6.38×10^{-3} at 94 MeV, with a global significance of about 1σ , and it propagated mass-resolution and radiative-fraction systematics explicitly into the final prompt limit [4]. Figure 1 collects those published HPS prompt benchmarks in the compact form most relevant for the present GPR note.

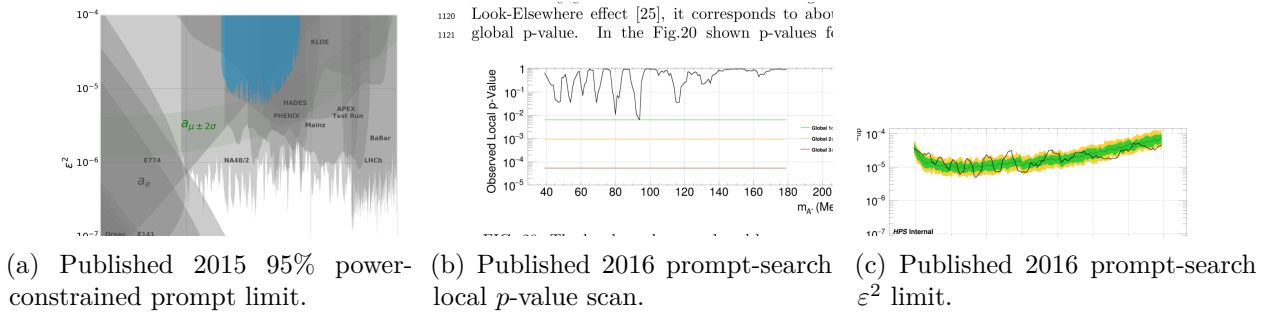


Figure 1: Published HPS prompt-resonance benchmarks. These re-exported crops are used as high-level context for the current GPR workflow rather than as a replacement for the original publications [3, 4].

1.2 Complementary prompt-visible constraints

The relevant comparison is not only which experiments reach the same mass window, but also how their exclusions populate the $(m_{A'}, \epsilon^2)$ plane over the broader low-mass visible-dark-photon range. In that space HPS occupies a distinct fixed-target niche between very-low-mass beam-dump and meson-decay constraints and the higher-mass collider reach. Figure 2 shows that context using legacy HPS phase-space contour inputs, with the old lepton-anomaly overlays deliberately omitted. The 2015 and 2016 HPS engineering-run prompt contours are highlighted because they are the directly relevant published HPS benchmarks for the present GPR analysis; both are displayed at an approximate 90% CL using the historical reach-notebook rescaling. This display-only approximation is not a recalculation of either published likelihood and is not used in the v3 upper limit.

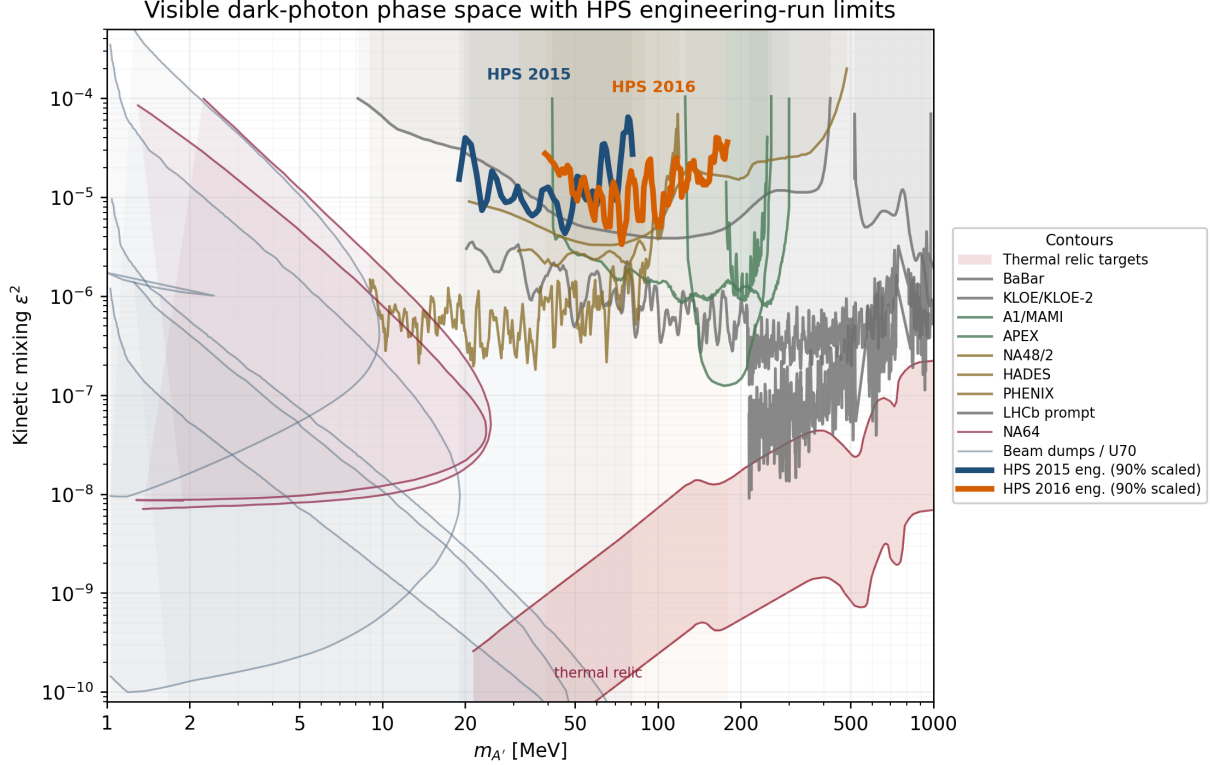


Figure 2: Visible dark-photon phase space used to contextualize the HPS prompt engineering-run results. The world-data contours include BaBar, KLOE/KLOE-2, A1/MAMI, APEX, NA48/2, HADES, PHENIX, NA64, LHCb, and the older beam-dump or low-mass constraints from E137, E141, E774, Orsay, KEK, and U70 [5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21]. The thermal relic target band is shown as a benchmark for sub-GeV dark matter motivation [2]. The HPS 2015 and 2016 engineering-run prompt limits are highlighted using the legacy display convention that multiplies the published 95% ϵ^2 contours by 1.64/1.96 [3, 4]; the NA48/2 and BaBar inputs are treated in the same way. This approximate Gaussian-ratio rescaling is retained only for historical context and is not a recalculation of a 90% CL upper limit with the CL_s construction. Older $g - 2$ contours from the source folder are intentionally omitted.

1.3 Motivation for the present GPR analysis

Gaussian-process regression is best viewed here as a modern form of covariance-based interpolation. The historical starting point is kriging, developed in mine-valuation and geostatistical applications by Krige and formalized by Matheron: given observations of a smooth field at some locations, the task is to predict the field elsewhere while carrying the uncertainty induced by finite sampling and spatial correlation [22, 23]. In a resonance search the coordinate is not geographic position but invariant mass. The same statistical idea is useful because the prompt trident continuum is smooth on the scale of the HPS mass resolution, while a prompt A' would appear as a narrow localized excess. GPR replaces an explicit closed-form background curve with a probability distribution over smooth functions, specified by a mean and covariance kernel and conditioned on the observed training bins outside the mass-dependent excluded window [24]. For each test mass, the practical output is therefore not only a background prediction in the blinded window, but a correlated uncertainty on that prediction.

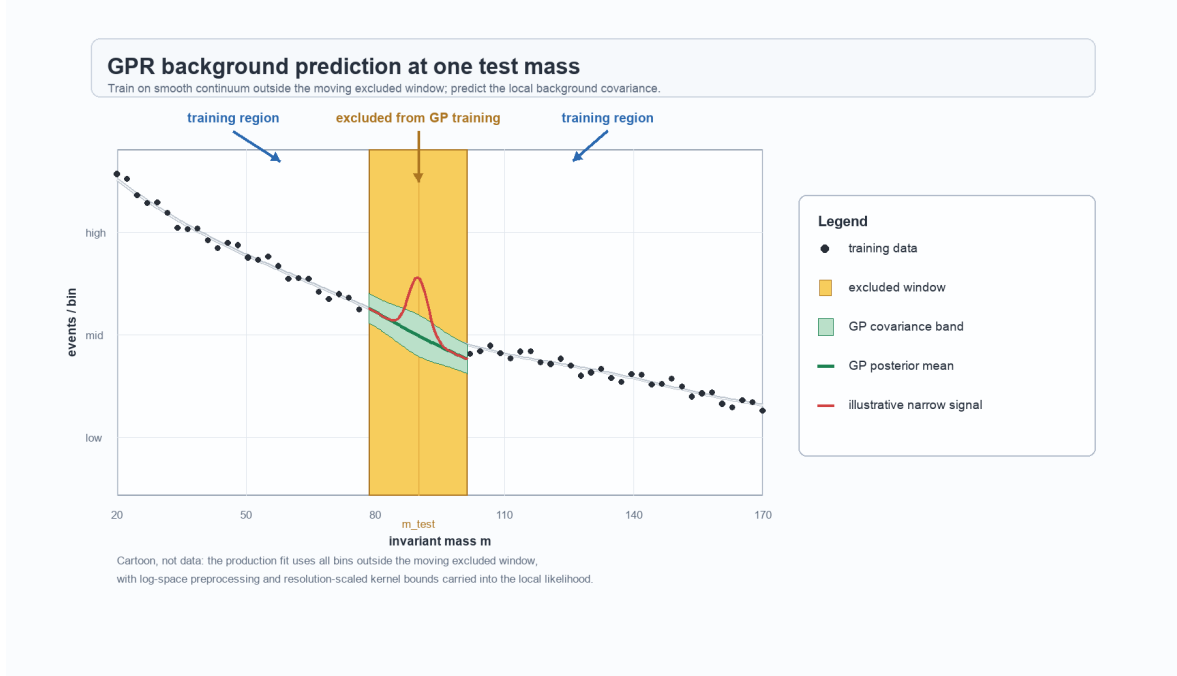
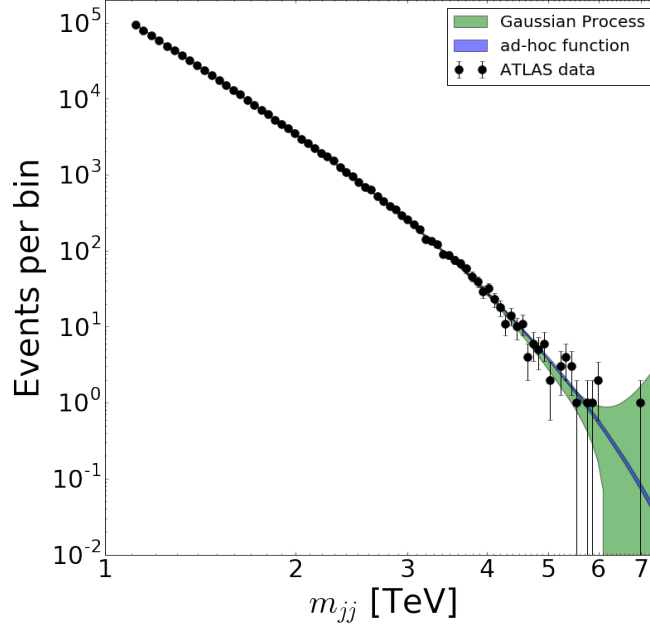
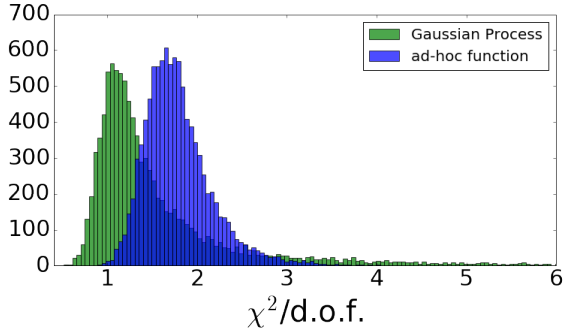


Figure 3: Cartoon of the per-test-mass GPR workflow. For each test mass, the GP is trained on bins across the full mass range except for a narrow excluded window centered on that mass. The trained GP predicts the posterior mean and covariance inside the excluded window, where a narrow A' -like signal hypothesis is overlaid only for illustration.

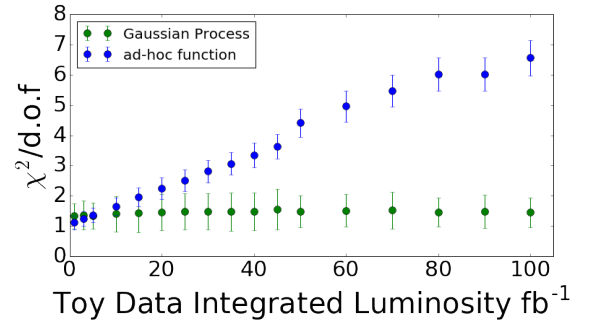
This viewpoint is attractive for HPS for the same reason emphasized in the 2017 high-energy-physics study of Frate *et al.*: fixed empirical functions restrict the background to a low-dimensional family, and discrepancies that are hidden by statistical fluctuations can become limiting as luminosity grows or as the scan region is widened [25]. Figure 4 reproduces three of their comparison plots. In toy spectra built from ATLAS dijet data, the GP background model maintains a near-unit goodness-of-fit scale and remains stable as the effective luminosity increases, while the ad-hoc functional form degrades. The lesson for HPS is not that a GP is assumption-free: the kernel, training exclusion, transform, and length-scale bounds are all analysis choices. The benefit is that the smoothness assumption is made explicit and propagated as a covariance, rather than being hidden in a hand-selected analytic form.

The later CMS-oriented GPR signal-extraction study by Gandrakota *et al.* provides the closest methodological bridge to the present implementation: choose a kernel class, condition the GP on binned count data, compare candidate kernels against the smooth background structure, and then separate a localized signal component from the inferred background in a reproducible statistical workflow [26]. The HPS analysis adopts that framework in a form matched to a fixed-target prompt-resonance search. It trains the GP in log-mass/log-count space, removes the test-mass window from training, ties the allowed length scale to the detector resolution, converts the posterior prediction back into count space, and carries the resulting mean vector and covariance matrix into the local profile likelihood. As detailed in Section 4, GPR turns smooth-background interpolation into an explicit probabilistic model that can be validated, profiled, and combined across datasets.

Earlier HPS prompt searches used mass-window-specific background models, window-by-window optimization, and a resonance extraction whose numerical behavior depended on the local fit boundaries. Here, every mass hypothesis is assigned its own GP fit on a fixed, dataset-specific

(a) Toy-spectrum mean and $\pm 1\sigma$ background bands.

(b) Toy goodness-of-fit distributions.



(c) Goodness of fit versus effective luminosity.

Figure 4: Methodological comparison between Gaussian-process and ad-hoc functional-form background modeling reproduced from Ref. [25]. The panels show that, in the 2017 dijet toy study, the GP retained an accurate smooth-background description while the fixed functional form became increasingly strained as the effective statistical precision improved.

support, with a narrow region around that hypothesis excluded from training. The fit supplies a mean vector and covariance matrix for the blinded bins. Signal extraction, local significance, modified-frequentist CL_s limits, radiative-fraction conversion, and the shared- ϵ^2 combination then follow within one statistical framework.

This change is especially valuable for the present HPS program because one statistical workflow must describe three campaigns with different mass ranges and detector configurations. In v5 the tested signal hypotheses span 19 MeV to 90 MeV for 2015, 39 MeV to 180 MeV for 2016, and 50 MeV to 250 MeV for 2021, while the GP fit supports extend beyond those tested intervals to 14 MeV to 135 MeV, 30 MeV to 210 MeV, and 36 MeV to 300 MeV, respectively. A stable GP model therefore matters both for numerical smoothness and for carrying the same radiative-fraction and uncertainty treatment across datasets without sacrificing narrow-resolution sensitivity. It also motivates the combined analysis: the minimal visible A' model predicts one common coupling ϵ^2 at a given mass, while each HPS run maps that coupling into a different expected signal yield through its own integrated luminosity, acceptance, resolution, and radiative normalization. The statistical combination therefore treats ϵ^2 as the shared parameter of interest and multiplies the independent dataset likelihoods, profiling the dataset-specific background nuisance parameters separately. In this common-parameter construction, datasets with better local signal-to-background leverage carry more information, while weaker channels still contribute correctly; the gain in sensitivity comes from adding independent likelihood information for the same minimal-model signal rather than from averaging separate limits after the fact.

1.4 Scope of this note

Version 5 consolidates the detector description, statistical method, validation history and observed results for the proposed full-2021 unblinding review. The current data comprise full 2015, full 2016 and the released 2021 10% sample. No additional 2021 events enter this document. The 2019 run provides detector context but does not contribute an observed spectrum.

The current seven-scope release uses the frozen 14–135, 30–210 and 36–300 MeV GP supports for 2015, 2016 and 2021. Earlier configurations are identified as historical when they are needed to explain an analysis choice. Observed limits use the bounded, piecewise-asymptotic 90% CL_s construction with profiled backgrounds. Pointwise expected-limit bands, empirical limit-tail diagnostics, and the later conditional global-probability studies answer different questions and are presented separately.

The full-2016 result retains a disclosed cross-process state-replay exception. Later calibration and global studies also expose sensitivity to the assumed background spectrum. This note records those limitations as issues for the unblinding review; it does not certify their resolution or authorize opening additional data. Appendix A records how the recent studies changed the analysis and the presentation.

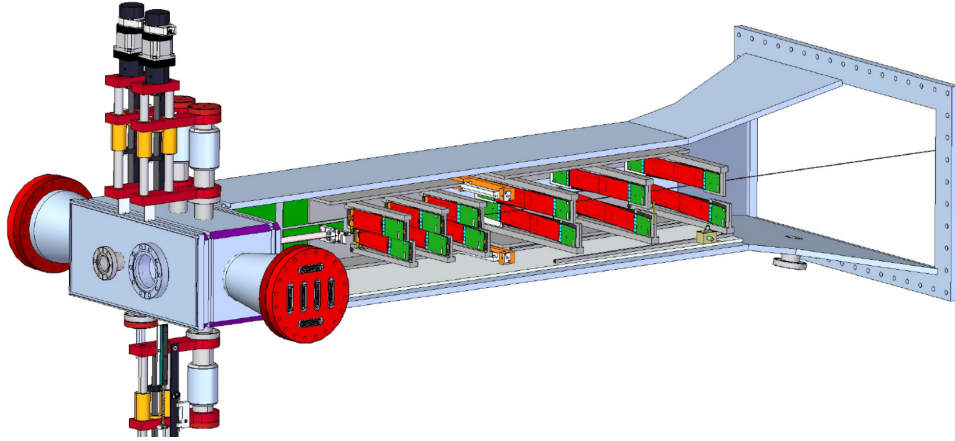
2 HPS apparatus and data sets

2.1 Detector configurations for the prompt search

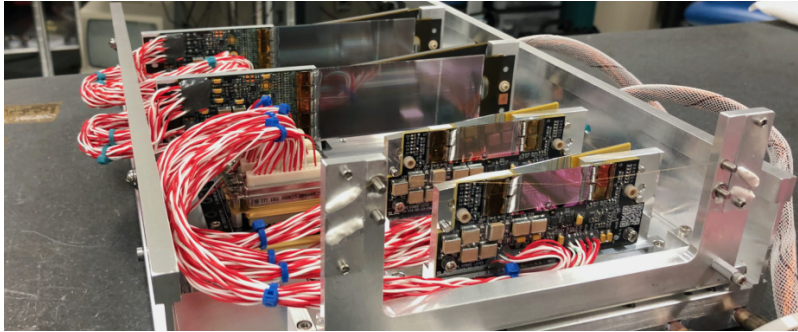
HPS is a forward spectrometer optimized for narrow e^+e^- resonances and displaced vertices in the high-rate environment immediately downstream of a thin tungsten target. Its prompt $A' \rightarrow e^+e^-$ sensitivity is set primarily by three subsystems: the Silicon Vertex Tracker (SVT), which determines the mass and vertex resolution; the electromagnetic calorimeter (ECal), which provides the trigger and track-cluster matching; and the beamline/target geometry, which defines the forward acceptance and the occupancy environment [27, 4].

The 2015 and 2016 engineering runs used the baseline six-station SVT. In that configuration, six axial/stereo tracking stations were distributed between $z = 10$ cm to 90 cm downstream of the target, with the first active silicon only about 1.5 mm from the beam plane [28, 27]. The ECal geometry used in the engineering runs was already close to the final HPS layout: two PbWO_4 crystal halves with the highest-rate crystals nearest the beam removed, leaving the beam to pass between the two halves [28, 27].

Before the first physics run in 2019, HPS upgraded the front tracker. The 2019 and 2021 configurations added a seventh double layer at $z = 5$ cm and used thin, slim-edge sensors with short split triplets [27, 29]. Moving the first precision measurements closer to the beam reduced material and occupancy while improving acceptance for low-mass, small-opening-angle pairs. The ECal crystal layout was unchanged, but a positron hodoscope was installed before the 2019 run to support a single-arm trigger for events near the ECal beam gap [27]. Figure 5 compares the tracker layouts used in the engineering and physics runs.



(a) Baseline six-layer SVT used in the 2015 and 2016 engineering runs [28].



(b) Front-layer upgrade used in the 2019 and 2021 physics runs, including the added seventh double layer and slim-edge front sensors [27, 29].

Figure 5: SVT configurations used in the prompt-resonance analyses. The 2015/2016 baseline tracker and the 2019/2021 tracker differ in both layer count and the design of the innermost stations. The ECal crystal layout was unchanged; the physics-run configuration added a positron hodoscope.

2.2 Data-taking campaigns and analysis scope

The current analysis uses full 2015 and 2016 engineering-run samples and the released 2021 10% physics-run sample. The same frozen campaign configuration enters both standalone and shared-coupling results. The engineering data share the baseline tracker used in the published prompt searches; the 2021 sample uses the upgraded tracker and positron hodoscope. No 2019 data enter the fits.

Table 1 summarizes the run periods, analyzed samples, search intervals, and GP fit supports for the three-campaign combination. A mass hypothesis is tested only inside the search interval, whereas the GP is trained on the wider support after removing the local $2.25\sigma_m$ exclusion. The 2016 support begins at 30 MeV: with the five-bin rebinning used here, a nominal 35 MeV edge would leave no training-bin center below the exclusion at the 39 MeV search endpoint.

Run period	Beam energy	Sample	Search interval	GP fit support
2015 engineering run	1.056 GeV	1170 nb ⁻¹	19 MeV to 90 MeV	14 MeV to 135 MeV
2016 engineering run	2.3 GeV	10 608 nb ⁻¹	39 MeV to 180 MeV	30 MeV to 210 MeV
2021 physics run	3.74 GeV	Native 10% sample from a ~ 160 pb ⁻¹ parent run	50 MeV to 250 MeV	36 MeV to 300 MeV

Table 1: Data sets and mass domains in the three-campaign combination. The 2015 and 2016 inputs are the full engineering-run samples; the 2021 input is the distributed native 10% sample. The wider GP supports provide training sidebands beyond both ends of each search interval. The corresponding published prompt searches covered 19 MeV to 81 MeV in 2015 and 39 MeV to 179 MeV in 2016 [3, 4].

The 36–300 MeV support was first developed in a separate 2021 study and was subsequently used by the v4.9.12 final-sample combination. The older 40–300 MeV support belongs to the historical combination and is not the configuration of the current result curves.

Integrated luminosity and analyzed fraction do not enter the GP interpolation. They enter later, with the target, acceptance, efficiency, and radiative fraction, when an extracted prompt yield is converted to an equivalent ϵ^2 limit.

The 2015 and 2016 samples are fully unblinded. The 2021 input is distributed across the run rather than drawn from a contiguous block, so it samples varied run conditions while leaving most of the exposure unopened. The ROOT file does not contain enough exposure metadata to infer an integrated luminosity; “10%” describes how the sample was constructed. The three-campaign result combines the full 2015 and 2016 samples with this 2021 sample and is not the final result for the 2021 run.

Figure 6 compares the unit-normalized invariant-mass shapes. Figure 7 shows the same spectra in events per MeV and marks the fit supports and search intervals used in the three-campaign combination.

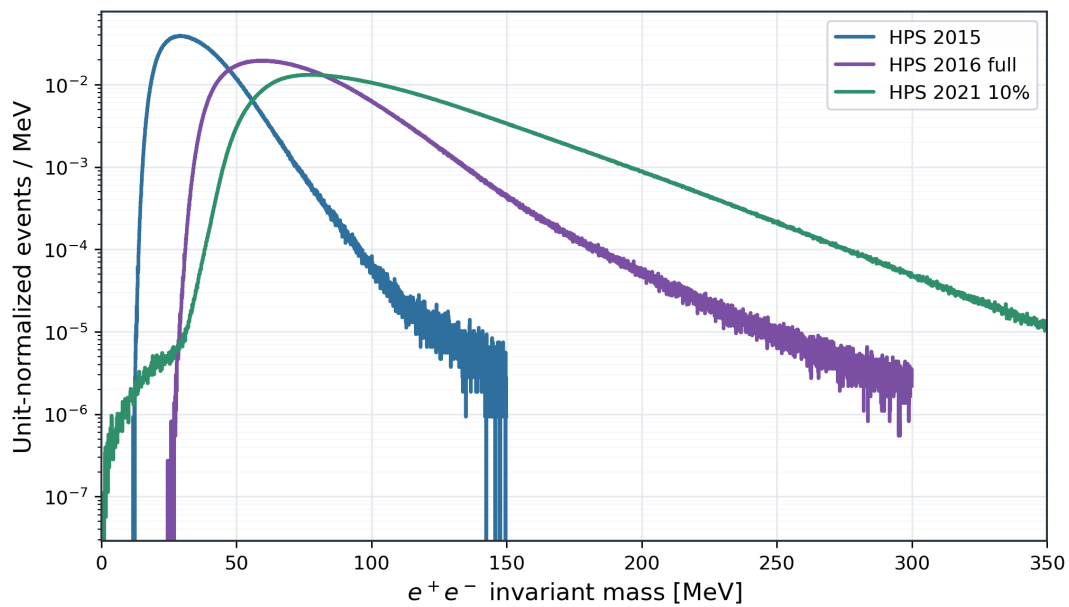


Figure 6: Unit-normalized invariant-mass shapes for the full 2015 and 2016 samples and the native 2021 10% sample. Each histogram is divided by its event count and bin width, so the comparison shows spectral shape rather than sample size.

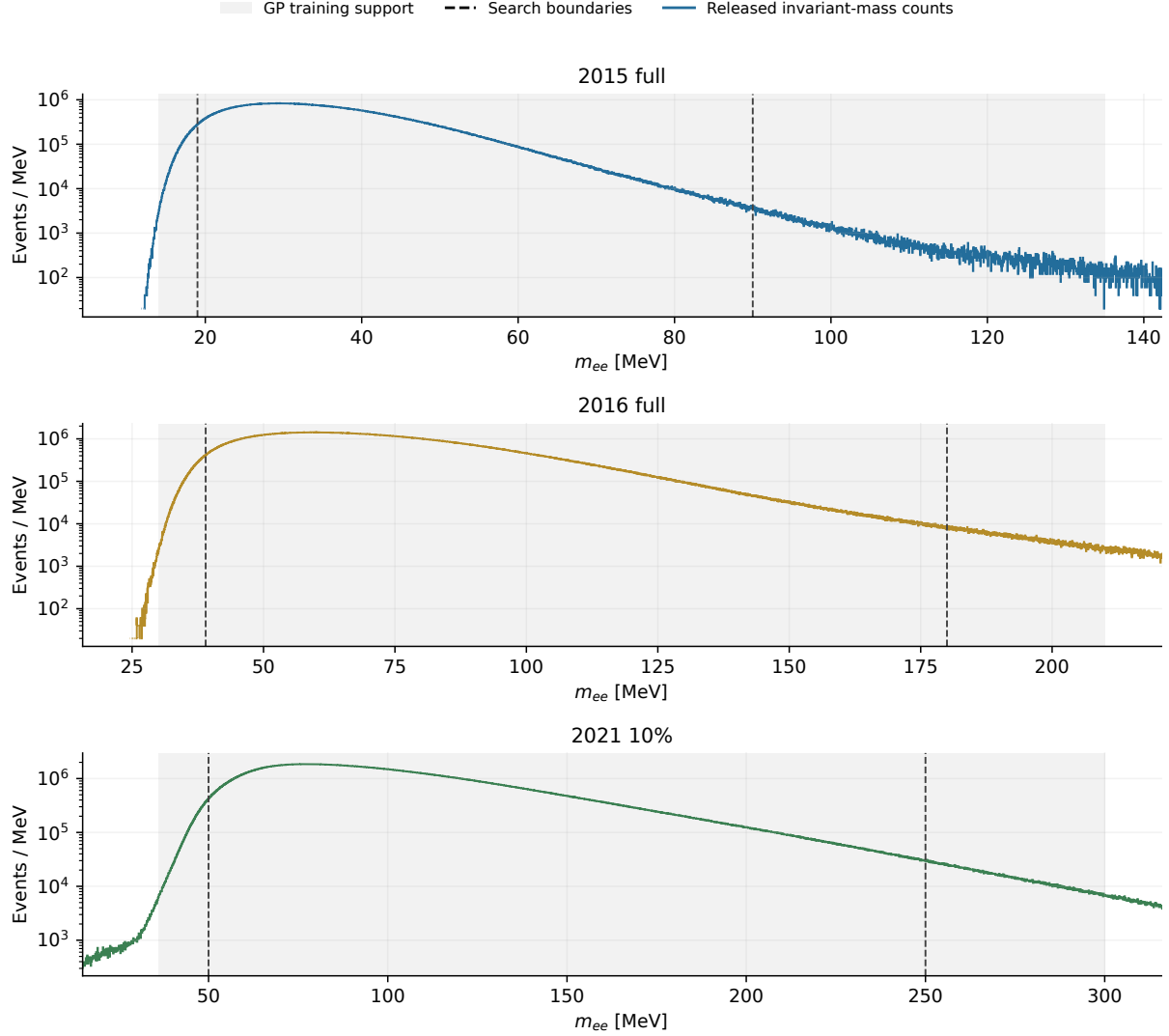


Figure 7: Invariant-mass distributions from the hash-verified released inputs, shown in events per MeV at the native input widths of 0.05, 0.05 and 0.125 MeV. Gray shading marks the current GP supports, including 36–300 MeV for 2021, and dashed lines bound the narrower interval in which signal hypotheses are tested. The mass-dependent $2.25\sigma_m$ exclusion is removed from the GP support separately at every tested mass.

Although the three campaigns used different detector configurations and exposures, they enter the search through the same observable: a selected prompt e^+e^- invariant-mass spectrum. The next section explains the selections and the two inputs that connect a fitted resonance yield to the detector—mass resolution and radiative fraction.

3 Event selection, mass resolution, and radiative fraction

Each HPS input is an invariant-mass spectrum formed after a run-specific event selection. That selection determines the composition of the prompt e^+e^- sample; the detector response sets the width of a possible resonance, and the radiative fraction sets its normalization. Together, these ingredients connect the recorded events to the signal model used in the search.

3.1 Prompt event selection

Across the three HPS campaigns, the prompt selection follows the same experimental logic: oppositely charged tracks matched to ECal clusters, tight cluster-time coincidence, track-quality and vertex-reconstruction requirements, and momentum cuts that retain prompt radiative tridents while suppressing accidentals and converted wide-angle bremsstrahlung. The 2015 and 2016 selections are inherited from the published prompt searches. For 2021 we use the prompt target-constrained (prompt-TC) sample summarized in Table 2. It requires $p_{\text{sum}} > 2.8 \text{ GeV}$ and applies no additional vertex-fit χ^2 requirement. The stored hit count is three-dimensional and does not document a separate eight-hit two-dimensional electron requirement, so we do not include one here.

In Table 2, V_0 denotes a reconstructed e^+e^- vertex, GBL is the track-fit algorithm, FEE denotes the full-energy-electron veto, and TC means target constrained. Pair1, Single-2, and Single-3 are the trigger names used during the corresponding data-taking periods.

Table 2: Comparison of the prompt-resonance selections. The 2015 and 2016 columns follow the published and internal engineering-run summaries. For 2021, the preselection entries follow the full analysis note [30, Sec. 4.8 and App. A]; the prompt entries list the additional requirements documented for the native 10% prompt-TC sample.

Cut type	2015 (Eng. run 1)	2016 (Eng. run 2)	2021 (native 10%)
Trigger / topology	Pre: pairs-1 trigger; ECal halves; $y_{e^+}^{\text{clu}} y_{e^-}^{\text{clu}} < 0$. Prompt: same.	Pre: Pair1 trigger bit; opposite ECal halves; the trigger does not encode track charge. Prompt: same.	Pre: Single-2 or Single-3 positron single-cluster trigger; Single-3 also requires the matched hodoscope condition used by the 2021 trigger menu. Prompt: no additional prompt-TC requirement.
Track momenta	Pre: $p(e^-) < 0.75 E_{\text{beam}}$. Prompt: $p_{\text{sum}} > 0.8 E_{\text{beam}}$ and $p_{\text{sum}} < 1.2 E_{\text{beam}}$.	Pre: $p(e^\pm) < 2.15 \text{ GeV}$; $p_{\text{sum}} < 1.2 E_{\text{beam}}$. Prompt: $p_{\text{sum}} > 1.90 \text{ GeV}$ and $p_{\text{sum}} < 2.4 \text{ GeV}$.	Pre: $p_{\text{trk}} > 0.4 \text{ GeV}$ for each track; FEE veto $p(e^-) < 2.9 \text{ GeV}$; vertex momentum $p_{\text{vtx}} < 4.0 \text{ GeV}$. Prompt: the prompt-TC sample requires $p_{\text{sum}} > 2.8 \text{ GeV}$.
Track fit / hits	Pre: loose track-cluster match: $\chi_{\text{trk-clu}}^2 < 10$. Prompt: GBL track $\chi_{\text{trk}}^2 < 40$.	Pre: $\chi_{\text{trk}}^2 < 12$; PID goodness < 10 . Prompt: no additional prompt requirement is documented.	Pre: $\chi_{\text{trk}}^2/\text{ndf} < 20$ and $N_{2\text{D hits}} \geq 10$ for the tracks. Prompt: no additional 2D-hit requirement is established; the available field counts 3D hits.

Cut type	2015 (Eng. run 1)	2016 (Eng. run 2)	2021 (native 10%)
Track-cluster timing	Pre: — Prompt: $ t_{\text{clu}} - t_{\text{trk}} - 43 \text{ ns} < 4.5 \text{ ns}$.	Pre: $ t_{\text{trk}} - t_{\text{clu}} < 6 \text{ ns}$. Prompt: no additional value stated.	Pre: full-note data timing windows: $\Delta(t_{e^-, \text{trk}}, t_{e^+, \text{clu}}) < 6.0 \text{ ns}$, $\Delta(t_{e^+, \text{trk}}, t_{e^+, \text{clu}}) < 5.1 \text{ ns}$, and $\Delta(t_{e^-, \text{trk}}, t_{e^+, \text{trk}}) < 7.8 \text{ ns}$. Prompt: no additional prompt-TC timing requirement.
Cluster-cluster coincidence	Pre: — Prompt: $ \Delta t_{\text{clu-clu}} < 2 \text{ ns}$.	Pre: $ \Delta t_{\text{clu-clu}} < 2 \text{ ns}$. Prompt: $ \Delta t_{\text{clu-clu}} < 1.43 \text{ ns}$.	Pre: no separate offline cluster-cluster coincidence appears in the full preselection table; the Pair-0 hardware trigger has a cluster-pair timing window $ \Delta t < 12 \text{ ns}$ and VTP prescale 100. Prompt: no additional prompt-TC offline coincidence requirement.
Vertex / V_0 quality	Pre: if multiple V_0 remain, choose the best χ^2_{vtx} . Prompt: $\chi^2_{\text{vtx}} < 75$ and $p(V_0) < 1.2 E_{\text{beam}}$.	Pre: target-constrained V_0 candidates. Prompt: no additional prompt χ^2_{vtx} requirement is documented.	Pre: full-note upgraded-detector values: $p_{\text{vtx}} < 4.0 \text{ GeV}$ and $\chi^2_{\text{vtx}} < 20$. Prompt: no additional prompt-TC vertex-fit χ^2 requirement.
Positron cluster energy	Pre: — Prompt: —	Pre: — Prompt: —	Pre: full-note sanity cut $E_{e^+}^{\text{clu}} > 0.2 \text{ GeV}$. Prompt: no additional prompt-TC positron-cluster threshold.
cWAB suppression	Pre: — Prompt: e^+ must have L1 and L2 hits; $d_0(e^+) < 1.1 \text{ mm}$; and $(p_T^- - p_T^+) / (p_T^- + p_T^+) < 0.47$.	Pre: — Prompt: not applied in the final 2016 prompt selection summary.	Pre: no explicit cWAB-suppression cut is listed in the full 2021 preselection table. Prompt: no additional prompt-TC cWAB requirement.
Candidate multiplicity	Pre: unique- V_0 policy: keep the best χ^2_{vtx} . Prompt: enforced by the unique- V_0 logic.	Pre: — Prompt: require a single surviving V_0 candidate.	Pre: exactly one selected good vertex: $N_{\text{vtx}} > 0$ and $N_{\text{vtx}} < 2$. Prompt: no additional prompt-TC multiplicity requirement.

The 2021 preselection entries follow the full analysis note [30, Sec. 4.8 and App. A]. The native 10% spectrum comes from the prompt-TC sample. The GPR fit adds no event-level cuts. Each campaign enters the analysis through its selected invariant-mass histogram and the detector-response and normalization inputs for the same event category.

The engineering-run notes use *preselection* for the broad reconstruction sample and *prompt*

selection for the tighter resonance-search sample. The 2021 input is closer to the latter because the prompt-TC sample retains the high- p_{sum} side. Table 2 keeps that distinction without adding an unrecorded two-dimensional hit requirement.

3.2 Kinematic distributions after the 2021 preselection

The five groups in Figures 8–12 show the reconstructed quantities used to inspect the 2021 loose preselection. Figure 8 displays the mass spectrum and vertex-fit quality; Figures 9 and 10 show the electron and positron momenta and impact parameters. The two-dimensional displays in Figures 11 and 12 make their mass dependence visible and distinguish broad kinematic structure from the narrow simulated mass hypotheses. The signal samples are scaled for display, so these figures compare reconstruction shapes rather than absolute yields or efficiencies. The vertex-fit χ^2 remains a diagnostic: no additional vertex-fit χ^2 requirement is applied to the regenerated 2021 GPR input. These archived plots document the preselection context and do not constitute a new event selection or a validation of the final likelihood.

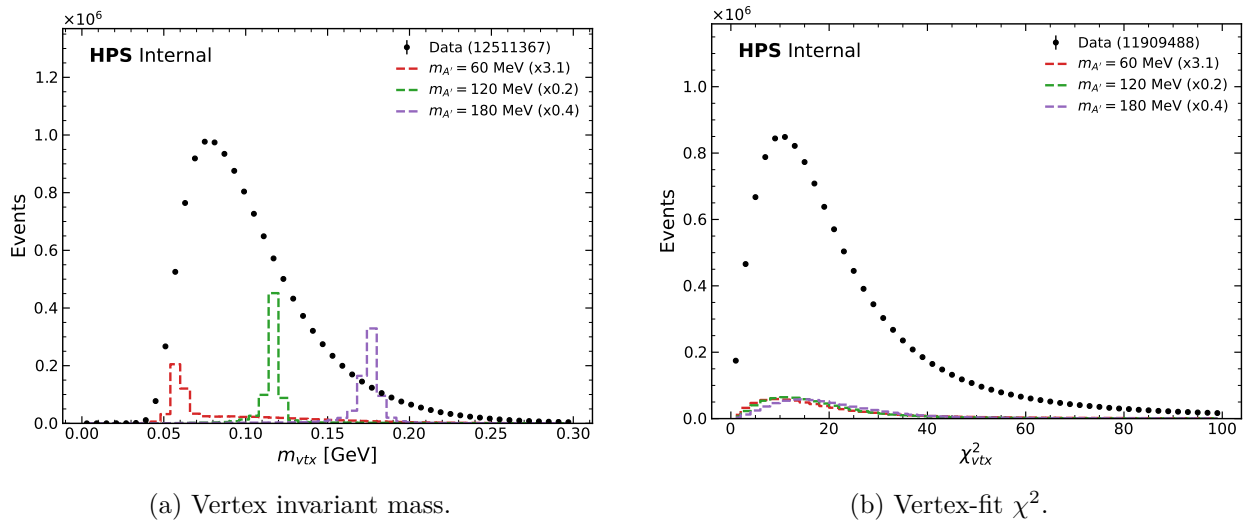


Figure 8: Reconstructed-vertex distributions after the loose preselection, comparing data (points) with the signal MC (filled histogram). The vertex-fit χ^2 is shown as a diagnostic; no vertex-fit χ^2 requirement is applied to the regenerated 2021 GPR input.

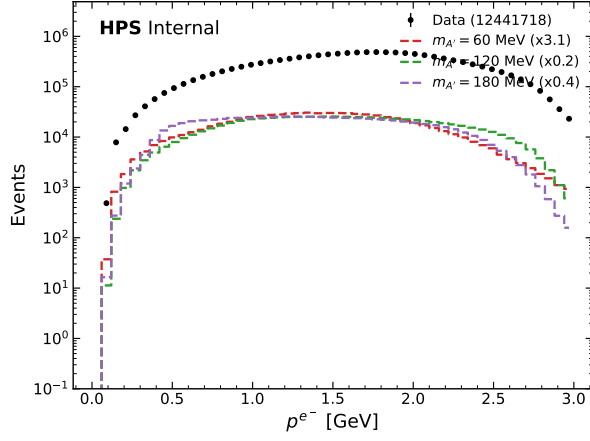
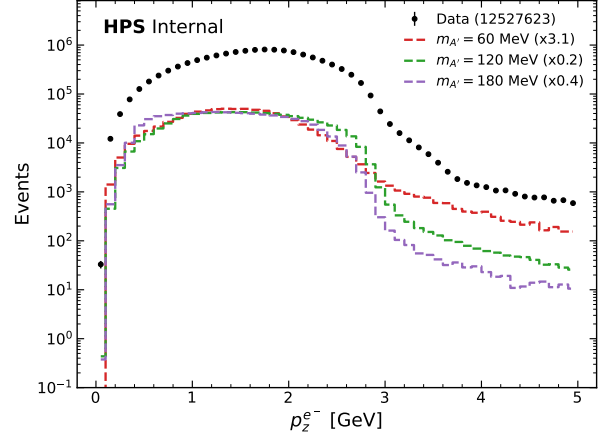
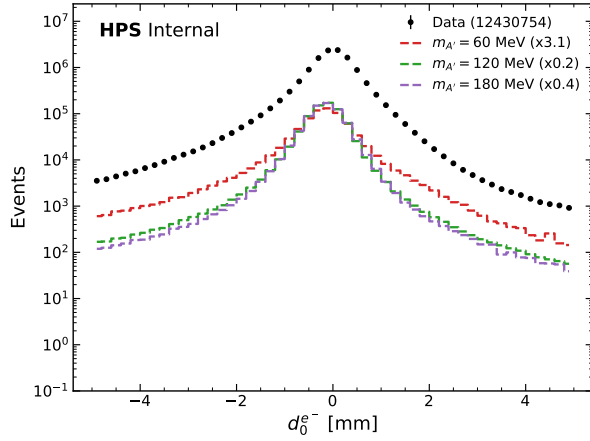
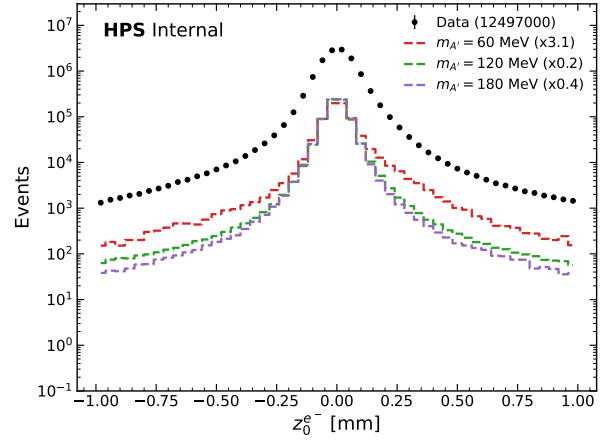
(a) Electron momentum p .(b) Electron longitudinal momentum p_z .(c) Electron transverse impact parameter d_0 .(d) Electron longitudinal impact parameter z_0 .

Figure 9: Electron-track kinematic distributions after the loose preselection, comparing data (points) with the signal MC (filled histogram).

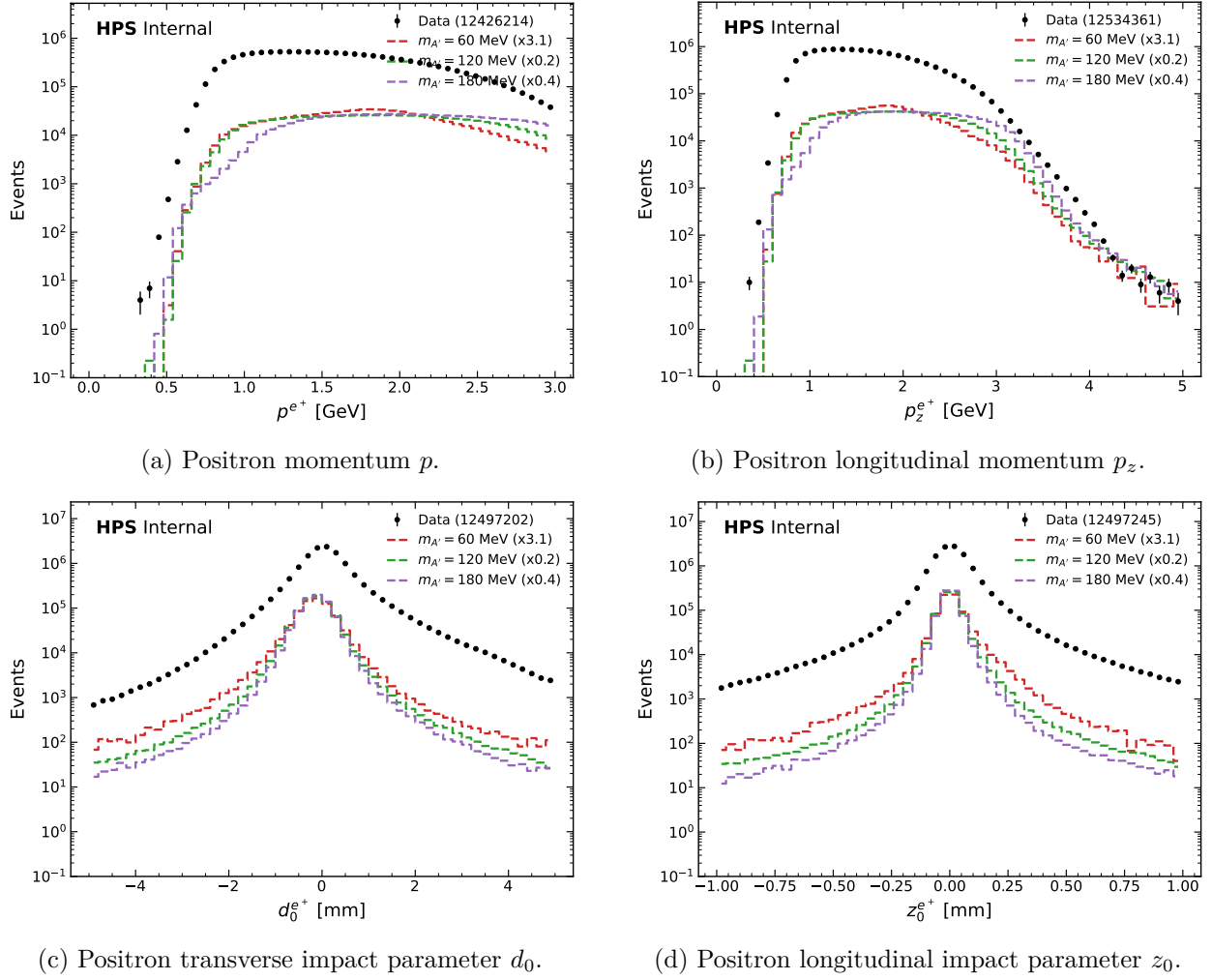


Figure 10: Positron-track kinematic distributions after the loose preselection, comparing data (points) with the signal MC (filled histogram).

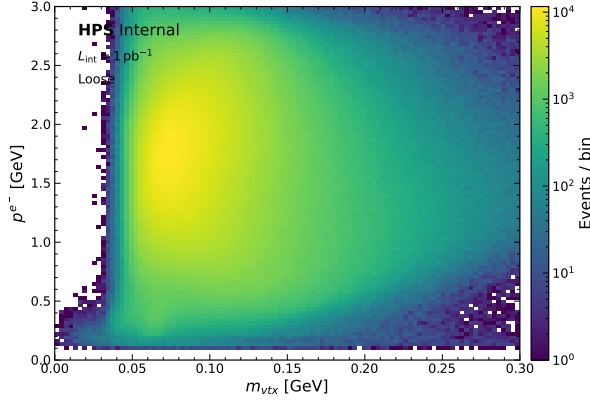
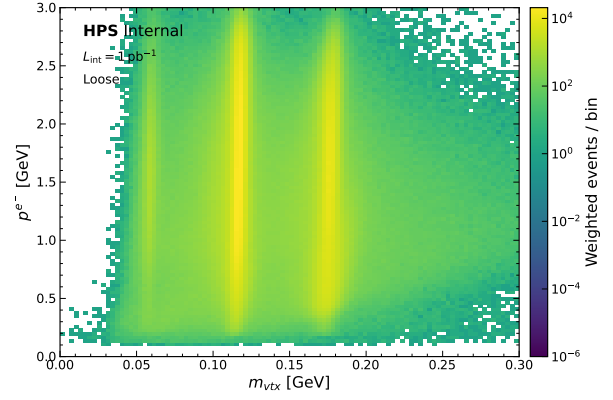
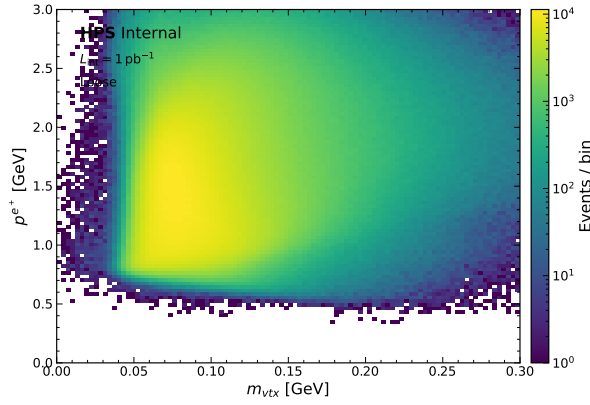
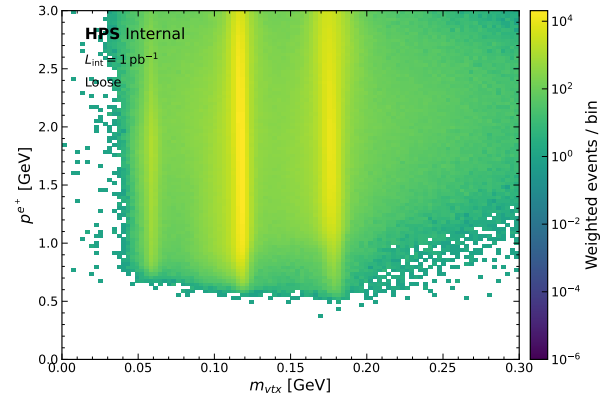
(a) Mass vs. electron p (data).(b) Mass vs. electron p (signal).(c) Mass vs. positron p (data).(d) Mass vs. positron p (signal).

Figure 11: Reconstructed invariant mass versus momentum observables, for data (left) and signal MC (right), after the loose preselection.

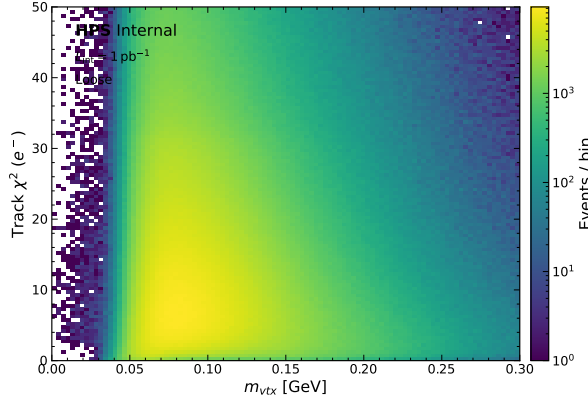
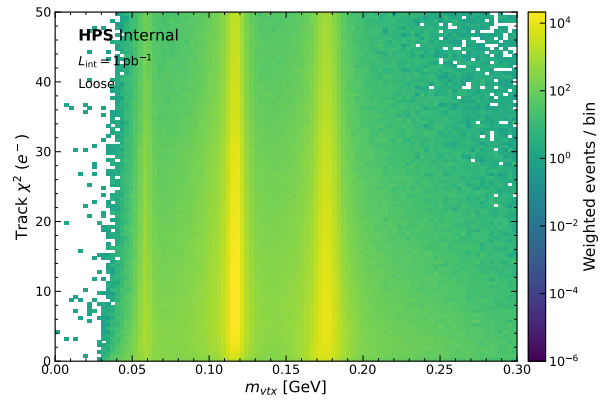
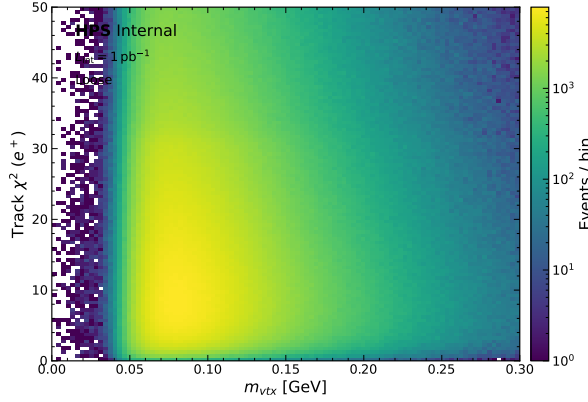
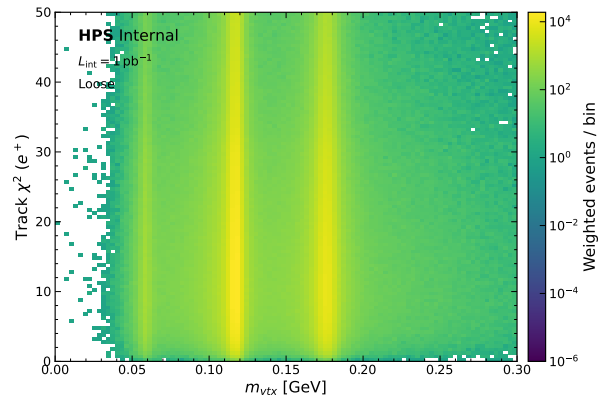
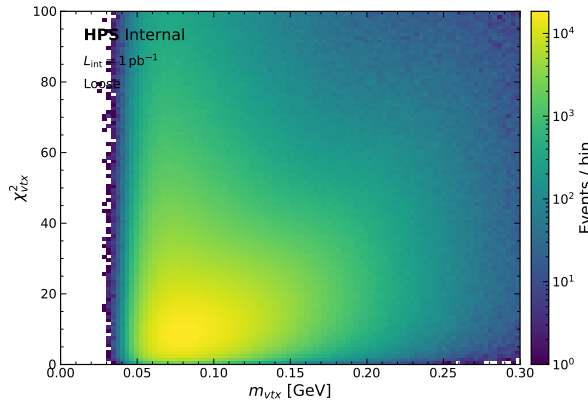
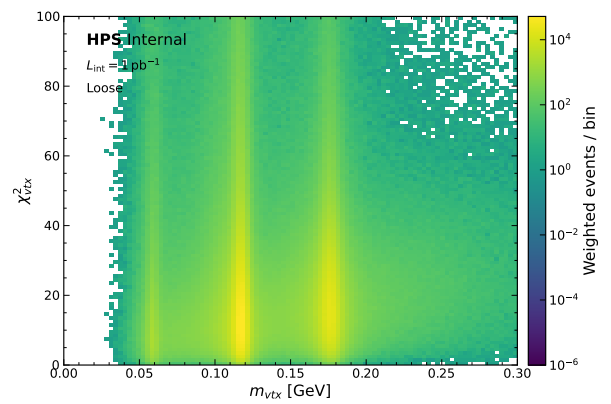
(a) Mass vs. electron χ^2 (data).(b) Mass vs. electron χ^2 (signal).(c) Mass vs. positron χ^2 (data).(d) Mass vs. positron χ^2 (signal).(e) Mass vs. vertex χ^2 (data).(f) Mass vs. vertex χ^2 (signal).

Figure 12: Reconstructed invariant mass versus track- and vertex-fit quality, for data (left) and signal MC (right), after the loose preselection.

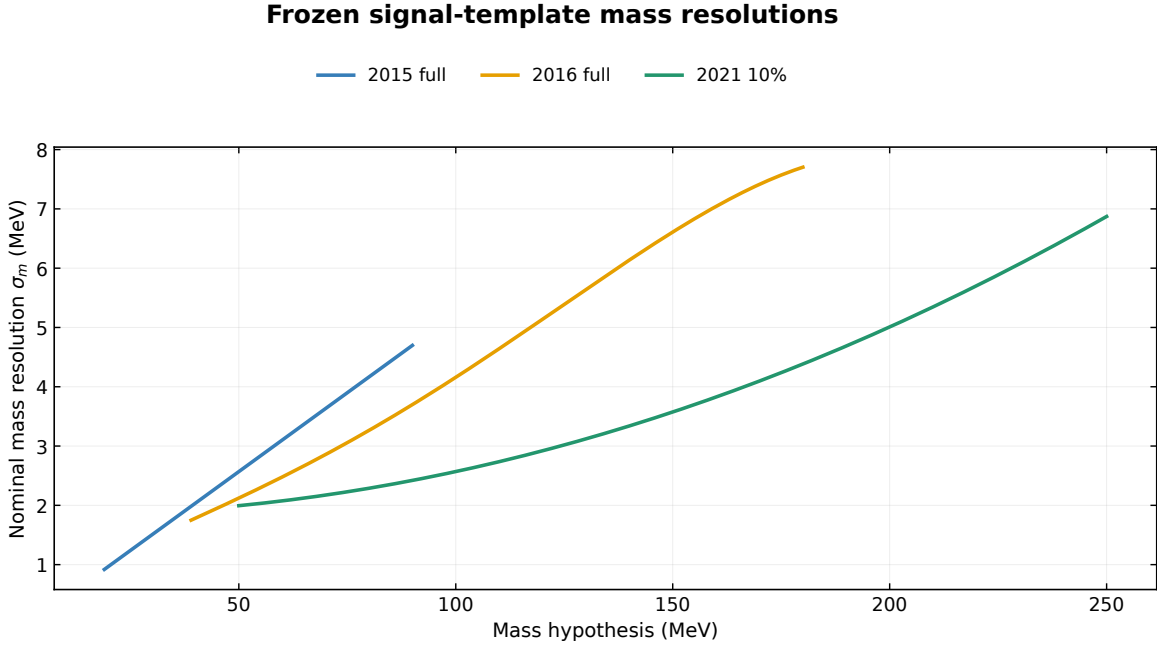


Figure 13: Frozen nominal mass-resolution parameterizations over the current search intervals. These inputs set the signal-template widths, moving fit windows and resolution-scaled kernel constraints. Their signal-width variations are examined in Section 6.8.

3.3 2015 mass resolution

The 2015 signal width is calibrated with Møller data and propagated across mass with simulated A' samples [31]. The Møller peak fixes the detector resolution at a known mass, and the data–MC momentum-resolution difference determines the additional smearing applied to the simulation. Rescaling the fitted A' widths to that data anchor gives

$$\sigma_{2015}(m) = -9.22 \times 10^{-5} + 5.322 \times 10^{-2} m, \quad (3)$$

with m and σ_{2015} expressed in GeV.

The Møller measurement supplies the absolute response, while the simulated mass grid supplies its mass dependence. The associated uncertainty therefore reflects the Møller calibration, the target position, and the residual momentum-response mismodelling [3, 31]. At each test mass this curve sets both the Gaussian signal width and the width of the excluded training interval.

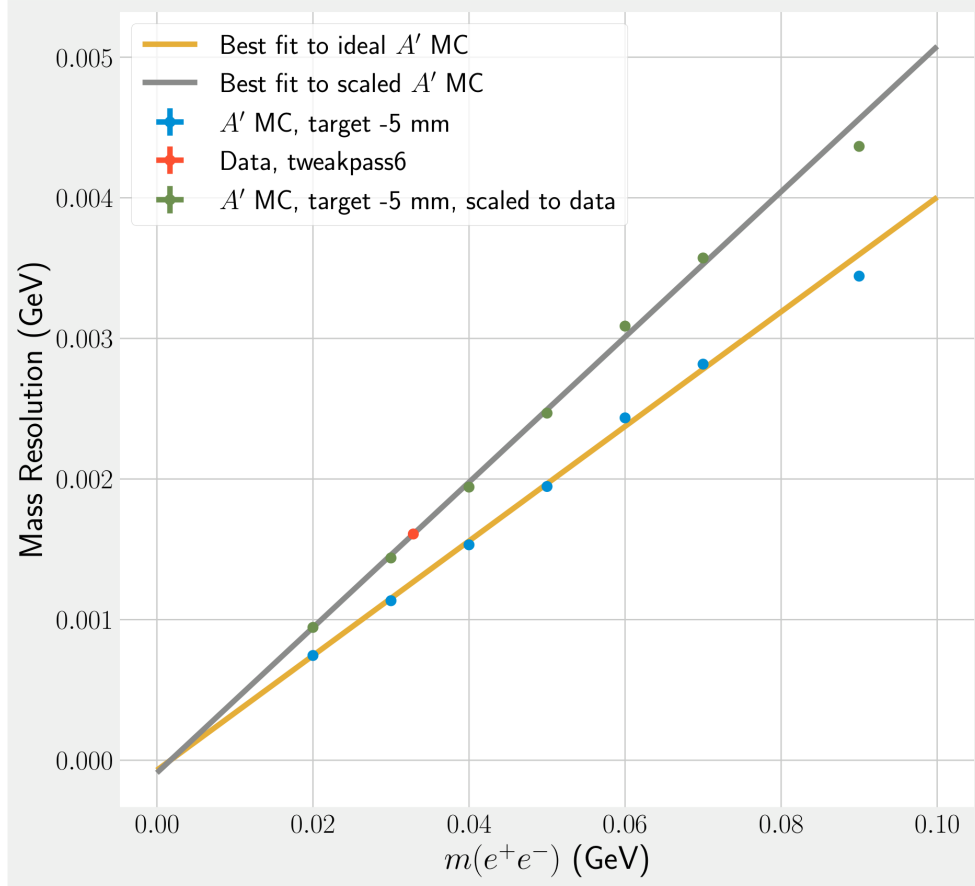


Figure 14: 2015 mass-resolution parameterization from scaled A' MC and Møller data [31]. The Møller point anchors the absolute detector response in data, while the scaled A' simulation supplies the mass dependence across the search window. Because the prompt dark-photon width is negligible, this curve directly sets the resonance width assumed in the search and therefore the local background entering the limit.

3.4 2016 mass resolution

The 2016 calibration follows the same strategy over a broader mass range [28, 4]. Møller events fix the absolute response at 48.5 MeV; full-energy-electron momentum peaks determine the extra smearing needed to match data; and smeared A' samples carry that calibration across the search range. A fourth-order polynomial fit to those samples defines the 2016 signal model,

$$\begin{aligned} \sigma_{2016}(m) = & 3.8 \times 10^{-4} + 4.1 \times 10^{-2}m - 2.7 \times 10^{-1}m^2 \\ & + 3.49m^3 - 11.11m^4, \quad m \leq 0.18 \text{ GeV}, \end{aligned} \quad (4)$$

Earlier extended-range studies continued the polynomial linearly above $m_0 = 0.18 \text{ GeV}$ with slope 0.0239 to avoid its high-mass turnover. The present 2016 scan ends at 180 MeV, close to the published 39 MeV to 179 MeV interval, so that continuation is not used. The 2015 scan similarly

remains near its published 19 MeV to 81 MeV interval, with only the upper endpoint extended to 90 MeV.

The smearing correction reconciles the Møller and simulated A' widths with the observed tracking response [28, 4]. Because the resolution worsens with mass, both the signal template and the excluded training window broaden, increasing the prompt background beneath a narrow signal hypothesis.

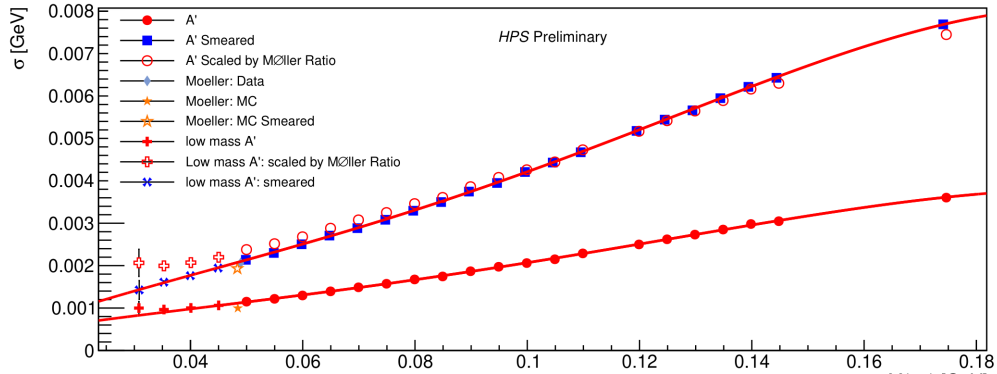


Figure 15: 2016 smeared- A' and Møller-based mass-resolution summary [28]. The smeared A' points show how the data-driven momentum-smearing correction is propagated away from the single Møller calibration point. The worsening resolution with mass broadens the signal window and is one of the main detector effects limiting prompt sensitivity at higher mass.

3.5 2021 mass resolution

The upgraded 2019/2021 tracker requires a separate resolution parameterization. We use the inclusive target-constrained V0 fit without hit requirements shown in Figure 16. Before the final data-MC broadening, the red curve is

$$\sigma_{2021}^{\text{TC,raw}}(m_{\text{MeV}}) = \left(1.4786 - 1.1 \times 10^{-3} m_{\text{MeV}} + 6.87 \times 10^{-5} m_{\text{MeV}}^2\right) \text{ MeV}, \quad (5)$$

where m_{MeV} is the numerical mass value in MeV. The analysis then applies the target-constrained data/MC scale derived from the 2021 smearing study,

$$s_{\text{TC}} = \frac{2.186}{1.745} \simeq 1.25, \quad \sigma_{2021}^{\text{TC,smeared}}(m) = s_{\text{TC}} \sigma_{2021}^{\text{TC,raw}}(m). \quad (6)$$

The scale comes from the target-constrained row after hit and curvature (Ω) smearing are applied to the simulation. Equation 6 then sets the 2021 signal width and the corresponding excluded training interval at each mass.

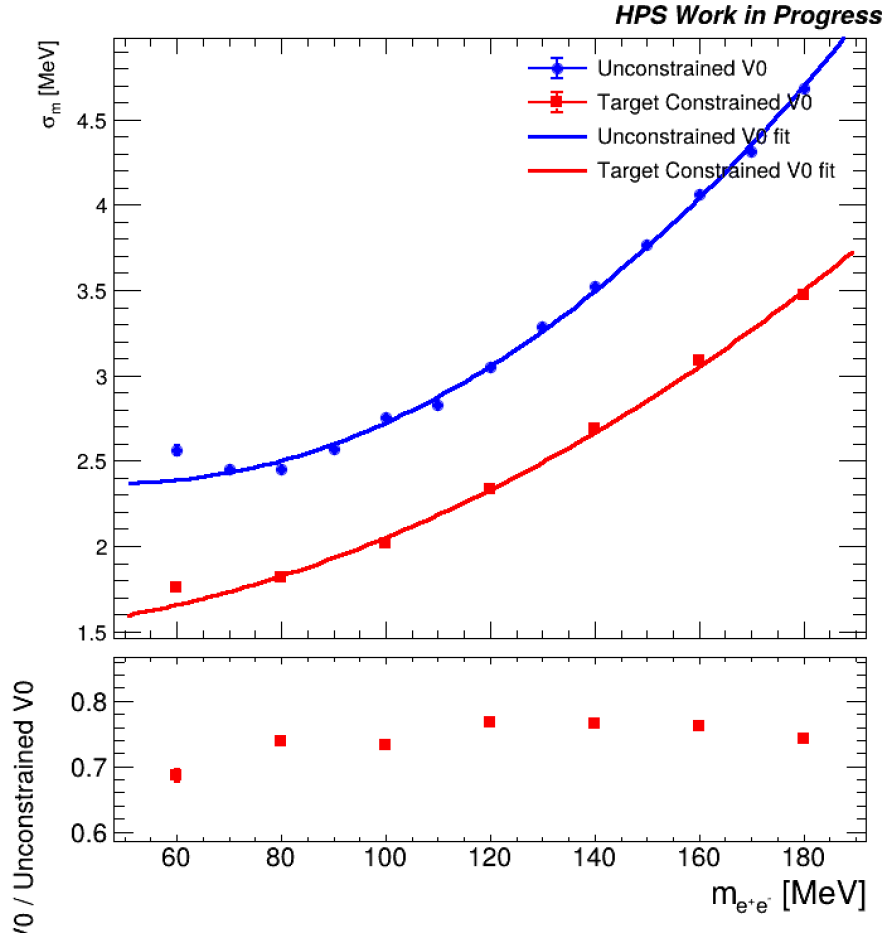


Figure 16: 2021 mass-resolution summary used for the prompt analysis. The red curve is the inclusive target-constrained V0 quadratic fit without hit requirements before the TC data/MC broadening in Eq. 6. The blue curve shows the corresponding unconstrained V0 reference fit included in the underlying study for comparison.

Vertex Type	Hit Smear Only (MeV)	Omega + Hit Smear (MeV)	Data (MeV)
UC	2.329 ± 0.013	2.463 ± 0.014	2.946 ± 0.036
BSC	2.092 ± 0.012	2.193 ± 0.012	2.569 ± 0.035
TC	1.590 ± 0.009	1.745 ± 0.010	2.186 ± 0.031

Figure 17: 2021 Møller run data/MC mass-resolution comparison used to set the smearing scale. Here UC denotes the unconstrained vertex sample, BSC denotes the beamspot-constrained vertex sample, and TC denotes the target-constrained vertex sample. The TC row is the most relevant row for the prompt analysis; comparing the data value 2.186 MeV to the Ω +hit-smear Monte Carlo value 1.745 MeV gives the applied scale factor $2.186/1.745 \simeq 1.25$.

Figure 18 compares the three resolution curves used to construct the signal templates and excluded training windows over their respective scan ranges.

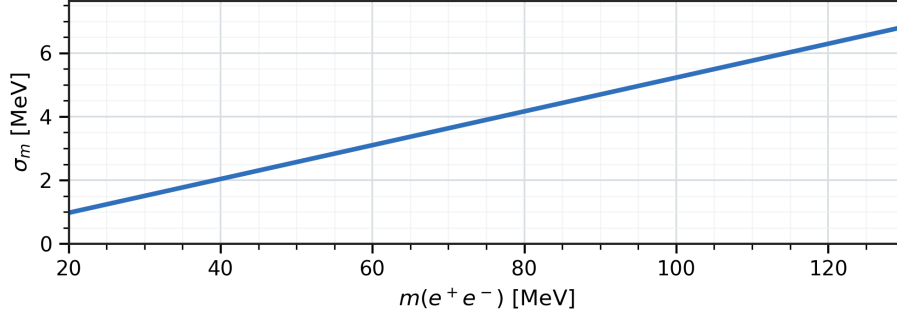
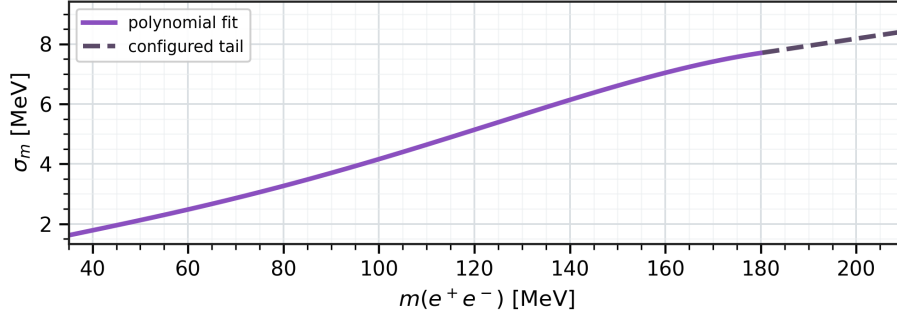
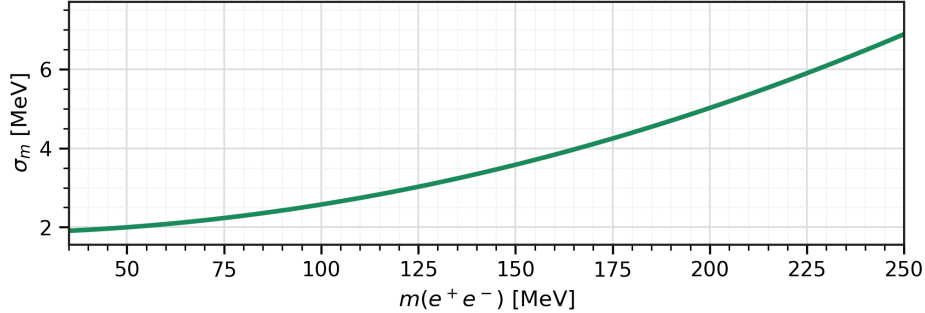
Mass-resolution parameterizations over current GPR scan ranges**2015 scaled A-prime MC parameterization****2016 smeared A-prime MC parameterization****2021 TC-smeared V0 parameterization**

Figure 18: Mass-resolution parameterizations used by the analysis over the adopted scan ranges. The 2016 scan ends at the polynomial boundary of 180 MeV, so its configured high-mass continuation is not used. The 2021 panel shows the target-constrained V0 parameterization after multiplication by the TC smearing scale $s_{\text{TC}} = 2.186/1.745 \simeq 1.25$ and extended to 250 MeV.

The effect of varying this nominal signal width is examined in Section 6.8. That comparison changes the extraction template while retaining the archived GP prediction; it is a model-dependence study and does not refit the detector resolution from the observed candidates.

3.6 Radiative fraction and the ε^2 normalization

The radiative fraction encodes the fraction of the selected prompt background arising from the purely radiative trident process,

$$f_{\text{rad},d}(m) = \frac{dN_{\gamma^*,d}/dm}{dN_{\text{tri},d}/dm + dN_{\text{wab},d}/dm}, \quad (7)$$

where the numerator is the selected radiative-trident component and the denominator is the full selected prompt background for dataset d [3, 4]. Because heavy-photon production is normalized to the radiative trident rate, f_{rad} is the physics quantity that converts a fitted prompt signal yield into a limit on the coupling. With this notation,

$$K_d(m) \equiv A_d(\varepsilon^2 = 1) = \frac{3\pi m f_{\text{rad},d}(m) \rho_d(m)}{2\alpha_{\text{EM}}}, \quad A_d = K_d(m) \varepsilon^2, \quad (8)$$

where $\rho_d(m)$ is the selected prompt-spectrum density in dataset d . The engineering-run inputs follow the corresponding prompt searches, while the 2021 normalization remains provisional for the native 10% sample.

3.6.1 2015 radiative fraction

The 2015 prompt search used a constant approximation,

$$f_{\text{rad},2015}(m) = 0.085, \quad (9)$$

which the analysis retains as its 2015 baseline. The dedicated 2015 radiative-fraction study compares the selected radiative-trident component with the total prompt background and adopts the red horizontal line at 8.5% as a conservative description across the search interval. The constant comes from the internal HPS normalization study rather than from the GP fit and is retained for consistency with the published 2015 analysis.

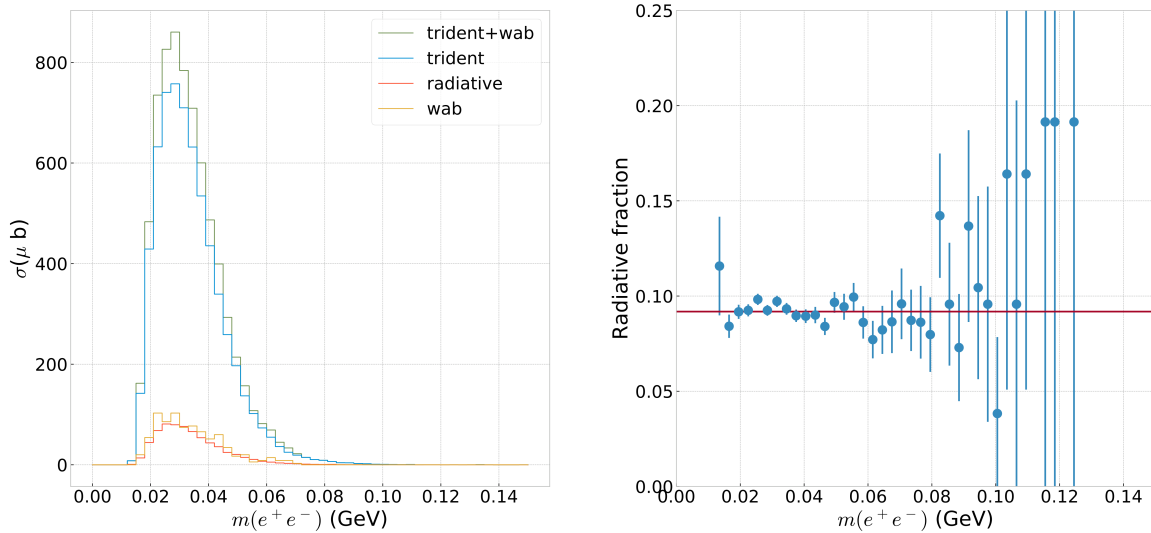


Figure 19: Dedicated 2015 radiative-fraction inputs. The left panel shows the selected cross-section composition, while the right panel shows the radiative fraction used for normalization. The red horizontal line denotes the 8.5% constant adopted as the conservative fit across the 2015 prompt-search dataset. This quantity converts the measured prompt spectrum into the expected A' normalization, so it affects the final ε^2 interpretation rather than the local spectral fit itself.

3.6.2 2016 radiative fraction

The 2016 prompt search used a fifth-order polynomial derived from radiative, trident, and converted-WAB MC after the full prompt selection. Those components determine the mass-dependent con-

version from fitted signal yield to ε^2 over the 2016 search range. For the 2016 input, we instead use the constant approximation

$$f_{\text{rad},2016}(m) = 0.05, \quad (10)$$

which is numerically stable over the wider scan range. Figure 20 shows the mass-dependent simulation used in the published determination; the constant used here is a working numerical approximation to that richer model.

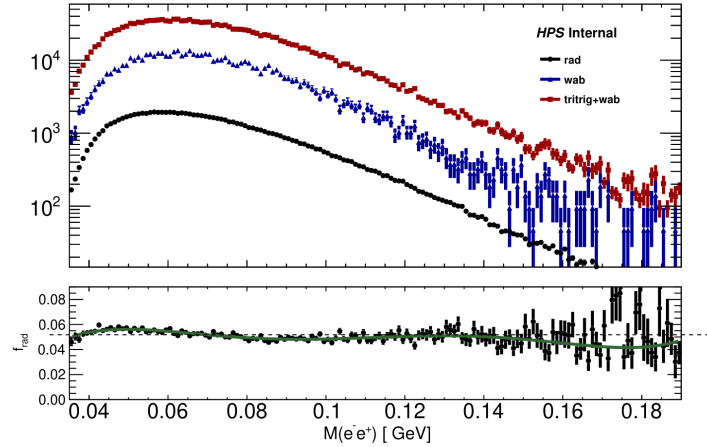


Figure 30: The differential rates in units of MeV^{-1} for all the MC samples which go into the radiative fraction. The rad component is a function of the true mass of the generator level gamma* for the event. The WAB and tritrig components are a function of the invariant mass of the selected V0 candidate. The bottom panel shows the radiative fraction, the green line is the fifth order polynomial fit, and the dotted line is the zeroth order polynomial fit.

Figure 20: 2016 radiative-fraction study from radiative, trident, and converted-WAB MC [28]. The simulation gives an intrinsically mass-dependent normalization, although this analysis uses the constant $f_{\text{rad},2016} = 0.05$ for numerical stability.

3.6.3 2021 radiative fraction

For the native 2021 10% sample, we use the provisional value

$$f_{\text{rad},2021}(m) = 0.05. \quad (11)$$

The 2021 full analysis note constructs the ratio from selected radiative-trident, tritrig, and WAB samples after preselection and a high-momentum requirement quoted as $p_{\text{vtx}} > 2.8$ GeV. A fit over $0.05 < m < 0.25$ GeV gives the constant average $f_{\text{rad}}^{\text{const}} = 0.0496$ [30, Sec. 6.1, Figs. 92–93, Eqs. 25–26]. Thus 0.05 is a reasonable approximation over most of the 2021 search range. Before it is used for a final 2021 result, this normalization must be checked directly against the prompt-TC p_{sum} selection and its threshold convention; the mass-dependent parameterization provides a useful cross-check.

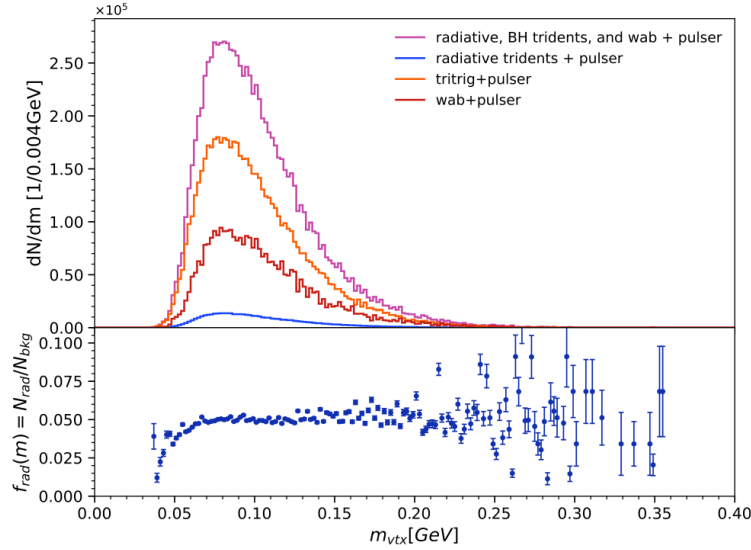


Figure 92: Invariant mass distributions of background samples. The lower panel shows the ratio of radiative tridents to the combined tritrig and WAB samples. This is referred to as the radiative fraction.

Figure 21: 2021 selected-mass distributions used to form the radiative fraction in the full-analysis note [30]. The upper panel shows the selected radiative, tritrig, and WAB components, while the lower panel gives $f_{\text{rad}}(m) = N_{\text{rad}}/N_{\text{bkg}}$.

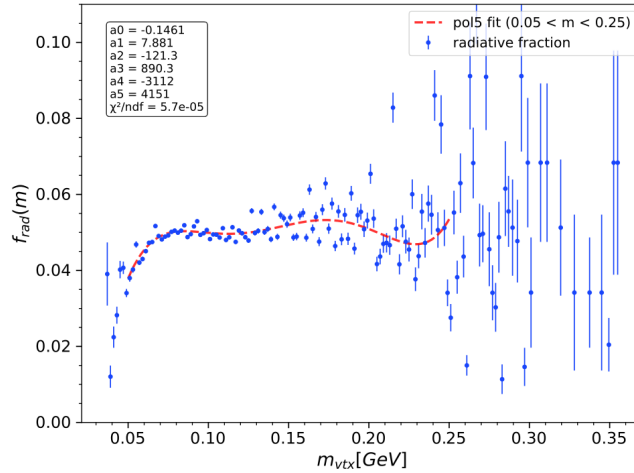


Figure 93: 5th-order polynomial fit to the radiative fraction. The fit range and results are indicated in the figure.

Figure 22: Fifth-order 2021 full-analysis-note fit to the binned radiative fraction [30]. The fitted curve is close to the constant value 0.05 across most of the range used by the 2021 analysis.

3.7 Systematic penalties from the radiative fraction

The analysis applies a data-set-dependent radiative-fraction penalty,

$$f_{\text{rad},d}^{\text{eff}}(m) = (1 - \delta_{f,d}) f_{\text{rad},d}(m). \quad (12)$$

The penalty enters only when signal yield is converted to ε^2 . It changes the normalization of the inferred coupling without altering the blind-window likelihood.

The nominal configuration uses

$$(\delta_{f,2015}, \delta_{f,2016}, \delta_{f,2021}) = (0.070, 0.070, 0.046). \quad (13)$$

For a fixed signal-yield limit, these shifts weaken the coupling-squared limit by factors $1/0.93$ for 2015 and 2016 and $1/0.954$ for 2021. In the combined fit each $K_d(m)$ is reduced by its own amount, so a larger shared ε^2 is required to produce the same yields. The factors are fixed, one-sided normalization shifts; they do not enter the GP interpolation and are not profiled as nuisance parameters.

The 7% engineering-run penalties reflect the several-percent uncertainty in the 2015 radiative and total e^+e^- composition and the corresponding 2016 uncertainty from the selected radiative, trident, and converted-WAB inputs [31, 4]. We retain 7% as a conservative first-order treatment for both engineering runs. The 2021 term uses the upgraded-detector estimate below.

The 2021 full analysis note assigns 3.5% from the RAD and WAB cross sections shown in Fig. 23, using their quoted widths and the relative normalization $R = 0.244$ [30, Sec. 9.1, Fig. 161]. It assigns a further 3% for preselection and radiative-fraction modeling, motivated by WAB-selective mismodeling in the selected background [30, Sec. 9.2]. Treating the two terms as independent gives

$$\delta_{f,2021}^{\text{note}} = \sqrt{(0.035)^2 + (0.030)^2} = 0.046, \quad f_{\text{rad},2021}^{\text{eff}} = 0.954 f_{\text{rad},2021}. \quad (14)$$

The 2015 and 2016 penalties remain at 7%.

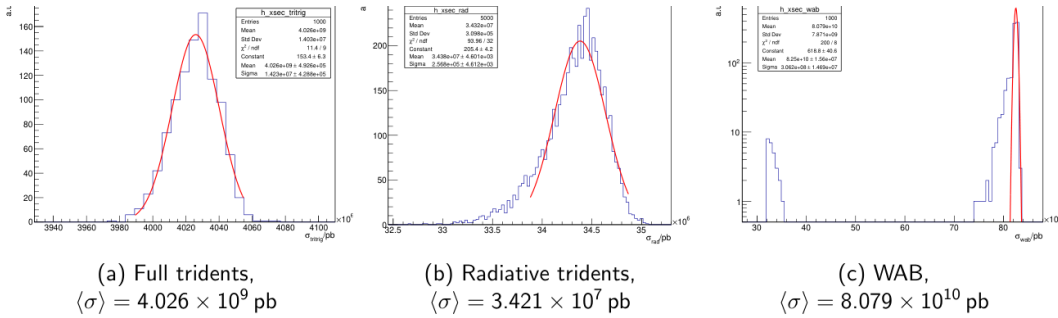


Figure 16: Distribution of generated cross sections for background MC samples.

Figure 23: Generated cross-section distributions for the 2021 tritrig, radiative, and WAB background MC samples [30]. These distributions set the cross-section normalization inputs for the 2021 radiative-fraction calculation.

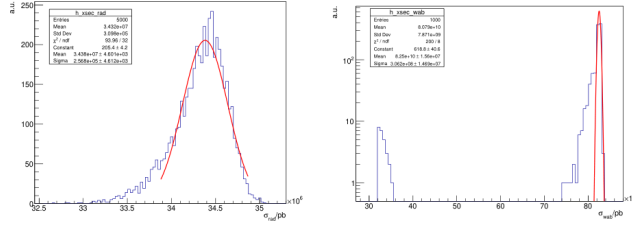


Figure 161: **Left:** Distribution of cross sections for radiative events, **Right:** Distribution of cross sections for WAB events.

Figure 24: RAD and WAB cross-section distributions used in the 2021 note’s radiative-fraction cross-section uncertainty propagation [30]. Together with the relative normalization $R = 0.244$, these inputs lead to the 3.5% MC cross-section contribution.

The 4.6% term covers only the radiative-fraction normalization. The broader 2021 displaced-search uncertainty model in Fig. 25 also includes mass resolution, target position, and tight-selection effects that must be revisited before applying it to the full data set [30, Sec. 9, Table 58]. Those signal-yield and acceptance uncertainties are not included in Eq. (12).

Source	Percentage
MC Cross Section	3.5%
Pre-selection Uncertainty	3%
Invariant Mass Resolution	6.5%
Target Position Uncertainty	9%
MC-Data Tight Selection Uncertainty (Intuition)	2%
MC-Data Tight Selection Uncertainty (ANN)	4%

Table 58: Table of (expected) uncertainties for the displaced vertex search

Figure 25: Broader 2021 full-analysis-note systematic summary [30]. Only the MC cross-section and preselection entries are folded into the radiative-fraction-only penalty in Eq. (14); the other entries remain separate signal-yield or acceptance effects.

The selected spectra, resolution curves, and radiative fractions now specify the three data-dependent ingredients of the search: the observed mass distribution, the shape of a narrow signal, and the conversion from fitted signal yield to coupling. The next section describes how the GP predicts the smooth background beneath each tested mass.

4 Methodology

4.1 Background interpolation around a blinded mass window

At each mass hypothesis, the search must predict a smooth trident background without allowing a possible narrow resonance to influence that prediction. We therefore fit the GP on a fixed, dataset-specific mass range and exclude a small region around the test mass from training. The GP interpolates across the excluded region, and those same bins are then used in the signal-plus-background likelihood. Earlier HPS studies redefined the sidebands in separate piecewise windows. The fixed-support construction gives smoother behavior across the scan and allows the three data sets to enter a joint likelihood with a shared coupling.

4.2 Search interval and GP fit support

The search interval specifies where a resonance is tested; the wider GP fit support specifies which bins can constrain its background. At each mass, bins inside the moving $\pm 2.25\sigma_m$ exclusion are omitted from GP training. Keeping training data beyond the search endpoints avoids unnecessarily one-sided interpolation. The current settings are given in Table 3 and apply to both standalone and combined fits.

Data	Search [MeV]	GP support [MeV]	Upper length factor
2015 full	19–90	14–135	8
2016 full	39–180	30–210	12
2021 10%	50–250	36–300	15

Table 3: Frozen settings of the current v4.9.12 input release inherited by v5. The 2016-inclusive results remain conditional on the recorded state-replay exception. The 2021 lower edge follows the conditional support study; its earlier 40 MeV value is retained only in historical comparisons.

The 2016 lower edge at 30 MeV also reflects the actual bin geometry. At the 39 MeV search endpoint, the blind interval begins at 35.063 MeV, whereas a 35 MeV support starts at a bin center of 35.125 MeV and supplies no lower-side training center. Training masks are defined using bin centers, not only continuous interval endpoints. The 2021 support study is described in Section 5.10; its post-selection practical rule and remaining 55 MeV under-recovery remain part of the qualification of these settings.

4.3 Observable, scan grid, and blinding

The input observable is the reconstructed e^+e^- invariant-mass histogram for each dataset. Throughout this section, d labels the dataset, m denotes the reconstructed e^+e^- invariant mass, m_0 is the test-mass hypothesis, and $\sigma_d(m)$ is the dataset-specific mass resolution introduced in Section 3. The mass-hypothesis spacing is

$$\Delta m = 1 \text{ MeV}, \quad (15)$$

and the blind window for a mass hypothesis m_0 is centered on the detector resolution,

$$\mathcal{B}_d(m_0) = [m_0 - n_{\text{blind}}\sigma_d(m_0), m_0 + n_{\text{blind}}\sigma_d(m_0)]. \quad (16)$$

The extraction half-width and GP training exclusion are independently configurable. Early injection studies used $n_{\text{blind}} = 1.64$ with the same $1.64\sigma_d$ training exclusion. The nominal analysis uses a

wider $2.25\sigma_d$ blind/extraction window with the GP training exclusion also set to $2.25\sigma_d$. For an ideal Gaussian line shape, the $1.64\sigma_d$ interval contains

$$f_{\text{blind}}^{\text{ideal}}(1.64) = \text{erf}\left(\frac{1.64}{\sqrt{2}}\right) \simeq 0.899, \quad (17)$$

while the $2.25\sigma_d$ interval contains

$$f_{\text{blind}}^{\text{ideal}}(2.25) = \text{erf}\left(\frac{2.25}{\sqrt{2}}\right) \simeq 0.976. \quad (18)$$

We normalize the signal over the full Gaussian line shape and use only its blinded-bin slice in the local likelihood; the slice is not renormalized inside the window.

The notation $\mathcal{B}_d(m_0)$ denotes the set of histogram bins that lie inside the blinded interval for dataset d at test mass m_0 . In the nominal fit there is no additional gap between the extraction window and the GP training sample: when the extraction and training-exclusion half-widths are equal, the training exclusion starts immediately outside $\mathcal{B}_d(m_0)$. A wider exclusion, for example $1.96\sigma_d$ instead of $1.64\sigma_d$, reduces signal leakage into the training region but also increases the interpolation distance and therefore the GP predictive uncertainty. In the simple background-dominated proxy $S/\sqrt{B} \propto F(k)/\sqrt{k}$ with $F(k) = \text{erf}(k/\sqrt{2})$, the optimum lies near $k \simeq 1.4$, so the earlier $1.64\sigma_d$ extraction was close to the pure-sensitivity optimum. Section 5 then motivates the current $2.25\sigma_d$ blind/extraction choice as a signal-recovery improvement after full-refit pseudoexperiments showed appreciable training-region absorption at narrower exclusions.

We assign bins by their fixed centers rather than splitting bins that overlap a boundary. Let $c_i = (e_i + e_{i+1})/2$ be the center of bin i . The full bin is included in the blinded likelihood when

$$m_0 - n_{\text{blind}}\sigma_d(m_0) \leq c_i \leq m_0 + n_{\text{blind}}\sigma_d(m_0), \quad (19)$$

and it enters the sideband sample only when its center lies strictly outside the training-exclusion interval. The same center mask is applied to the counts, GP prediction, covariance, and signal template, so no observed count is divided between the two regions.

The Gaussian signal template is integrated over the full bin edges in Eq. (41), but the selected set of bins changes discretely when a boundary passes a bin center. Consequently the quoted local significance and ε^2 limit are not meaningful to a precision finer than the mass binning and scan grid used to define these masks. The same discrete procedure is used for the data and pseudoexperiments.

4.4 Log-space preprocessing and the choice $\alpha = 1/y$

Following recent HEP GPR background-modeling studies [32, 26], we fit the GP in transformed coordinates:

$$x = \log m, \quad z = \begin{cases} \log y, & y > 0, \\ 0, & y = 0, \end{cases} \quad (20)$$

where y is the observed bin count. The log transformation is used for two related reasons. First, the prompt trident spectrum spans a wide dynamic range, and modeling z rather than y turns multiplicative shape variations into approximately additive ones. Second, $x = \log m$ makes equal distances in the GP coordinate correspond to equal fractional changes in mass, which is the relevant scale once the detector resolution is written in the transformed coordinate as $\sigma_x(m) \simeq \sigma_d(m)/m$.

Barr and Liu emphasize that applying logarithms to both axes improves convergence and performance for wide, steeply falling spectra [32]. The GP regression step uses scikit-learn’s GaussianProcessRegressor estimator [33]; the surrounding HPS analysis code supplies the histogram handling, mass-dependent sideband masks, signal templates, profile likelihood, and CL_s limit construction.

The GP is therefore used to model an unobserved smooth mean function $z_{\text{true},d}(x)$ for the log-count spectrum. In that language, “latent” means only that the smooth log-rate is not observed directly; it is inferred from the histogram counts through the transformed coordinates and the heteroscedastic variance term. The GP mean vector is the vector of latent log-count means evaluated at the selected training-bin coordinate. For Poisson counts $Y_i \sim \text{Pois}(\lambda_i)$, the delta method gives $\text{Var}[\log Y_i] \approx \text{Var}[Y_i]/E[Y_i]^2 \approx 1/\lambda_i$, which motivates the plug-in choice $\alpha_i \approx 1/y_i$ for occupied bins. A heteroscedastic diagonal term α_i is then added to the training covariance,

$$\alpha_i = \begin{cases} 1/y_i, & y_i > 0, \\ 1, & y_i = 0, \end{cases} \quad (21)$$

$$K_{\text{obs}} = K(X, X) + \text{diag}(\boldsymbol{\alpha}), \quad (22)$$

so that each α_i plays the role of an observation-variance term for bin i . This is the quantity that GaussianProcessRegressor adds to the diagonal during fitting, so it is treated here as a variance model rather than an arbitrary tuning parameter.

Here α_i is the latent-space variance associated with the transformed counts; we do not add an extra $\log y$ weight. For empty bins the floor $\alpha_i = 1$ is a unit variance in log-count space, not a pseudo-count of one event. It simply prevents zero-count bins from receiving ill-defined or disproportionate weight after the log transformation. In this high-statistics search such bins are rare, so the floor acts mainly as a conservative numerical safeguard rather than as a dominant source of uncertainty.

The distinction from Ref. [32] is more specific than a difference in kernel choice. Barr and Liu also use a Constant $\times$ RBF kernel and optimize its amplitude and length scale. Their independent methodological study uses a spectrum derived from a published CMS search; it is not a CMS Collaboration GPR result. In addition to optimizing the kernel, they treat the diagonal array as tunable regularization. For occupied bins their nominal prescription is

$$\alpha_i^{\text{BL}} = \frac{\sqrt{y_i}}{y_i \log y_i} = \frac{1}{\sqrt{y_i} \log y_i}, \quad (23)$$

and fixed regional multipliers f_i are then applied to α_i^{BL} . Reducing the first eleven multipliers to 0.1 raised the fraction of 100 background-only pseudoexperiments passing their BumpHunter criterion from 51% to 87%, but localized residual structures remained; their high-luminosity benchmark subsequently extended the reduced-multiplier region to the first twenty bins [32]. The study therefore demonstrates that diagonal tuning can improve a particular benchmark, while also showing that the tuned region is sample- and regime-dependent.

The active HPS configuration applies no regional multiplier to the early bins and fixes Eq. (21) as the delta-method variance model. This choice matters because scikit-learn’s alpha parameter has both an observation-noise interpretation and a numerical Tikhonov-regularization interpretation [34]. HPS adopts the former and places its separately validated smoothness control in the allowed length-scale domain. These are complementary controls: changing α_i changes the leverage of bin i on the covariance diagonal, whereas changing ℓ changes the off-diagonal correlation range of the latent function. No matched study presented here establishes that one strategy is intrinsically superior to the other.

4.5 Kernel choice and distinct regularization roles

We model the background with a single ConstantKernel $\times$ RBF kernel,

$$k(x, x') = C \exp \left[-\frac{(x - x')^2}{2\ell^2} \right], \quad (24)$$

with broad amplitude bounds and a resolution-scaled RBF length-scale domain. This is the same simple kernel class used in the methodological studies of Refs. [26, 32], but the surrounding noise and validation choices need not be the same. The two quantities optimized in every HPS sideband fit are C and ℓ . The ConstantKernel value $C = k(x, x)$ is the pointwise prior variance of the latent log-rate, so $\sqrt{C}$ is its prior standard deviation; C is not a fitted event-count normalization. The length scale ℓ controls how quickly covariance falls with separation in $x = \log m$, and therefore how rapidly the background function may bend. The fixed α_i values are not a third kernel parameter and there is no optimized WhiteKernel in the baseline. They enter only through the observation-covariance diagonal.

Figure 26 separates these roles graphically. This separation is useful when discussing regularization: changing C rescales the covariance amplitude, changing ℓ changes the correlation geometry and smoothness of the latent function, and changing α_i changes how strongly one observation constrains that function.

$C \times \text{RBF}$ plus observation noise: three different roles

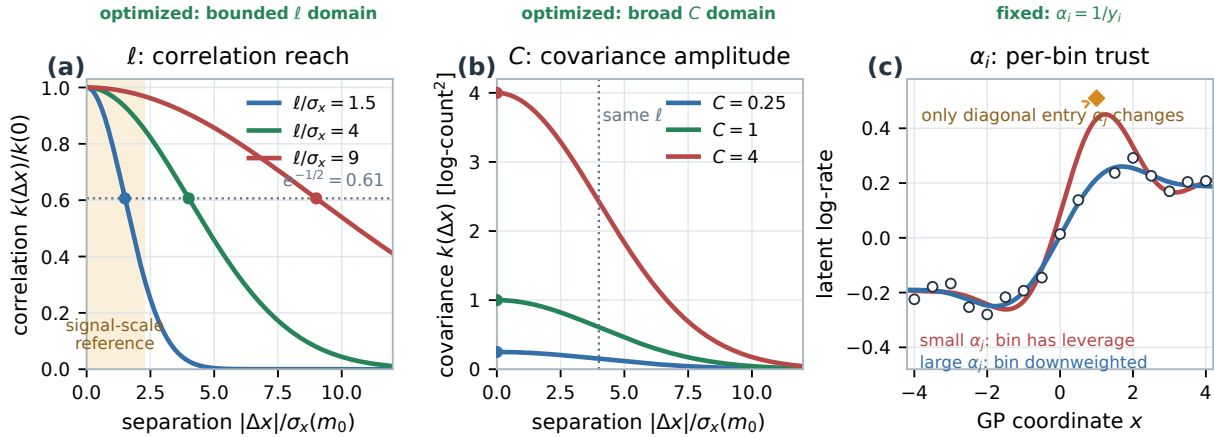


Figure 26: Roles of the three quantities in the GPR covariance. Panel (a) shows the normalized RBF correlation for three illustrative values of $\ell/\sigma_x(m_0)$; at a separation equal to ℓ , every curve has correlation $e^{-1/2} \simeq 0.61$. The length scale sets the correlation range. Panel (b) varies C at fixed ℓ . It changes the covariance amplitude, with pointwise prior standard deviation $\sqrt{C}$, without changing the normalized correlation range. Panel (c) holds the kernel fixed and changes one diagonal variance α_j , thereby changing that bin's statistical weight. We optimize C and ℓ on the sidebands and fix α_i from the count-based variance rule. The curves are schematic.

We use the Constant $\times$ RBF model. A Constant $\times$ Matérn kernel with $\nu = 3/2$ and piecewise kernels remain exploratory robustness checks; they have not been tested for coverage in the three-campaign configuration.

4.6 Resolution-scaled length-scale parameterization and bounds

Because the GP is trained in $x = \log m$, detector resolution must be translated into log-mass units. We define

$$\sigma_x(m) = \log(m + \sigma_d(m)) - \log m \simeq \frac{\sigma_d(m)}{m} \quad (\sigma_d \ll m), \quad (25)$$

and use this local physical scale to set the RBF bounds. Within the search intervals used here, the local term sets both bounds,

$$\ell_{\min,d}(m) = k_{\min,d}\sigma_x(m), \quad \ell_{\max,d}(m) = k_{\max,d}\sigma_x(m). \quad (26)$$

For the RBF covariance used here,

$$k_{\text{RBF}}(x, x') = \sigma_f^2 \exp\left[-\frac{(x - x')^2}{2\ell^2}\right], \quad \frac{k_{\text{RBF}}(x, x + \ell)}{k_{\text{RBF}}(x, x)} = e^{-1/2} \simeq 0.61. \quad (27)$$

Thus ℓ is a covariance-decay scale in the GP input coordinate: larger values favor smoother background functions, while smaller values admit more rapid variation [24, 25]. The ratio $\ell/\sigma_x(m)$ expresses that nuisance-model smoothness relative to the width of the narrow signal hypothesis. It is not, without an independently validated mapping, a detector resolution measurement or a physical background correlation length. The fitted ℓ_{opt} is the log-marginal-likelihood optimum conditional on the kernel, data support, sideband mask, noise model, and configured optimizer domain. Consequently an optimum at ℓ_{max} establishes only that the numerical constraint is active; it neither measures a physical scale at the boundary nor locates the unconstrained optimum. The implementation optimizes log-transformed kernel hyperparameters within these configured bounds [34].

For a fixed dataset d and mass hypothesis m_0 , let $\mathbf{z}_{\text{sb}}$ contain only the transformed sideband observations and define

$$K_y(C, \ell) = C R_\ell(X_{\text{sb}}, X_{\text{sb}}) + \text{diag}(\boldsymbol{\alpha}_{\text{sb}}). \quad (28)$$

The optimized pair maximizes

$$\log p(\mathbf{z}_{\text{sb}} \mid X_{\text{sb}}, C, \ell) = -\frac{1}{2} \mathbf{z}_{\text{sb}}^T K_y^{-1} \mathbf{z}_{\text{sb}} - \frac{1}{2} \log |K_y| - \frac{n_{\text{sb}}}{2} \log(2\pi). \quad (29)$$

The quadratic term rewards agreement with the sideband observations, while the determinant term accounts for the covariance volume used to obtain that agreement. The implementation searches in the logarithms of C and ℓ with multiple optimizer restarts. This is an empirical-Bayes, or type-II maximum-likelihood, selection of the covariance parameters; it is not a profile of ℓ as a frequentist nuisance parameter in the subsequent local CL_s likelihood.

Figure 27 shows why the optimum is specific to a dataset and mass hypothesis. Moving m_0 changes which bins are excluded from training and changes the resolution-scaled domain through $\sigma_x(m_0)$. The same dataset can therefore return a different $(C_{\text{opt}}, \ell_{\text{opt}})$ at neighboring hypotheses even though the kernel family is unchanged. An interior maximum means only that the configured range is nonbinding for that fit. A maximum on a boundary means that the range is active, while a long shallow likelihood ridge means that the parameters are weakly identified. None of these cases turns ℓ_{opt} into a direct measurement of detector resolution or a physical background correlation length.

How the GP selects its smoothness at each test mass

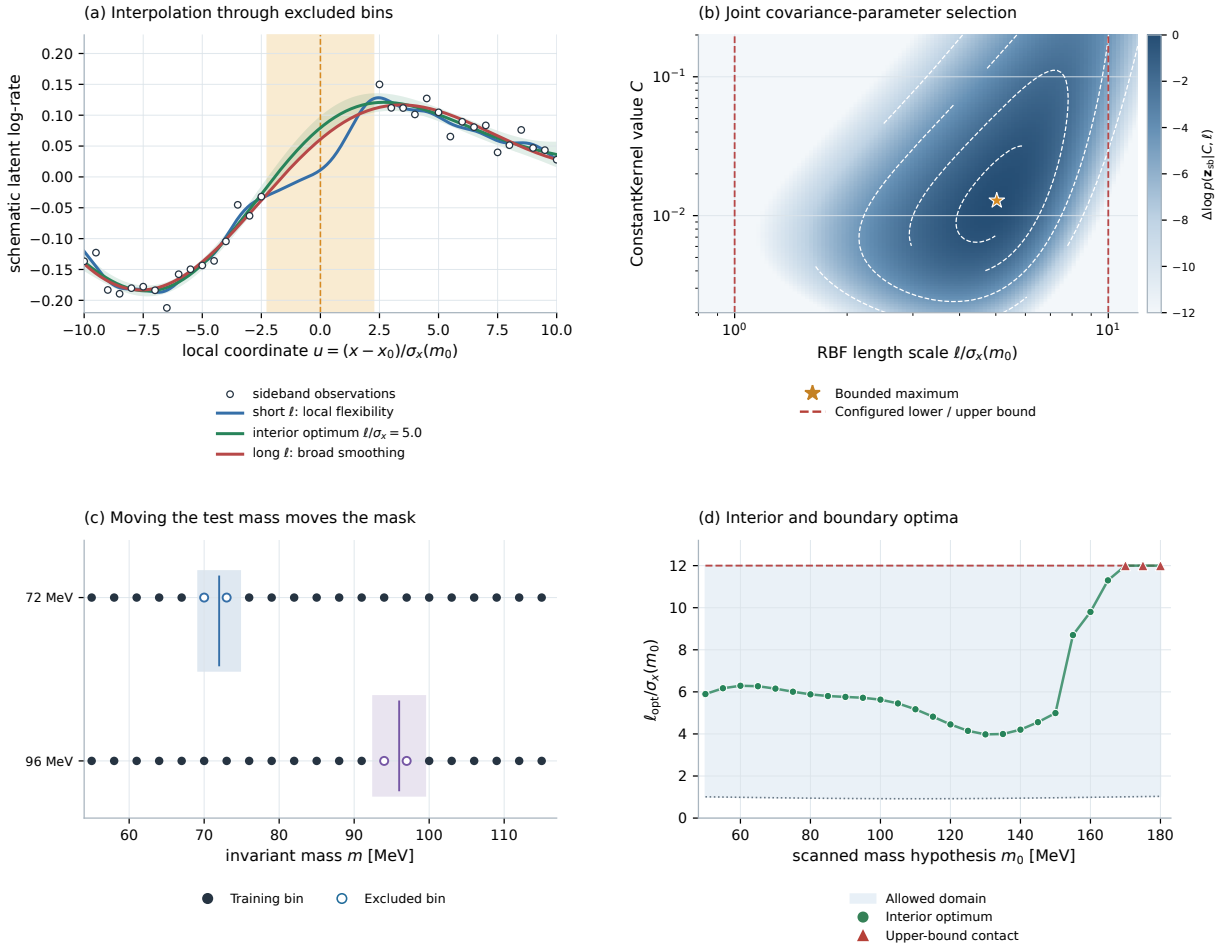


Figure 27: Meaning of per-mass-hypothesis hyperparameter optimization. Panel (a) shows one schematic sideband problem in $u = (x - x_0)/\sigma_x(m_0)$. The shaded $2.25\sigma_x$ interval is excluded from training; the three curves illustrate the different interpolation behavior allowed by short, interior, and long candidate length scales. Panel (b) shows the joint log-marginal-likelihood surface for the same synthetic sidebands. The star is the bounded maximum in (C, ℓ) , and the elongated contours expose their possible tradeoff. Panel (c) shows that moving m_0 changes both the sideband mask and the local resolution scale that defines the allowed ℓ interval. Panel (d) illustrates how a scan records an interior optimum differently from repeated upper-bound contact: the former is a conditional best fit within a nonbinding range, whereas the latter diagnoses an active numerical domain and motivates a predeclared range and closure study. The figure is explanatory only; it contains no HPS observations or fit results.

This distinction determines how the hyperparameter study enters a physics result. If the allowed range forces ℓ too low, the GP retains unnecessary signal-scale flexibility and can absorb part of a localized excess; if the fitted background is forced too smooth, broad mismodeling can instead leave a biased local residual. Raising only the upper bound permits, but does not require, a smoother solution. A larger data set also need not prefer a larger ℓ : additional statistics can resolve background structure and move the optimum downward. The upper range is therefore

selected from boundary occupancy, likelihood ordering, optimizer stability, and fixed-amplitude signal-recovery diagnostics, never from a favorable observed limit or local p_0 . These checks can justify a bounded candidate range, but they do not by themselves establish frequentist coverage.

The three-campaign analysis uses the dataset-specific factors in Table 4. The lower factors come from full-refit closure studies; changing the 2016 upper factor does not requalify those studies at full statistics.

Dataset	$k_{\min,d}$	$k_{\max,d}$	Basis
2015 full	1.0	8	full-refit closure study
2016 full	0.9	12	closure plus post-selection range study
2021 10%	1.1	15	matched-refit closure study

Table 4: Resolution-scaled RBF length-scale factors used in the three-campaign analysis. The factors multiply $\sigma_x(m)$ in Eq. (26). The 2016 upper factor was chosen after a range limitation appeared in the observed scan; it is therefore a post-selection choice, not a pre-unblinding or coverage qualification.

Figure 28 displays the allowed GP length-scale regions in log-mass coordinates. The shaded region is a configured domain, not an error band on the fitted smoothness. The implementation retains the upper-bound floor $0.8 k_{\max,d} \text{median}_{m \in \mathcal{M}_d} \sigma_x(m)$, with the median evaluated on its configured 300-point search grid; that floor is nonbinding over the three plotted search ranges.

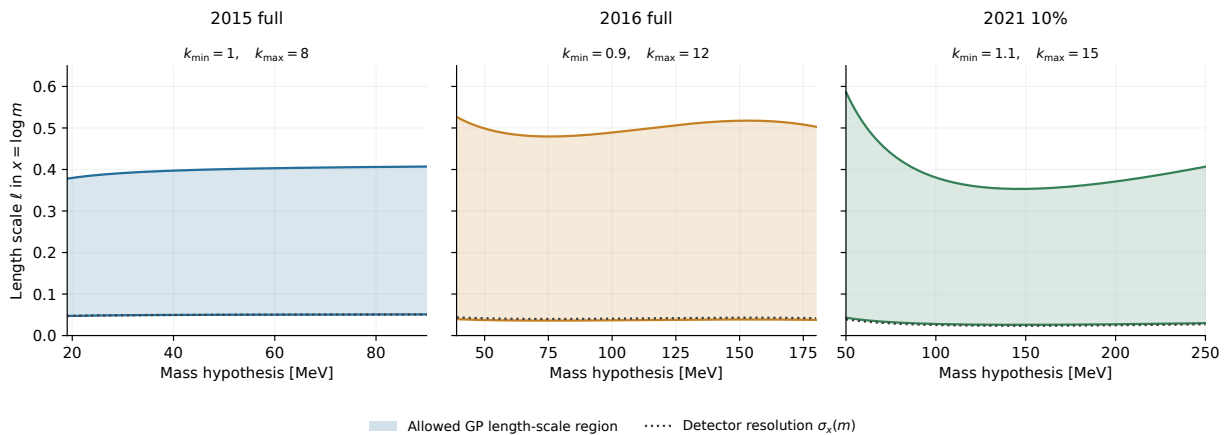


Figure 28: Allowed RBF length scales in $x = \log m$ across the three search ranges. Shading lies between the lower and upper bounds evaluated from the frozen analysis card; the dotted curves show the local detector-resolution scale $\sigma_x(m)$. The lower/upper factors are 1.0/8, 0.9/12, and 1.1/15 for 2015, 2016, and 2021, respectively. These are numerical domains for the sideband fit, not fitted length scales or detector-resolution uncertainties.

The lower factors are based on full-refit functional-form closure scans, with the observed-spectrum result kept out of the selection criterion. For 2015, the 100-toy scan slightly prefers $k_{\min} = 1.1$ in its aggregate score, but $k_{\min} = 1.0$ has similar pull-width calibration and a weaker expected-reach proxy. We retain 1.0 as the more conservative choice. For 2016, the 25-toy continuation supports the change to $k_{\min} = 0.9$: its nonzero-injection RMS pull mean is 0.321 and its median pull width is 0.948, compared with 0.403 and 0.889 for $k_{\min} = 1.1$. It gives the smallest aggregate

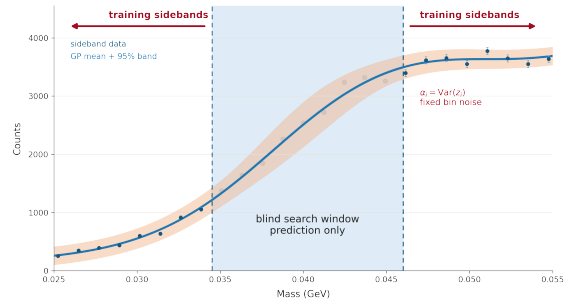
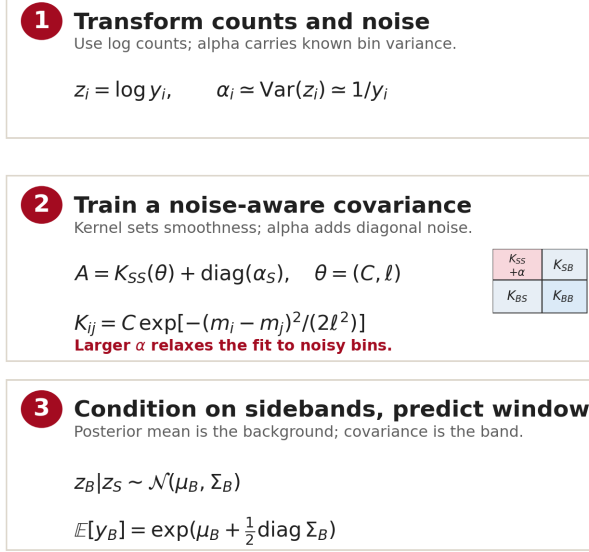
calibration score among the scanned 0.9, 1.0, and 1.1 rows and reduces the coherent apparent signal-recovery bias that motivated the follow-up. For 2021, the corrected matched-reference scan finds small rank separations among the same three settings. The $k_{\min} = 1.1$ row has the median pull width closest to unity and the best extraction-level ε^2 proxy, while the 0.9 and 1.0 rows have slightly smaller aggregate scores. We choose 1.1 as a calibration–reach tradeoff, not because it wins every diagnostic. These studies used the data fractions stated in Section 5 and do not establish full-statistics closure.

The initial upper factors were 8, 8, and 15 for 2015, 2016, and 2021. In the reviewed observed scan, the 2016 optimum reached its upper bound at all 142 hypotheses. We therefore varied only $k_{\max,2016}$, evaluating factors 8, 10, 12, 15, and 20 on the same 39–180 MeV mass grid. A fit is counted as upper-bound occupied when $\ell_{\text{opt}}/\ell_{\max} \geq 0.999$. The respective occupancies are 142, 56, 0, 0, and 0 of 142. Factor 12 is the first scanned setting with no occupied hypotheses. It is also followed by a numerical plateau: from factor 12 to 15, the largest pointwise change in the 2016 observed yield upper limit is 0.083%, the largest $|\Delta Z|$ is 0.00255, and the log-marginal-likelihood differences lie between -4.5×10^{-6} and 8.0×10^{-6} . The factor-15 to factor-20 comparison is similarly stable. The selection is based on removal of the artificial boundary and the next-setting plateau, not on which observed limit is tighter or which local p_0 is smaller.

Because the observed spectrum revealed the range limitation, factor 12 is a post-selection choice for the three-campaign analysis. It is not a pre-unblinding choice or a coverage-calibrated setting; full-statistics functional-form closure and direct coverage remain separate questions. The configurations continue to retain a dataset-wide median-resolution floor on $\ell_{\max,d}$ so that the ceiling does not become artificially small where the local resolution is exceptionally tight. This is a guard based on a representative σ_x value, not a floor derived from luminosity or event statistics. Alternative upper bounds, fixed-background fits, and the common-lower-bound overlays remain robustness studies collected in the appendices.

An initial 2021 source/exposure study used a nested-Poisson construction. It motivated the later matched-refit studies but is not part of the procedure used here.

Figure 29 summarizes the same construction in the language used by the implemented likelihood. The sideband bins set the log-count training sample and its heteroscedastic bin-noise assignment, the kernel hyperparameters are optimized only on those sidebands, and the trained GP is then conditioned to predict the mean and covariance in the blinded mass window. That predicted count-space mean and covariance are the background inputs passed to the local profiled signal extraction.



(a) Log-count transform, noise-aware covariance training, and window prediction. The filled covariance boxes indicate the sideband-sideband, sideband-blind, and blind-blind kernel blocks used when conditioning the GP.

(b) Sideband-only training and blinded-window prediction. Filled markers show binned count data, vertical error bars show the fixed per-bin statistical noise, and the horizontal reference lines are count-axis guides used to compare sideband and blinded-window scales.

Figure 29: Kernel and blinded-window construction. The GP is trained on sideband bins with fixed bin-noise variances, not on the blinded bins. The posterior prediction in the blind window supplies the background mean and covariance that enter the local likelihood; the shaded band in the schematic denotes the GP posterior interval and the dashed vertical lines mark the blinded-window boundaries.

4.7 Blind-window prediction in count space

For each mass point, we train the GP only on the bins outside the training exclusion window and predict the posterior mean and covariance of the latent smooth log-rate in the blinded bins. Since the fit is performed in log-count space, that posterior is mapped back to count space through the standard lognormal moment relations:

$$\mu_i = \exp\left(\mu_i^{(\log)} + \frac{1}{2}\Sigma_{ii}^{(\log)}\right), \quad (30)$$

$$\text{Cov}(\lambda_i, \lambda_j) = \mu_i \mu_j \left[\exp\left(\Sigma_{ij}^{(\log)}\right) - 1 \right]. \quad (31)$$

The resulting blind-window mean vector $\mathbf{b}_d(m)$ and latent-background covariance $\mathbf{C}_d^{\text{GP}}(m)$ supply the local extraction. Poisson variation of the observed counts is handled separately in the likelihood. Thus this covariance describes uncertainty in the predicted smooth mean, not the counting variance of the histogram.

Before factorization, the numerical implementation symmetrizes the covariance and applies the small diagonal conditioning recorded by the solver checks. Below, $\mathbf{C}_d$ denotes that conditioned covariance, with

$$\mathbf{C}_d \equiv \mathbf{C}_{d,\text{eff}} = L_d L_d^\top. \quad (32)$$

The lower-triangular factor L_d maps independent unit Gaussian coordinates into correlated count-space deformations. The regularization permits stable factorization; it is not an added physics uncertainty.

4.8 Profiled background implementation

At one tested mass, write the local Poisson mean as

$$\lambda_d(A_d, \theta_d) = \mathbf{b}_d + L_d \theta_d + A_d \mathbf{w}_d. \quad (33)$$

Here A_d is the total Gaussian signal yield and $\mathbf{w}_d$ is the bin-integrated template restricted to the likelihood window, as defined in Section 4.9. The nuisance vector θ_d has a unit Gaussian constraint, so a displacement $L_d \theta_d$ preserves the correlations predicted by the GP. At fixed A_d , the fit minimizes

$$\ell_d(A_d, \theta_d) = \sum_i \left[\lambda_{d,i} - n_{d,i} + n_{d,i} \log \left(\frac{n_{d,i}}{\lambda_{d,i}} \right) \right] + \frac{1}{2} \theta_d^\top \theta_d, \quad \lambda_{d,i} > 0. \quad (34)$$

The $n \log n$ contribution is zero at $n = 0$. Apart from constants independent of the fit, ℓ_d is the negative log likelihood. Its first term measures the Poisson mismatch; the second penalizes background deformations in units of the correlated GP uncertainty. The profiled objective is

$$\hat{\theta}_d(A_d) = \arg \min_{\theta_d} \ell_d(A_d, \theta_d), \quad (35)$$

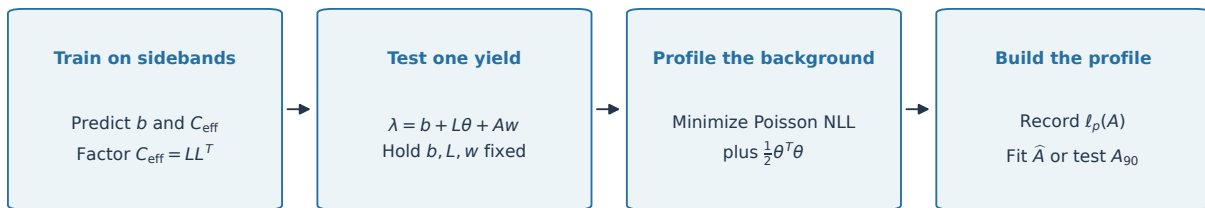
$$\ell_{p,d}(A_d) = \ell_d \left(A_d, \hat{\theta}_d(A_d) \right). \quad (36)$$

For an interior solution, the nuisance gradient and Hessian are

$$\nabla_{\theta} \ell_d = L_d^\top \left(\mathbf{1} - \frac{\mathbf{n}_d}{\lambda_d} \right) + \theta_d, \quad (37)$$

$$H_{\theta\theta} = L_d^\top \text{diag} \left(\frac{n_{d,i}}{\lambda_{d,i}^2} \right) L_d + I, \quad (38)$$

where the vector division is elementwise. These expressions make the correlated adjustment explicit: the fit changes a set of nuisance modes, rather than choosing unrelated background shifts in each bin.



$L\theta$ moves bins coherently; the Gaussian penalty controls the size of that deformation.

Figure 30: At each tested signal yield, the background deforms within the fixed GP constraint and the minimized objective is recorded. Repeating this optimization constructs the profile curve used for extraction and limits. The diagram describes the calculation; it contains no fitted data.

During this local profile, $\mathbf{b}_d$, L_d , the bin selection and the signal template remain fixed. Profiling θ_d does not retrain the GP kernel. A complete pseudoexperiment may first retrain the sideband prediction under its declared kernel policy, then apply this same local fit. Positivity applies to the total Poisson mean $\lambda_{d,i}$; a separate positivity constraint is not imposed on $\mathbf{b}_d + L_d \theta_d$. The production solver uses numerical safeguards near zero means; accepted interior solutions are checked against the objective above. Signed auxiliary signal fits describe both excesses and deficits, while physical discovery and upper-limit tests use a nonnegative signal.

This Gaussian-constrained construction follows the profile-likelihood framework of Refs. [35, 36, 25]. The constraint comes from sideband interpolation and should not be described as an independent auxiliary measurement. Its qualification therefore includes the background and signal-recovery tests presented later.

4.9 Signal model and conversion to ε^2

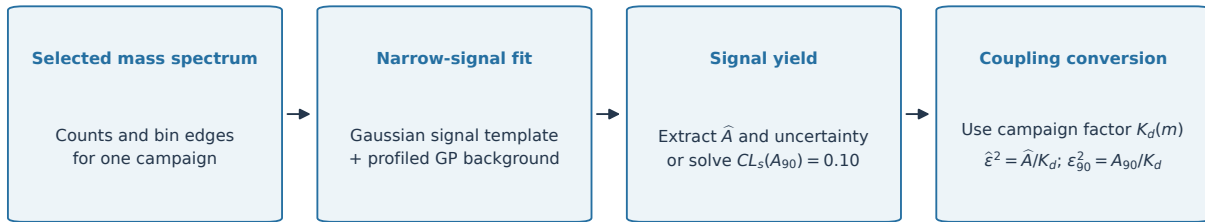


Figure 31: From the selected mass spectrum to a fitted signal yield or upper bound, then to the corresponding coupling. The campaign factor contains the declared radiative-fraction and prompt-density normalization. The applicable visible branching factor is included when interpreting the result above the dimuon threshold.

For each dataset d and mass hypothesis m_0 , we first construct a full-range Gaussian line-shape template over the histogram binning,

$$g_{d,i}(m_0) = \int_{e_i}^{e_{i+1}} \frac{1}{\sqrt{2\pi}\sigma_d(m_0)} \exp\left[-\frac{(m - m_0)^2}{2\sigma_d(m_0)^2}\right] dm, \quad (39)$$

where e_i and e_{i+1} are the lower and upper edges of histogram bin i . The template is normalized on the stored histogram support:

$$w_{d,i}^{\text{full}}(m_0) = \frac{g_{d,i}(m_0)}{\sum_{j \in \text{full range}} g_{d,j}(m_0)}, \quad \sum_{i \in \text{full range}} w_{d,i}^{\text{full}}(m_0) = 1. \quad (40)$$

The template entering the local likelihood is then the blinded-bin slice of this full-range object,

$$w_{d,i}(m_0) = \begin{cases} w_{d,i}^{\text{full}}(m_0), & i \in \mathcal{B}_d(m_0), \\ \text{not used in the local fit,} & i \notin \mathcal{B}_d(m_0). \end{cases} \quad (41)$$

By the blinded-bin slice we mean exactly this restriction of the full normalized Gaussian template to the subset of bins in $\mathcal{B}_d(m_0)$; the template is not renormalized after the restriction. Its blinded-bin

sum is therefore

$$f_{\text{blind},d}(m_0) = \sum_{i \in \mathcal{B}_d(m_0)} w_{d,i}(m_0) < 1, \quad (42)$$

with a value near 0.90 for a $1.64 \sigma_d$ half-width and near 0.98 for the current $2.25 \sigma_d$ observed-limit half-width. The signal amplitude A_d is defined as the *total* local Gaussian signal yield associated with the mass hypothesis, not merely the yield inside the blind window. The expected signal contribution entering bin i of the local likelihood is therefore

$$s_{d,i}(A_d; m_0) = A_d w_{d,i}(m_0), \quad (43)$$

so the total expected signal actually visible inside the blind window is $f_{\text{blind},d}(m_0) A_d$.

This convention preserves the fact that a finite blind region never contains the entire Gaussian line shape. It also keeps the signal parameter used in the fit aligned with the full-histogram injection model used in the procedural toy studies.

The conversion between signal yield and dark-photon coupling follows the standard fixed-target relation between A' production and radiative trident production [2, 3, 4]. In the present notation,

$$A_d \equiv N_{A',d} = \frac{3\pi}{2N_{\text{eff}}^{\text{BR}} \alpha_{\text{EM}}} m_{A'} \varepsilon_{\text{phys}}^2 \left. \frac{dN_{\gamma^*}}{dm} \right|_{m=m_{A'}}, \quad (44)$$

where α_{EM} is the fine-structure constant and $3\pi/2$ is the narrow-width coefficient relating on-shell production to radiative tridents. Here $\varepsilon_{\text{phys}}^2$ is the physical kinetic-mixing parameter and $N_{\text{eff}}^{\text{BR}} = \Gamma_{\text{tot}}/\Gamma_{ee} = 1/\mathcal{B}_{ee}$ includes the phase-space dependence of all open decay channels. In the minimal model it equals one below the dimuon threshold. For the electron-pair yield, define

$$\varepsilon_{ee}^2 \equiv \frac{\varepsilon_{\text{phys}}^2}{N_{\text{eff}}^{\text{BR}}}. \quad (45)$$

The following fit and combination equations abbreviate this electron-channel coordinate as ε^2 . The displayed minimal-visible interpretation multiplies it by $N_{\text{eff}}^{\text{BR}}$ above $2m_\mu$, as specified in the dimuon-threshold subsection. Writing the radiative differential rate as

$$\frac{dN_{\gamma^*}}{dm} = f_{\text{rad},d}(m) \rho_d(m), \quad (46)$$

where $\rho_d(m)$ is the differential prompt background rate in the selected invariant-mass spectrum, we convert between A_d and ε^2 through

$$\varepsilon^2 = \frac{2\alpha_{\text{EM}} A_d}{3\pi m f_{\text{rad},d}(m) \rho_d(m)}, \quad (47)$$

where $\rho_d(m)$ is estimated from the observed histogram as a local counts-per-GeV density in the separately declared prompt-density interval,

$$\rho_d(m) \equiv \frac{N_d[m - n_{\text{blind}}\sigma_d(m), m + n_{\text{blind}}\sigma_d(m)]}{2 n_{\text{blind}}\sigma_d(m)}, \quad (48)$$

with $n_{\text{blind}} = 1.64$ retained as the current prompt-density normalization window even when the extraction blind window is widened to $2.25 \sigma_d$. This keeps the yield-to- ε^2 conversion tied to the established local density scale while the likelihood itself can recover the wider signal tails. The

density-window width remains configurable so that alternative conventions can be reproduced when needed. It is convenient to define the dataset-specific conversion factor

$$K_d(m) \equiv A_d(\varepsilon^2 = 1) = \frac{3\pi m f_{\text{rad},d}(m) \rho_d(m)}{2\alpha_{\text{EM}}}, \quad (49)$$

so that a shared coupling ε^2 predicts a dataset-specific total signal yield $A_d = K_d(m)\varepsilon^2$.

Unless stated otherwise, the methodology section uses the nominal radiative fraction $f_{\text{rad},d}(m)$. The optional penalty that propagates normalization uncertainty into ε^2 is treated as a systematic variation rather than as part of the core GP fit and is discussed separately in Section 3.7. The numerical penalty values are therefore result-configuration choices, not ingredients of the background interpolation or local extraction fit. When that systematic option is enabled for dataset d , the effective radiative fraction is

$$f_{\text{rad},d}^{\text{eff}}(m) = (1 - \delta_{f,d}) f_{\text{rad},d}(m), \quad (50)$$

so the same signal-yield limit maps to a weaker ε^2 limit.

4.10 Statistical inference in a single dataset

For a fixed dataset d and mass hypothesis m , let $\mathbf{n}_d$ denote the vector of observed blinded-bin counts, $\mathbf{b}_d$ the GP-predicted blinded-bin mean, $\mathbf{C}_d = L_d L_d^\top$ the corresponding covariance of Section 4.7 and Eq. (32), and $\mathbf{w}_d(m)$ the non-renormalized blinded-bin slice of the full Gaussian template defined in Eq. (41). The profiled local likelihood is

$$\mathcal{L}_d(A_d, \boldsymbol{\theta}_d) = \left[\prod_{i=1}^{N_d(m)} \text{Pois}(n_{d,i} | b_{d,i} + (L_d \boldsymbol{\theta}_d)_i + A_d w_{d,i}(m)) \right] \exp\left(-\frac{1}{2} \boldsymbol{\theta}_d^\top \boldsymbol{\theta}_d\right), \quad (51)$$

where the Gaussian nuisance vector $\boldsymbol{\theta}_d \sim \mathcal{N}(0, I)$ induces the correlated background-deformation field $L_d \boldsymbol{\theta}_d$ defined below Eq. (32), and $N_d(m)$ is the number of bins in the local blinded set $\mathcal{B}_d(m)$. The signal contribution entering the Poisson mean is exactly the blinded-bin restriction of Eq. (43).

At fixed m , profiling Eq. (51) with no physical sign constraint on A_d yields the unconstrained extraction estimator $\hat{A}_d^{\text{unc}}$ and its local uncertainty $\sigma_{A,d}$. This unconstrained fit is used for closure, linearity, and pull studies because it allows downward background fluctuations to be captured by the estimator itself rather than being clipped at zero. For discovery and upper limits, however, the parameter of interest is restricted to the physical region $A_d \geq 0$, and the corresponding physical best fit is

$$\hat{A}_d = \max(0, \hat{A}_d^{\text{unc}}). \quad (52)$$

We denote by $\hat{\boldsymbol{\theta}}_d$ the nuisance-parameter maximizer at the unconditional best fit and by $\hat{\boldsymbol{\theta}}_d(A_d)$ the conditional maximum likelihood estimator of $\boldsymbol{\theta}_d$ for fixed A_d .

4.10.1 Local significance

The single-dataset discovery statistic is the bounded profile-likelihood ratio

$$q_{0,d}(m) = -2 \ln \frac{\mathcal{L}_d(A_d = 0, \hat{\boldsymbol{\theta}}_d(0))}{\mathcal{L}_d(\hat{A}_d, \hat{\boldsymbol{\theta}}_d)}, \quad (53)$$

with $\hat{A}_d \geq 0$. The local one-sided discovery p -value and the corresponding Gaussian significance are then obtained with the asymptotic Cowan-et-al. mapping [36]. The resulting local-significance

definition is explicitly one-sided and discovery facing: downward fluctuations are allowed in the unconstrained fit used for closure studies, but they do not generate a positive local discovery significance. We use this profiled multi-bin test throughout.

4.10.2 CL_s upper limits

To test a fixed non-negative signal amplitude A_d , the analysis uses the bounded one-sided profile-likelihood statistic

$$\tilde{q}_{A_d}(m) = \begin{cases} 0, & \hat{A}_d^{\text{unc}} > A_d, \\ -2 \ln \frac{\mathcal{L}_d(A_d, \hat{\theta}_d(A_d))}{\mathcal{L}_d(\hat{A}_d^{\text{unc}}, \hat{\theta}_d)}, & 0 \leq \hat{A}_d^{\text{unc}} \leq A_d, \\ -2 \ln \frac{\mathcal{L}_d(A_d, \hat{\theta}_d(A_d))}{\mathcal{L}_d(0, \hat{\theta}_d(0))}, & \hat{A}_d^{\text{unc}} < 0, \end{cases} \quad (54)$$

which is the standard bounded Cowan construction for upper limits [36]. For a positive observed statistic, the corresponding modified-frequentist tail areas are

$$\text{CL}_{s+b}(A_d; m) = P(\tilde{q}_{A_d} \geq \tilde{q}_{A_d}^{\text{obs}} | A_d + \text{bkg}), \quad \text{CL}_b(A_d; m) = P(\tilde{q}_{A_d} \geq \tilde{q}_{A_d}^{\text{obs}} | \text{bkg}), \quad (55)$$

and

$$\text{CL}_s(A_d; m) = \frac{\text{CL}_{s+b}(A_d; m)}{\text{CL}_b(A_d; m)}. \quad (56)$$

Dividing by CL_b prevents a downward background fluctuation from producing an artificially aggressive exclusion in a region with little intrinsic sensitivity [37, 38]. At confidence level $1 - \alpha$, the single-dataset upper limit is

$$A_{\text{UL},d}(m; \alpha) = \min\{A_d : \text{CL}_s(A_d; m) \leq \alpha\}. \quad (57)$$

The analysis uses $\alpha = 0.10$, corresponding to a 90% CL upper limit; comparison studies at 95% use $\alpha = 0.05$. We report the limit both in signal-yield space and, after applying the dataset-specific conversion factor of Eq. (49), in ε^2 space. These are fixed-mass statements. The correlation-aware global-significance calculation is introduced separately in Section 4.20; it does not change the pointwise CL_s upper-limit definition.

4.11 Asymptotic and toy calibration

The local-significance mapping introduced above is treated asymptotically throughout the scan. For upper limits, the same bounded statistic $\tilde{q}_{A_d}$ is calibrated in one of two ways. In asymptotic mode, the background-only Asimov data set is used to evaluate the positive reference quantity $q_A \equiv \tilde{q}_{A_d,A} > 0$ for the tested nonzero amplitude. Writing $q_{\text{obs}} \equiv \tilde{q}_{A_d}^{\text{obs}}$, the normal deviates entering the two tail areas are

$$z_{s+b} = \begin{cases} \sqrt{q_{\text{obs}}}, & q_{\text{obs}} \leq q_A, \\ \frac{q_{\text{obs}} + q_A}{2\sqrt{q_A}}, & q_{\text{obs}} > q_A, \end{cases} \quad z_b = \begin{cases} \sqrt{q_A} - \sqrt{q_{\text{obs}}}, & q_{\text{obs}} \leq q_A, \\ \frac{q_A - q_{\text{obs}}}{2\sqrt{q_A}}, & q_{\text{obs}} > q_A. \end{cases} \quad (58)$$

The corresponding tail areas are $\text{CL}_{s+b} = 1 - \Phi(z_{s+b})$ and $\text{CL}_b = \Phi(z_b)$, with $\text{CL}_s = \text{CL}_{s+b}/\text{CL}_b$. At $q_{\text{obs}} = 0$ the statistic has a point mass, so inclusive toy tails and the continuous asymptotic endpoint need a declared tie convention. The implementation retains the continuous endpoint of the displayed mapping, $\text{CL}_{s+b} = 1/2$ and $\text{CL}_b = \Phi(\sqrt{q_A})$; inclusive empirical tails at zero would both equal one. The reported upper-limit roots lie on the positive-statistic branch.

The name *Wald approximation* refers to treating the profiled objective as locally quadratic, with an approximately normal unconstrained estimator:

$$2[\ell_p(A) - \ell_p(\hat{A}^{\text{unc}})] \simeq \frac{(A - \hat{A}^{\text{unc}})^2}{\sigma_A^2}. \quad (59)$$

Both branches of Eq. (58) use this asymptotic idea. The second branch accounts for the physical boundary when the unconstrained best fit is negative. Its reference denominator is then the fit at zero, so $\tilde{q}_A \simeq (A^2 - 2A\hat{A}^{\text{unc}})/\sigma_A^2$, rather than $(A - \hat{A}^{\text{unc}})^2/\sigma_A^2$. This corresponds to $q_{\text{obs}} > q_A$ in the approximation. The implementation evaluates the observed and Asimov profile statistics numerically and then applies this piecewise probability mapping; it does not replace the underlying Poisson fit with least squares. The approximation itself is not a coverage test.

In toy mode, the same statistic $\tilde{q}_{A_d}$ is calibrated directly with background-only and signal-plus-background pseudoexperiments rather than by switching to a different ordering variable. The shared-coupling combination described below uses the identical logic after replacing the single-dataset amplitude A_d with the common parameter of interest ε^2 .

The pseudoexperiment program remains important even with the asymptotic formulae in place because the HPS likelihood combines a multi-bin Poisson term, a GP-conditioned blind-window covariance, a non-renormalized Gaussian signal template, and a mass-dependent yield-to- ε^2 conversion. The pull, spurious-signal, and injection-recovery diagnostics of Section 5 test this machinery under the declared conditional backgrounds. They are not direct CL_s coverage studies or a scan-wide calibration of Z .

4.12 Matched-reference injection–extraction diagnostics

We define the injection scale with a background-only reference fit that uses the same support, binning, exclusion mask, kernel bounds, preprocessing, and GP refit as the corresponding pseudoexperiment. If $\mathcal{F}[\mathbf{n}; \mathbf{w}]$ denotes that full procedure for counts $\mathbf{n}$ and the blind-window slice of the unit-normalized full-range signal template $\mathbf{w}$, then

$$(\hat{A}_{\text{ref}}, \sigma_{A,\text{ref}}) = \mathcal{F}[\mathbf{n}_{\text{ref}}; \mathbf{w}], \quad A_{\text{inj}}(z) = z \sigma_{A,\text{ref}}, \quad z \in \{0, 1, 3, 5\}. \quad (60)$$

The reference uncertainty is a fixed calibration of the injection coordinate at a given source, toy index, and mass. It is not the post-injection uncertainty returned for an individual extraction.

For pseudoexperiment t , the signed pull and paired response are

$$p_t(z) = \frac{\hat{A}_t(z) - A_{\text{inj},t}(z)}{\sigma_{A,t}(z)}, \quad (61)$$

$$R_t(z) = \frac{\hat{A}_t(z) - \hat{A}_t(0)}{A_{\text{inj},t}(z)}, \quad z > 0. \quad (62)$$

The zero-injection mean pull diagnoses a conditional spurious-signal offset under the declared generating truth. Its width probes the returned uncertainty scale, while R_t separates injection response

from a background-specific zero-signal offset. These quantities use the toy-level $\sigma_{A,t}$ in the denominator. A finite 25-background sample supports two-sided 90% Student- t intervals on the mean and descriptive widths, not a direct coverage statement.

Full refitting and fixed-background displays answer different questions. Holding the GP fixed isolates the local profile-likelihood extraction. Refitting the GP on every pseudo-spectrum additionally tests background adaptation and possible signal absorption. We use the latter matched-refit procedure. Numerical acceptance is decided from likelihood, reproducibility, covariance, and kernel-coordinate diagnostics before amplitudes or pulls are summarized.

4.13 Observed-limit toy diagnostics

The analysis uses two distinct families of p -values for different purposes. The first is the discovery statistic of Eq. (53), summarized through $q_0(m)$, $p_0(m)$, and $Z_{\text{local}}(m)$. The second consists of diagnostics derived from the background-only CL_s limit ensemble at each mass. For N_{toys} background-only pseudoexperiments and an observed limit $\varepsilon_{\text{UL,obs}}^2(m)$ at the confidence level under study, those quantities are

$$p_{\text{strong}}(m) = \frac{1}{N_{\text{toys}}} \sum_{t=1}^{N_{\text{toys}}} \mathbf{1} \left[\varepsilon_{\text{UL},t}^2(m) \leq \varepsilon_{\text{UL,obs}}^2(m) \right], \quad (63)$$

$$p_{\text{weak}}(m) = \frac{1}{N_{\text{toys}}} \sum_{t=1}^{N_{\text{toys}}} \mathbf{1} \left[\varepsilon_{\text{UL},t}^2(m) \geq \varepsilon_{\text{UL,obs}}^2(m) \right], \quad (64)$$

$$p_{\text{two}}(m) = \min[1, 2 \min(p_{\text{strong}}(m), p_{\text{weak}}(m))]. \quad (65)$$

Here p_{strong} quantifies how unusually *strong* the observed limit is relative to the background-only toy ensemble, while p_{weak} quantifies how unusually *weak* it is. The two-sided quantity p_{two} is retained only as a symmetric diagnostic; because it responds to both unusually strong and unusually weak limits, it is less targeted than the one-sided discovery or exclusion tests and is not used as a search statistic. Figure 32 gives the compact fixed-mass definition of these quantities, while Figure 33 shows representative strong-limit, weak-limit, and bounded median-case examples from the same toy-limit logic.

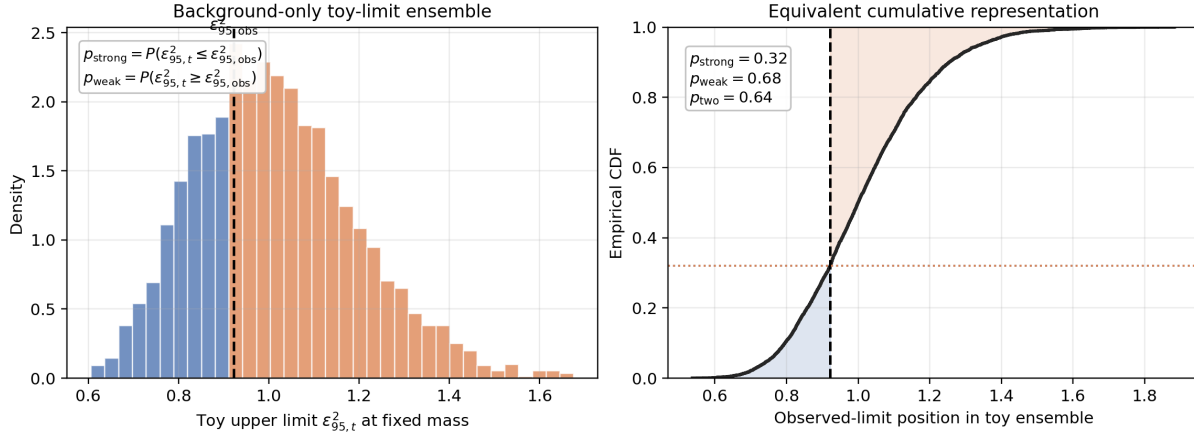


Figure 32: Illustrative toy-limit diagnostic at fixed mass. The left panel shows a background-only ensemble of observed CL_s upper limits, and the right panel shows the corresponding empirical cumulative distribution. The vertical line marks the observed limit. The lower-tail area defines p_{strong} , the upper-tail area defines p_{weak} , and $p_{two} = \min[1, 2 \min(p_{strong}, p_{weak})]$ is used only as a symmetric diagnostic summary.

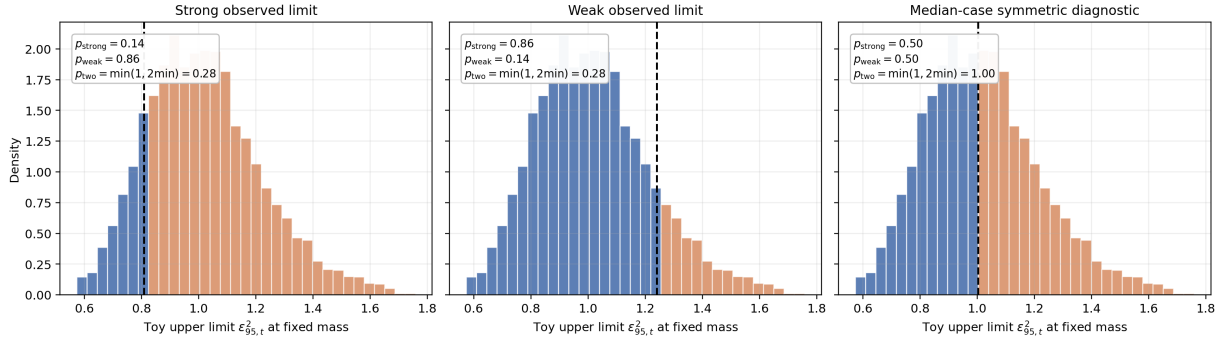


Figure 33: Representative fixed-mass observed-limit toy diagnostics. The left panel shows an unusually strong observed limit relative to the background-only toy ensemble, the middle panel shows an unusually weak observed limit, and the right panel shows a near-median case where the bounded symmetric summary saturates at $p_{two} = 1$. These are limit diagnostics, not discovery statistics.

These quantities are therefore limit diagnostics rather than discovery-significance measures in the sense of p_0 . If the observed limit lies near the median of the background-only ensemble, then roughly half of the toys lie above it and half lie below it, making both p_{strong} and p_{weak} naturally $\mathcal{O}(0.5)$. Their finite resolution is $1/N_{toys}$. In the three-campaign combined ensemble, $N_{toys} = 300$, so a one-count change is $1/300 \simeq 0.0033$ and a zero-count tail is reported as 0 of 300 rather than as an exact zero probability. These toy-limit diagnostics cannot by themselves establish or refute a high-significance local excess in the discovery statistic defined by the one-sided local p_0 mapping above.

4.14 Shared- ε^2 combination across datasets

A dark photon has one coupling, while each campaign has its own luminosity, selected prompt rate and mass resolution. At a fixed mass, a common $\psi \equiv \varepsilon^2$ therefore predicts different signal yields in the active datasets, as shown in Figure 34.

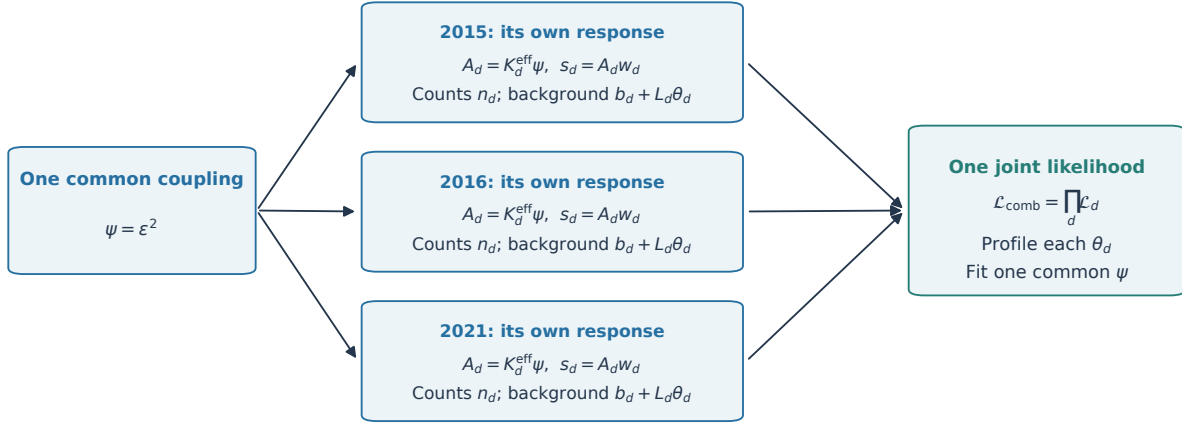


Figure 34: One coupling predicts a campaign-specific signal through its normalization and resolution. Each campaign retains its own counts and correlated background constraint; the likelihoods are multiplied and fitted with one common coupling. Only datasets whose declared search contains the tested mass contribute.

Let $\mathcal{D}(m)$ be that active set. We retain the signal template $\mathbf{w}_d$, count mean $\mathbf{b}_d$ and covariance factor L_d defined above. In the electron-channel convention of Eq. (49), the radiative-fraction penalty gives

$$K_d^{\text{eff}}(m) = (1 - \delta_{f,d})K_d(m), \quad (66)$$

$$\mathbf{s}_{\text{unit},d}(m) = K_d^{\text{eff}}(m)\mathbf{w}_d(m), \quad (67)$$

$$\boldsymbol{\lambda}_d(\psi, \boldsymbol{\theta}_d) = \mathbf{b}_d + L_d \boldsymbol{\theta}_d + \psi \mathbf{s}_{\text{unit},d}. \quad (68)$$

The visible-branching correction used above the dimuon threshold is stated separately below. The campaign likelihood and its product are

$$\mathcal{L}_d(\psi, \boldsymbol{\theta}_d) = \left[\prod_i \text{Pois}(n_{d,i} \mid \lambda_{d,i}) \right] \exp\left(-\frac{1}{2} \boldsymbol{\theta}_d^\top \boldsymbol{\theta}_d\right), \quad (69)$$

$$\mathcal{L}_{\text{comb}}(\psi, \{\boldsymbol{\theta}_d\}) = \prod_{d \in \mathcal{D}(m)} \mathcal{L}_d(\psi, \boldsymbol{\theta}_d). \quad (70)$$

The background nuisance vectors are independent between campaigns. No shared luminosity, radiative-fraction or resolution nuisance is profiled in this baseline combination. This is the usual common-parameter structure of a multi-channel likelihood [35, 39].

The implementation concatenates the count, background and unit-signal vectors and uses a

block-diagonal covariance. In compact notation,

$$\mathbf{n} = \text{col}_{d \in \mathcal{D}} \mathbf{n}_d, \quad \mathbf{b} = \text{col}_{d \in \mathcal{D}} \mathbf{b}_d, \quad (71)$$

$$\mathbf{s}_{\text{unit}} = \text{col}_{d \in \mathcal{D}} \mathbf{s}_{\text{unit},d}, \quad \boldsymbol{\theta} = \text{col}_{d \in \mathcal{D}} \boldsymbol{\theta}_d, \quad (72)$$

$$L_{\text{comb}} = \text{blockdiag}_{d \in \mathcal{D}} L_d, \quad \mathbf{C}_{\text{comb}} = L_{\text{comb}} L_{\text{comb}}^\top. \quad (73)$$

Here col stacks column vectors without combining their bins. The single vector mean is

$$\boldsymbol{\lambda}(\psi, \boldsymbol{\theta}) = \mathbf{b} + L_{\text{comb}} \boldsymbol{\theta} + \psi \mathbf{s}_{\text{unit}}. \quad (74)$$

Thus all campaigns are fitted at the same ψ , while the nuisance blocks respond separately to their spectra.

It is the *profiled negative log likelihoods* that add at fixed coupling:

$$\ell_{p,\text{comb}}(\psi) = \sum_{d \in \mathcal{D}(m)} \ell_{p,d}(\psi), \quad (75)$$

$$\hat{\psi}^{\text{unc}} = \arg \min_{\psi \in \mathbb{R}} \ell_{p,\text{comb}}(\psi), \quad \hat{\psi}^+ = \max(0, \hat{\psi}^{\text{unc}}). \quad (76)$$

The auxiliary signed fit is used for extraction diagnostics. For physical upper limits the bounded statistic is

$$\tilde{q}_\psi = \begin{cases} 0, & \hat{\psi}^{\text{unc}} > \psi, \\ 2 \sum_{d \in \mathcal{D}(m)} [\ell_{p,d}(\psi) - \ell_{p,d}(\hat{\psi}^+)], & \hat{\psi}^{\text{unc}} \leq \psi, \end{cases} \quad \psi \geq 0. \quad (77)$$

Every denominator is evaluated at the *same common best fit* $\hat{\psi}^+$. Adding likelihood ratios normalized to separate campaign best fits would define a different statistic. Likewise, neither averaging upper limits nor adding local significances performs this fit. The shared-coupling upper limit satisfies

$$\varepsilon_{\text{UL}}^2(m; \alpha) = \min\{\psi \geq 0 : \text{CL}_s(\psi; m) \leq \alpha\}, \quad \alpha = 0.10. \quad (78)$$

The combined local excess test uses this same profiled objective at zero and at $\hat{\psi}^+$. Its gain comes from the information supplied by independent campaigns about one physical parameter, while retaining each campaign's normalization and background uncertainty.

4.15 Global significance and the look-elsewhere treatment

The most extreme local excess in a declared scan is summarized by its minimum one-sided p_0 or maximum q_0 ,

$$p_{0,\min}^{\text{obs}} = \min_m p_0^{\text{obs}}(m), \quad q_{0,\max}^{\text{obs}} = \max_m q_0^{\text{obs}}(m). \quad (79)$$

The cleanest global-significance definition is then scan based:

$$p_{\text{global}}^{\text{toy}} = \frac{1}{N_{\text{toys}}} \sum_{t=1}^{N_{\text{toys}}} \mathbf{1}[q_{0,\max}^{(t)} \geq q_{0,\max}^{\text{obs}}], \quad (80)$$

with

$$Z_{\text{global}}^{\text{toy}} = \Phi^{-1}(1 - p_{\text{global}}^{\text{toy}}). \quad (81)$$

The normal-quantile conversion requires a resolved tail probability. With zero exceedances, the result is the one-sided 95% sampling bound $p_{\text{global}} < 1 - 0.05^{1/N_{\text{toys}}}$; an infinite significance is

not assigned. Section 4.20 applies this convention to the conditional full-field calculation. Earlier versions used an effective-trials approximation based on the scan span and detector resolution as an analytic look-elsewhere reference [40]:

$$p_{\text{global}}^{\text{Sidak}} = 1 - \left(1 - p_{0,\text{min}}^{\text{obs}}\right)^{N_{\text{eff}}}, \quad Z_{\text{global}}^{\text{Sidak}} = \Phi^{-1}\left(1 - p_{\text{global}}^{\text{Sidak}}\right). \quad (82)$$

For small $p_{0,\text{min}}^{\text{obs}}$, Eq. (82) reduces to the familiar $p_{\text{global}} \simeq N_{\text{eff}} p_{0,\text{min}}$ form.

This effective-trials expression is retained as a historical heuristic. A scan span and resolution scale do not by themselves model the nonstationary correlations caused by the moving GP exclusion or changing campaign membership. The conditional full-field calculation in Section 4.20 evaluates those correlations explicitly. Neither the heuristic nor additional Gaussian field sampling replaces validation of the generating background and the accessible tails with coherent complete Poisson scans.

4.16 Signal absorption and alternative background-profile treatments

We describe blind-window background uncertainty with additive Gaussian-constrained nuisance parameters derived from the GP mean and covariance. Normalizing the signal over its full line shape removes the bias from blind-window renormalization, but GP refitting or the nuisance model can still absorb part of an injected signal.

The most likely residual sources are the refitting freedom of the GP hyperparameters, especially the effective length scale, and the flexibility of the Gaussian-profile background nuisance itself. We therefore retain fixed-background and alternative Poisson treatments as targeted robustness tests if matched pseudoexperiments reveal unacceptable absorption.

4.17 Dimuon-threshold interpretation

The yield-to-coupling conversion in Eq. (47) is an electron-channel normalization: the fitted resonance amplitude is the $A' \rightarrow e^+e^-$ yield visible in the prompt invariant-mass spectrum. For the minimal kinetically mixed dark photon, this is also the full visible branching interpretation below the dimuon threshold, $m_{A'} < 2m_\mu = 211.3167 \text{ MeV}$, where the $\mu^+\mu^-$ final state is kinematically closed. This is the same fixed-target convention used in the published HPS prompt searches [2, 3, 4]. Above threshold, the same electron-channel yield corresponds to a weaker minimal-model limit on ε^2 because the total width includes the newly open $A' \rightarrow \mu^+\mu^-$ mode [1, 41].

Neglecting the electron mass and remaining below the two-pion hadronic threshold, the charged-lepton width ratio is

$$R_{\mu/e}(m_{A'}) \equiv \frac{\Gamma(A' \rightarrow \mu^+\mu^-)}{\Gamma(A' \rightarrow e^+e^-)} = \sqrt{1 - \frac{4m_\mu^2}{m_{A'}^2}} \left(1 + \frac{2m_\mu^2}{m_{A'}^2}\right), \quad m_{A'} > 2m_\mu, \quad (83)$$

with $R_{\mu/e} = 0$ below threshold. The inverse electron branching fraction used for the minimal-model reinterpretation is therefore

$$N_{\text{eff}}^{\text{BR}}(m_{A'}) \equiv \frac{1}{\mathcal{B}(A' \rightarrow e^+e^-)} = 1 + R_{\mu/e}(m_{A'}), \quad (84)$$

and an electron-channel coupling limit is mapped to the minimal visible-dark-photon convention by

$$\varepsilon_{\text{min}}^2(m_{A'}) = N_{\text{eff}}^{\text{BR}}(m_{A'}) \varepsilon_{ee}^2(m_{A'}). \quad (85)$$

This operation does not change the local fit, the signal-yield upper limit, the CL_s value, or the local p_0 scan; it changes only the coupling interpretation of an electron-channel result. Figure 35 shows the factor applied in the 90% CL result obtained with CL_s . The factor is unity below $2m_\mu$, rises continuously above threshold, and reaches $N_{\text{eff}}^{\text{BR}} = 1.725$ at 250 MeV ($\mathcal{B}_{ee} = 0.580$).

The mass range of a scan-global discovery statement must equal its declared search range [40]. A branching-fraction conversion changes the physical interpretation of the coupling, not that search definition. Pointwise CL_s exclusion curves are not multiplied by a look-elsewhere trials factor.

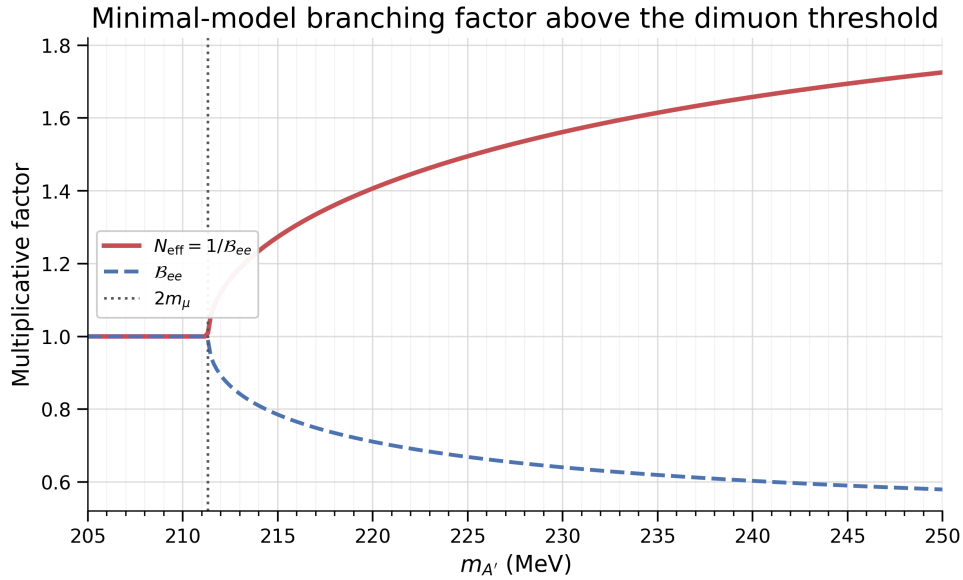


Figure 35: Minimal-model branching factor used to reinterpret electron-channel ε^2 limits above the dimuon threshold. The red curve shows $N_{\text{eff}}^{\text{BR}} = 1/\mathcal{B}(A' \rightarrow e^+e^-)$ and the blue dashed curve shows $\mathcal{B}(A' \rightarrow e^+e^-)$.

With the observable, background prediction, signal model, and likelihood specified, the remaining question is empirical: does this procedure create a false narrow signal, misstate its uncertainty, or absorb a signal that is actually present? We answer that question with pseudoexperiments in the next section.

4.18 Why the background uncertainty is profiled

The background underneath an excluded mass window is inferred from its sidebands. Even with large event counts, that interpolation has uncertainty and can respond coherently across several bins. The signal fit therefore allows the background to change within the correlated uncertainty predicted by the GP. Treating the predicted mean as exactly known would attribute all remaining variation to counting statistics or to the signal.

Section 4.8 defines the correlated Gaussian deformation and the objective minimized at every signal yield. Its Gaussian penalty propagates the sideband prediction uncertainty while the reviewed GP kernel coordinates remain fixed during the local fit. The comparisons below test the effect of this choice; qualification of the support, mask and kernel policy remains a separate part of the analysis.

The controlled 2021 comparison also fitted a positive latent log-GP intensity, using the same data and signal model. Its comparison with the nominal correlated Gaussian profile tests the

approximation to the background distribution. A third fit fixed the GP count mean, setting $L\theta = 0$. Figure 36 retains these three choices. A smaller fixed-background observed limit does not establish an improvement in sensitivity: the treatment has also removed a source of uncertainty.

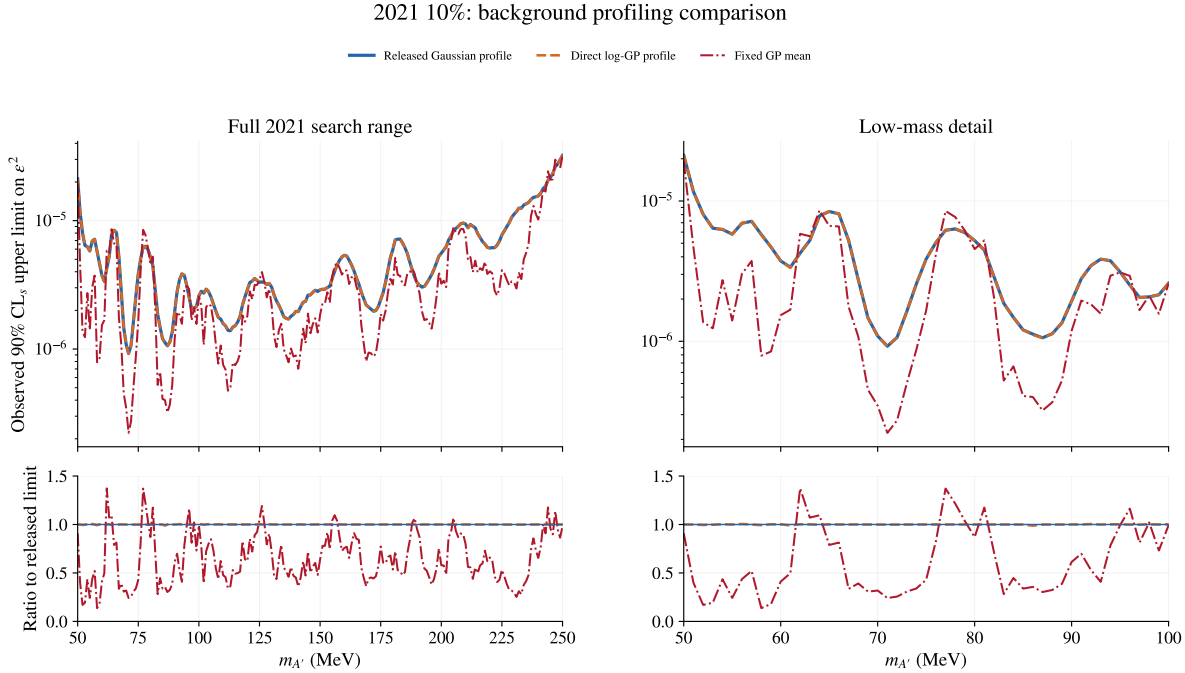


Figure 36: Controlled comparison of observed 2021 10% upper limits using the nominal correlated Gaussian background profile, a positive log-GP profile, and a fixed GP mean. All curves use the same observed counts and frozen analysis settings. The dimuon branching correction is applied once above threshold. The fixed-mean curve omits background-estimation uncertainty; its smaller uncalibrated limit is assessed against the injection and calibration studies below. Adapted from Fig. 1 of the v4.9.13 calibration report.

4.19 What the signal injections establish

The dedicated comparison generated 500 spectra at each of six masses, 65, 71, 78, 100, 182 and 231 MeV, for each tested background construction and injection $A_{\text{true}} = 0, 2\sigma_{\text{ref}}, 5\sigma_{\text{ref}}$. Here σ_{ref} is the profiled Fisher reference error fixed before generation. Thus a five-reference-error injection means a fixed physical event yield within that ensemble, rather than a guarantee of a measured 5σ significance. Both methods fit the same spectra and physical signals; different injection strengths use separate ensembles.

Three constructions separate different questions. A known-background control tests the fitting machinery when the fixed assumption is exact. A second construction varies the background with the nominal GP covariance and tests whether that uncertainty is correctly propagated. A third fluctuates a complete smooth spectrum and retraining the sidebands, including signal tails outside the moving mask. The latter spectrum was obtained from data outside 60–86 MeV using the frozen 66 MeV kernel. It is a selected response diagnostic, and does not independently define the physical background family.

When the nominal GP uncertainty is present in generation, the fixed fit has background-only pull widths of 1.85–2.65 and excludes the true stronger injection in 22.0–32.2% of trials. The

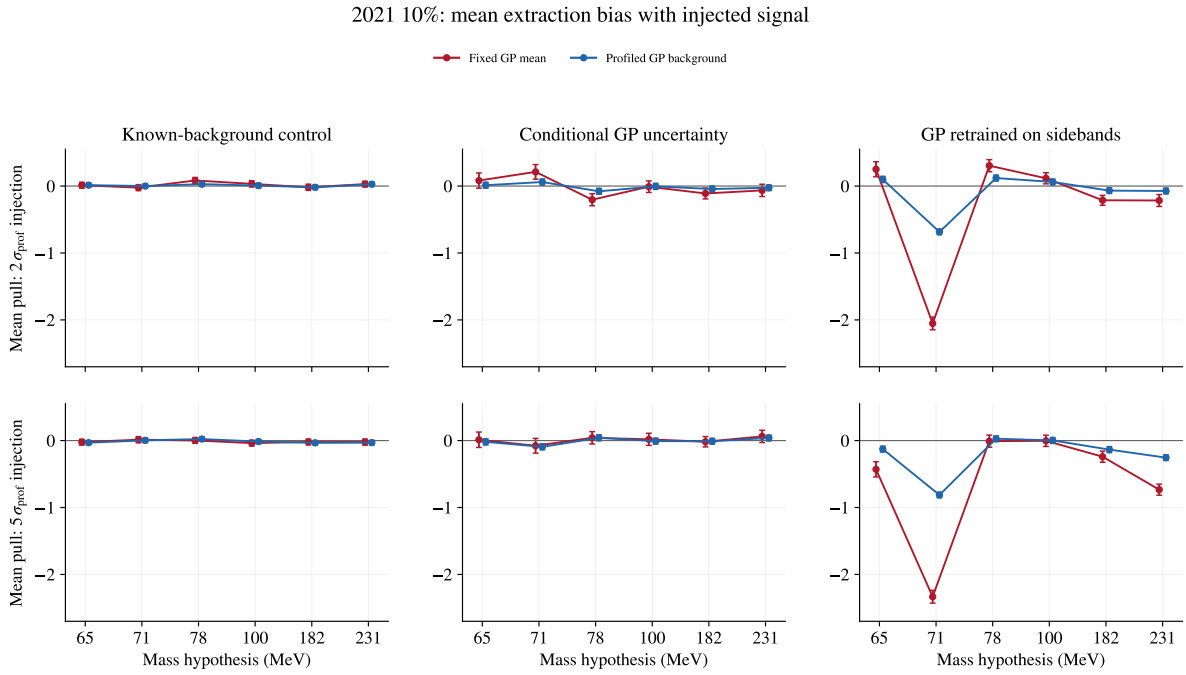


Figure 37: Mean signal-extraction pull for injections of $2\sigma_{\text{ref}}$ and $5\sigma_{\text{ref}}$, where the profiled Fisher reference error fixes the event yield before generation. Each mass–strength–background cell contains 500 spectra, shared by the fixed and profiled methods. Pull denotes $(\hat{A} - A_{\text{true}})/\hat{\sigma}_A$; bars are Monte Carlo errors on its mean. At 71 MeV, a signal injected at 71 MeV into the retrained smooth-background construction has mean pull -2.33 for the fixed fit and -0.81 for the profiled fit at the stronger injection. Profiling reduces this conditional offset below one fitted standard error, but the associated exclusion test still fails. Adapted from Fig. 9 of the v4.9.13 calibration report.

corresponding profiled exclusion fractions are 8.2–12.2%. Profiling is therefore the better motivated baseline for 2021. The retrained 71 MeV example also identifies a limitation: the fixed and profiled fits exclude the true stronger yield in 349/500 and 145/500 trials, respectively. A mean pull smaller than one does not by itself ensure a valid 90% exclusion rule.

The proposed primary presentation retains the profiled asymptotic CL_s construction, whose practical signal-recovery tests motivate its use. Its final qualification must also account for the failures exposed by repeated-analysis calibration. Appendix B explains this alternative probability calculation and its observed effects, which are especially large in part of the 2016 scan. Retaining a primary presentation is a methodological proposal for the unblinding review; it does not turn conditional recovery into uniform coverage or resolve the outstanding 2016 numerical exception.

4.20 A GP calculation of scan-global significance

A local probability describes a fluctuation at a fixed mass. A global probability asks how often a complete background experiment would give an equally extreme result anywhere in the declared search. Neighboring hypotheses reuse events and sidebands, so a fluctuation can affect several fits. The method of Ananiev and Read [42] estimates the correlation of the local-significance field from controlled background perturbations, then generates many inexpensive correlated fields.

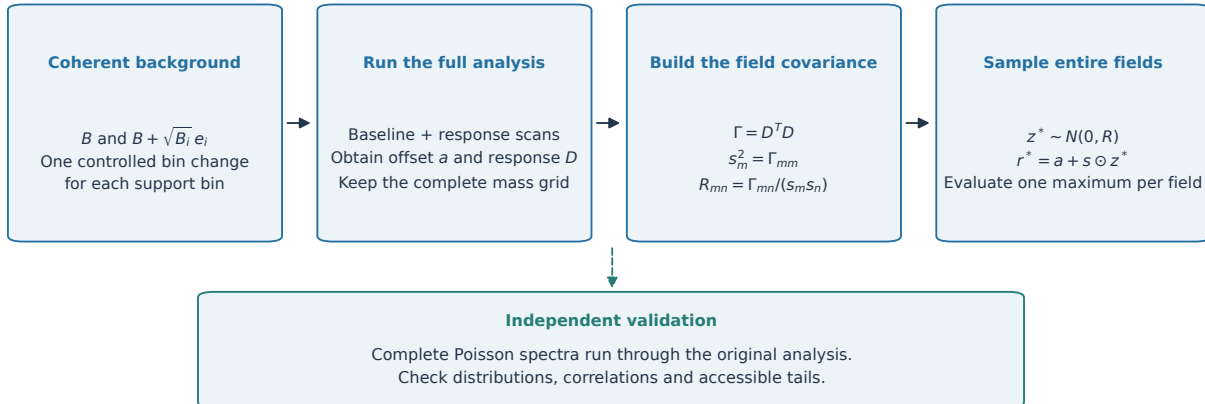


Figure 38: The computational gain in the significance-GP method. Full analysis scans of controlled perturbations estimate the field covariance once. Gaussian vector draws then produce many correlated scans without repeating those fits. A separate ensemble of complete Poisson experiments tests the approximation. This is an explanatory diagram of the archived HPS implementation, following Ref. [42]; the offset and width are retained explicitly.

The background-interpolation GP predicts counts inside a mass window. The significance GP instead approximates how the *fit result varies across tested masses* in repeated background experiments. Let ψ be the single-campaign yield or common coupling and let ℓ_p be its profiled negative log likelihood. The signed local coordinate is

$$r(\mathbf{n}; m) = \text{sgn}(\hat{\psi}^{\text{unc}}) \sqrt{2[\ell_p(0; m) - \ell_p(\hat{\psi}^{\text{unc}}; m)]}. \quad (86)$$

It retains both excesses and deficits. The conventional one-sided asymptotic display uses

$$Z_{\text{local}} = \max(0, r), \quad p_0 = 1 - \Phi(Z_{\text{local}}). \quad (87)$$

A negative r therefore has zero excess significance, with $p_0 = 0.5$ under this display convention.

From the analysis response to a covariance. For one coherent expected spectrum $\mathbf{B}$, first evaluate the unfluctuated scan. Then increase one support bin by its Poisson standard deviation and repeat the complete scan:

$$a_m = r(\mathbf{B}; m), \quad (88)$$

$$D_{im} = r(\mathbf{B} + \sqrt{B_i} \mathbf{e}_i; m) - a_m. \quad (89)$$

Here $\mathbf{e}_i$ changes only bin i . The offset a_m is the response to the deterministic reference spectrum; it need not equal the exact mean of the Poisson-fit distribution. Each row of D represents one independent noise direction, not one random Monte Carlo experiment. The covariance, marginal width and correlation are

$$\Gamma_{mn} = \sum_i D_{im} D_{in}, \quad (90)$$

$$s_m = \sqrt{\Gamma_{mm}}, \quad (91)$$

$$R_{mn} = \frac{\Gamma_{mn}}{s_m s_n}. \quad (92)$$

There is no division by the number of support bins. To first order, $D_{im} \simeq \sqrt{B_i} \partial r_m / \partial n_i$, so the sum propagates independent Poisson-bin noise through the full fitting response. The matrix R is estimated from that response; no RBF kernel is imposed on the significance field.

Correlated simulations and one declared ordering. A simulated field has

$$\mathbf{z}^* \sim \mathcal{N}(\mathbf{0}, R), \quad r_m^* = a_m + s_m z_m^*. \quad (93)$$

The archived extension retains a nonzero offset and nonunit width rather than assuming an exactly centered, unit-normal marginal distribution. Subtracting a_m when constructing the response centers the covariance; it does not subtract events from the observed spectrum.

The principal ordering standardizes each coordinate and keeps only positive raw signal fits:

$$z_m = \frac{r_m - a_m}{s_m}, \quad (94)$$

$$t_m = \begin{cases} z_m, & r_m > 0, \\ -\infty, & r_m \leq 0, \end{cases} \quad p_m^{\text{cond}} = \begin{cases} 1 - \Phi(z_m), & r_m > 0, \\ 1, & r_m \leq 0. \end{cases} \quad (95)$$

For the fixed full search grid $\mathcal{M}$, define

$$T = \max_{m \in \mathcal{M}} t_m, \quad p_{\text{global}} = \Pr(T^* \geq T_{\text{obs}}). \quad (96)$$

If no fitted signal is positive, $T = -\infty$. For N complete simulated fields, the estimated tail is

$$\hat{p}_{\text{global}} = \frac{k}{N}, \quad k = \sum_{j=1}^N \mathbf{1}[T_j^* \geq T_{\text{obs}}]. \quad (97)$$

At zero exceedances the one-sided 95% sampling bound is $1 - 0.05^{1/N}$; no zero probability is assigned. A pointwise global curve uses each observed t_m as the threshold against the *same full-grid* maximum distribution.

A separate archived ordering uses $U = \max_m \max(0, r_m)$. It tests the largest raw fluctuation, whereas T ranks positive fits relative to the chosen background’s offset and width. These probabilities cannot be interchanged after inspecting their values. The positive-fit gate also explains why the conditional convention gives one for nonpositive fits, while the nominal asymptotic plot gives 0.5.

Each individual study began with ten complete Poisson scans, followed by 1,000 independently generated complete scans and 200,000 GP fields. Every mass in one scan uses the same full spectrum. Sideband targets, count-dependent errors and posterior predictions are updated; the reviewed kernel coordinates remain fixed. Independently generated pointwise toy collections cannot be joined by their row numbers to reconstruct this correlation. The 2015, 2016 and 2021 grids contain 72, 142 and 201 integer mass hypotheses over 19–90, 39–180 and 50–250 MeV, respectively.

The combined study evaluates the actual shared-coupling likelihood over 232 integer masses from 19 to 250 MeV. The active datasets are 2015 at 19–38 MeV, 2015+2016 at 39–49 MeV, all three at 50–90 MeV, 2016+2021 at 91–180 MeV, and 2021 at 181–250 MeV. Independent constituent random streams are paired into 1,000 joint experiments, with each whole experiment preserved across the grid. The response basis uses 1,626 support bins and 1,627 deterministic scans. Drawing one joint field preserves correlations across membership boundaries; combining separate p-values or treating segments as independent would define another test.

Profiled scan	Selected mass [MeV]	GP exceedances out of 200,000	GP global p	Direct exceedances out of 1,000
2015	51	3,797	0.018985	13
2016	42	0	$< 1.498 \times 10^{-5}$	0
2021 10%	92	5,038	0.02519	26
Combined union	76	0	$< 1.498 \times 10^{-5}$	0

Table 5: Existing conditional significance-GP diagnostics under the archived coherent stress background, using the principal ordering in Eq. 96. Zero GP exceedances give the displayed one-sided 95% sampling bound; zero direct exceedances give only $p < 0.002991$. These bounds condition on their respective simulation models. The table reports neither a validated rare particle tail nor a global calibration of the two-truth envelope in Appendix B.

The latest probability audit gives a particularly instructive combined example. At 76 MeV the observed $r = 0.166$ corresponds to an asymptotic local $p_0 = 0.434$. The stress background predicts $a_m = -8.700$ with width 0.979, so its offset-adjusted score is 9.053. This large score measures a mismatch with that selected background reference. It cannot be reported as a 9σ particle signal. Under the alternative raw ordering, every simulated scan exceeds the observed maximum and the estimated global probability is one. Neither comparison is a general goodness-of-fit test.

Direct scans check central distributions, widths, correlations and accessible maximum tails. In the 2016 profiled study one unadjusted bulk-distribution comparison has nominal $p = 0.034$, reinforcing the need to inspect approximation discrepancies. Agreement elsewhere and the absence of adjusted marginal flags do not validate the unresolved rare tails. The 2016 stress shape also retains source-fit and numerical qualifications discussed in Appendix B.

Remaining requirements. Final particle-global interpretation requires a justified background family and kernel policy, independent validation of local probabilities, and adequate simulation of the complete mass and model search. Pending those checks, Table 5 reports conditional diagnostics and sampling bounds.

5 Background model validation with pseudo-experiments

The full-refit ensembles described in this section are historical validation studies of source and exposure choices. Later calibration and global studies retain the observed kernel coordinates while retraining the GP mean and count-dependent errors; those are distinct ensemble definitions.

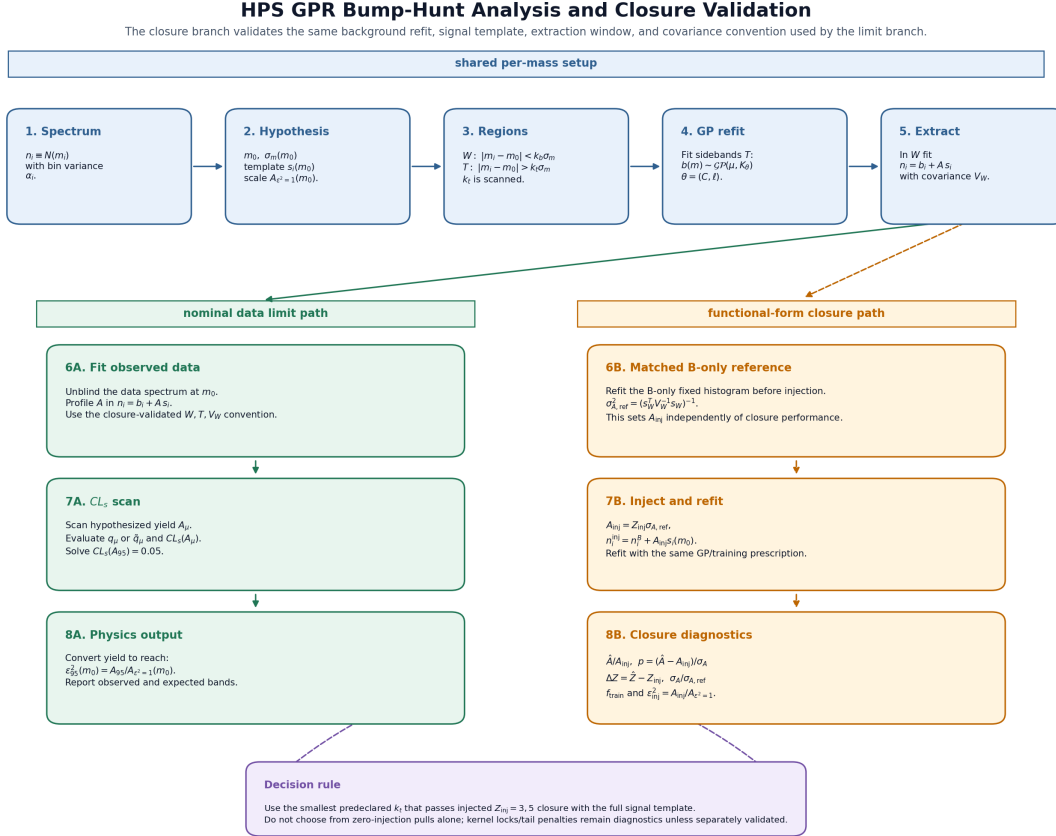


Figure 39: Analysis and validation flow. In the data path, sidebands around each mass hypothesis train the GP and the blinded bins enter the signal-plus-background likelihood. In the validation path, a smooth source-fitted mean is Poisson fluctuated, a known signal may be added, and the complete GP fit and extraction are repeated. The zero-injection ensemble tests spurious signal; nonzero injections test signal recovery and uncertainty calibration. These are extraction-closure tests, not confidence-limit coverage or a scan-wide significance calibration.

5.1 What the validation asks

Two questions drive the validation. If a smooth, data-like spectrum contains no resonance, does the GPR sideband fit return a spurious signal? If we add a known signal, does the same procedure recover it with an uncertainty of the right scale? For each pseudoexperiment we Poisson-fluctuate a smooth function fitted independently of the GP, refit the GP, and extract the signal. These tests exercise the count-space prediction, the mass-dependent blind and training masks, the full-range Gaussian signal template, the matched-reference injection convention, and a fresh GP optimization

for every pseudo-spectrum. The statistical definitions are given in Section 4.12.

Each generating function is a smooth stress model fitted to an HPS source spectrum; we do not treat it as a unique physical parameterization of the trident continuum. Any closure statement is therefore conditional on that model and the stated extraction settings, and says nothing by itself about bias in the observed data. We use four exposure ensembles:

- $1\% \times 10$: the fitted native-1% expectation scaled by ten, giving 10%-equivalent statistics;
- native 10%: an independently fitted expectation from the available 10% source sample;
- $1\% \times 100$: the native-1% expectation scaled by one hundred, giving full-sample-equivalent statistics;
- $10\% \times 10$: the native-10% expectation scaled by ten, also giving full-sample-equivalent statistics.

The last two ensembles test whether a proposed background model and its numerical settings remain suitable at the statistical scale of the intended 100% analysis. They inform the model that must be fixed before final unblinding; they do not use the unreleased 2021 events or by themselves complete that qualification. Because the two source samples differ in selection and shape, equal nominal exposure does not make them interchangeable.

5.2 Optimizing the blind window

The GP must not learn from the core or near tails of the signal it is later asked to test. A Gaussian signal retains non-negligible weight outside a $1.64\sigma_m$ central interval, so training immediately beside that interval can let the background fit absorb signal-like counts and reduce the recovered amplitude. We varied the sideband boundary without using an observed limit to guide the choice.

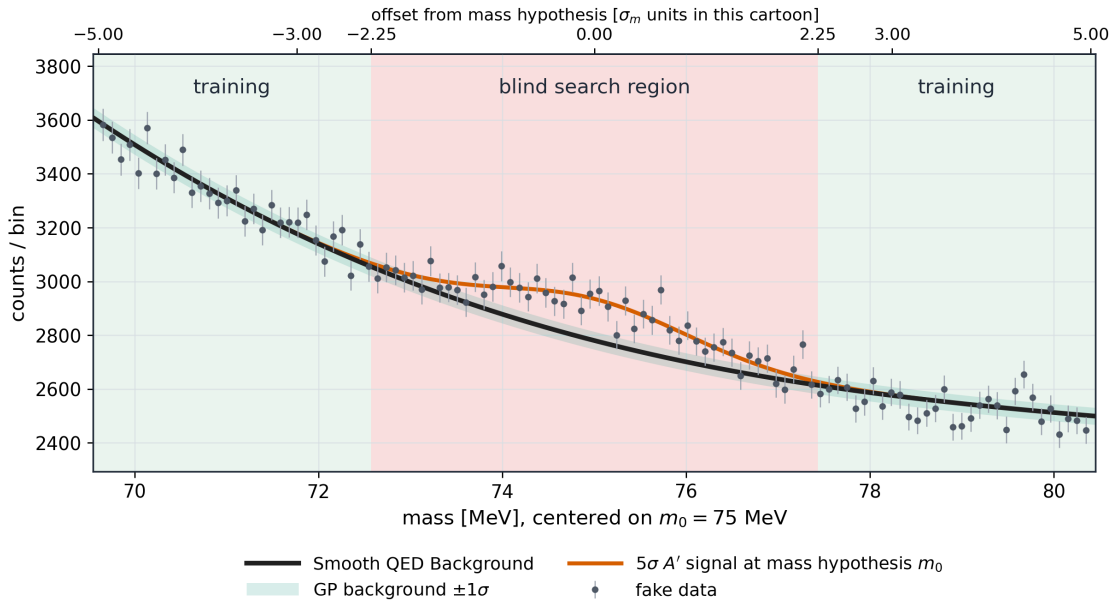


Figure 40: Selected $2.25\sigma_m$ blind and training-exclusion geometry. The GP is trained outside the widened interval, while all bins inside it enter the local signal-plus-background likelihood. This retains the signal tails for extraction without allowing those same bins to influence the sideband fit.

k_{train}	median expected ε_{95}^2 proxy	$\langle \hat{A}/A_{\text{inj}} \rangle_{z=3}$	$\langle \hat{A}/A_{\text{inj}} \rangle_{z=5}$	$\max f_{\text{train}}$
1.96	7.92×10^{-5}	0.730	0.715	0.0530
2.25	9.53×10^{-5}	0.961	0.943	0.0258
2.58	1.08×10^{-4}	1.052	1.036	0.0106
3.00	1.19×10^{-4}	1.049	1.037	0.00288

Table 6: Historical 2015 matched-reference study with a $1.64\sigma_m$ extraction window and variable training exclusion. The train-2.25 row substantially reduces the fraction of the signal template available to the GP while recovering the injected amplitude and retaining useful local precision. The median ε_{95}^2 entries are Gaussian-equivalent comparison proxies, not final CL_s limits.

The widened $2.25\sigma_m$ interval is used for both the training exclusion and the extraction likelihood. Compared with a $1.64\sigma_m$ extraction plus a separate 1.64 – $2.25\sigma_m$ guard band, this construction gives comparable closure while retaining the near signal tails in the likelihood. The background and signal fits can then use the same mass-dependent geometry.

The historical comparisons in Figs. 41–44 show the broader evidence behind that choice. Recovery and residual significance measure signal absorption; pull means and widths test the estimated yield and uncertainty. After the training exclusion was fixed, Fig. 45 compared the narrow extraction with a separate guard against the widened extraction. The widened window kept similar conditional closure while retaining signal tails in the fit.

5.3 Generating spectra and the role of functional forms

We generate the background pseudoexperiments independently of the GP used for their extraction. A positive analytic intensity is fitted to a native HPS mass spectrum, normalized on the specified toy support, integrated over every histogram bin, and Poisson fluctuated. A complete signal template is then added when $z > 0$. Every pseudo-spectrum is subsequently analyzed with a new GP sideband fit. Generating the toys with one model and fitting them with another is essential here: toys drawn from the same GP used in the extraction would provide a much weaker test of background misdescription.

The functional forms are explicit so that the generating assumptions can be distinguished from the GP used to analyze the toys. Let m be in GeV, $S(m; t, w) = [1 + \exp(-(m - t)/w)]^{-1}$, and $u_d = (m - m_{\text{mid},d})/m_{\text{scale},d}$. The 2015 and 2016 year-matched seeds ($d = 2015, 2016$) and the 2021 seed are

$$B_d(m) = N_d S(m; t_d, w_d) (m - m_{0,d})^{a_d} \exp \left[-\frac{m - m_{0,d}}{\theta_d} + c_{1,d} u_d + c_{2,d} u_d^2 \right], \quad (98)$$

$$B_{2021}(m) = N_{21} S(m; t_{21}, w_{21}) m^{a_{21}} \exp \left[-\frac{m}{\theta_{21}} + c_{1,21} m + c_{2,21} m^2 \right]. \quad (99)$$

Equation 98 is zero for $m \leq m_{0,d}$; Eq. 99 is defined for $m > 0$. The widths and decay scales are positive. The midpoint and scale in u_d are fixed by the source-fit interval; normalization N_d , turn-on parameters, powers and curvature coefficients are fitted separately to each source. These are the archived **fShiftSigPowTail** family for 2015 and 2016 and **fSigPowExpQ** family for 2021 1%. They are empirical smooth intensities, not unique continuum models.

The four-exposure study instead used independently fitted members of **fGenGammaThresh**,

$$G_d(m) = N_d S(m; t_d, w_d) (m - m_{0,d})^{a_d} \exp \left[-\left(\frac{m - m_{0,d}}{\lambda_d} \right)^{p_d} \right], \quad (100)$$

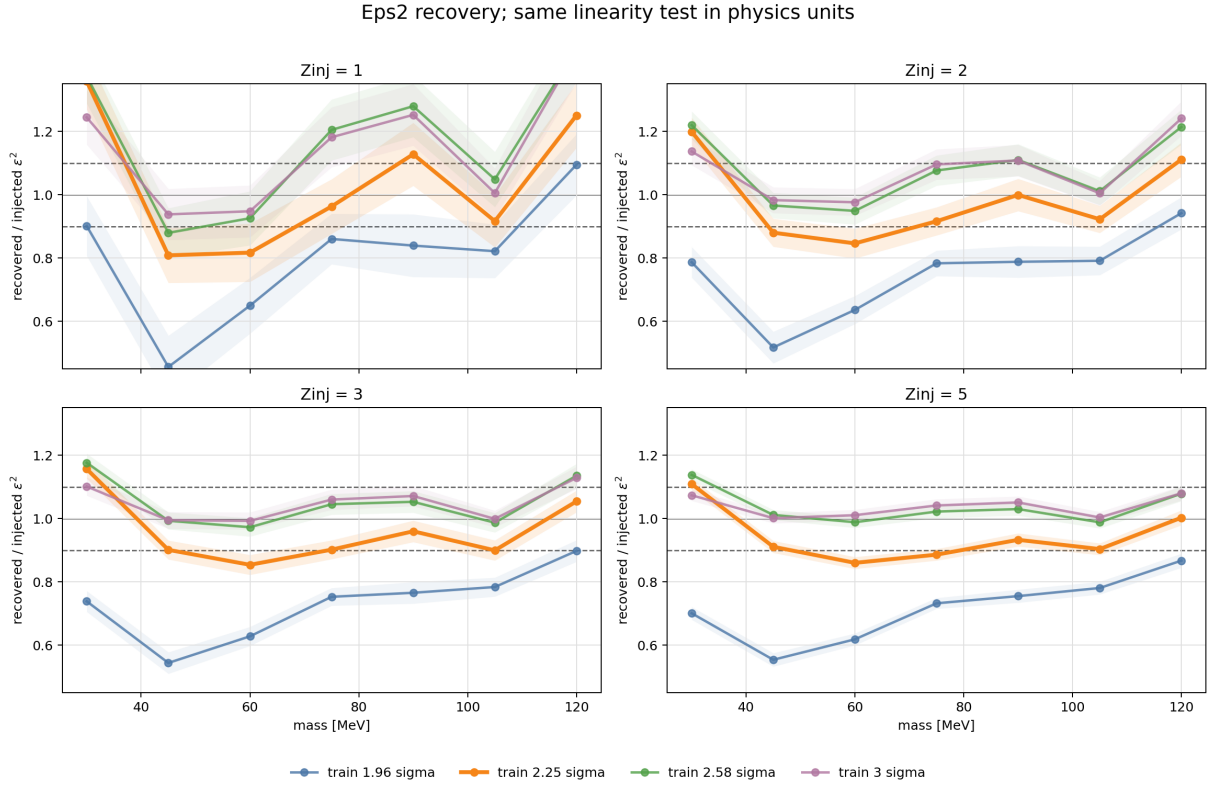


Figure 41: Historical 2015 matched-reference training-exclusion comparison, restored from the v4.9.7 validation archive. The extraction width was $1.64\sigma_m$, while the training exclusion was varied through 1.96, 2.25, 2.58 and $3.00\sigma_m$. Recovered coupling coordinates are divided by the injected values. The $2.25\sigma_m$ choice substantially reduces under-recovery relative to the narrow guard, while avoiding the larger uncertainty cost of wider exclusions. The plotted spreads belong to the archived ensembles, not a new confidence-limit calibration.

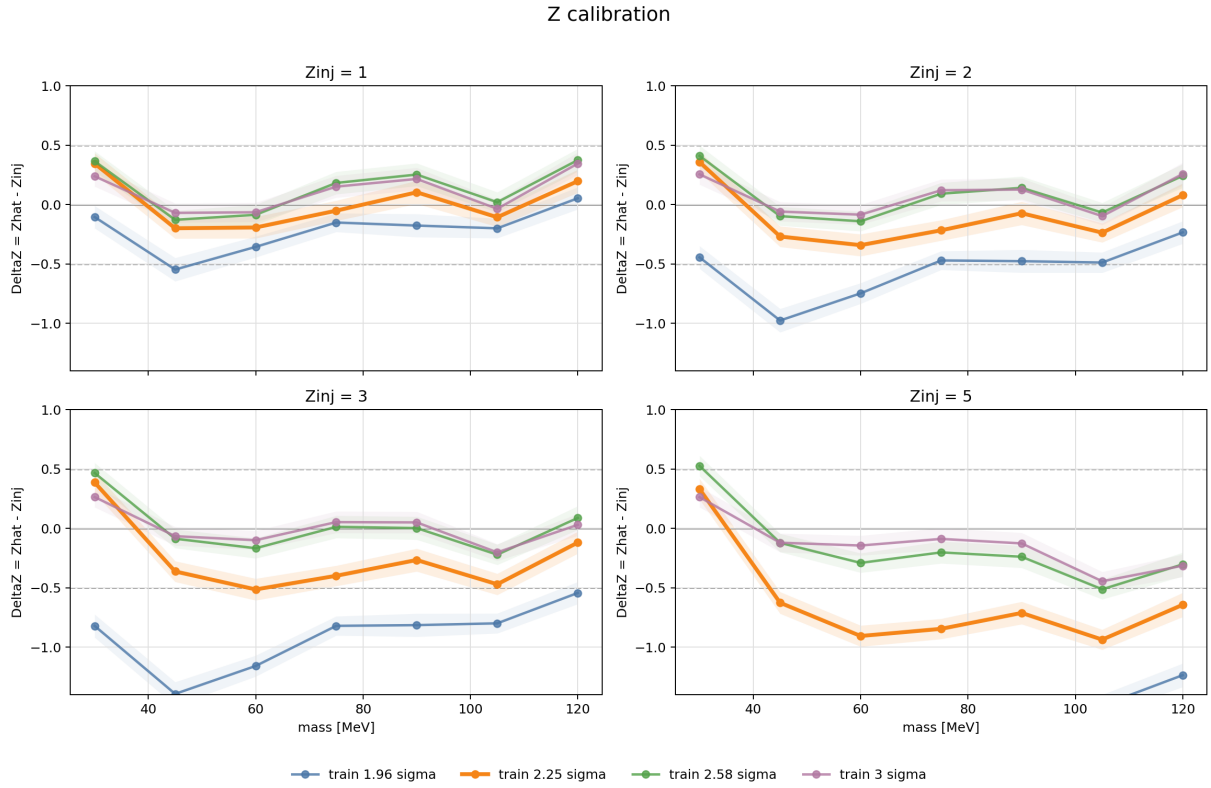


Figure 42: Residual fitted significance in the same historical 2015 training-exclusion study as Fig. 41. The common mass and injection grid compares the response after the signal strength has been set by its matched reference. Together with amplitude recovery, this diagnoses the signal-absorption cost of letting the GP train too close to the extraction window.

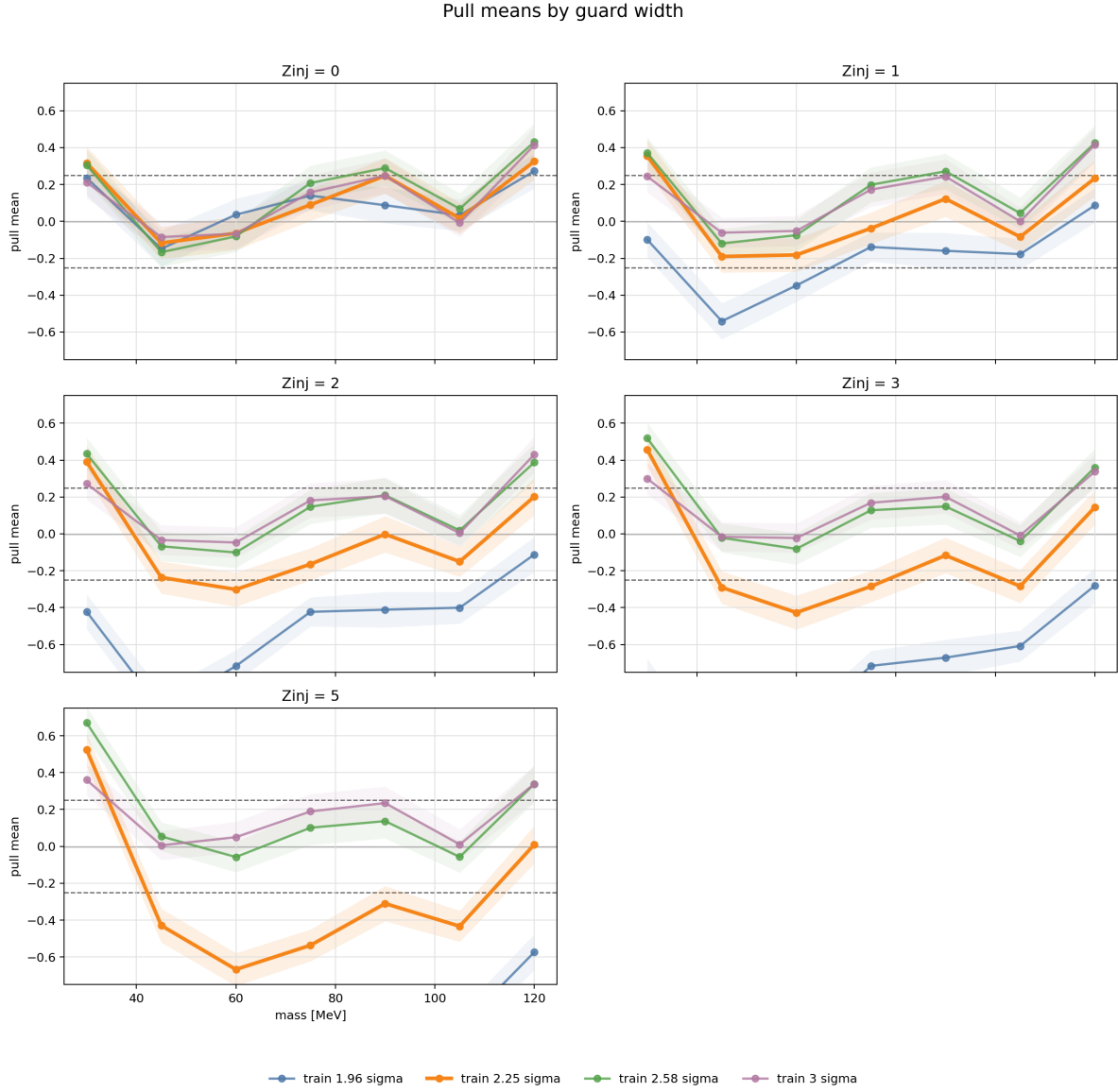


Figure 43: Signal-extraction pull means from the same historical training-exclusion comparison. The mass-dependent offsets supplement the recovery diagnostics in Figs. 41 and 42. These are historical conditional response tests for the stated generating truth and masks.

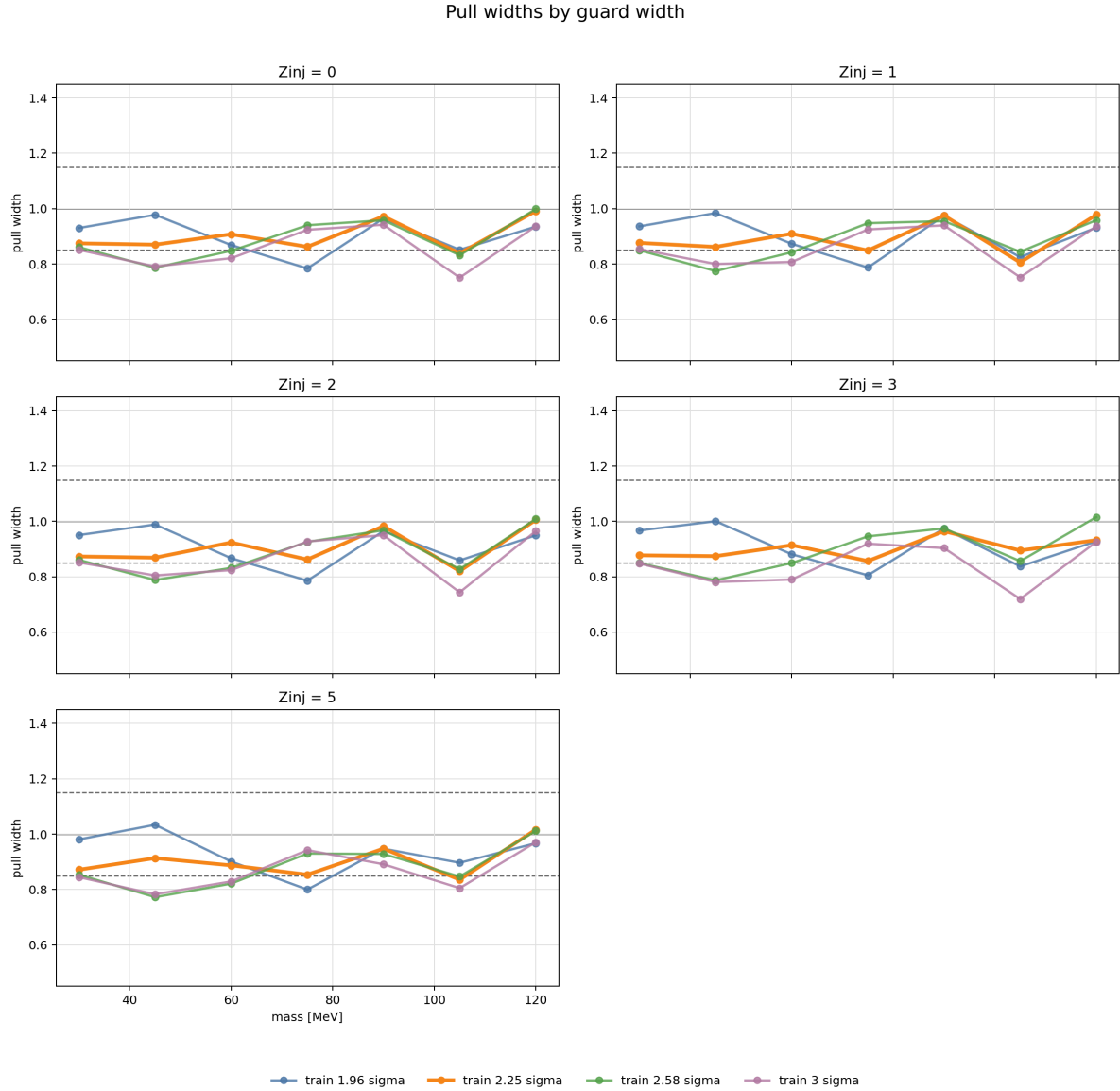


Figure 44: Signal-extraction pull widths from the same historical training-exclusion comparison. They test the reported uncertainty scale alongside the pull means in Fig. 43. The recovery, mean and width diagnostics informed the $2.25\sigma_m$ training choice; the subsequent extraction-geometry comparison is shown separately below.

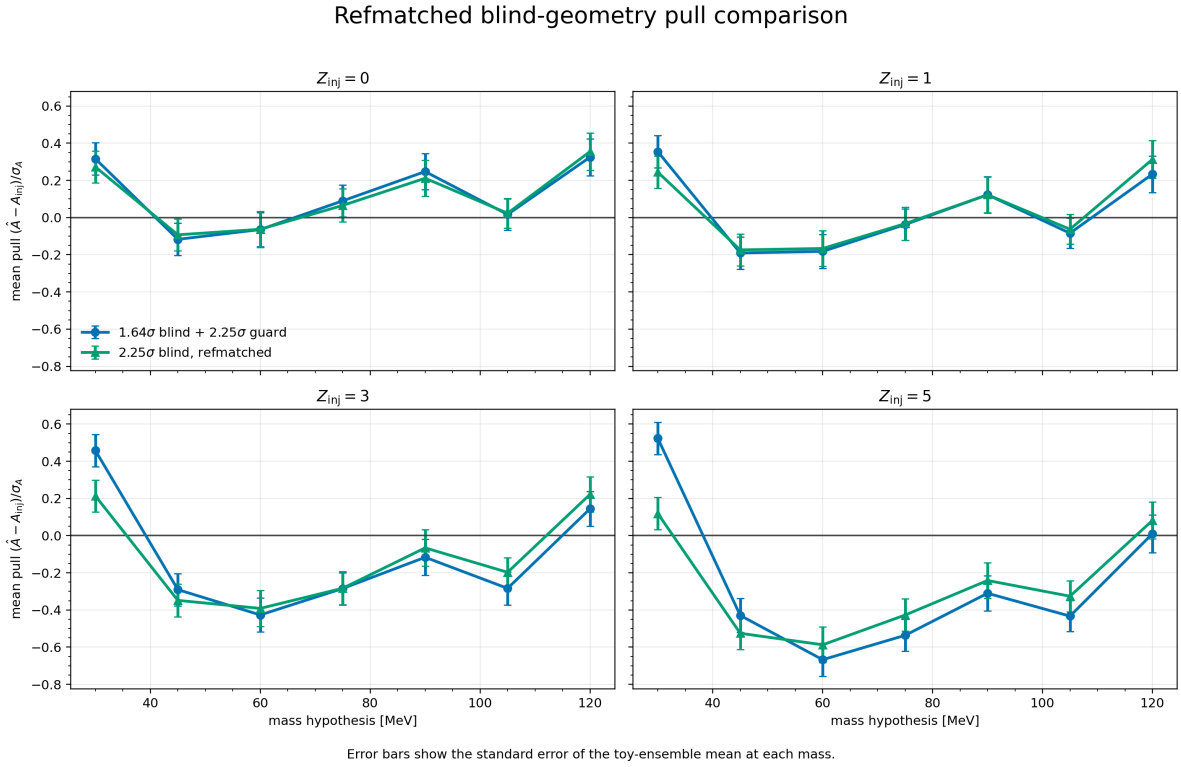


Figure 45: The subsequent matched-reference comparison of two extraction geometries in the same 2015 functional-form study. Both exclude training bins out to $2.25\sigma_m$; one retains a $1.64\sigma_m$ extraction and a separate guard, while the other fits all bins inside $2.25\sigma_m$. The $z = 0, 1, 3, 5$ pull means and widths show similar conditional behavior. Together with the retained signal tails, this supported the widened blind/extraction window. Historical numerical values are unchanged.

defined for $m > m_{0,d}$ and $\lambda_d, p_d, w_d > 0$, with zero intensity below $m_{0,d}$. Its parameters are refitted independently to the native 1% and native 10% sources, then held fixed while their expected counts are exposure-scaled. Bin means are integrals of the selected function, normalized over the declared toy support; they are not merely the function evaluated at bin centers. Figure 46 shows the year-matched source fits and toy means. The distinct degree-five and degree-six threshold replacements at 65 MeV are defined below and in Appendix F; they do not retroactively change the original four-exposure ensemble.

The mass-resolved injection scans selected the lower RBF length-scale factors without using the observed limit curve. For the resulting settings, the aggregate zero-signal mean pulls are +0.029, −0.016, and −0.005 for the 2015, 2016 10%, and 2021 1% studies, respectively; the corresponding nonzero-injection signed averages are −0.223, −0.262, and −0.034. Mass-resolved mean, width, significance-residual, and recovery checks support these summaries. They motivate the chosen lower factors but do not replace direct mass-by-mass closure.

5.4 2021 validation ensembles at four effective exposures

The 2021 study uses 100 pseudoexperiments in each of four exposure ensembles and five mass hypotheses. The baseline extraction uses 40–300 MeV GP support, 50–250 MeV search range, $k_{\min} = 1.1$, $k_{\max} = 15$, five-bin local rebinning, log-mass/log-count preprocessing, the $2.25\sigma_m$ blind and training exclusions, two required sidebands, and twelve randomized optimizer restarts in addition to the nominal start. The four baseline ensembles use the same generalized-gamma threshold family, fitted independently to the native 1% and native 10% sources. Their source Pearson χ^2/ndf values are 1.108 and 2.781, respectively. The latter misdescription is largest through the steep threshold turn-on. We therefore use a threshold-refined model at 65 MeV while leaving the other mass points unchanged.

Displayed ensemble	Generating model	GP support	Role in the test
1% $\times$ 10, except 65 MeV	generalized-gamma threshold model	40–300 MeV	baseline smooth-threshold test
1% $\times$ 10, 65 MeV	degree-five threshold-turn-on continuation	40–300 MeV	threshold-refined 65 MeV test
1% $\times$ 100, all masses	generalized-gamma threshold model	40–300 MeV	baseline smooth-threshold test
native 10%, except 65 MeV	generalized-gamma threshold model	40–300 MeV	baseline smooth-threshold test
native 10%, 65 MeV	threshold-focused sigmoid–power–exponential mean	30–300 MeV	extended-support 65 MeV test of the measured turn-on
10% $\times$ 10, all masses	generalized-gamma threshold model	40–300 MeV	baseline smooth-threshold test

Table 7: Models used in the 2021 background-validation figures. At 65 MeV, the two threshold-focused tests replace the baseline points rather than being pooled with them. Distinct markers identify the change of generating model and, for native 10%, the extension of the GP support.

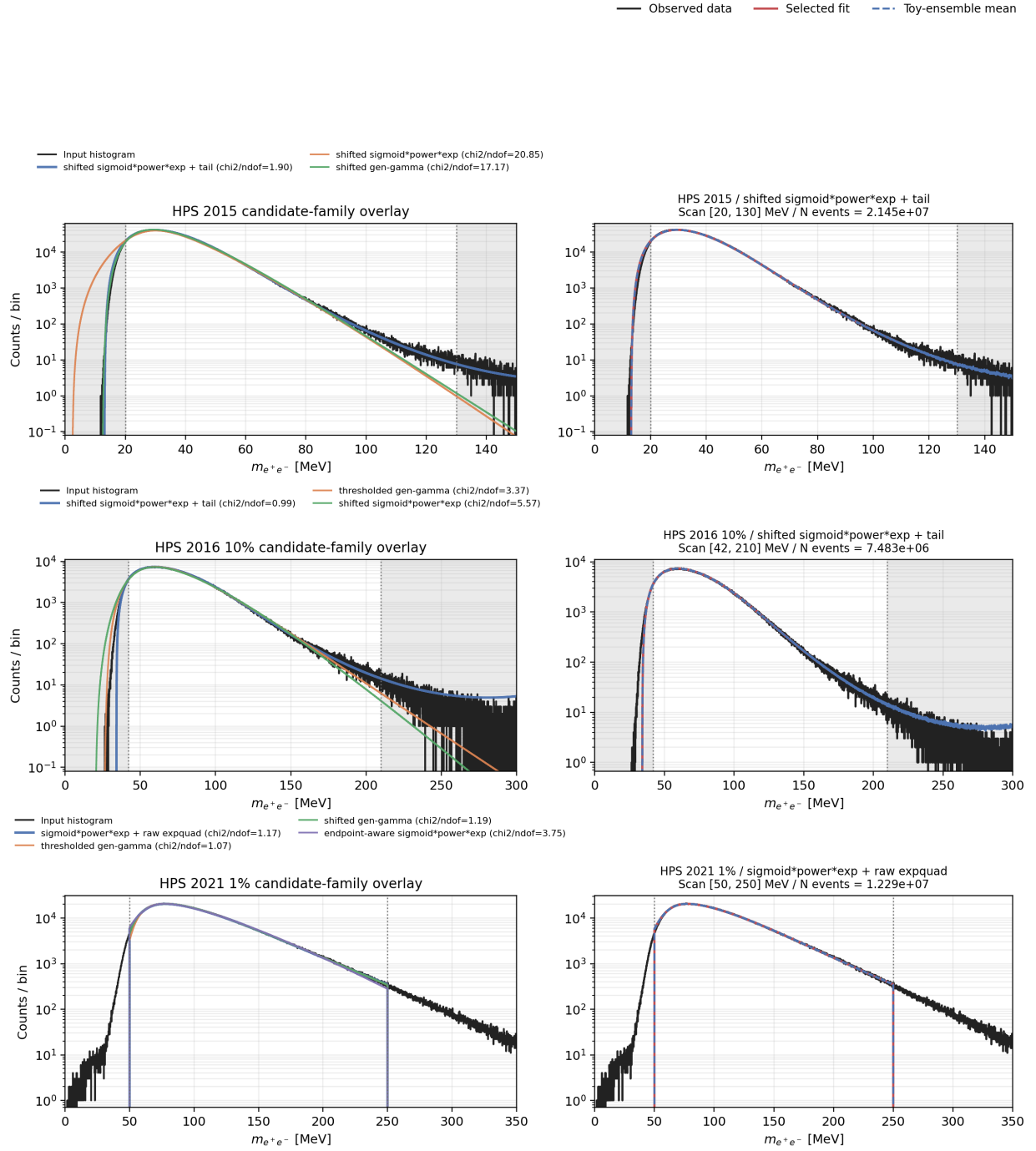


Figure 46: Analytic generating means used in the year-matched functional-form studies. Each dataset row compares the source spectrum with the candidate families and then shows the selected mean against the generated toy ensemble. The functions define smooth positive pseudo-data means and are not asserted to be unique physical models of the continuum.

5.5 Targeted validation of the rising threshold

1% $\times$ 10 threshold-refined model. The first 65 MeV test uses a source-fitted near-threshold model. Over 35–80 MeV it is a logistic turn-on multiplied by an exponentiated degree-five Chebyshev expansion and is joined smoothly to the specified high-mass tail. The degree is selected before any GP extraction: the native-bin deviance per degree of freedom is 0.994 for the native 1% source and the five-bin value is 0.850, with no parameter-bound contact. Its disclosed 80–300 MeV tail Poisson deviance per degree of freedom is 1.122, so this selected 1%-source truth is broadly data-like over the complete 40–300 MeV support rather than only in the fitted turn-on interval. An initial 20-background development subset gave a mean pull of 0.300 and a width of 1.109. We then analyzed an independently specified continuation, bringing the total to 100 backgrounds and allowing 90% intervals to be reported.

Native-10% extended-support model. The second 65 MeV test models the measured 30–40 MeV shoulder and analyzes it with the complete 30–300 MeV GP support. Its threshold component is a logistic turn-on times a degree-six exponentiated Chebyshev form fitted over 30–80 MeV, joined with a C^2 taper to a smooth high-mass continuation. In the native 10% source, the local native-bin and five-bin deviances per degree of freedom are 1.140 and 1.421. Thus the generating spectrum is data-like in the threshold-sensitive region even though the baseline family is not. Its 85–300 MeV tail deviance per degree of freedom is 6.220; this native-10% ensemble therefore relies on the weak-correlation and tail-influence qualification below rather than on a support-wide goodness-of-fit claim.

This study tests the 65 MeV extraction with 30–300 MeV support; it is separate from the lower-edge study that selected 36–300 MeV for the 2021-only analysis. The question here is whether a data-like generating mean that passes the local source-fit tests can be processed by the complete extraction without material signal bias. With log preprocessing, the direct RBF correlation between masses m and m' is

$$\rho(m, m') = \exp \left[-\frac{\log^2(m'/m)}{2\ell^2} \right]. \quad (101)$$

At 65 MeV, the optimized length scales in the two targeted ensembles span 0.2784–0.3185 for 1% $\times$ 10 and 0.2468–0.2691 for native 10%, with medians 0.2963 and 0.2592. The corresponding direct correlations with 150 MeV span 1.10–3.19% and 0.321–0.800%, respectively. Across the complete 55–70 MeV set the largest direct correlations with 150 MeV occur at 70 MeV and are 5.71% and 1.81%. The correlation decreases rapidly above 150 MeV. These results quantify why a threshold-focused source fit can provide a meaningful local test while the generated spectrum and extraction still retain the full GP support.

The correlation calculation is a fixed-hyperparameter statement. Section 6.10 shows what the scan correlations can imply for signal extraction, including neighboring responses induced by injected features. This calculation is not a claim that high-mass bins have mathematically zero influence. All retained bins still enter the global marginal likelihood and can shift the optimized length scale or kernel constant. A same-input diagnostic makes the aggregate effect explicit: after fixing the selected hyperparameters, removing the training bins above 150 MeV shifts the 65 MeV pull by a mean of -0.130 with a paired 90% interval $[-0.151, -0.110]$ in 1% $\times$ 10, and by -0.195 $[-0.211, -0.179]$ in native 10%. The maximum absolute shifts are 0.376 and 0.400. In the first ensemble the mean changes from 0.139 to 0.009; in the second it changes from -0.246 to -0.441 , whose 90% interval $[-0.611, -0.271]$ crosses the descriptive $|\bar{p}| = 0.5$ boundary used below. Counts above 150 MeV therefore have modest, statistically resolved collective influence even though every

pairwise RBF correlation is small. This diagnostic holds the chosen hyperparameters fixed and does not test their global-likelihood response to tail removal.

The optimized kernels explain why a threshold-focused source fit is informative for the full extraction: the low-mass result is directly coupled most strongly to the turn-on region, while the generated spectrum, specified smooth continuation, and full extraction retain the actual support. The restricted source fit thus provides a targeted conditional validation of local 65 MeV behavior with the stated support. It does not establish tail-independence or universal GP closure for every possible high-mass truth.

5.6 Matched refits and fit-quality checks

For every background, mass, and generating model, an independently optimized background-only fit supplies the matched reference uncertainty. Signals with $z = 1, 3, 5$ are generated with mean amplitude $A_{\text{inj}} = z\sigma_{A,\text{ref}}$, their full Gaussian templates are added to the same background counts, and the GP is refitted after applying the same training mask. The injected amplitude is therefore calibrated to the local background-only fit, whereas the pull denominator is always the uncertainty returned by the injected fit.

We never decide whether to keep a fit by looking at its fitted signal or pull. Repeated attempts use the same pseudo-data with independent optimizer starts, and we choose among them using the GP log marginal likelihood, reproducibility of the length scale and kernel constant, covariance validity, uncertainty-ratio checks, and contact with the kernel bounds. The choice never uses $\hat{A}$, the pull, agreement with the injected value, ε^2 , or a local p value. A statistical outlier is retained whenever it passes these numerical checks.

The baseline study retained 7994 of 8000 fits; every ensemble contains 99 or 100 usable fits. The six rejected fits failed reproducibility checks. Each threshold-focused study contains 400 usable fits—100 backgrounds at each of $z = 0, 1, 3, 5$ —with no rejection and no kernel-bound contact. Covariance, count, and pull checks agree, and the numerical selection is made without reference to the extracted physics result.

5.7 How closure is judged

At zero injection, the signed pull $p = \hat{A}/\hat{\sigma}_A$ is a spurious-signal statistic. The ensemble mean tests whether the extraction is centered and the sample standard deviation tests whether the returned uncertainty has the appropriate scale. At nonzero injection, the pull of Eq. (61) tests centering about the known amplitude and the paired response of Eq. (62) separates recovery of the added signal from a background-specific zero-signal offset. Strengths $z = 0, 1, 3, 5$ reuse the same background pseudo-spectrum and are consequently correlated; they are one response curve, not four independent confirmations.

For each 100-background ensemble at one mass and signal strength, we report the usable count n , the pull mean and its two-sided 90% Student- t interval, and the pull width with its exact normal-theory 90% chi-square interval. An ensemble is reported only if at least 95 of its 100 fits pass the numerical quality checks. We use a Holm correction for the two planned threshold-focused points at $z = 0$, and a separate correction for the twenty displayed four-exposure, five-mass points. The correlated nonzero injections are response diagnostics and do not enter either family. We flag a mean only when its Holm-adjusted two-sided p value is below 0.05 and $|\bar{p}| \geq 0.2$.

The practical centering tolerance $|\bar{p}| < 0.5$ was chosen after the 100-background results were available. It is a descriptive benchmark, not a predeclared equivalence test. We report it alongside the t /Holm results, which are more sensitive to small but statistically resolved offsets. For the two

targeted points we also ask whether the full 90% interval on the mean lies inside the practical band. A pull width is consistent with one only when its 90% interval contains one. None of these tests measures CL_s coverage.

5.8 Spurious-signal results from 100 pseudoexperiments

Table 8 gives the two targeted 65 MeV results. The baseline rows show why the threshold-focused tests were needed; the two sets of pulls are not combined. Both targeted samples exceed the 95-of-100 reporting threshold and pass the numerical checks above. We compare the results under the original statistical test, the post-result practical $|\bar{p}| < 0.5$ benchmark, and the separate unit-width diagnostic.

65 MeV ensemble and model	n	$\bar{p}$ [90% CI]	s_p [90% CI]	centering
1% $\times$ 10, baseline	100	0.7890 [0.6293,0.9488]	0.9622 [0.8625,1.0907]	outside band
1% $\times$ 10, threshold-refined, full sample	100	0.1394 [−0.0499,0.3288]	1.1404 [1.0222,1.2927]	inside band
1% $\times$ 10, threshold-refined, independent continuation	80	0.0993 [−0.1150,0.3135]	1.1514 [1.0196,1.3265]	inside band
native 10%, baseline	100	0.7160 [0.5452,0.8867]	1.0284 [0.9218,1.1657]	outside band
native 10%, extended support, full sample	100	−0.2463 [−0.4165,−0.0761]	1.0253 [0.9190,1.1622]	inside band [†]
native 10%, extended support, independent continuation	75	−0.2840 [−0.4943,−0.0737]	1.0934 [0.9646,1.2661]	inside band

Table 8: Background-only closure at 65 MeV. Intervals are two-sided 90% intervals: Student- t for the mean and exact normal-theory chi-square for the population standard deviation. The two full-sample targeted intervals lie inside the post-result $[-0.5, 0.5]$ practical band, whereas the baseline means do not. [†]The native-10% full sample nevertheless has targeted Holm-2 $p = 0.03632$ and $|\bar{p}| > 0.2$; statistical resolution of a small offset and practical centering are distinct statements. The 1% $\times$ 10 width interval excludes one in both the full sample and independent continuation.

Both threshold-focused samples fall inside the post-result practical centering band in the full sample and in the independent continuation. The 1% $\times$ 10 pull width is mildly high, while the native-10% mean retains a statistically resolved negative offset in the targeted two-point test. We therefore report practical centering, evidence for a nonzero offset, and pull-width calibration as separate results.

Across the full display, all twenty zero-signal mean pulls lie inside $[-0.5, 0.5]$; the largest magnitude is 0.338 for 10% $\times$ 10 at 180 MeV. That point is the only flag after the separate twenty-point Holm correction. With the correlated nonzero injections included, all eighty mean pulls remain inside the practical band and 75 of their 90% intervals are wholly contained. Exact unit-width calibration is less uniform: three of twenty zero-signal width intervals and fifteen of eighty all-strength intervals exclude one.

Figure 48 directly compares the baseline and targeted 65 MeV ensembles, showing why the threshold-specific study was needed.

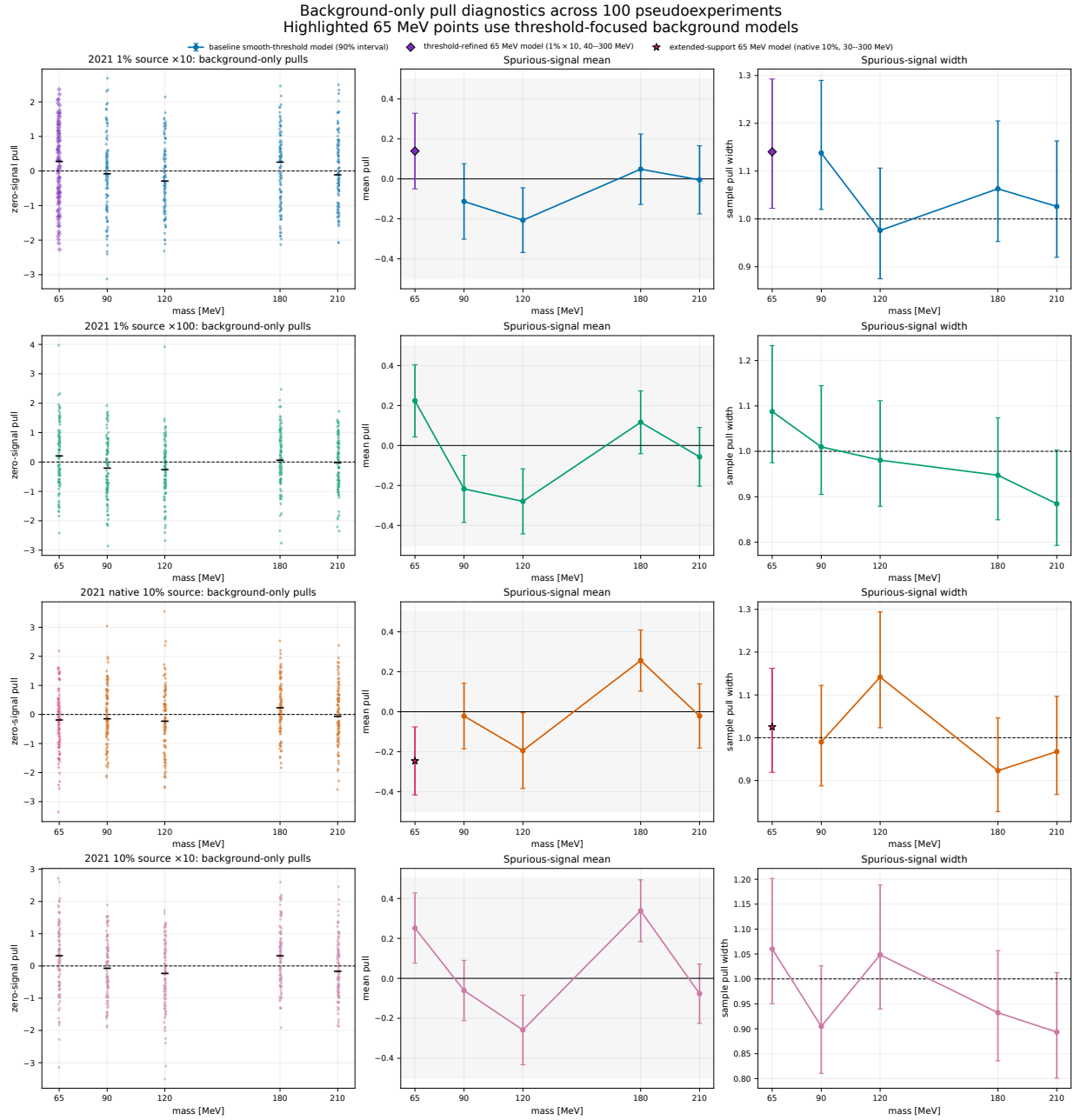


Figure 47: Zero-signal diagnostics for four 2021 exposure ensembles, with 100 pseudoexperiments per point. Circular markers show the baseline smooth-threshold model. At 65 MeV, diamonds and stars show the two threshold-focused tests defined in Table 7; they replace, rather than pool with, the baseline points. Bars are two-sided 90% intervals.

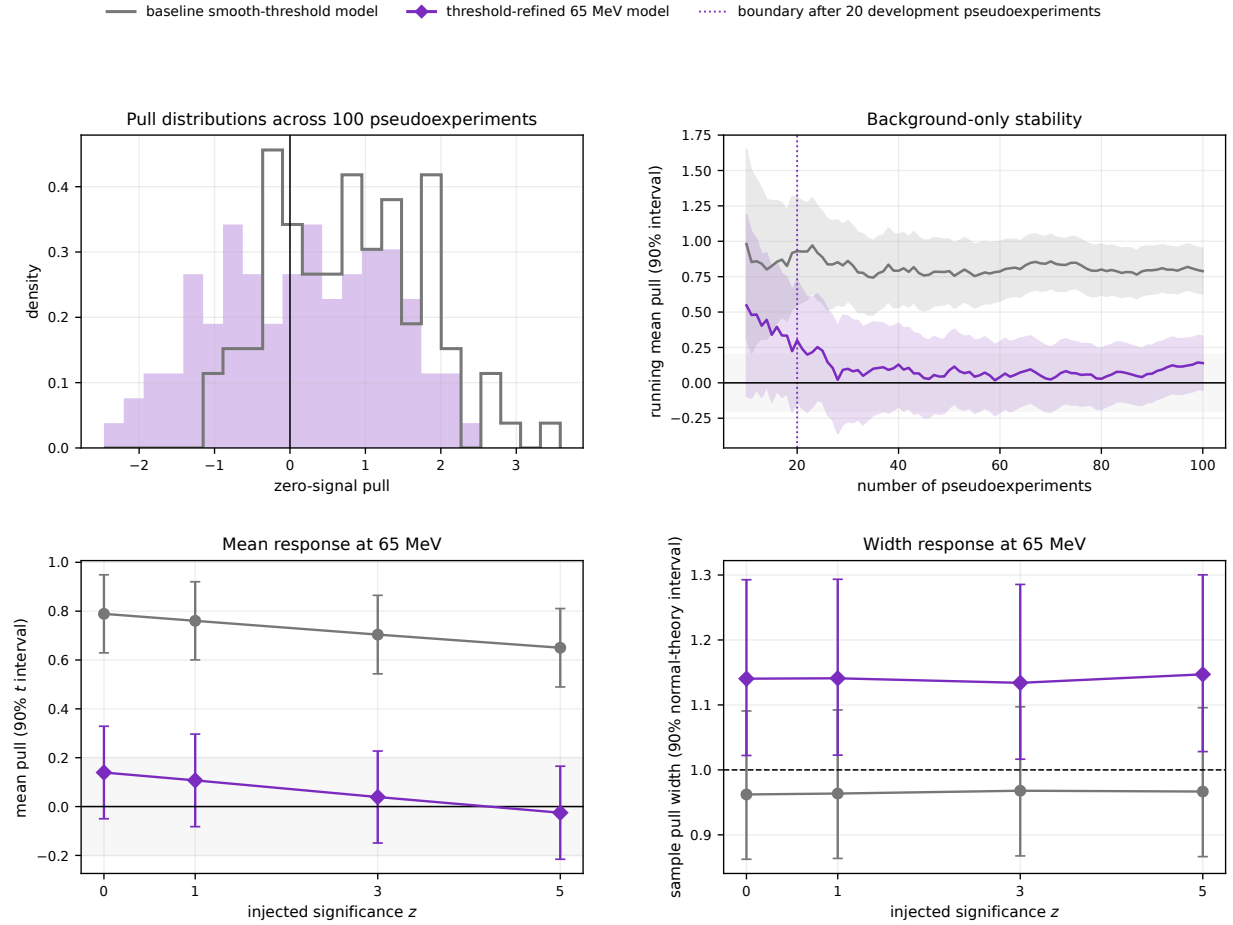
Threshold-model comparison for the $1\% \times 10$ sample at 65 MeV

Figure 48: Targeted $1\% \times 10$ test at 65 MeV, comparing the threshold-refined and baseline smooth-threshold models. The upper panels show the zero-injection pull distribution and the running mean; the lower panels show the mean and width across the correlated $z = 0, 1, 3, 5$ extractions. Mean intervals are two-sided 90% Student- t intervals and width intervals are exact normal-theory 90% chi-square intervals. The full sample contains the 20-background development subset; the independent continuation in Table 8 gives the same qualitative result.

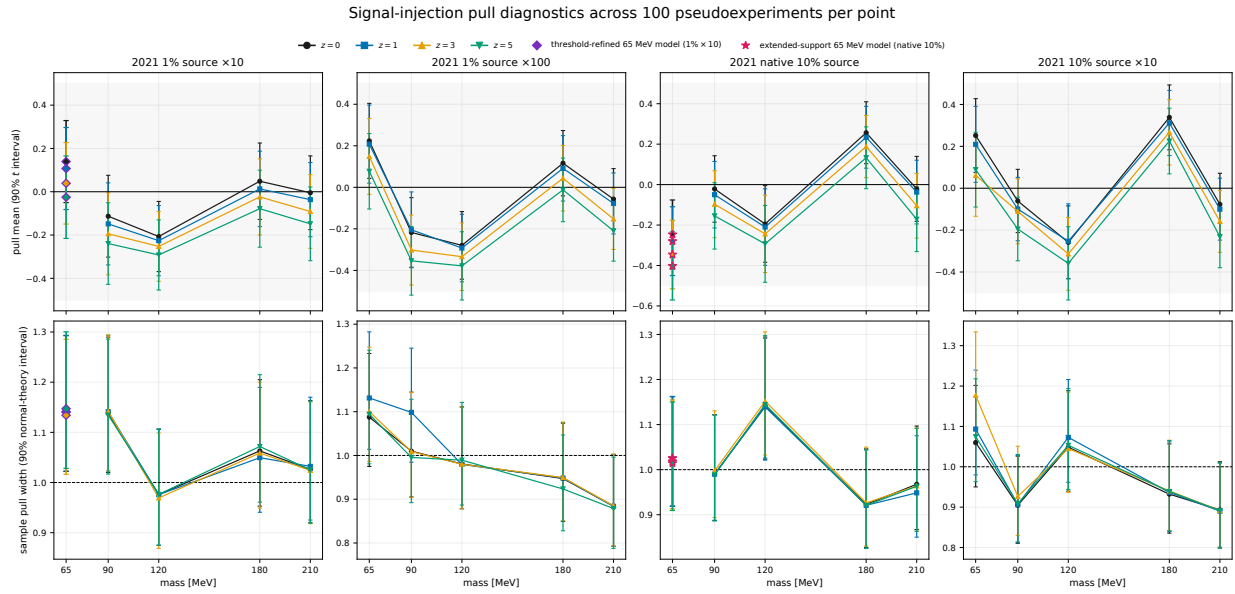


Figure 49: Background-validation summary. Columns correspond to $1\% \times 10$, $1\% \times 100$, native 10%, and $10\% \times 10$; the upper row shows pull means and the lower row pull widths as functions of mass. The four series are $z = 0, 1, 3, 5$, with zero injection visually emphasized and the correlated nonzero response series drawn more lightly. Circular markers show the baseline model; at 65 MeV, diamonds and stars show the targeted studies defined in Table 7. Mean bars are two-sided 90% Student- t intervals and width bars are exact normal-theory two-sided 90% chi-square intervals. The four strengths share backgrounds.

5.9 Recovery of injected signals

A centered zero-signal ensemble is necessary but not sufficient: a background model could remove the false offset and still absorb a real narrow signal. The nonzero rows therefore repeat the full GP fit after adding matched one-, three-, and five-reference-sigma signals. Figure 49 displays the mean and width at every strength and mass; Table 9 gives the numerical 65 MeV replacement results, including the paired response $R = (\hat{A}_z - \hat{A}_0)/A_{\text{inj}}$.

The paired response separates signal recovery from the additive background offset. In the two threshold-focused tests, the mean response is stable at 0.966–0.969 over $z = 1, 3, 5$, with mutually consistent 90% intervals and no worsening with strength. The full procedure therefore recovers 96.6–96.9% of the tested amplitude, corresponding to a stable 3.1–3.4% attenuation. We retain that attenuation in the interpretation. Agreement across strengths is not counted multiple times because the same 100 background realizations are reused.

65 MeV model	z	$\bar{p}$ [90% CI]	s_p [90% CI]	$\bar{R}$ [90% CI]
1% $\times$ 10, threshold turn-on	0	0.139 [−0.050,0.329]	1.140 [1.022,1.293]	—
	1	0.107 [−0.082,0.297]	1.141 [1.023,1.293]	0.9678 [0.9643,0.9713]
	3	0.039 [−0.149,0.227]	1.134 [1.016,1.285]	0.9665 [0.9646,0.9684]
	5	−0.025 [−0.216,0.165]	1.147 [1.028,1.300]	0.9670 [0.9654,0.9686]
native 10%, 30–300 support	0	−0.246 [−0.417,−0.076]	1.025 [0.919,1.162]	—
	1	−0.280 [−0.450,−0.109]	1.025 [0.919,1.162]	0.9664 [0.9629,0.9700]
	3	−0.346 [−0.515,−0.177]	1.019 [0.913,1.155]	0.9666 [0.9649,0.9684]
	5	−0.402 [−0.571,−0.234]	1.015 [0.909,1.150]	0.9686 [0.9669,0.9703]

Table 9: Full-sample 65 MeV extraction at all matched-reference strengths for the two threshold-focused models. Mean intervals are two-sided 90% Student- t intervals and width intervals are exact normal-theory two-sided 90% chi-square intervals. The response summaries are paired background-by-background and use two-sided 90% Student- t intervals. All strengths share the same backgrounds and are correlated. The 1% $\times$ 10 widths are mildly high at every strength; the native-10% widths are compatible with one at this precision.

The validation above assumes fixed GP supports. For the 2021-only scan, we must also choose where that support begins.

5.10 Choosing the 2021 GP support

The signal search tests masses from 50 to 250 MeV, but the GP needs data beyond that interval to constrain the background near either endpoint. At low mass, the native 2021 spectrum has a sparse shoulder followed by a rapid turn-on: its 0.125 MeV bins contain only a few hundred events around 30–32 MeV and several thousand by 40 MeV. We compared several choices for the lower edge of the GP support, holding the upper edge at 300 MeV and leaving the search interval unchanged. The goal was to leave out the least stable part of the turn-on without throwing away useful sideband data.

5.10.1 Source shape and candidate supports

Figure 50 shows the source spectrum and the tested lower edges. The 30 MeV setting is retained as a control but was not eligible for selection; the candidate edges were 32, 34, 36, 38, and 40 MeV.

2021 native-10% threshold spectrum and scanned GP-support edges

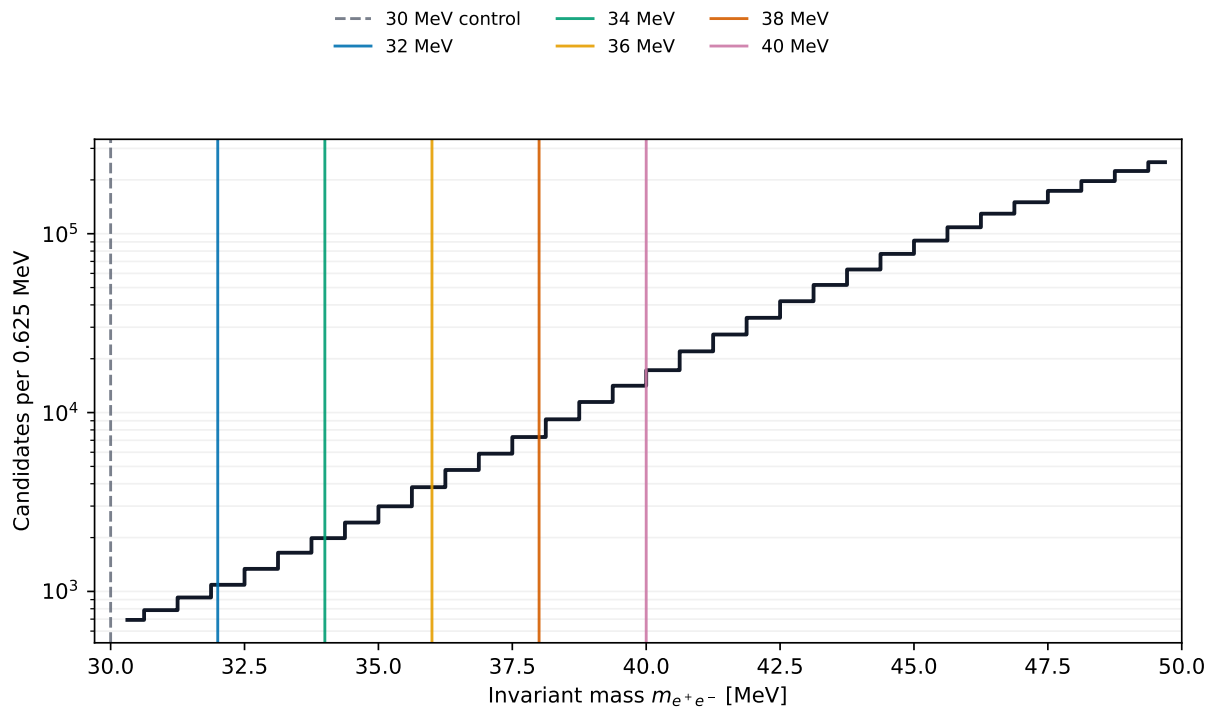


Figure 50: Native 2021 10% source histogram near threshold. Vertical lines mark the 30 MeV control and the five candidate lower edges of the GP support. The logarithmic vertical scale makes the low-count shoulder and rapid turn-on visible.

5.10.2 Initial scan

The first stage reuses the 100 background-only pseudoexperiments generated from the fitted source spectrum in the four-exposure study; it does not generate new pseudo-data. At each support edge, we perform matched-reference extractions at 55, 60, 65, and 70 MeV with 0-, 2-, and $5\text{-}\sigma_A$ injections. The scan changes only the lower support edge. Binning, preprocessing, blind and training masks, kernel bounds, signed extraction, and optimizer checks remain fixed as described in Sections 4 and 5. For repeated optimizer starts, we retain the fit using reproducibility, covariance validity, kernel diagnostics, and GP log marginal likelihood; neither the pull nor any limit or p -value enters that choice.

The first 25 backgrounds were processed at all six edges. All 1800 fits passed the numerical checks, with no invalid covariance matrix or contact with a kernel bound. The original rule rejected an edge if any mass-injection combination had $|\bar{p}| \geq 0.5$ and a two-sided 90% Student- t interval that excluded zero. Every eligible edge failed. Their worst mean-pull magnitudes ranged from 1.072 to 1.530, and the ineligible 30 MeV control reached 1.624. Even the best first-stage candidate, the 38 MeV edge, was therefore well outside the original 0.5 bound.

5.10.3 A practical rule and an independent check

After the first stage, we adopted a less stringent working rule: at least 9 of the 12 mass-injection means and at least 3 of the 4 zero-signal means had to satisfy $|\bar{p}| < 0.75$, and no mean could reach 1.25 in magnitude. The numerical checks, the minimax score, and the 0.10 tie margin were left unchanged. Among candidates within that margin, the lower edge was preferred so that the fit retained more data. Because this rule was formulated after the first 25 backgrounds had been examined, it is a post-selection screen, not a predeclared equivalence test or a test of coverage. It was fixed before the remaining 75 backgrounds were processed.

Only the 36 and 38 MeV candidates pass the practical rule in the first stage. The 38 MeV edge has the smaller worst pull, but 36 MeV lies within the retained tie margin and is therefore selected for confirmation. The independent continuation uses backgrounds 25–99 at 34, 36, and 38 MeV. At 36 MeV, 10 of 12 means and all four zero-signal means satisfy the 0.75 condition in the continuation alone; the worst magnitude is 0.901. In the complete 100-background sample, the corresponding counts are 9 of 12 and 3 of 4, with a worst magnitude of 0.960. Every fit passes the numerical checks. With the rule already fixed, we chose 36–300 MeV for the subsequent 2021-only analysis without further tuning. The later v4.9.12 release carries this same support into the standalone and combined current-data results.

At the selected edge, the full-sample means at 55 MeV are -0.773 , -0.841 , and -0.960 for the 0-, 2-, and $5\text{-}\sigma_A$ injections. Their 90% intervals exclude zero, so the low-mass under-recovery is retained as a real limitation of this test. Across the nine mass-injection combinations from 60 to 70 MeV, the means range from -0.110 to $+0.325$ and the pull widths from 0.939 to 1.034. The 65 MeV zero-signal mean is 0.253, with a 90% interval of $[0.086, 0.421]$.

Each numerically valid pseudoexperiment also provides a bounded Wald $\tilde{q}_\mu$ 90% CL_s yield limit. We use those values only to check that the limit is stable under nearby support choices; the smallest limit never determines the selected edge. This study therefore supports the 36–300 MeV choice for the declared fitted-source ensemble, with the 55 MeV caveat above. It does not establish direct coverage or truth-independent background closure.

Support comparison using 100 pseudoexperiments

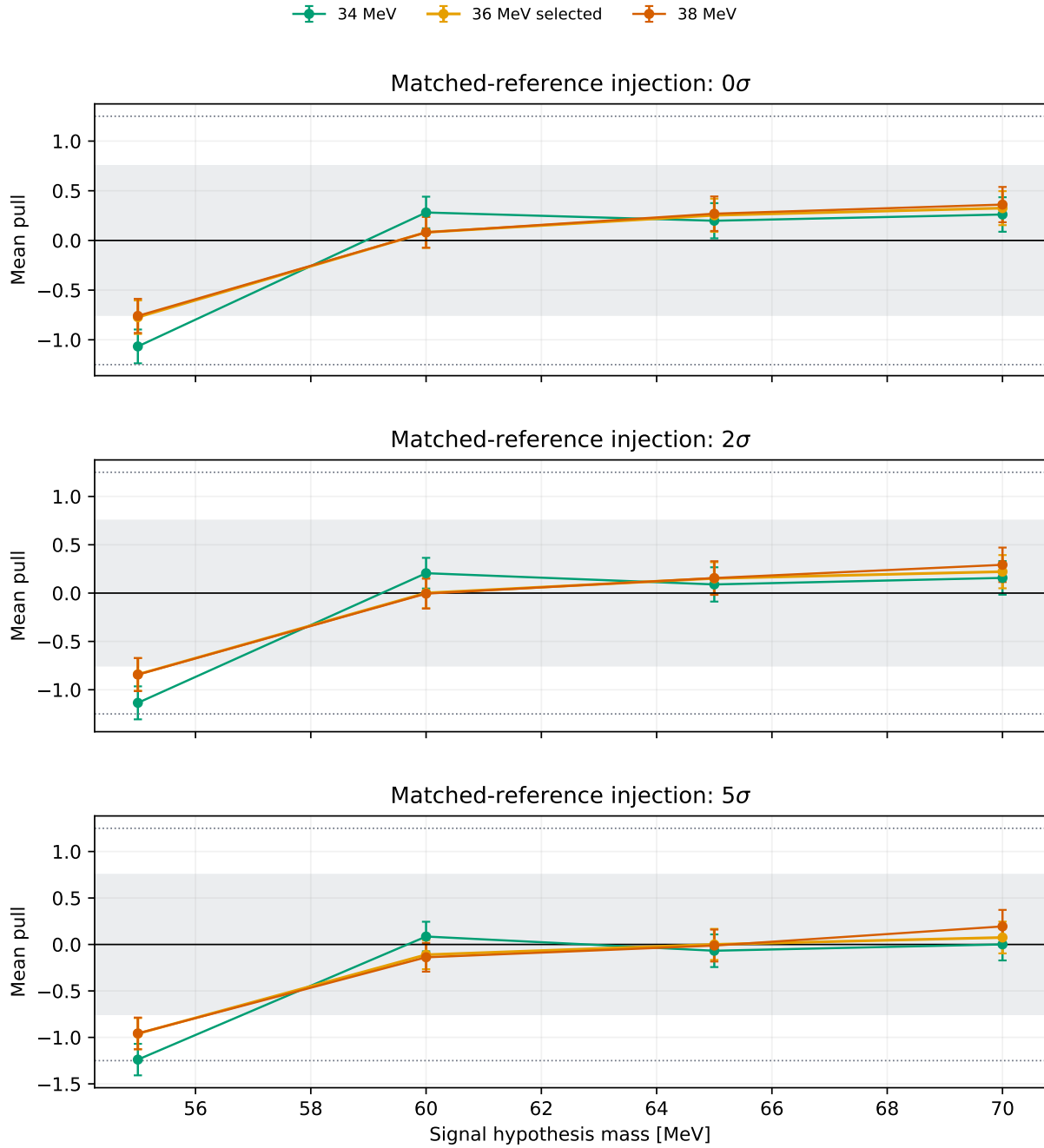


Figure 51: Mean pulls in the complete 100-background sample for the selected edge and its two neighboring candidates. Error bars are two-sided 90% Student- t intervals on the mean. The shaded band denotes the post-selection $|\bar{p}| < 0.75$ practical region; dotted lines show the magnitude-1.25 guard. The remaining exceptions are confined to the 55 MeV hypothesis.

5.11 Validation evidence and resulting decisions

Table 10 restores the decision summary from v4.9.7 Section 5.10 and names the figures that answer each question. The decisions concern the stated historical ensembles; the later calibration findings remain part of the qualification of the proposed final procedure.

Study	Evidence in this note	Recorded decision and limitation
Blind window	Figs. 41– 45; Table 6	Use $2.25\sigma_m$ for training exclusion and extraction; chosen from leakage and recovery, not the strongest limit proxy.
Generating shapes	Fig. 46; Table 7	Retain independent smooth-source tests; identify the two 65 MeV replacements and their different supports.
Spurious fitted signal	Fig. 47; Table 8	All 20 displayed background-only means meet the post-result $ \bar{p} < 0.5$ tolerance; this is practical conditional centering.
Pull widths	Fig. 49	Retain departures from unit width: 3/20 zero-signal and 15/80 all-strength 90% intervals exclude one.
Injected-signal response	Fig. 49; Table 9	Record 96.6–96.9% paired response; the remaining loss is visible and is not direct CL_s coverage.
Numerical stability	Matched-refit checks in this section; Appendix F	Use repeatable maximum-LML branches, independent of fitted signal; all 800 fits in the two replacement ensembles are usable.
Tail and scan influence	Fixed-hyperparameter tail-removal test; Section 6.10	Local pairwise correlations can be small while collective influence is nonzero; do not claim tail independence.
Final inference	Appendix B	Keep observed result states separate. Practical recovery alone neither establishes coverage nor completes unblinding qualification.

Table 10: Which study tested each analysis choice and what was carried forward. The 800 replacement fits are distinct from the 8,000-state historical four-exposure study restored in the appendix.

For these source-like spectra, all twenty zero-signal mean pulls and all eighty correlated injection-strength mean pulls lie inside the post-result ± 0.5 band. The paired response is 96.6–96.9%, with the remaining loss and non-unit pull widths retained in the reported diagnostics. Each pseudoexperiment uses an independent generating model and a fresh sideband fit; numerical choices do not use the fitted signal or pull.

This establishes conditional extraction recovery for the tested sources and supports. The native-10%, 65 MeV offset and the 55 MeV under-recovery remain visible limitations. Later rejection-rate calibration in Appendix B tests questions that these pull summaries do not answer. The historical recovery alone does not establish coverage, a scan-global significance calibration, or readiness for full-data unblinding.

6 Results for the released data

This is a snapshot of a partially unblinded analysis. It includes the fully unblinded 2015 and 2016 samples and the 10% sample currently available from 2021. The remaining 2021 data are outside the reviewed sample. The curves below describe the observed result in hand and do not project the full-2021 reach.

Each mass point uses the corrected, bounded, piecewise-asymptotic CL_s construction with the common signal-strength coordinate constrained to $\varepsilon^2 \geq 0$. The Gaussian-process supports, hyper-parameter states, and physics normalizations are frozen before the observed limits and p_0 values are evaluated. The inference is conditional on those frozen choices and on the asymptotic approximation. For 2016, the frozen GP support is 30–210 MeV and the upper resolution-scaled length-scale factor is 12. The 2015 and 2021 configurations retain their separately qualified supports. One numerical caveat applies to 2016. An independent cross-process replay passed the specified 10^{-6} state-reproduction check at 87 of 142 masses. A downstream replay with the optimizer disabled, together with the other release checks, passes; even so, every 2016-inclusive curve remains preliminary and conditional on this exception.

The three current inputs can be analyzed in seven distinct combinations: each data set alone, each of the three pairs, and all three together.

Scope	Active data sets	Statistical coordinate
Standalone	full 2015	campaign-specific ε^2
Standalone	full 2016	campaign-specific ε^2
Standalone	2021 10%	campaign-specific ε^2
Pair	full 2015 + full 2016	one shared ε^2
Pair	full 2015 + 2021 10%	one shared ε^2
Pair	full 2016 + 2021 10%	one shared ε^2
All three	full 2015 + full 2016 + 2021 10%	one shared ε^2

This subsection reproduces the observed-only release. The pointwise expected bands and empirical limit-tail study are added in the next subsection. The background-only p_0 values are fixed-mass, one-sided asymptotic quantities. They have no look-elsewhere correction and must not be interpreted as global significances. Nor do they constitute a direct coverage calibration of the upper-limit construction.

6.1 The complete observed search and conditional expected bands

The full search runs from 19 to 250 MeV. At each mass it includes every campaign whose declared search interval covers that point: 2015 alone at 19–38 MeV, 2015+2016 at 39–49 MeV, all three at 50–90 MeV, 2016+2021 at 91–180 MeV, and 2021 alone at 181–250 MeV. This membership rule is fixed by the search ranges; it never selects the dataset giving the smallest observed limit. Figure 52 places the nominal local asymptotic p_0 directly below the corresponding observed limit and conditional expected bands.

The bands retain 300 background-only draws per mass from the archived mass-local GP prediction. Each draw first samples a correlated latent background and then Poisson counts; constituent draws are shared among the combinations at that mass. The GP state is held fixed. Different masses do not form one coherent full-spectrum experiment. The resulting quantiles describe pointwise variation under that conditional model; they do not calibrate the CL_s sampling distributions or supply a global significance.

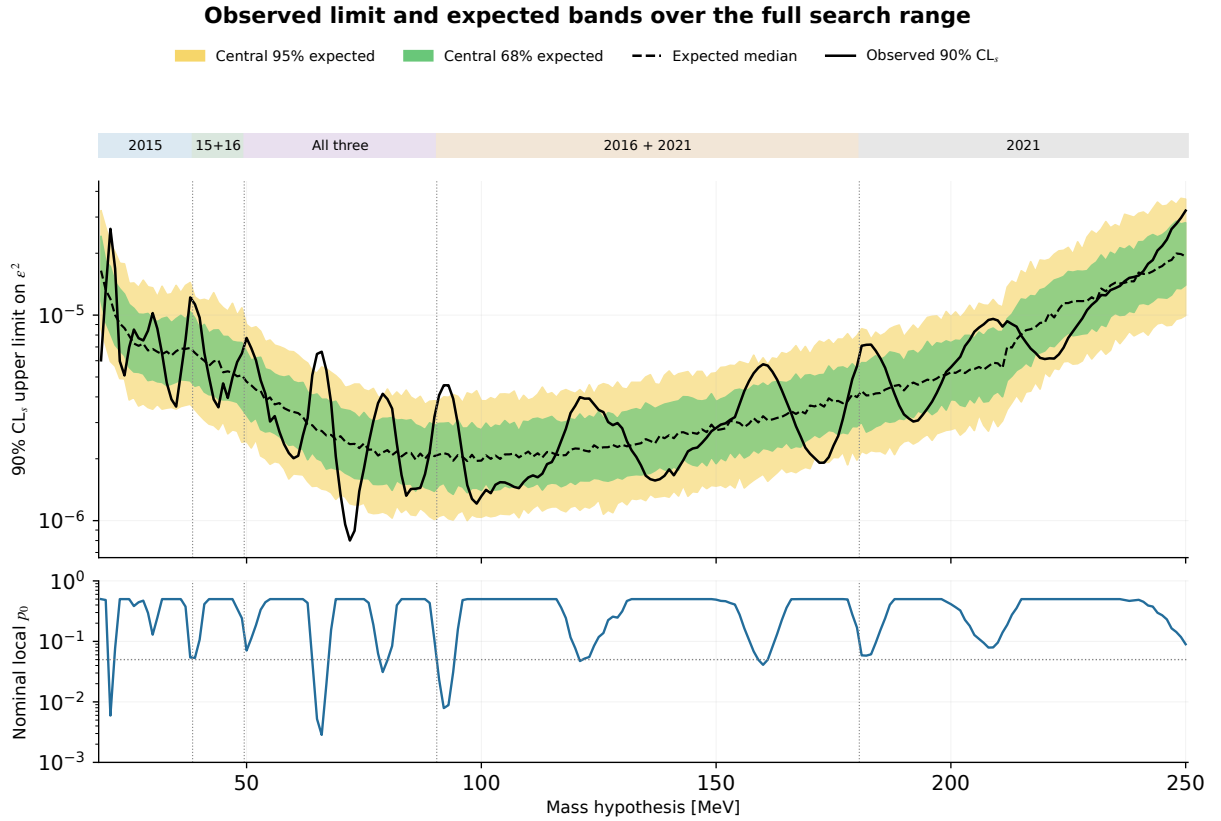


Figure 52: Complete 19–250 MeV observed search, with the preassigned active datasets shown above the curves. The upper panel shows 90% CL_s limits and the median, central 68% and central 95% ranges of the original 300 conditional background-only toys per mass. The lower panel adds the unchanged fixed-mass asymptotic p_0 . The observed polyline connects every adjacent tested mass, including membership transitions; no numerical endpoint is changed by those connections. The physical visible-branching correction is applied once above the dimuon threshold. These are pointwise bands; 2016-inclusive values retain the numerical exception.

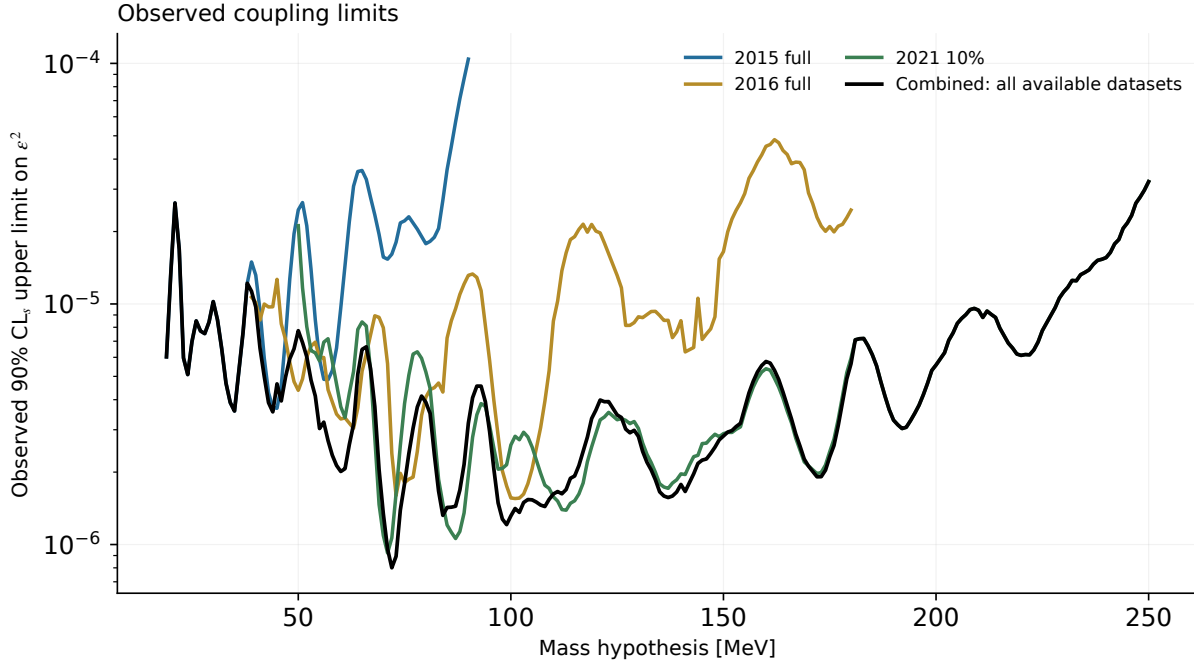


Figure 53: Observed coupling limits from the three individual released samples and the full maximal-available-dataset combination of Fig. 52. Only observed limits are shown. The combination is obtained from a shared-coupling likelihood with the fixed membership rule; it is not the envelope of the individual curves.

6.2 Conditional full-2021 response compared with BaBar

To show how the current result responds to a larger 2021 sample, we repeat the restricted density-response construction used in Figure 84 of v4.9.7. Let $\rho_d(m)$ be the observed prompt normalization density in the physical window $m \pm 1.64\sigma_m$. Holding the full 2015 and 2016 samples fixed and increasing only the 2021 contribution by a factor of ten gives

$$(\epsilon_{90}^2)_{\text{proxy}}(m) = (\epsilon_{90}^2)_{\text{current}}(m) \sqrt{\frac{\sum_{d \in \mathcal{D}(m)} \rho_d(m)}{\sum_{d \in \mathcal{D}(m)} s_d \rho_d(m)}}, \quad (s_{2015}, s_{2016}, s_{2021}) = (1, 1, 10). \quad (102)$$

Here $\mathcal{D}(m)$ is the active dataset set. The scale is unity below 50 MeV, density weighted where samples overlap, and exactly $1/\sqrt{10}$ where only 2021 contributes. The current and proxy minima are 8.0043×10^{-7} and 3.1898×10^{-7} , both at 72 MeV. These are properties of the displayed curves, not expected future outcomes.

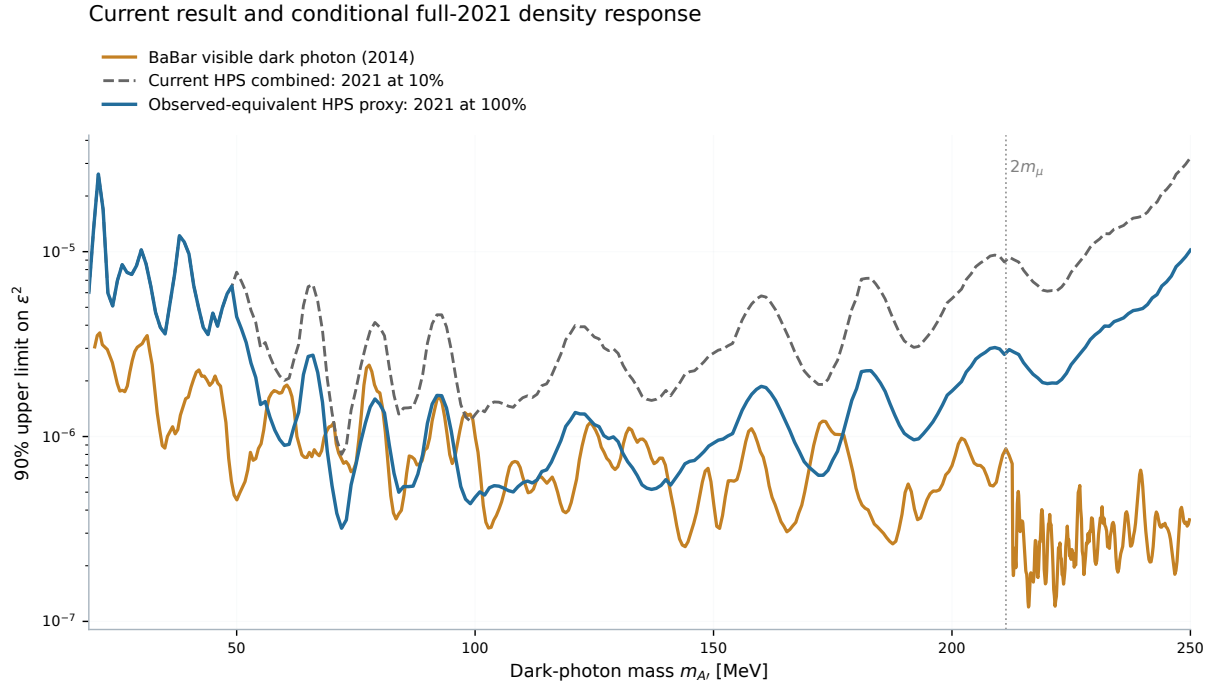


Figure 54: Current observed 90% upper limits and the conditional full-2021 density-response proxy compared with the published BaBar visible-dark-photon observed 90% contour [5]. The gray dashed curve is exactly the current maximal-available-dataset HPS result over 19–250 MeV. The blue curve applies Eq. (102) to its current observed normalization densities. It retains present fluctuations and is neither expected sensitivity nor a future observed limit or a fit to full-2021 data. HPS and BaBar use different statistical constructions. The inherited dimuon correction and 2016 qualifications remain applicable.

6.3 Signal-yield and coupling limits for each dataset

Figures 55–57 separate the observed yield limit from its conversion to ϵ^2 . The upper panels use the full Gaussian-template yield, while the likelihood uses the portion inside the fitted window. The different mass dependence of the yield and coupling limits follows from the dataset’s mass-dependent normalization. The nominal local minima occur at 51, 90, and 78 MeV in 2015, 2016, and 2021, with $p_0 = 8.46 \times 10^{-4}$, 3.09×10^{-4} , and 2.47×10^{-3} , respectively. They are fixed-mass asymptotic quantities; the coherent global calculation in Section 4.20 uses a separately declared scan statistic and conditional background model.

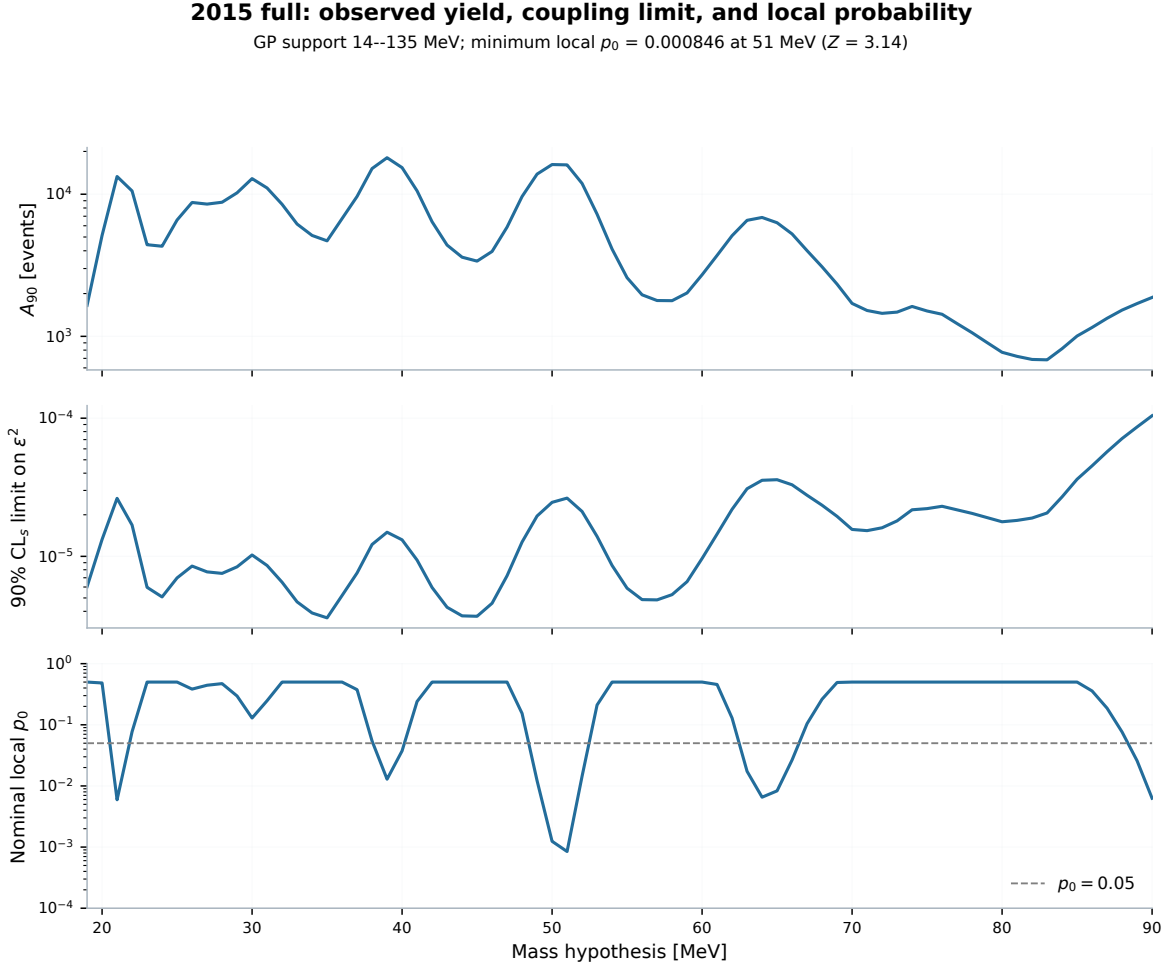


Figure 55: Observed 2015 full-sample scan on 19–90 MeV with 14–135 MeV GP support. From top to bottom: the 90% CL_s full-template signal-yield upper limit A_{90} , the corresponding ϵ^2 upper limit, and nominal local asymptotic p_0 . The same frozen released fits supply all panels. The dashed line marks $p_0 = 0.05$; no expected bands or look-elsewhere correction are included.

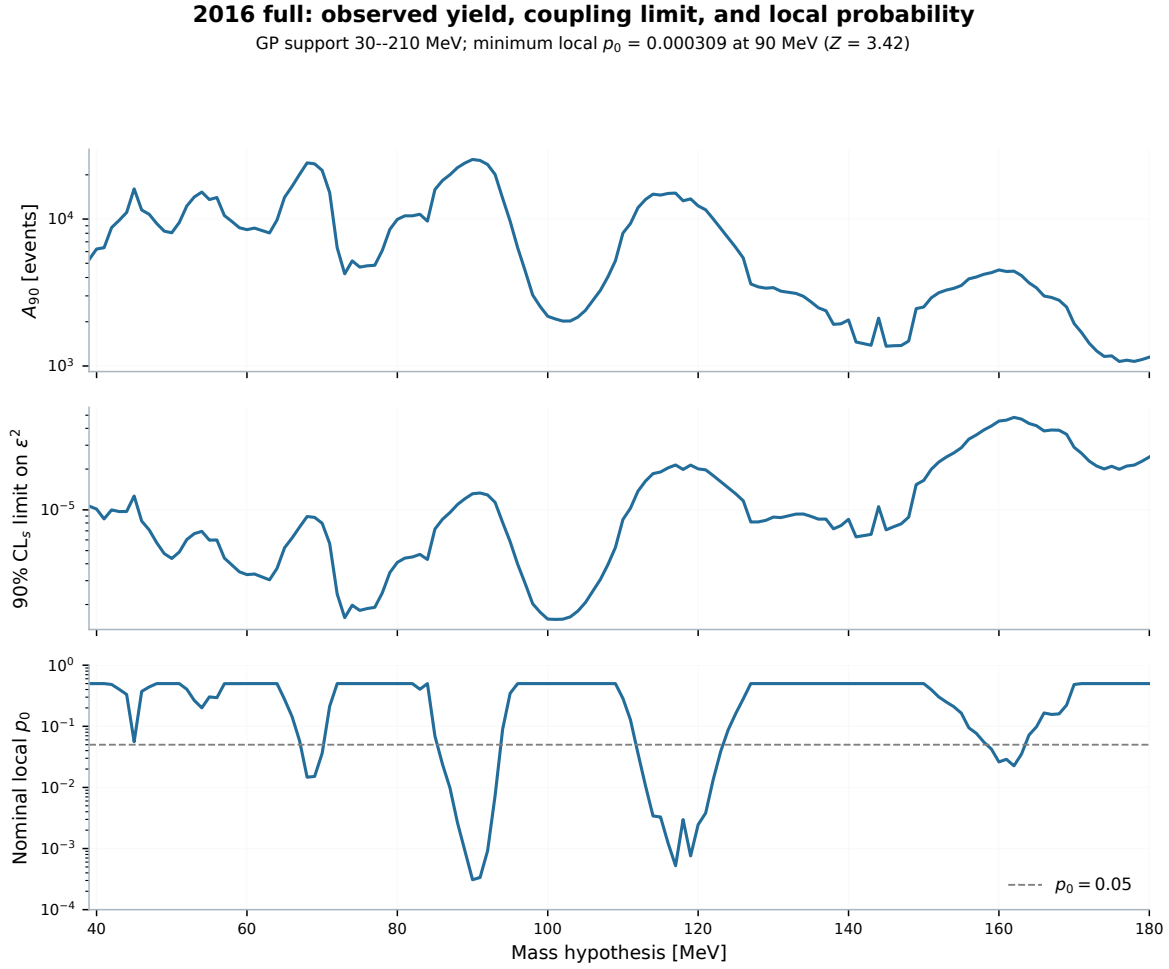


Figure 56: Corresponding observed 2016 full-sample scan on 39–180 MeV with 30–210 MeV GP support. The panels show the full-template A_{90} , ϵ^2 limit, and nominal local asymptotic p_0 . The existing 2016 numerical and generating-source qualifications remain applicable.

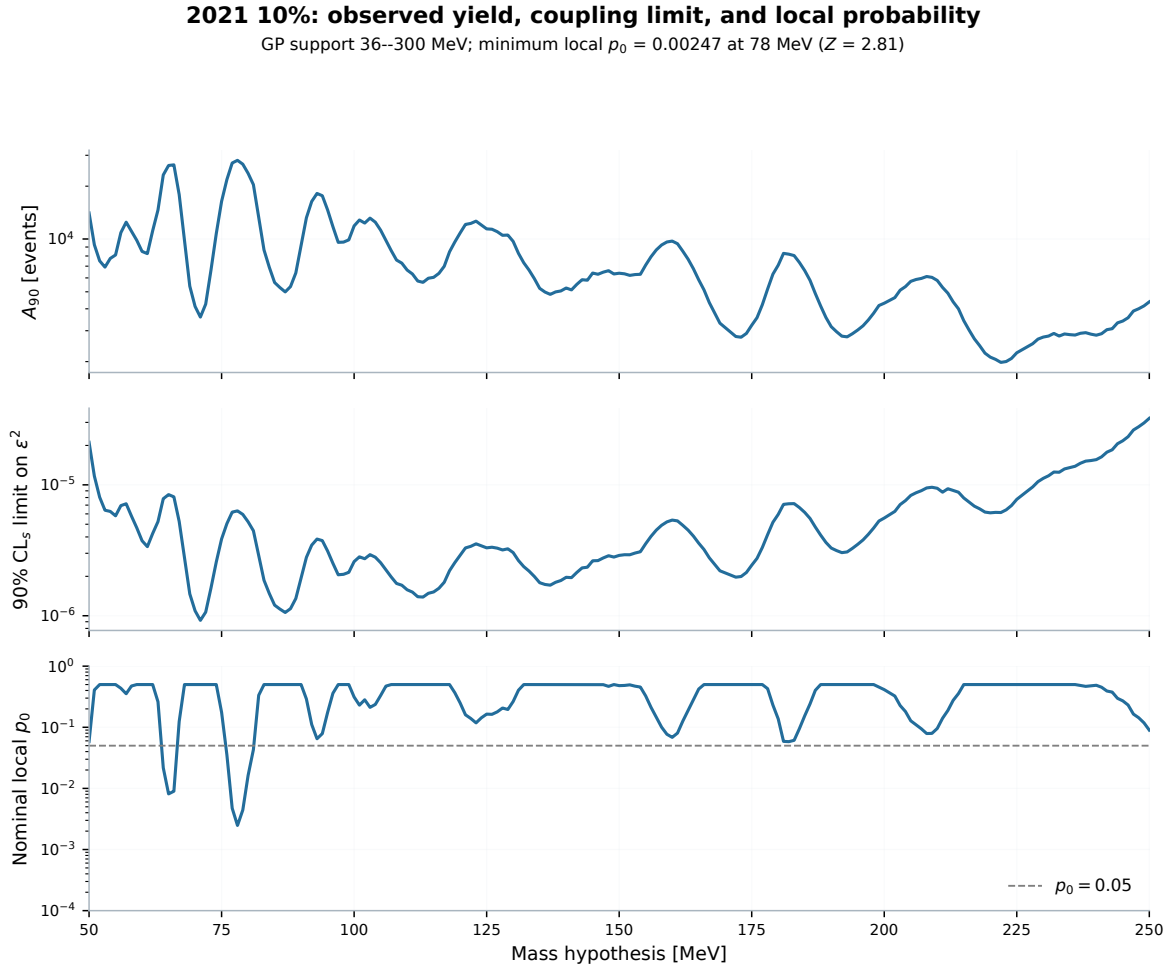


Figure 57: Corresponding observed 2021 10% scan on 50–250 MeV with 36–300 MeV GP support. The panels show the full-template A_{90} , minimal-visible ϵ^2 limit, and nominal local asymptotic p_0 . Above the dimuon threshold the inherited branching correction enters the coupling conversion once; it does not change the visible electron-pair yield or local probability.

6.4 Combining data sets through a shared coupling

Figure 58 contains all three pairwise combinations and the all-three result. At any given mass, the simultaneous likelihood profiles one nonnegative ε^2 shared by the active campaigns. Each campaign nevertheless retains its own signal normalization, GP prediction, covariance, and observed spectrum. Combining the data sets is thus a new likelihood calculation, not an envelope, average, or choice among the standalone limits. Each curve is evaluated only on the mass interval common to all of its active inputs.

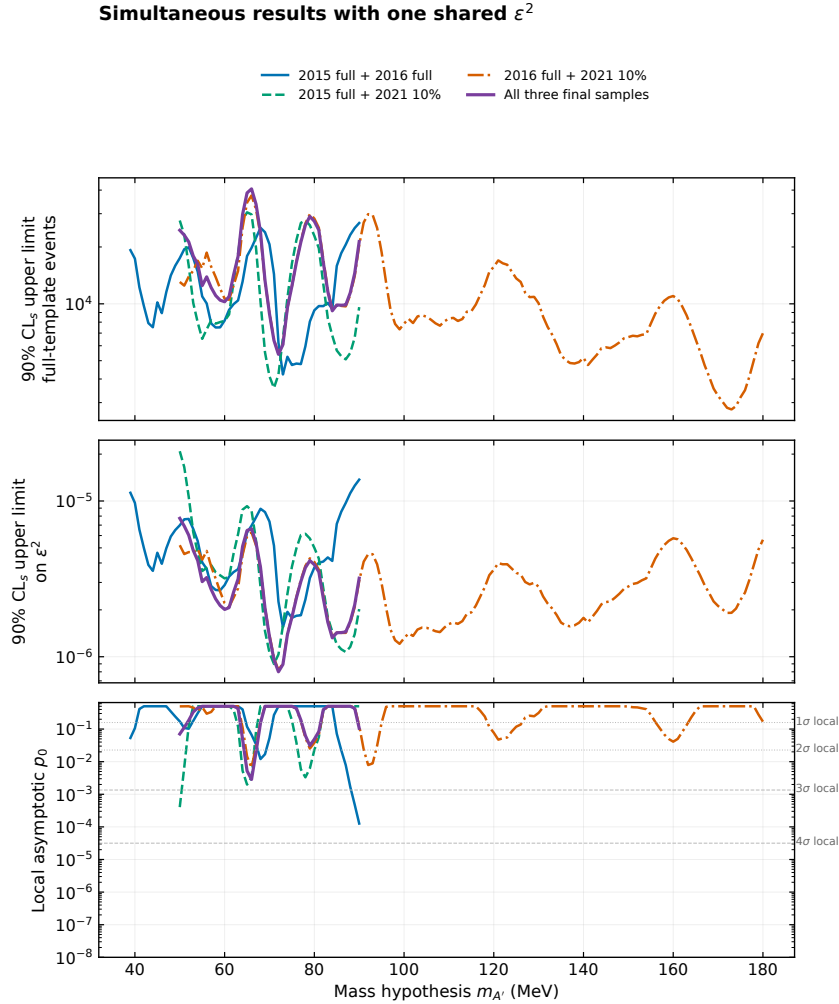


Figure 58: Observed shared- ε^2 results for the three pairwise combinations and the all-three overlap. For visual alignment, the top coordinate expresses the shared-coupling limit as the total compatible signal yield summed over the active campaigns. The middle panel gives the 90% bounded, piecewise-asymptotic CL_s upper limit on ε^2 , and the bottom panel gives the local asymptotic p_0 . No expected-limit bands are shown.

The shared coupling expresses the physical relationship among the campaigns: a putative dark photon has one ε^2 , while its expected event count changes with beam conditions, exposure, acceptance, and mass resolution. It also keeps the independent background information from each spectrum explicit.

6.5 Local asymptotic scans

Figure 59 places the local p_0 scans for all seven cases on the same axes, making structure at a common mass easier to compare across campaigns. Every plotted point is still a separate fixed-mass test. The conditional GP global studies are discussed separately in Section 4.20; they do not change these local asymptotic curves.

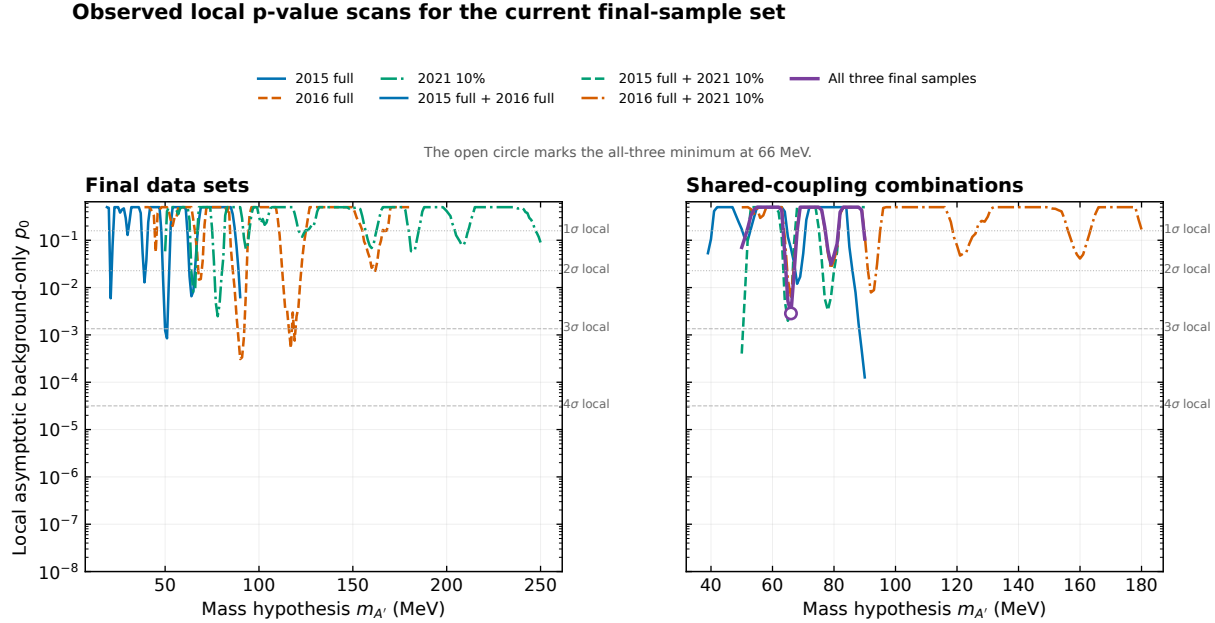


Figure 59: Local one-sided asymptotic background-only p_0 for the three standalone inputs, the three pairs, and the all-three combination. The values are conditional on the frozen GP supports and states. No look-elsewhere correction, global significance, or toy calibration is reported.

The common three-campaign interval has its smallest local asymptotic value at 66 MeV, with $p_0 = 2.845 \times 10^{-3}$ and $Z = 2.765$. The current detailed extraction displays appear in Section 6.9.

6.6 Conditional expected bands for the individual and overlapping scopes

The following panels resolve the combined overview into the all-three, pairwise and standalone searches. They use the same archived 300-toy pointwise quantiles described above.

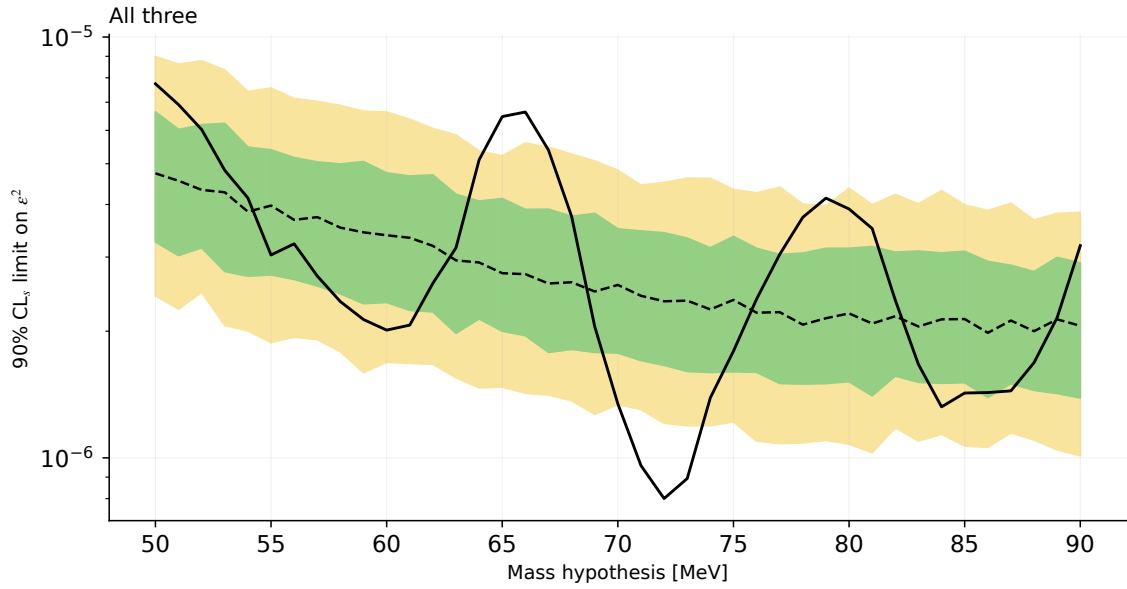


Figure 60: The all-three shared-coupling result on its common 50–90 MeV interval. Green and yellow show central 68% and 95% pointwise ranges, and the dashed curve is the median of 300 conditional background-only toys at each mass. The observed curve uses the same bounded asymptotic construction as the standalone fits.

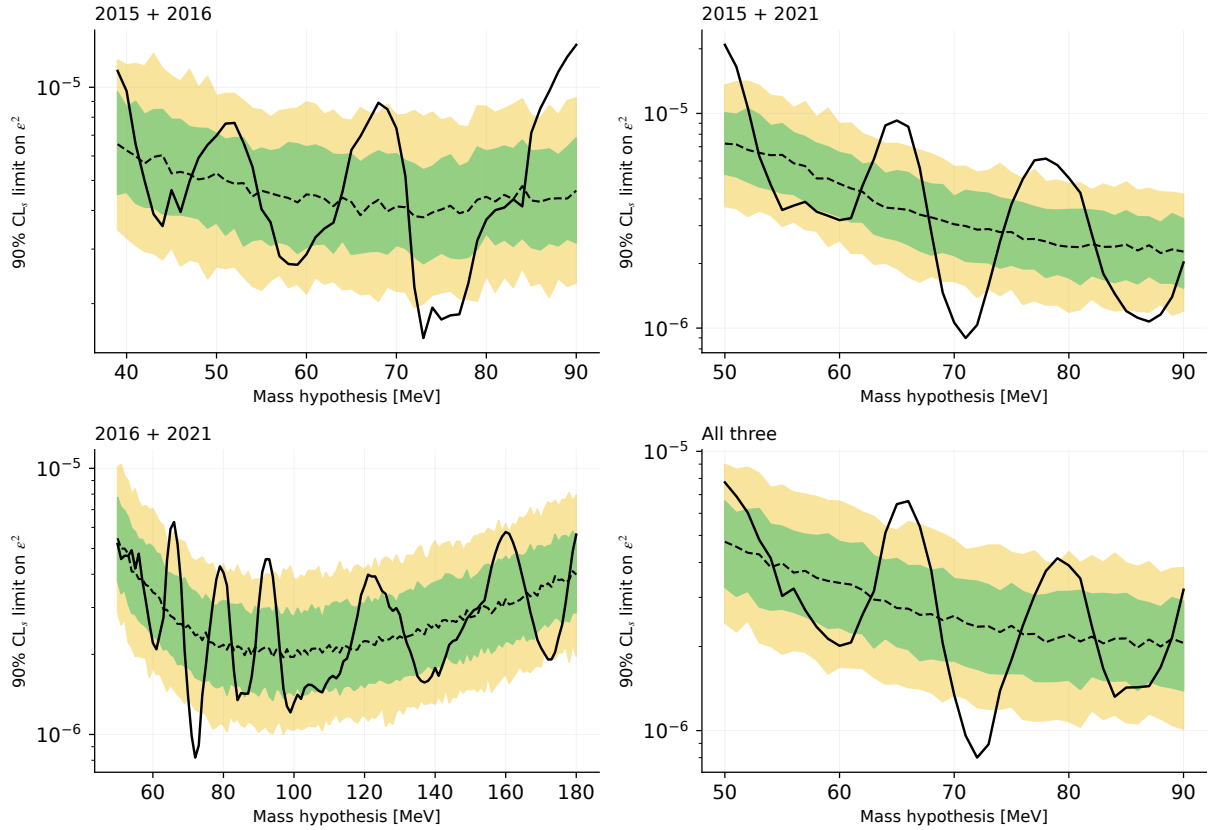


Figure 61: Pairwise and all-three results with the same conditional 300-toy pointwise bands. Comparisons at a given mass use paired constituent draws. Correlation between these panels does not supply correlation between separate mass hypotheses.

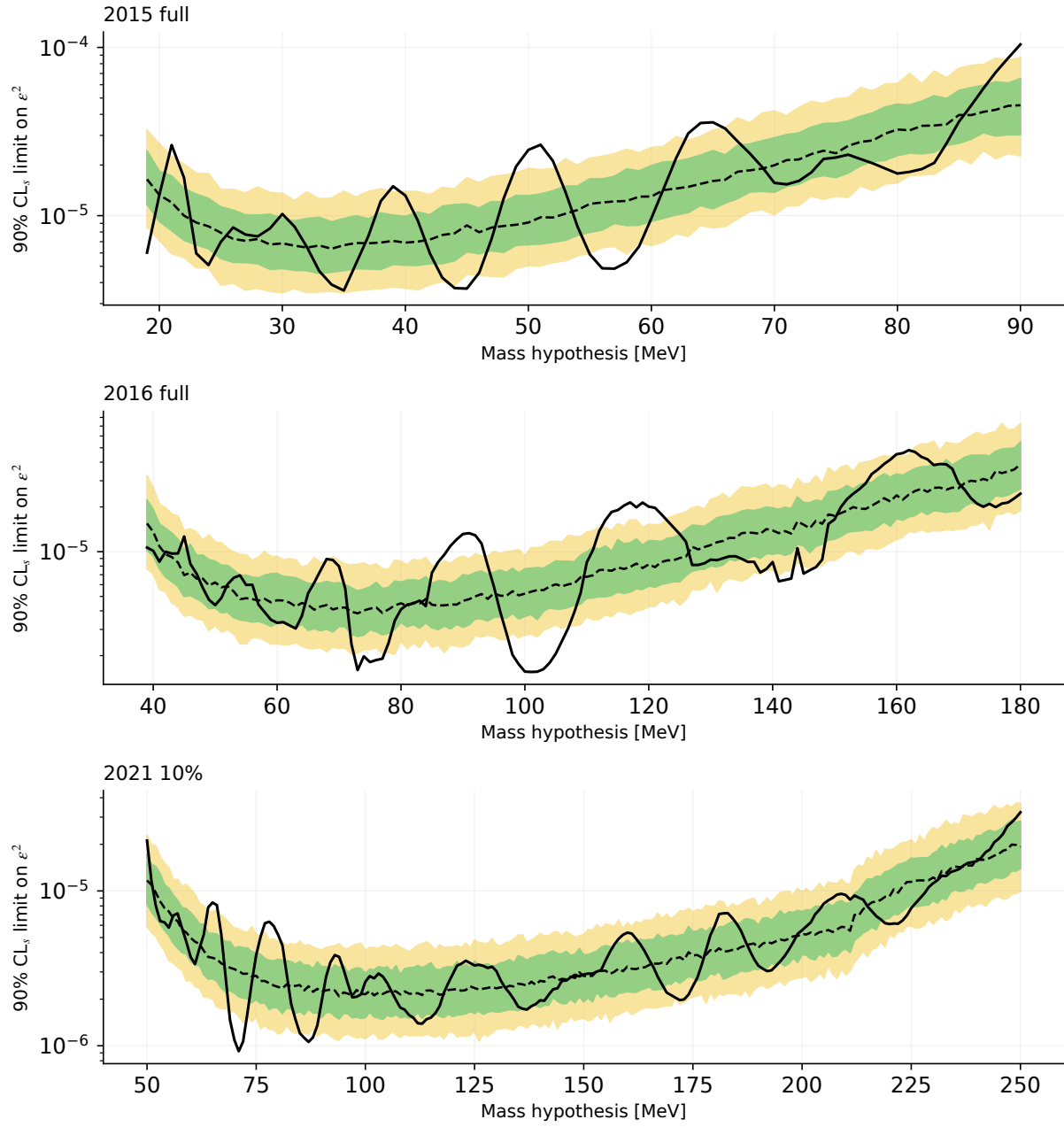


Figure 62: Standalone results for full 2015, full 2016 and the released 2021 10% sample, with the same 300-toy pointwise quantiles. The 2021 curve uses the frozen 36–300 MeV GP support.

6.7 Empirical limit-tail diagnostics and unresolved tails

A second set of probabilities locates an observed *upper limit* within its conditional background-only distribution. A low-limit tail describes an unusually strong observed constraint; a high-limit tail describes an unusually weak constraint. Neither is the local probability of a narrow positive signal. The two-sided diagnostic doubles the smaller tail, capped at one.

The targeted refinement added independent batches of 3,000 or 10,000 draws at 108 sparse scope–mass coordinates, for 660,000 additional scope evaluations. It retained the original 300-toy bands. Nine zero-count cells remained after the refinement. Zero exceedances therefore appear as one-sided 95% Monte Carlo upper bounds, not as zero probabilities. The requested further continuation is outstanding: this editorial session generated no new toys and does not replace unresolved tails with extrapolated probabilities.

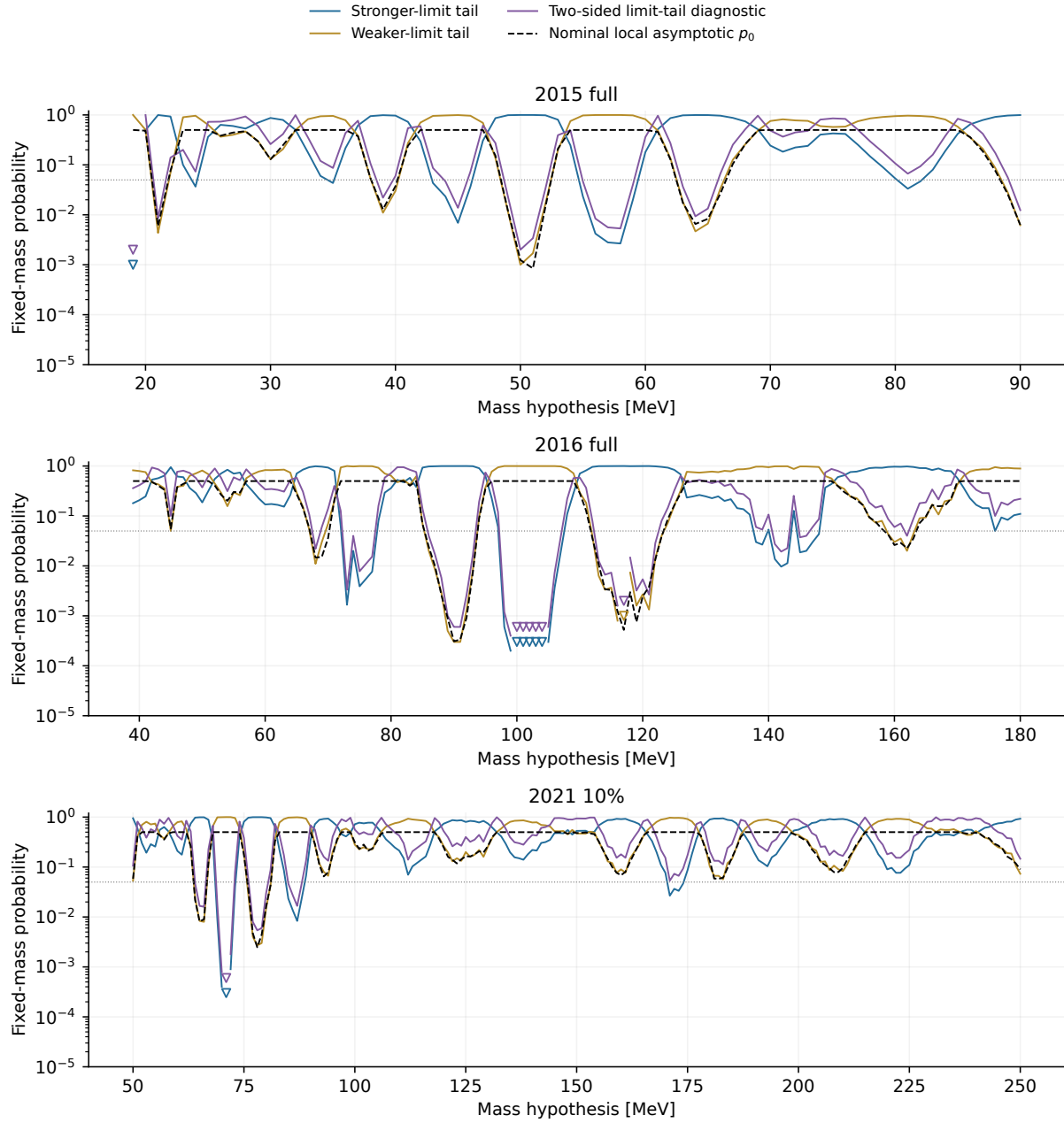


Figure 63: Empirical limit-tail diagnostics for each dataset. Unrefined points use 300 toys; selected coordinates use the saved independent 3,000- or 10,000-toy batches. Open downward triangles show one-sided 95% Monte Carlo upper bounds when no tail event is observed; for the two-sided diagnostic the bound is doubled and capped at one. These probabilities refer to the upper-limit statistic and are distinct from local asymptotic p_0 .

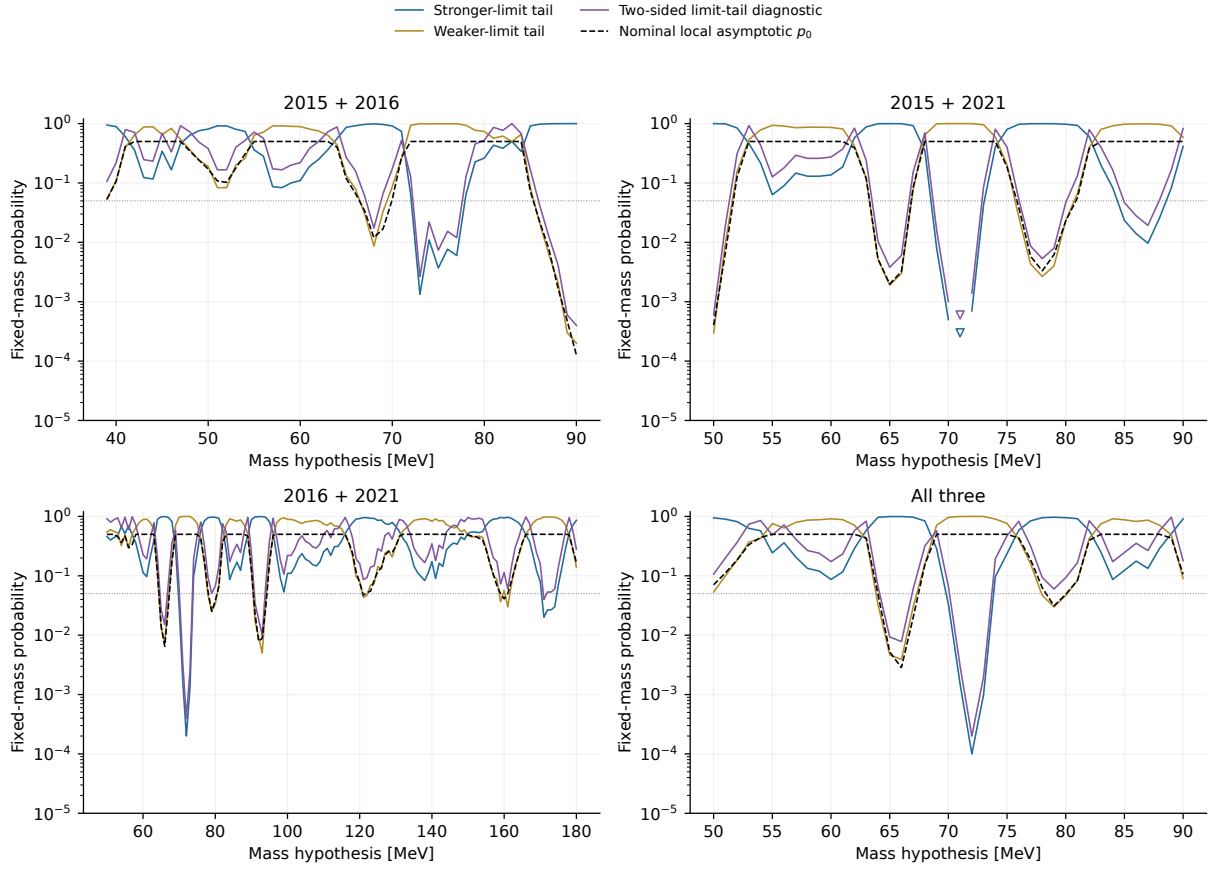


Figure 64: The same empirical limit-tail diagnostics for the pairwise and all-three combinations. Finite estimates and zero-count bounds remain visually distinct. Additional sampling can improve the precision of these conditional diagnostics, but cannot establish physical background validity or a discovery probability.

6.8 Varying the mass resolution in the signal template

The nominal resolutions were introduced with the detector inputs in Section 3. The saved width study repeats the 2021 extraction for templates with widths 0.8, 0.9, 1.0, 1.1 and 1.2 times nominal at all 201 tested masses. It holds the GP mean and covariance, training exclusion, fit bins and full-template yield-to-coupling conversion fixed. Thus the background is still profiled in the local likelihood, but the GP training is not repeated for a changed exclusion width.

All 1,005 saved mass-width limits pass their recorded numerical checks. The nominal calculation reproduces the production limits within a relative 1.28×10^{-6} . Across the grid, the median limit ratios at 0.8 and 1.2 times nominal are 0.799 and 1.202; at 78 MeV they are 0.687 and 1.329. The final result retains the nominal template. These variations describe sensitivity to the assumed shape; their envelope is not a systematic-error band or a likelihood profile of detector-resolution uncertainty.

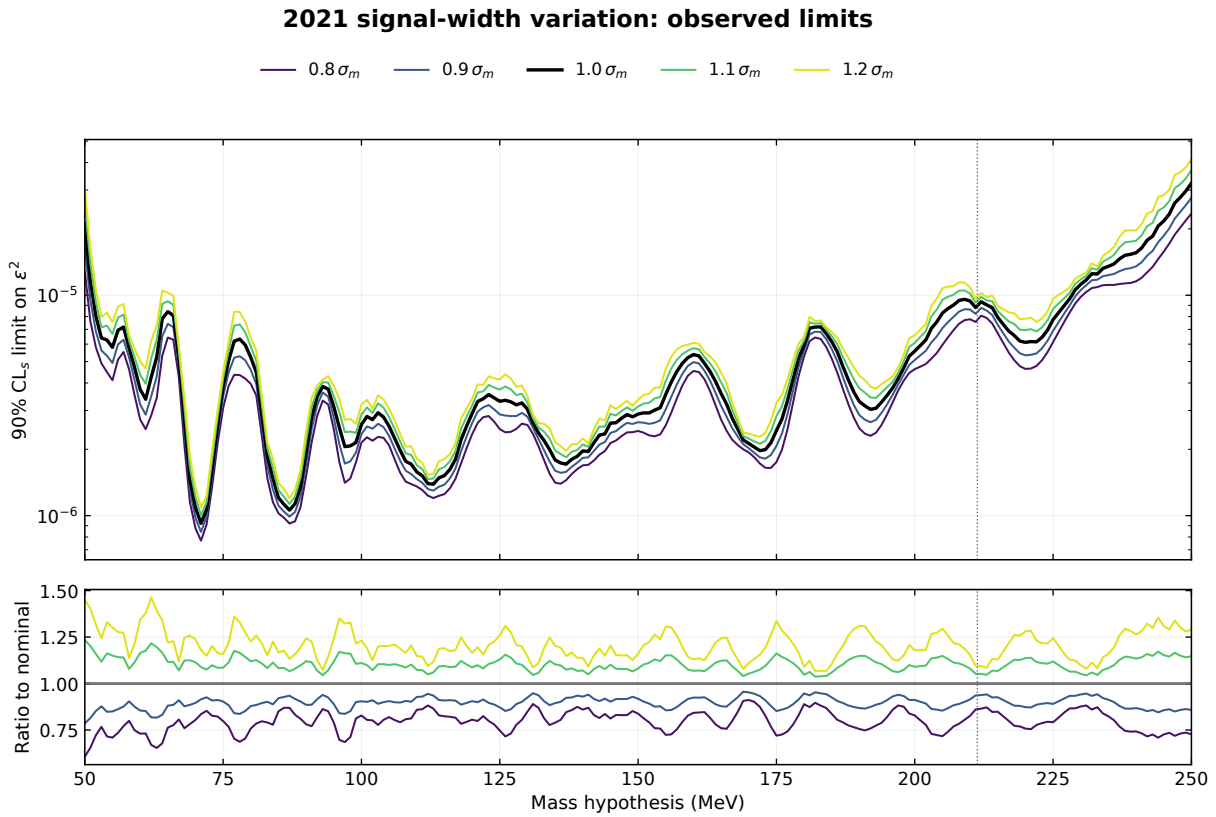


Figure 65: 2021 observed 90% CL_s limits for the five signal-template widths, with ratios to the nominal limit below. Every curve uses the same archived GP prediction and yield conversion. The visible-branching correction is common to all curves and cancels in the ratios. The plots retain all 1,005 saved limits.

Selecting a preferred mass or width after examining these curves does not add independent evidence for a resonance or measure a resolution miscalibration.

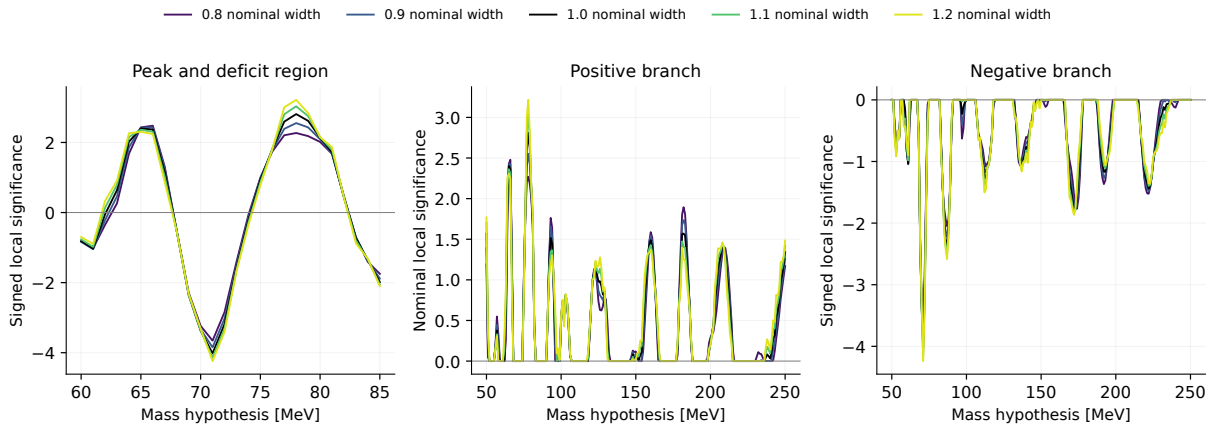


Figure 66: Local significance under the same template-width variations. The plotted signed local coordinate is $r = \text{sign}(\hat{A})\sqrt{2[\text{NLL}(0) - \text{NLL}(\hat{A})]}$; its negative branch describes deficits. The 1.2-width template changes the 78 MeV value from 2.810 to 3.216 and the 71 MeV value from -4.019 to -4.239 . The 66 MeV reference changes from 2.366 to 2.478 for the 0.8-width template.

6.9 Signal extraction from the observed data

The extraction displays show the leading separated positive fluctuations in each campaign, the common-coupling fits at 66 and 92 MeV, and the largest negative fitted contributions. Selection uses the dense observed profiled local statistic; it does not use the stress-centered global ordering. The 66 MeV common-coupling fit is the largest positive combined fluctuation, with nominal local significance about 2.76. The second leading positive feature of the full union is at 21 MeV and contains only 2015; 92 MeV is a second useful multi-campaign comparison, containing 2016 and 2021.

The plotted local coordinate is

$$r = \text{sign}(\hat{A})\sqrt{2[\text{NLL}(0) - \text{NLL}(\hat{A})]}. \quad (103)$$

It agrees with the usual nominal local excess significance on the positive branch. Negative values describe an auxiliary deficit fit, with positive Poisson expectations maintained, and do not represent a physical negative coupling. The probability mappings and their different sign conventions are defined in Section 4.20.

Data	Leading excess	Next separated excess	Deepest deficit
2015 full	51 (+3.14)	21 (+2.52)	19 (−3.21)
2016 full	90 (+3.42)	117 (+3.28)	102 (−4.50)
2021 10%	78 (+2.81)	65 (+2.40)	71 (−4.02)
Combined union	66 (+2.76)	21 (+2.52)	72 (−3.49)

Table 11: Selected mass in MeV, followed by signed nominal local significance. The 19 MeV deficit is at the 2015 search endpoint. These values rank selected local fits and are not global significances. The additional combined 92 MeV fit has nominal local significance 2.42.

Displays retain native analysis bins and add sideband context. Residuals subtract the same unprofiled GP mean throughout. Inside the likelihood window, separate curves show the profiled-background displacement, total displacement and signal. Lines join saved bin predictions, shown as counts per MeV. Multi-campaign sums use exact shared whole-bin edges and have no independent fit.

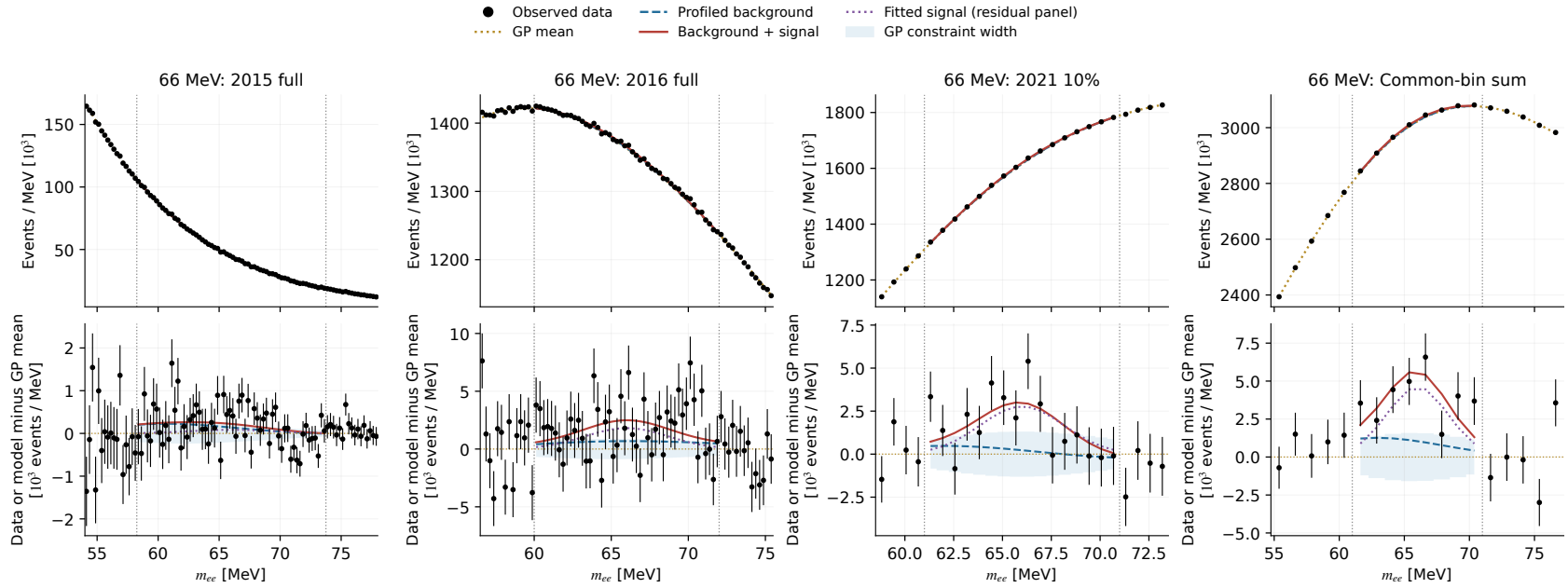


Figure 67: Common-coupling extraction at 66 MeV, with a shared fitted amplitude $\hat{\epsilon}^2 = 4.55 \times 10^{-6}$ and nominal local significance 2.76. Each campaign retains its own yield conversion and resolution. Bars show counting error only. Colored shading in the residuals is the zero-centered GP constraint width; it is neither fitted-background uncertainty nor total residual uncertainty.

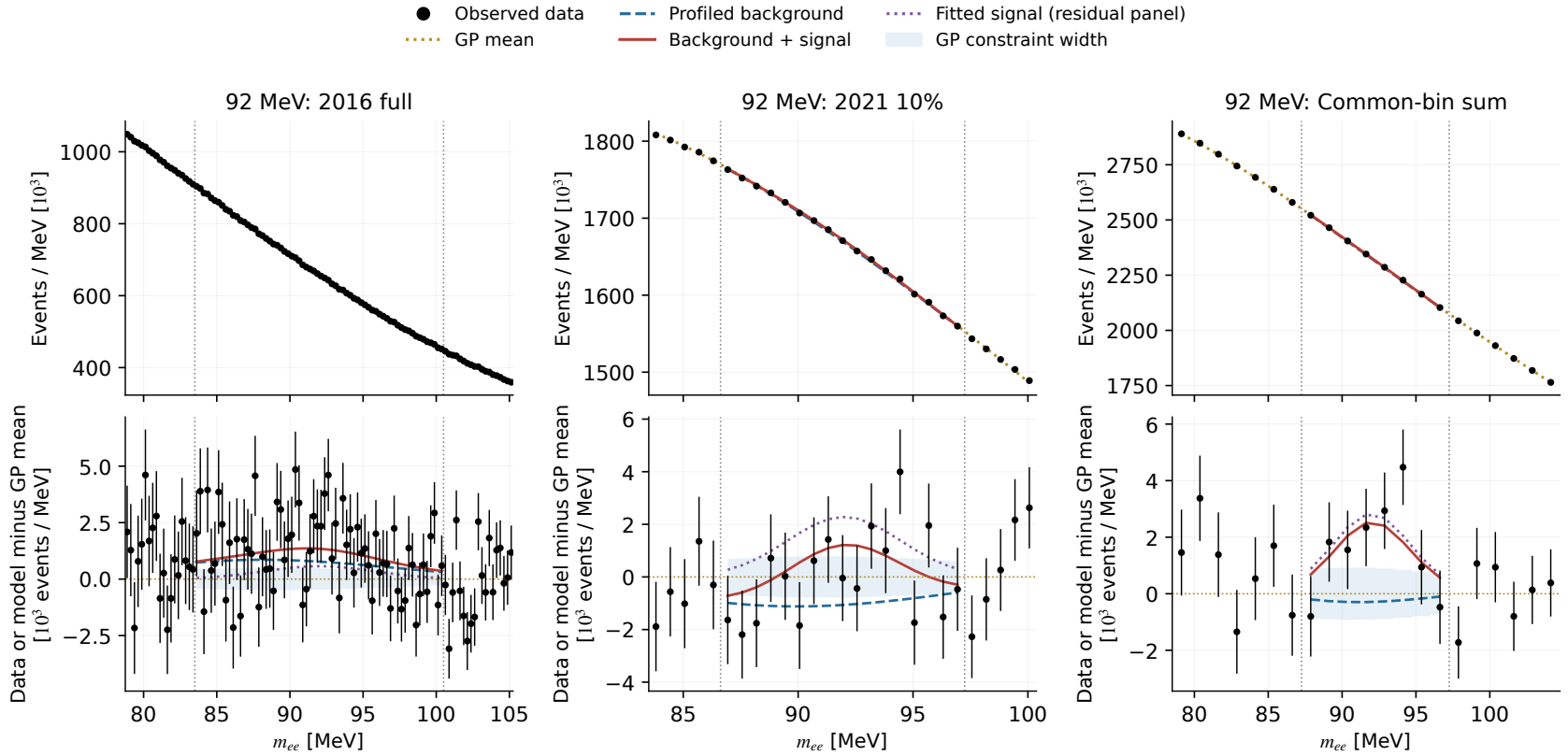


Figure 68: The 92 MeV common-coupling fit uses 2016 and 2021 and gives $\hat{\epsilon}^2 = 2.97 \times 10^{-6}$ with nominal local significance 2.42. The separate preferred rates differ, as shown at the end of this results section. Bars show counting error only; residual shading shows the zero-centered GP constraint width, not fitted-background or total residual uncertainty.

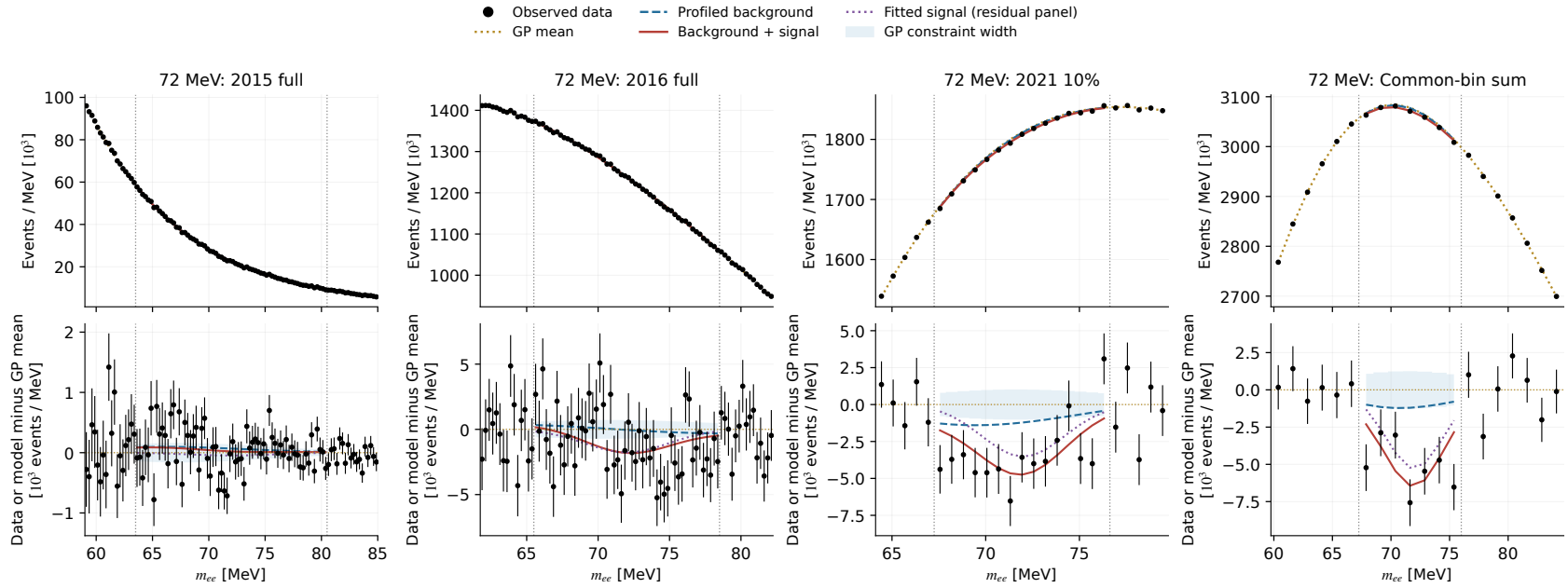


Figure 69: The deepest combined observed deficit at 72 MeV has signed local significance -3.49 and an auxiliary amplitude $\hat{\epsilon}^2 = -4.88 \times 10^{-6}$. The negative signal template describes missing counts relative to the fitted background. Bars show counting error only; residual shading shows the zero-centered GP constraint width, not fitted-background or total residual uncertainty.

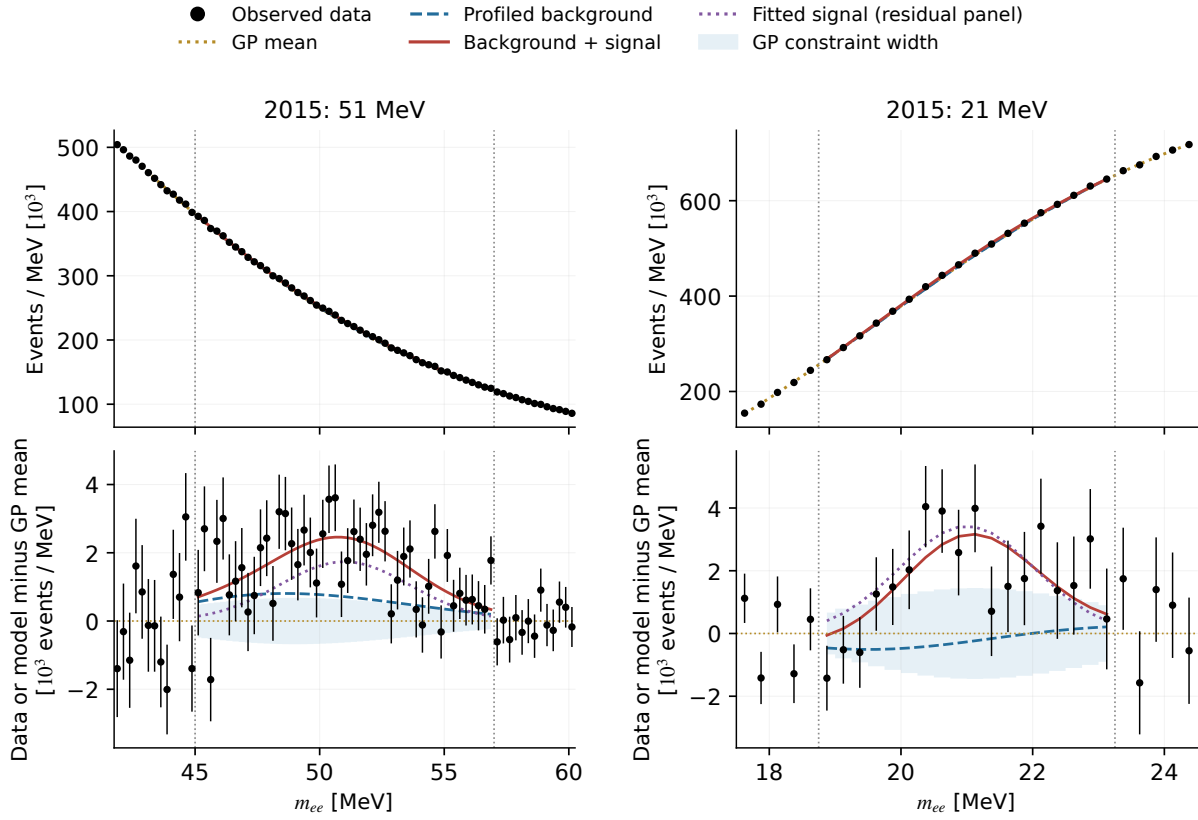


Figure 70: The two leading separated 2015 excesses at 51 and 21 MeV. The 21 MeV feature lies close to the search endpoint and has strong background-model dependence in the traditional-fit cross-check. Bars show counting error only; residual shading shows the zero-centered GP constraint width, not fitted-background or total residual uncertainty.

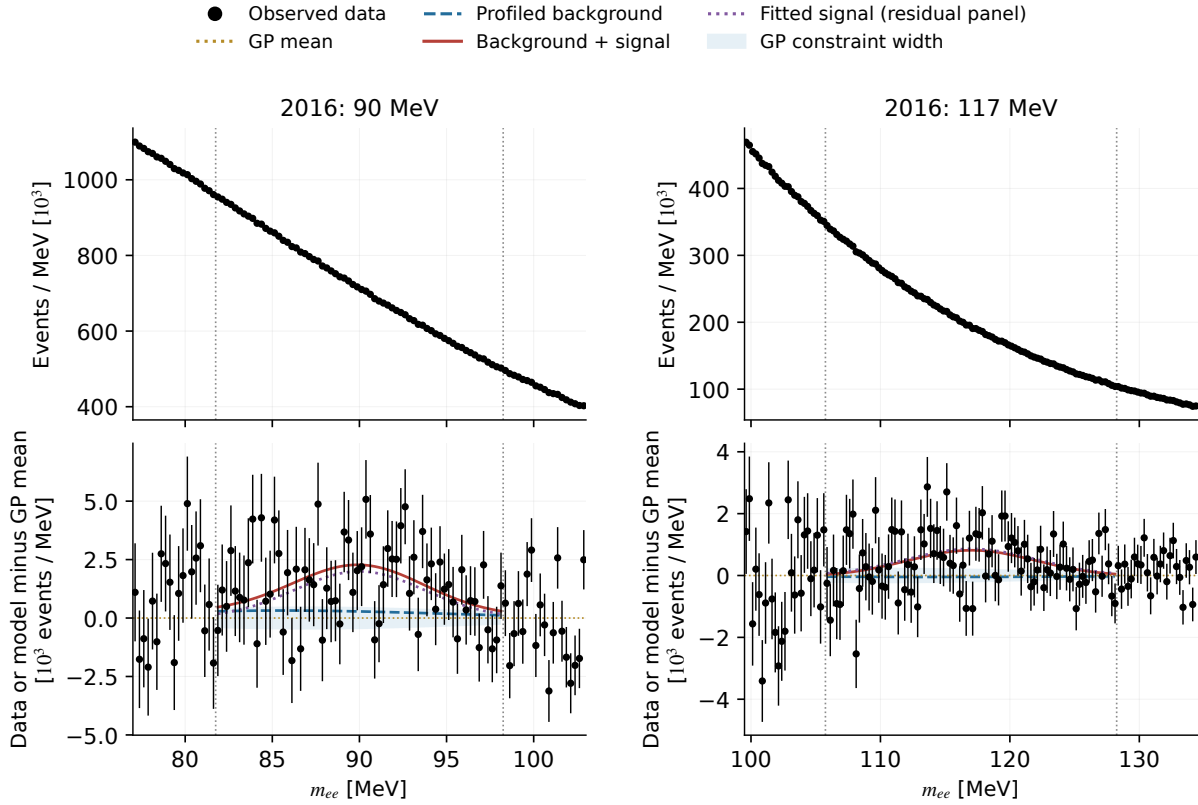


Figure 71: The leading separated 2016 excesses at 90 and 117 MeV. These fits retain the inherited 2016 numerical and background-source qualifications. Bars show counting error only; residual shading shows the zero-centered GP constraint width, not fitted-background or total residual uncertainty.

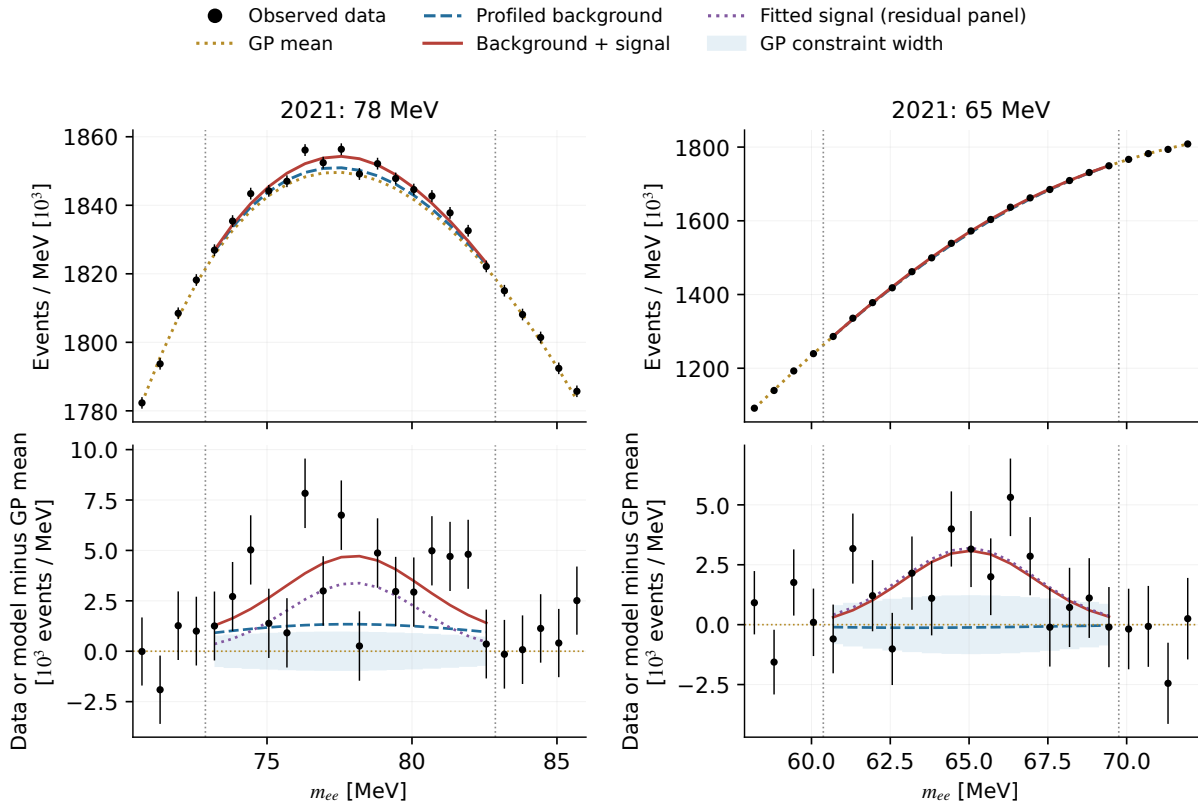


Figure 72: The leading separated excesses in the released 2021 10% sample at 78 and 65 MeV. These masses are fixed by the existing observed scan and are distinct from the combined 66 MeV coordinate. Bars show counting error only; residual shading shows the zero-centered GP constraint width, not fitted-background or total residual uncertainty.

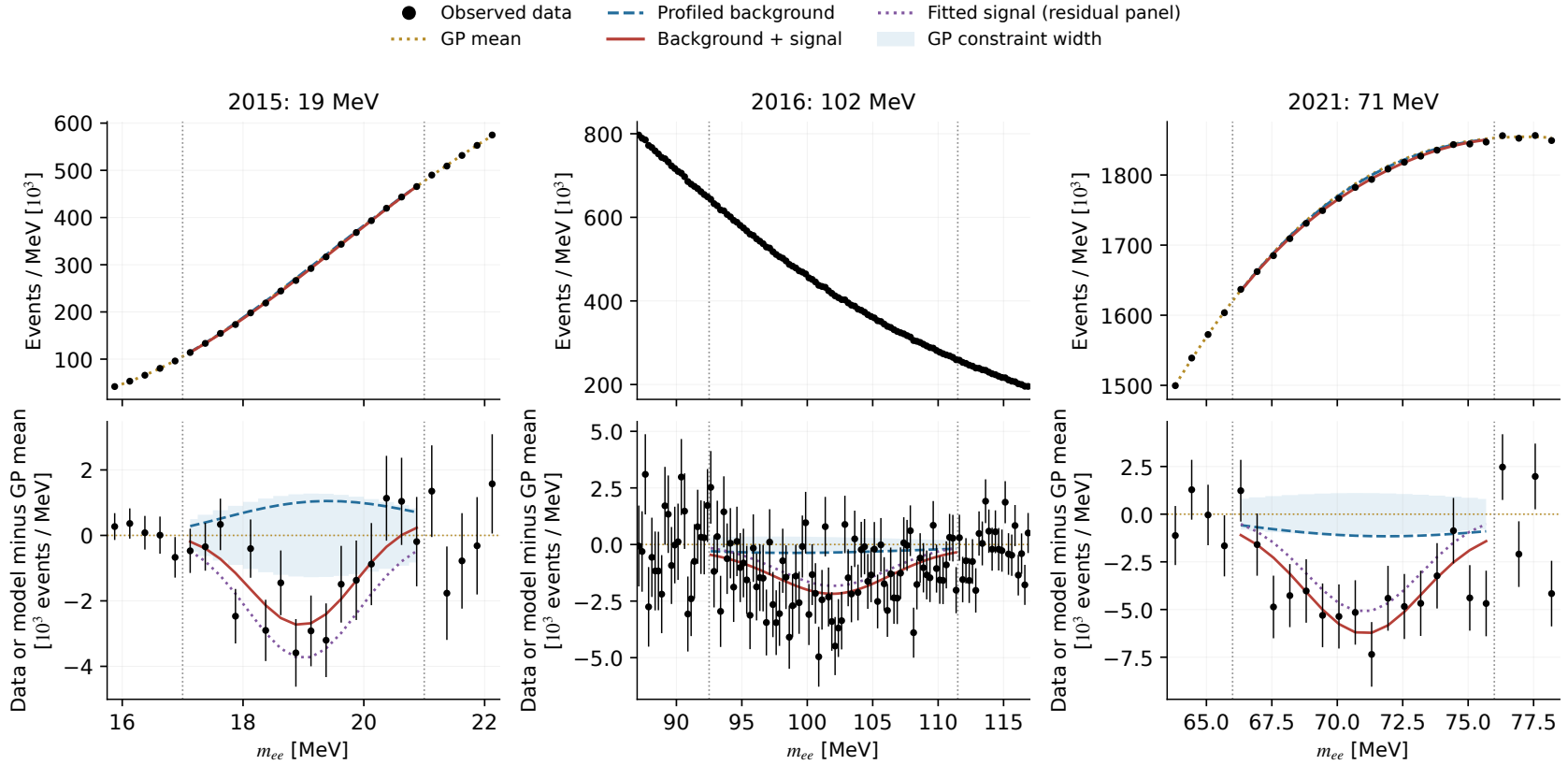


Figure 73: Individual deficit extractions at 19, 102 and 71 MeV in 2015, 2016 and 2021. The first is a search endpoint. Bars show counting error only; residual shading is the zero-centered GP constraint width, not fitted-background or total residual uncertainty. These are local background diagnostics. The joint deficit scan is retained in Appendix D.

6.10 Correlated fluctuations and signal-induced echoes

A positive signal can affect a neighboring mass test because the exclusion window moves. Counts protected from background training at the signal mass can enter the sidebands of a nearby hypothesis. The GP prediction then changes, and the neighboring signal fit can prefer a deficit. A negative fitted contribution can therefore arise without removing any physical events from the generating spectrum.

The deterministic 2021 replay in Figure 74 demonstrates this mechanism with saved positive templates and the current dense solver. Relative to its own background-only reconstruction, a 66 MeV injection changes the fitted local coordinate at 71 and 72 MeV by -1.646 and -1.740 ; a 78 MeV injection produces changes -1.825 and -1.968 . These changes are expressed in units of the nominal local-significance coordinate. They are not separately calibrated significances of an echo. The selected generating strengths are 17,142 events at 66 MeV, 19,273 at 78 MeV and 36,373 for the 65+78 MeV pair; the comparison is not an equal-strength experiment or a fitted two-particle hypothesis.

Figure 75 places the three individual correlation matrices beside the combined field. The off-diagonal positive and negative lobes show that adjacent fitted extrema need not be independent. In the combined panel, the response basis retains correlations where campaigns overlap and gives zero covariance where no data are shared. These are conditional covariance predictions; the independent complete-Poisson validation is summarized in Section 4.20.

A correlation slice can be expressed in the same local-significance units. With $z_m = (r_m - a_m)/s_m$, conditioning a Gaussian field on $z_{m_0} = 1$ gives $E[r_m - a_m \mid z_{m_0} = 1] = s_m R_{m,m_0}$. Figure 76 uses fixed anchors from the existing candidate catalogue. They illustrate the expected accompanying fluctuation under the background covariance; they are distinct from the positive-signal injection replay above.

Figure 77 gives a complementary observed-data check. Replacing the selected 2021 candidate regions changes the fitted 71 MeV deficit from -4.02 to -0.72 , yet the remote observed scan retains 92% of its original standard deviation. The corresponding retained fractions in 2015 and 2016 are 67% and 78%. Some neighboring structures are coupled to the replaced regions, while other variation remains. The replacements are conditional interventions and cannot identify a particle or create a new null ensemble.

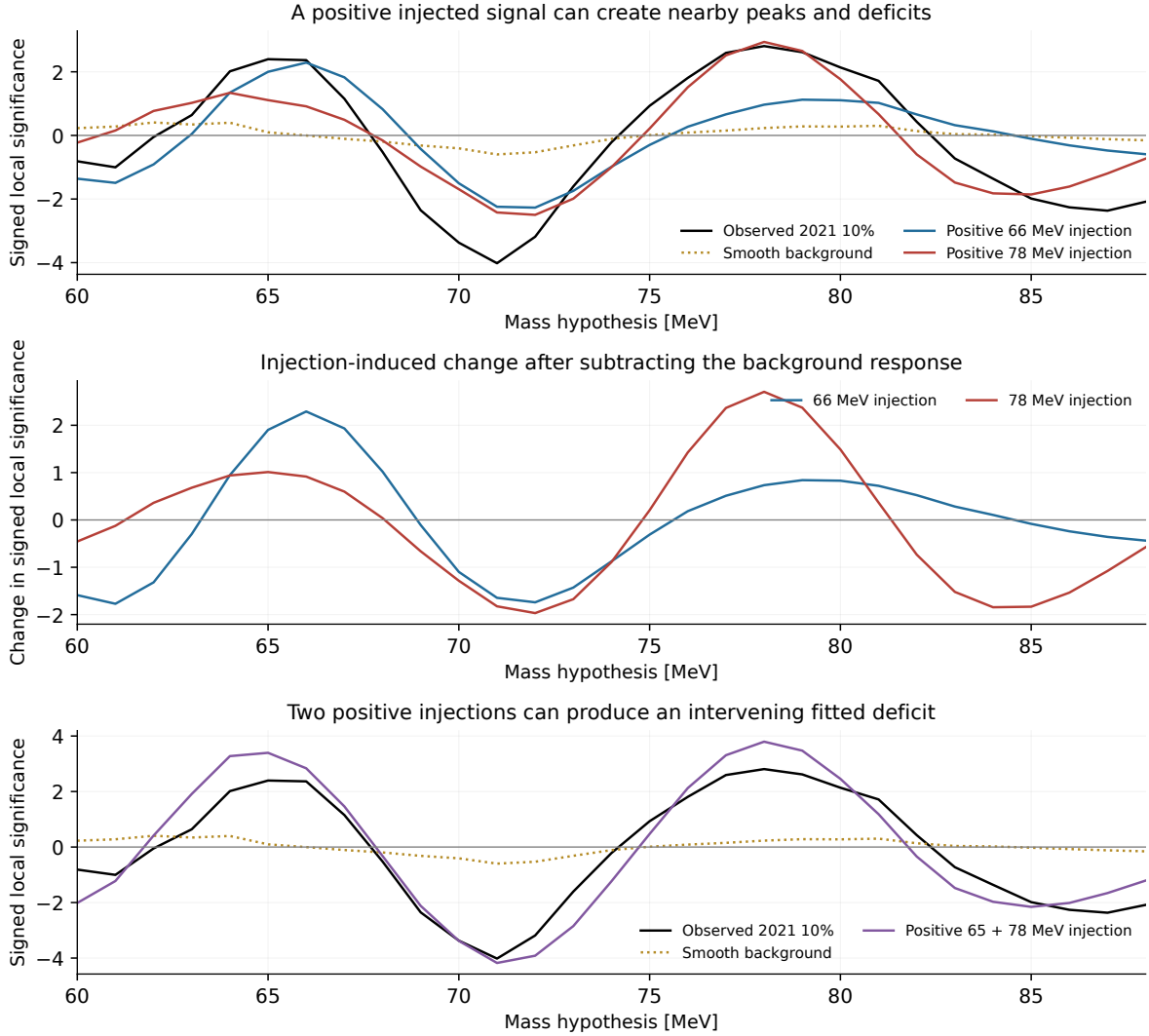


Figure 74: Positive injected signals can induce nearby fitted oscillations through the moving-mask background regression. The injection-induced change is $\Delta r(m) = r_m(B + S) - r_m(B)$ for the same smooth reference spectrum. The 65+78 MeV pair can induce a deep fitted deficit even though both generating signal components are nonnegative. These are conditional response illustrations; their agreement with one selected dip does not establish the identity of the observed parent peak.

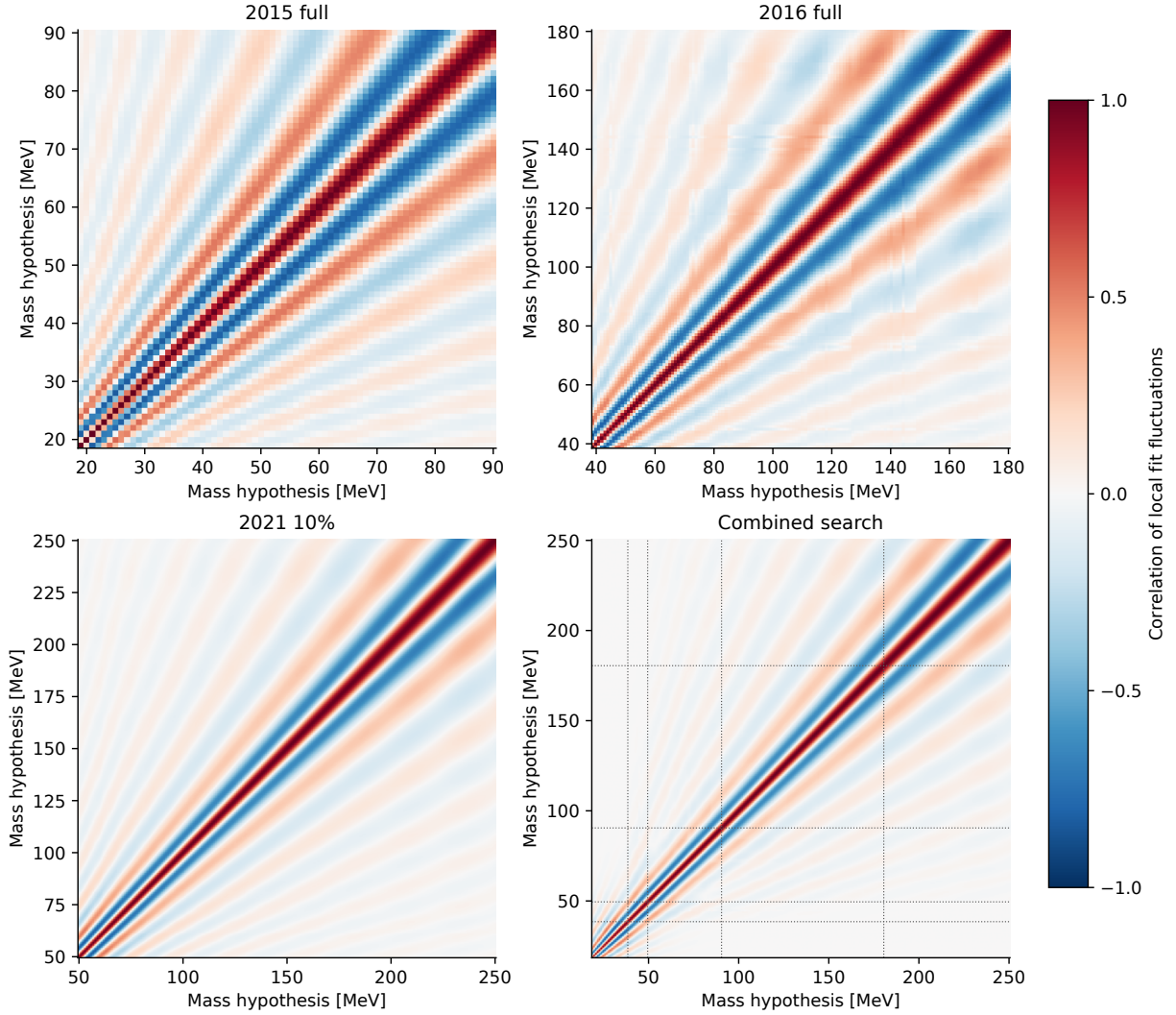


Figure 75: Profiled-response correlation matrices under the archived generating backgrounds: 2015 full at top left, 2016 full at top right, 2021 10% at bottom left, and the combined search at bottom right. All four panels use the same dimensionless color scale. Positive and negative entries describe same-direction and opposite-direction statistical fluctuations. Dotted lines in the combined panel mark changes in the active datasets. The matrices quantify dependence; they do not assign a probability to a selected observed echo or describe correlations of physical signal production.

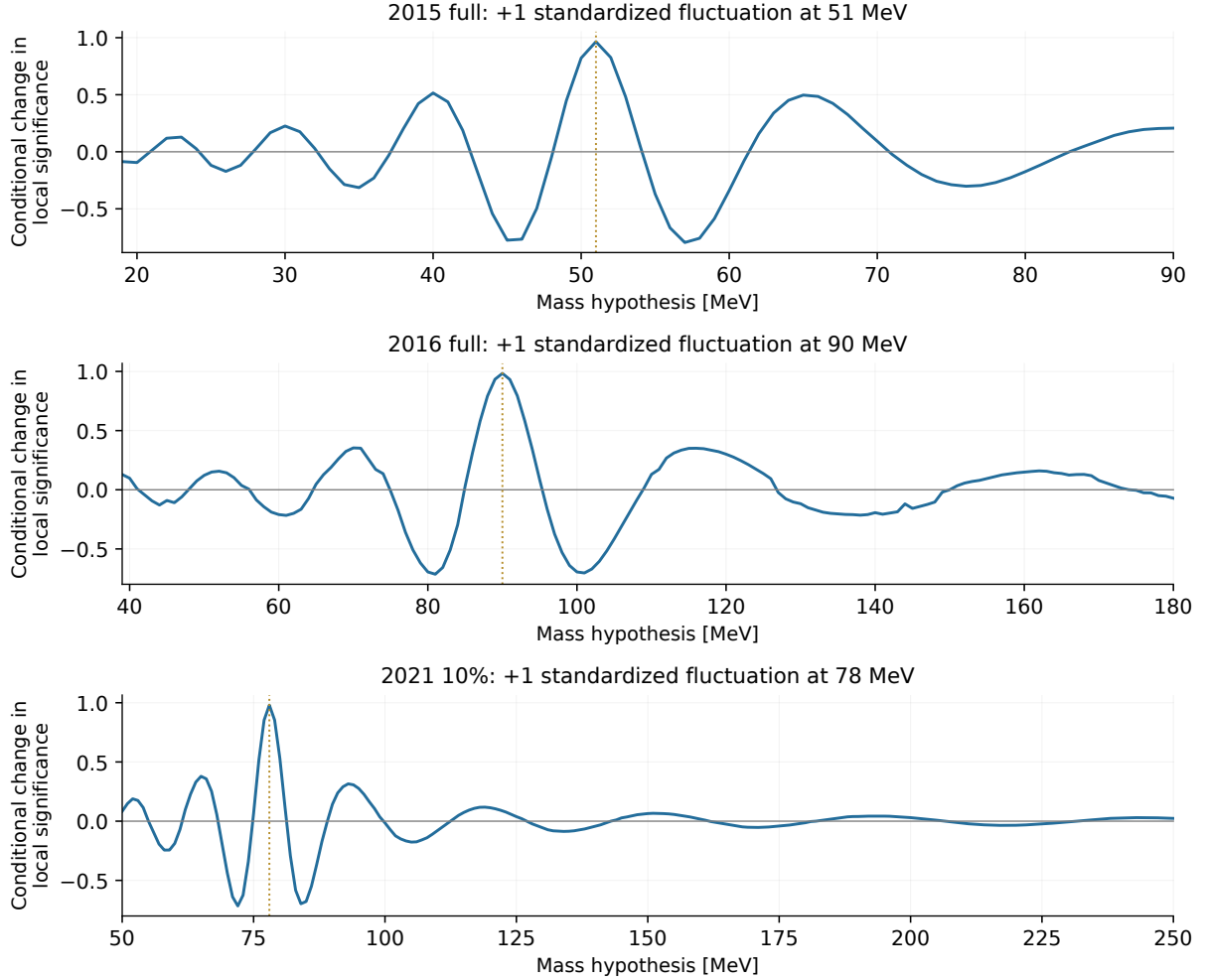


Figure 76: Conditional changes in the local-significance coordinate following a one-standard-deviation field fluctuation at 51 MeV in 2015, 90 MeV in 2016 and 78 MeV in 2021. The curves are $s_m R_{m,m_0}$, with each archived offset removed. Positive and negative lobes show the scale and sign of correlated fluctuations. These are Gaussian conditional means, not injected particles, probabilities of observed echoes, or independent significance gains.

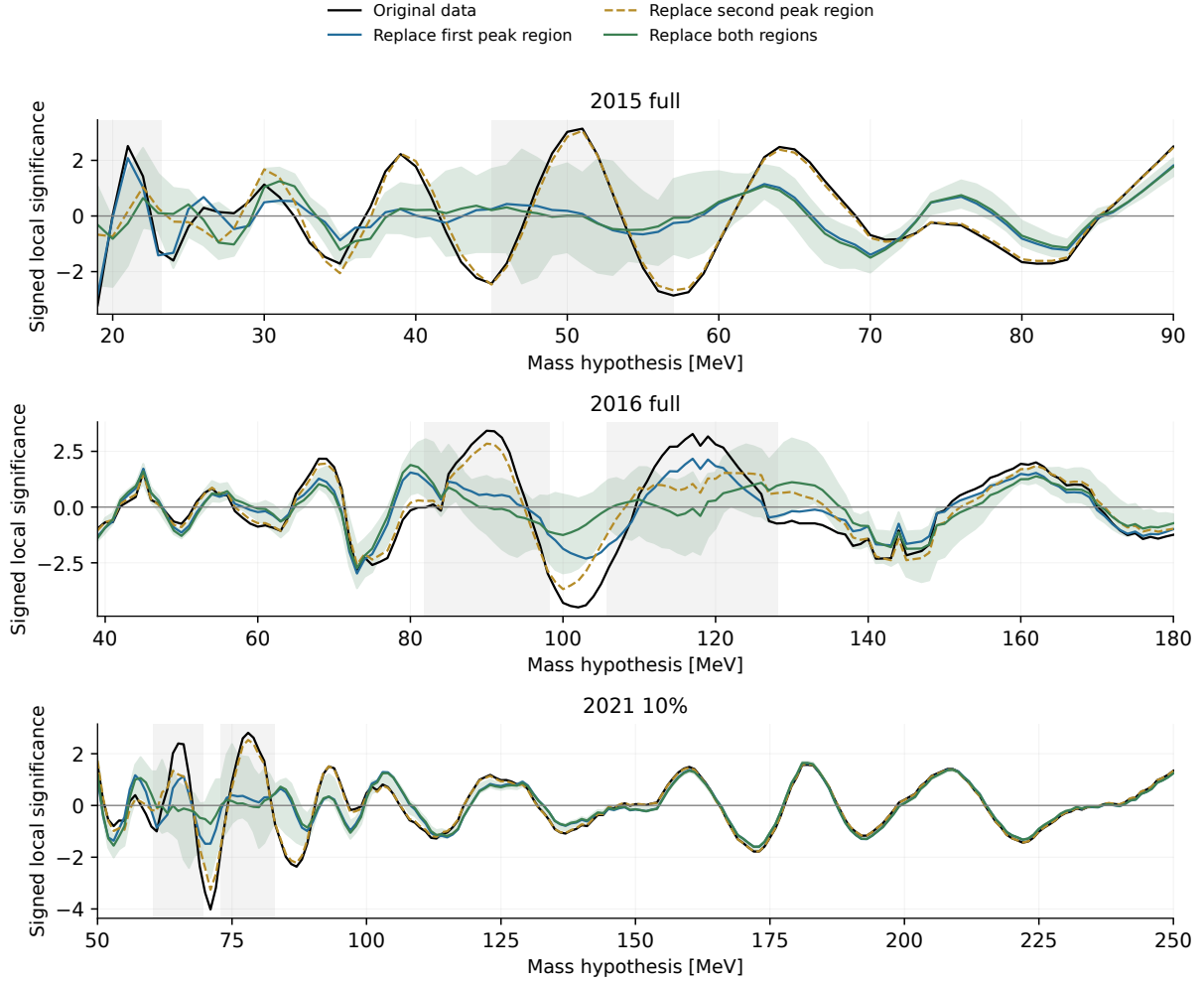


Figure 77: Observed scans before and after conditional replacement of the two selected candidate regions in each year. The first and second regions are, respectively, 45–57 and 18.75–23.25 MeV in 2015 (centers 51 and 21 MeV); 81.75–98.25 and 105.75–128.25 MeV in 2016 (90 and 117 MeV); and 72.875–82.875 and 60.375–69.75 MeV in 2021 (78 and 65 MeV). Boundaries follow the stored whole-bin selections around the $2.25\sigma_m$ masks. The envelope spans ten conditional replacements and is not a confidence band. Changes demonstrate how the local fit responds to altered candidate bins; the untouched remote variation is quantified in Appendix C. All curves retain their existing tested masses and the same nominal local-significance mapping.

6.11 Local probabilities and the conditional GP global studies

The following plots apply the method in Section 4.20 to each campaign's complete mass scan. They distinguish the raw local asymptotic reference, the local probability after accounting for the archived background response, and the scan-global probability under that same model. The global correction acts on a specified probability ordering; it does not rescale the observed upper limit.

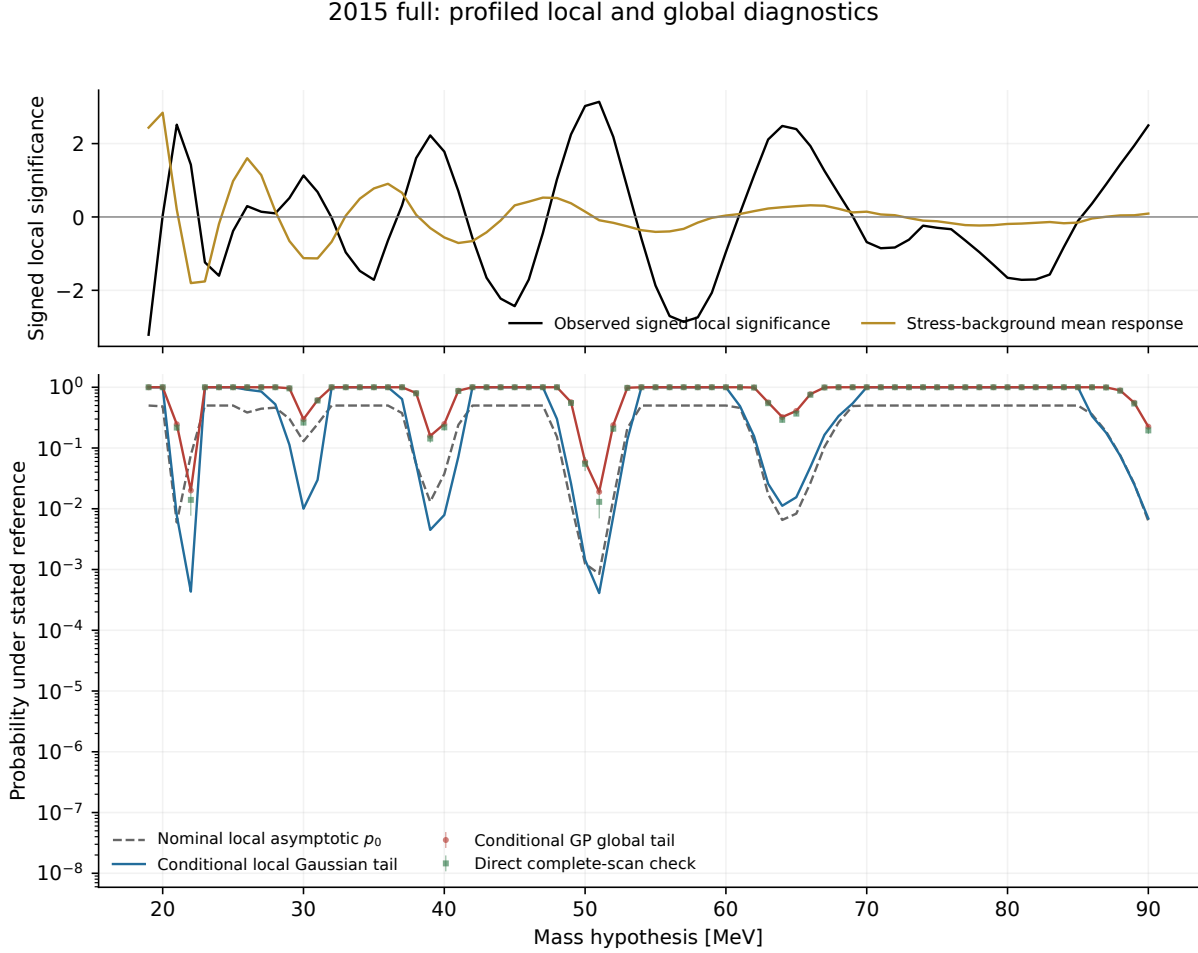


Figure 78: Profiled 2015 local and conditional GP global probabilities. The study used ten complete pilot scans, 1,000 independent validation scans and 200,000 GP fields. The principal global probability at the selected 51 MeV threshold is 0.018985 in the GP approximation; 13 of 1,000 direct scans exceed that threshold. The precision of either ensemble and the generating-background assumptions remain explicit.

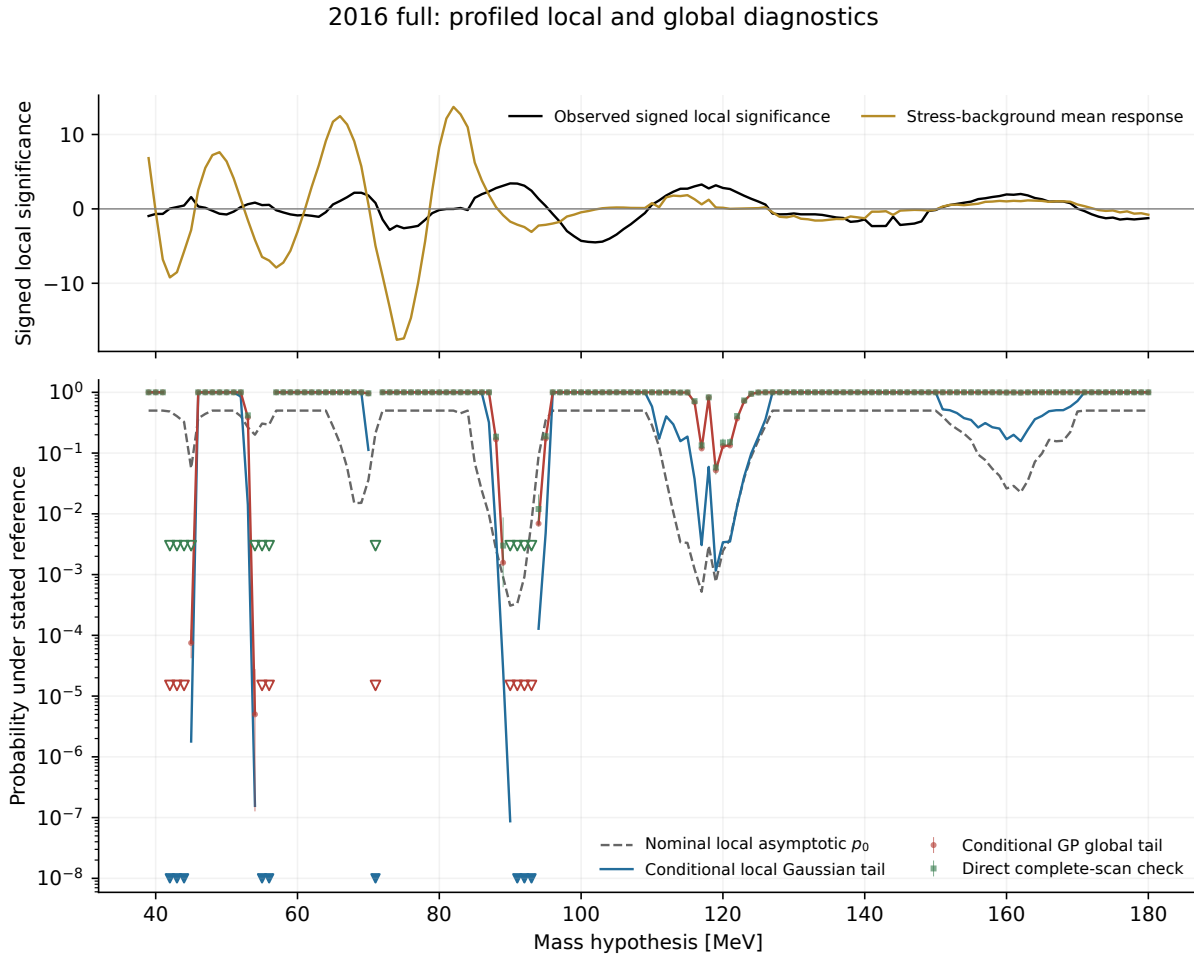


Figure 79: Profiled 2016 conditional global study. The leading principal-ordering threshold at 42 MeV has no exceedance in either ensemble. The corresponding one-sided 95% bounds are 1.498×10^{-5} for 200,000 GP fields and 0.002991 for 1,000 direct scans. They are finite-sampling bounds under the specified models, not a resolved rare particle probability. A qualified final 2016 global result remains outstanding.

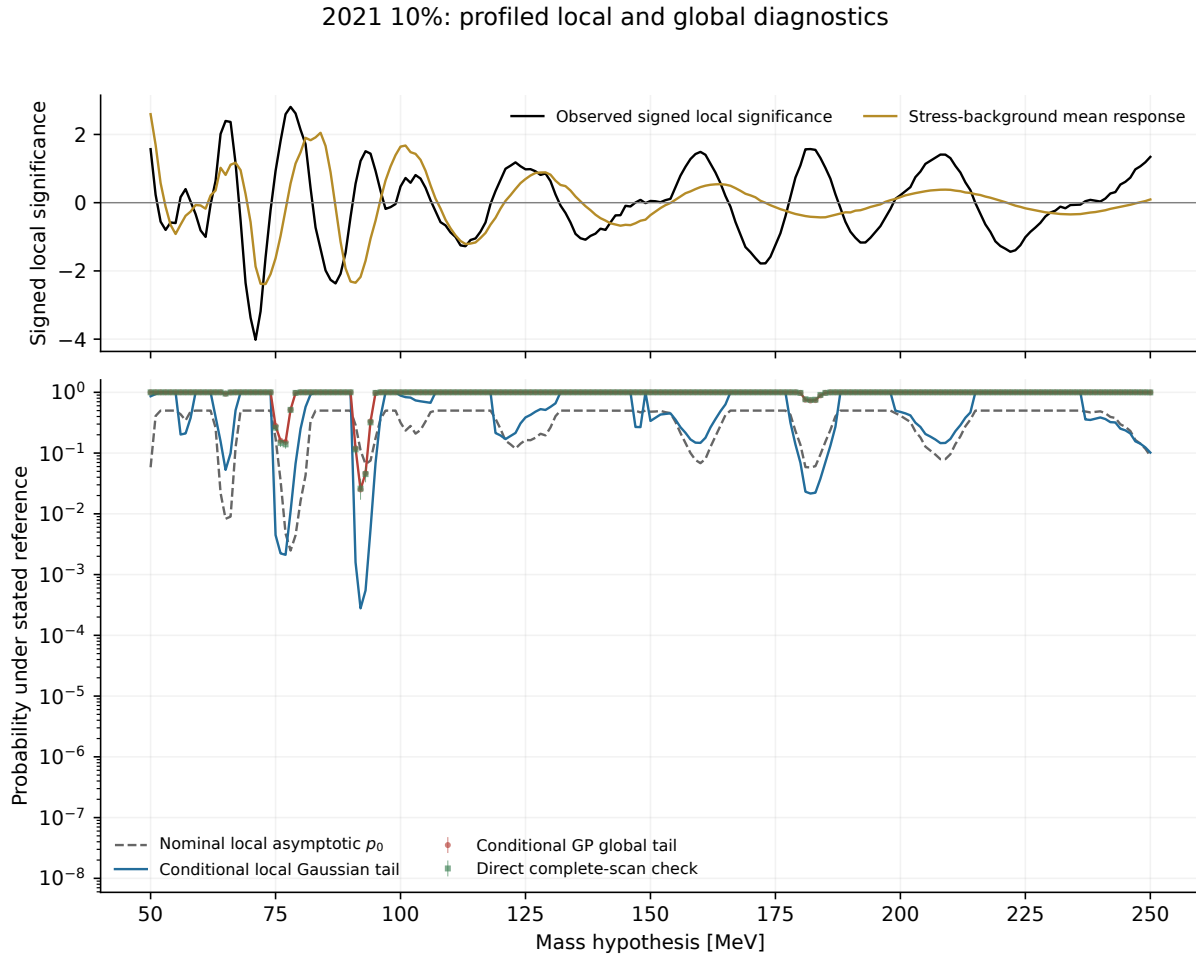


Figure 80: Profiled global study for the released 2021 10% sample. The principal ordering selects 92 MeV and gives GP global probability 0.02519, with 26 of 1,000 independent direct scans above the threshold. This selection differs from the largest raw positive fitted signal. Qualification of the generating background and the final analysis choices remains required for a final particle-global interpretation.

The combined study illustrates why the distinction matters. At 76 MeV the observed raw local coordinate is only 0.166, corresponding to asymptotic $p_0 = 0.434$. The archived stress spectrum has an offset of -8.700 and a response width of 0.979, giving an offset-adjusted score of about 9.05. This is a large disagreement with that reference response, not a 9σ observed resonance. Under the separate raw-maximum ordering, every simulated scan exceeds the observed maximum. The two tests cannot be interchanged according to which yields the more striking probability.

Combined full search: profiled local and global diagnostics

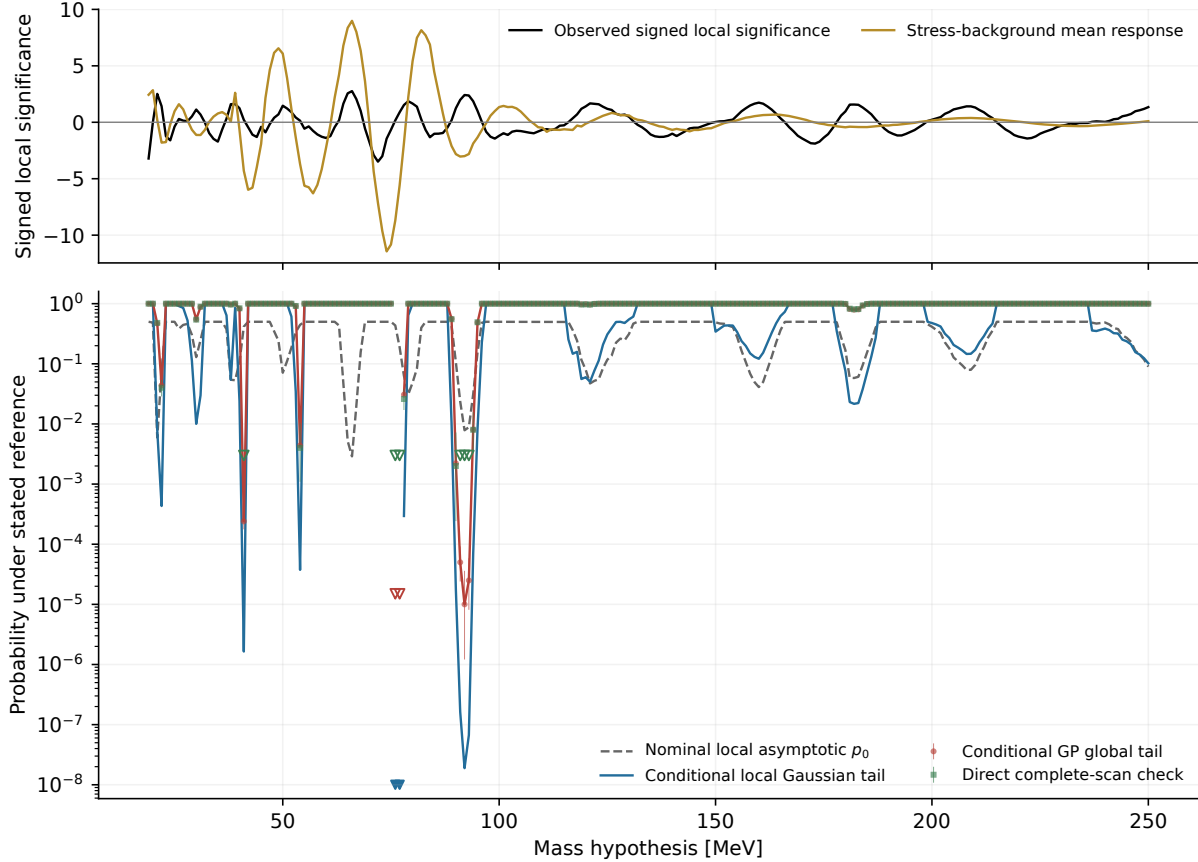


Figure 81: Full combined search with the same profiled local and conditional global probability conventions. The upper panel shows the observed local coordinate and the stress-background offset so that an extreme centered tail can be traced to its reference. Open downward triangles are one-sided 95% Monte Carlo upper bounds for zero counts; triangles at the local plotting floor mark analytic values below 10^{-8} . The raw and centered orderings are separate tests.

6.12 Agreement of fitted rates across campaigns

A shared coupling requires the signal yields in different campaigns to agree after their different normalizations are applied. Figure 82 compares the separate fitted coupling estimates with the common fit. The quantity formerly labelled ΔD is the loss in twice the log likelihood when separate campaign amplitudes are replaced by one shared amplitude:

$$\Delta D = 2 \left[\text{NLL}_{\text{common}} - \sum_d \text{NLL}_{d,\text{independent}} \right]. \quad (104)$$

A larger loss means the common rate describes the selected local spectra less well than separate rates. The number of freed amplitudes is one fewer than the number of active campaigns. Because the displayed masses were selected from the data, this descriptive comparison is not assigned an unadjusted compatibility probability.

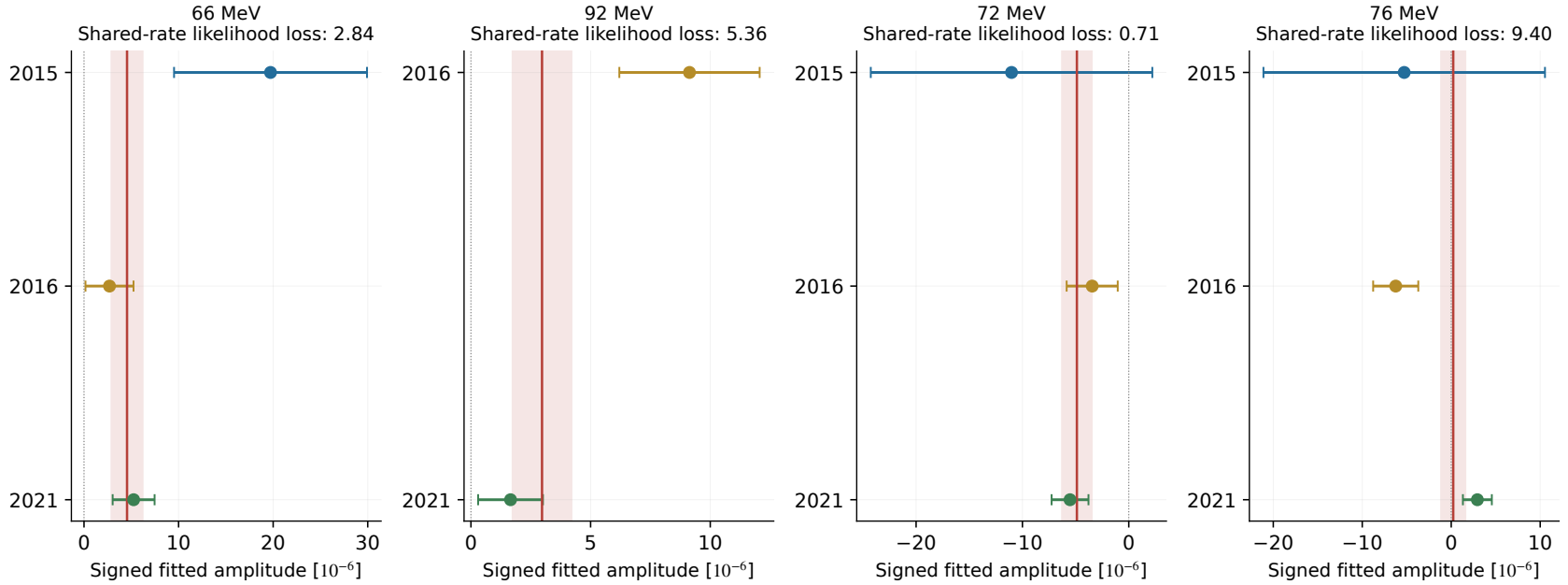


Figure 82: Separate campaign coupling estimates and local curvature errors compared with the common rate. The likelihood loss quantifies how much the local fit improves when each year can choose its own amplitude. The 66 MeV fit has positive contributions from all three years; the 92 MeV separate rates differ more visibly. These selected comparisons and local errors are not post-selection compatibility tests.

7 Status for the full-data unblinding review

The current analysis gives a reproducible set of observed limits and local asymptotic diagnostics for full 2015, full 2016 and the released 2021 10% sample. The common-coupling construction respects each campaign’s signal normalization, resolution and background constraint. Conditional injection studies support retaining the profiled background rather than fixing its predicted mean exactly. The later studies also make clear where the apparent precision of a local fit exceeds what the background validation establishes.

Three issues remain material to review. First, the 2016 state archive retains the disclosed cross-process reproduction exception; downstream prediction checks do not erase it. Second, the calibration study finds substantial rejection-frequency discrepancies for the raw asymptotic procedure under some declared generating backgrounds. Third, the most extreme GP global tails can be driven by large offsets in those background constructions, rather than by large positive observed signals. Those findings require a qualified background family and a declared final calibration choice before the corresponding probabilities acquire a final physics interpretation.

The additional data should test choices fixed using the currently released sample. Candidate masses, template widths, background controls, mass ranges, analysis variants and any sequential data looks need to be recorded before examining the new events. A staged increment can be assessed separately from its cumulative combination so that persistence is not inferred only from a sample containing the original fluctuation. This note does not prescribe or perform an additional data release; it provides the analysis record for that decision.

The remaining editorial and computational tasks are tracked in the accompanying handoff. In particular, unresolved Monte Carlo tails have retained bounds, unqualified 2016/2021 global results have explicit status statements, and no new toy ensemble or additional observed-data sample was introduced by the v5 consolidation.

A Change log and study history

This appendix begins with the history of the studies used to prepare the version 5.0.0 unblinding-request draft. The main text is a curated account of the current proposed procedure, while this chronology explains how the support choices, background treatment and interpretation changed. A later report does not automatically supersede every earlier result: the underlying data fraction, fit policy, numerical qualification and probability definition determine which comparison is valid.

A.1 Historical validation and the full-note foundation

The v4 analysis established the sideband GP interpolation, signal extraction and bounded asymptotic CL_s workflow. Subsequent v4.6–v4.8 studies examined trigger categories, injected-signal response, guard bands, residual structure and smooth alternative backgrounds. Their plots remain useful historical controls. They are not measurements of observed-data bias or global significance, and tests with different truth constructions or injected-yield definitions are not combined into a single ensemble.

The August 2026 v4.9 and v4.9.1 threshold studies established that 2021 signal recovery depends on the lower training edge and on the generating shape. The v4.9.5 support study, recorded on 20 August, compared six lower edges. No choice passed its original uniform mean-pull criterion. A documented amendment selected the 36 MeV edge under a practical criterion and confirmed it with independent backgrounds. The remaining 55 MeV offsets are retained in Appendix B. The older 40–300 MeV 2021 support is historical context; the current 2021 card uses the documented 36–300 MeV support.

The v4.9.7 study of 2 September attempted the 2016 support qualification needed for subsequent combined extraction. Its initial support scan failed the declared gate and stopped; its planned continuation and combined validation were not run. A later, separate observed-state replay retained an explicit 2016 numerical exception: 87 of 142 states were resolved, with 55 unresolved. These later qualifications take precedence over any historical statement that the validation alone established readiness for full-data unblinding. Version 5 retains the technical description of the detector inputs, signal normalization and fitting procedure from that full-note lineage, while stating the remaining review conditions explicitly.

The Harvard final-combinations writing sample, also prepared on 2 September, supplied the compact presentation of the available observed results and polished explanatory prose. Its inputs were full 2015, full 2016 and the released 2021 10% sample. This draft expands those sections with the subsequent statistical studies and separates the proposal to unblind from a claim that all calibration gates have passed. It does not open additional 2021 events.

A.2 Observed combinations, tails and resolution: v4.9.12

The 4–5 September v4.9.12 sequence consolidated observed combinations, expected-limit reference products, targeted statistical-tail sampling, and mass-resolution variations. Version 5 uses its observed individual and combined coupling limits, supplements the principal figure with asymptotic local probabilities, and retains the requested resolution comparisons as response studies. Figures from different releases are identified by their numerical source rather than assumed to be equivalent because their appearance is similar.

The targeted-tail study refined finite-simulation probabilities where possible. A zero exceedance count supplies a sampling bound and does not measure a zero probability; an unresolved endpoint remains explicitly unresolved. Resolution variations and selected candidate extractions assess how

the result responds to a changed template. They do not independently measure a resolution error or supply an additional significance gain.

The related 2021 peak–dip and reverse-injection studies showed how a positive localized feature can enter neighboring sidebands and induce negative or positive fitted responses as the mask moves. These studies motivate the correlation and extraction displays in the results section. Their generating smooth shapes and selected masses make them conditional response diagnostics.

A.3 Background profiling and empirical calibration: v4.9.13

The 5 September background-profiling study compared the nominal correlated Gaussian profile, a positive log-GP profile and a fixed GP mean. Known-background controls separated implementation behavior from uncertainty propagation; injections with fluctuating background and retrained sidebands demonstrated the limitations of the fixed approach and the remaining 71 MeV failure of the profile. Version 5 brings the controlled limit comparison and injected-bias figure into the methodology discussion, including the physical meaning of the two- and five-reference-error injections.

The accompanying calibration release evaluated 456 dataset/mass coordinates and independently checked exclusion and local-rejection rates. Its finite-set background envelope removed the adjusted failures within the declared validation family, while exposing large sensitivity to the 2016 stress construction. Appendix B reorganizes the full calibration discussion around a flowchart, background definitions, observed changes, interpolation offsets and validation. Its conclusion supports profiling but does not certify uncorrected asymptotic probabilities across every tested background.

A.4 Understanding calibration and extending global tests: v4.9.14–v4.9.15

The 6 September v4.9.14 interpretation report explained why calibration can turn an unusually tight limit into a broad feature, and why a large background-dependent offset differs from a wider statistical error. It also introduced the 2015 significance-GP pilot, using ten complete initial scans, 1,000 independent validation scans and 200,000 correlated fields. Version 5 adopts its distinction between the background GP and the GP describing correlations across the significance scan.

The v4.9.15 extension repeated this calculation for full 2016 and released 2021 10% data. It documented the coherent background construction, central-distribution checks and unresolved rare tails. These results motivate the method in Sec. 4.20, while the source-fit and convergence limitations of the 2016 stress spectrum prevent its very small formal local probabilities from becoming particle claims. The 2015 and 2021 finite global-tail estimates remain conditional method-development results.

A.5 Union scan, probability audit and candidate studies: v4.9.16

The 6 September combined-global study extended the actual shared-coupling likelihood across the complete 19–250 MeV union. Dataset membership changes with mass, and the new coherent calculation preserves correlations across those boundaries. It supplies 232 observed mass points and separates raw-local ordering from stress-offset-adjusted ordering. Its union limits remain pointwise asymptotic limits; the 41-point earlier calibration cannot be silently promoted to a calibration of the entire union.

The probability-audit and echo report checked the displayed extraction likelihoods against the saved scan. It showed that the apparently extreme 76 MeV stress-reference result accompanies a small observed positive local fit. Version 5 uses that distinction to label local significance and conditional background-reference probabilities consistently. Its deterministic signal-injection replay

also demonstrates that a real injected feature can induce nearby oscillations through the fitting procedure, without establishing that the observed features are particles.

The extraction-presentation report provided the candidate count and residual displays used in the results section. The deficit extension reused the same coherent experiments and Gaussian fields in the negative direction. It is retained as a background diagnostic, with its scan plot shown beside the extraction discussion and its detailed interpretation in the appendix. It adds no independent toy ensemble or direction-selection correction.

The candidate-removal and traditional-search report replaced selected regions with conditional expectations and compared fits to the original counts with conventional local backgrounds. These interventions changed the peak-dip pattern but did not eliminate all remote oscillation. The conventional fits were strongly dependent on window and polynomial choice. Version 5 keeps their extraction displays as model comparisons and their detailed study in the appendix. They are not independent confirmation of candidates selected from the same data.

Finally, the 2015 low-mass side study searched 15–20 MeV with local 12–28 MeV GP support and nearby support and polynomial controls. It checked the 17.25 MeV fluctuation and kernel-ceiling dependence, using small conditional toy ensembles. The study is placed at the end of the appendix, outside the qualified production range: the accepted signal shape and resolution below 19 MeV still require validation.

A.6 What version 5 changes, and the remaining review decision

The new note assembles these studies into one account: detector and statistical method, observed individual and combined results, candidate and resolution diagnostics, then organized technical appendices. The proposed primary fit profiles correlated background uncertainty; the calibration and significance-GP work remain separately identified alternatives and validation studies. Historical readiness language is replaced by the concrete conditions in Sec. 7. The statistical-tail gaps and background qualifications remain visible.

The 8 September editing pass curates existing results and improves their presentation. It does not supply new pseudoexperiments, a new global particle significance, or an authorization to inspect the remaining 2021 data. The appendix contents that follow provide a guide to the supporting studies, while the accompanying source ledger records the release directories and artifacts used for this draft.

A.7 The 5.0.1 presentation and historical restoration

Version 5.0.1 restores the preselection distributions, blind-window diagnostics, source/exposure studies, and published-search comparisons requested for review. It adds explicit profile-likelihood equations and diagrams, separate plots of each dataset’s yield and coupling limits, and a compact comparison of the mass-correlation matrices. The full-range observed polyline and all fitted results are inherited unchanged from 5.0.0. A conditional full-2021 density projection illustrates an exposure response; it is not an expected limit. The restored historical comparisons retain their original fit settings and confidence levels. These editorial and display changes do not close the remaining statistical-tail or background-validation requirements.

Appendix contents

A	Change log and study history	115
A.1	Historical validation and the full-note foundation	115
A.2	Observed combinations, tails and resolution: v4.9.12	115
A.3	Background profiling and empirical calibration: v4.9.13	116
A.4	Understanding calibration and extending global tests: v4.9.14–v4.9.15	116
A.5	Union scan, probability audit and candidate studies: v4.9.16	116
A.6	What version 5 changes, and the remaining review decision	117
A.7	The 5.0.1 presentation and historical restoration	117
B	Alternative background modeling through calibration	119
B.1	Which background shapes are being tested?	119
B.2	What is repeated, and what remains fixed?	120
B.3	From likelihood ratios to a calibrated limit	120
B.4	Why some observed limits and probabilities change strongly	121
B.5	Interpolation offsets and a useful diagnostic	128
B.6	Independent validation in plain language	129
B.7	The targeted 71 MeV follow-up and remaining qualification	130
C	Candidate removal and traditional background fits	131
C.1	What was removed and how it was replaced	131
C.2	What changes locally, and what persists farther away	133
C.3	Dependence on the replacement model	134
C.4	Traditional local fits on the original data	135
C.5	How to interpret the strongest-looking conventional outcomes	139
C.6	What this implies for presentation and the next data step	139
C.7	Reproducibility and numerical checks	140
D	Joint scan of deficits	141
E	Historical validation studies and their role in v5	142
E.1	Kernel, window and normalization development: June–August 2026	142
E.2	Source, exposure and threshold tests: August 2026	142
E.3	The 2016 support and numerical-state sequence: September 2026	142
F	Historical pseudoexperiment regimes and model decisions	143
F.1	Three toy-generation regimes	143
F.2	Historical full-100 2021 source-and-exposure closure	144
F.3	Why threshold modeling was tested at 65 MeV	158
F.4	Controlled 2016 upper length-scale range	161
G	Historical comparisons with the 2015 and 2016 resonance searches	163
G.1	The original and corrected 2015 polynomial analyses	163
G.2	The historical 2015 GPR comparison	163
G.3	The published 2016 method and the saved GPR comparison	164
H	Exploratory 2015 search on the 15–20 MeV rising edge	168
H.1	Why a separate support is useful	168
H.2	Local probabilities and dependence on the background	169
H.3	Signal extraction at fixed low-mass anchors	170
H.4	The leading fluctuation and the upper edge of the requested interval	171
H.5	Small conditional toy checks and what remains to establish	172

B Alternative background modeling through calibration

Calibration asks how the quoted limit would behave if the experiment were repeated under specified backgrounds. It changes the probability attached to the likelihood statistic, while retaining the observed counts and the fitted signal. In this study it is also a way to propagate a finite set of alternative background shapes through the same analysis. It cannot identify which shape is the true background.

There are two choices of fit and two choices of probability calculation. A fixed-background fit treats the estimated mean as known. A profiled fit allows correlated changes constrained by the GP covariance. Either fit can use asymptotic formulae or distributions obtained from repeated experiments. Figure 83 separates these choices.

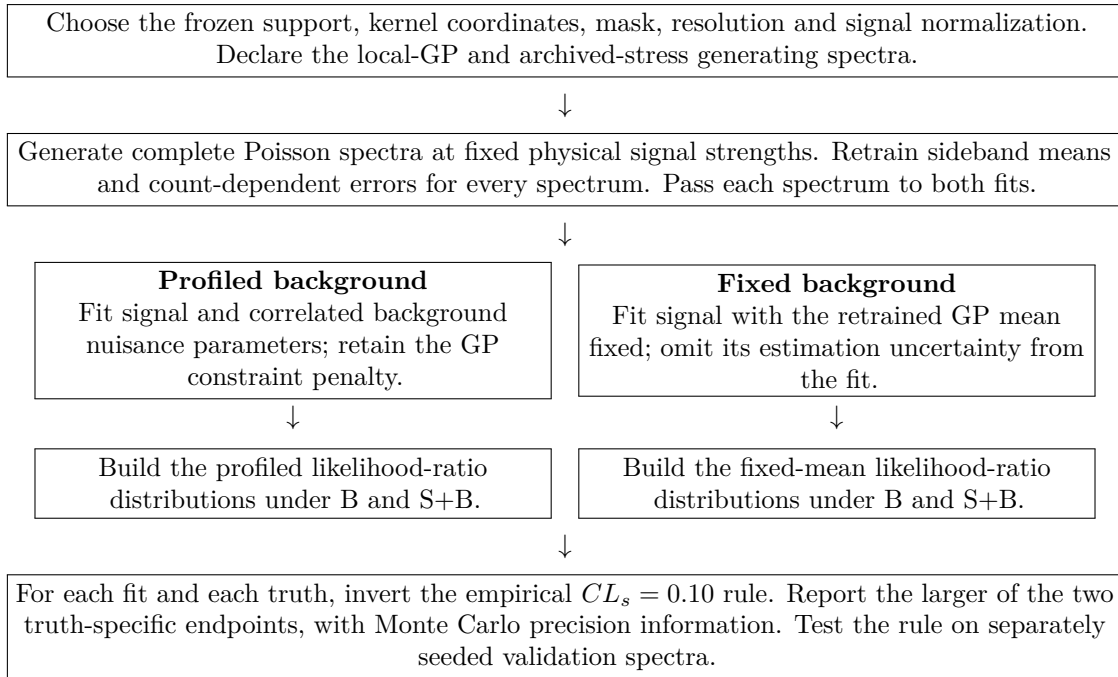


Figure 83: How the two background treatments are calibrated. Both branches receive identical spectra. The primary comparison uses two declared background truths for each method; it does not average the two methods or select whichever gives the tighter observed limit. B denotes background only and S+B denotes signal plus background.

B.1 Which background shapes are being tested?

The local-GP control extends the mass-local observed GP mean over the complete support. It tests repeated analysis of a shape close to the assumed interpolation. The archived stress input is a second smooth expected-count spectrum inherited from earlier source-fit studies. The word “stress” identifies its purpose: to expose sensitivity to a plausible construction or a known limitation. It does not certify that construction as a physical truth.

For 2015 the alternative is the saved `fShiftSigPowTail` expected-count spectrum. The 2016 alternative blends a low-mass fit to a 10% development subset with an archived broad continuation over 75–85 MeV, then uses a common normalization to the full sample. The continuation retains a failed fit-convergence/status flag and was allowed only as a conditional stress construction.

Independence from the full observed sample was not established. These facts limit the physical interpretation of the large effects it produces.

For 2021 the alternative is the native-10% `fSigPowExpQ`-anchored logistic–Chebyshev6 residual construction. Its threshold fit was useful for support studies, but the 85–300 MeV tail had deviance per degree of freedom 6.220. It is therefore not a globally qualified pure `fSigPowExpQ` background. Native bins are summed onto the production edges without an additional exposure factor or a new normalization after cropping.

The combined calibration uses two joint scenarios: every constituent has its local-GP truth, or every constituent has its archived stress truth. Taking an envelope over these scenarios does not cover every mixed assignment, nor all possible background misspecification. The separate 71 MeV reverse-injection smooth spectrum is outside this two-truth set. Its targeted follow-up is reported below.

B.2 What is repeated, and what remains fixed?

For truth t , a full pseudoexperiment has counts

$$N_{di} \sim \text{Pois} \left[b_{di}^{(t)} + \epsilon^2 s_{di}^{(1)} \right]. \quad (105)$$

The complete signal, including tails outside the moving mask, enters generation. The analysis recomputes log-count training targets, their count-dependent errors and the GP posterior for each spectrum. The production convention uses $\alpha_i = 1/N_i$ for positive counts and $\alpha_i = 1$ for zero counts. The signal conversion factors, resolution, kernel coordinates, support and mask policy remain frozen. Kernel optimization and the preceding support/model-selection procedure are not repeated.

This distinction matters when comparing with historical injection studies, which reoptimized kernels in their reference and injected fits. They also used each toy’s reference error to set injected yields. The present ensembles fix one physical signal strength before generation. A historical mean pull with a random fitted denominator and the mean yield error $E[\hat{A}] - A_{\text{true}}$ answer different questions.

The original support studies support practical recovery under their stated conditions. They do not imply zero offsets everywhere. At 65 MeV the native-10% v4.9.1 mean pull was -0.246 , with 90% interval $[-0.417, -0.076]$, and an independent continuation gave -0.284 . The accepted practical tolerance was adopted after those results. In the v4.9.5 support study no edge passed the original uniform $|\langle \text{pull} \rangle| < 0.5$ rule. The amended practical criterion selected 36 MeV and was independently confirmed, while the three 55 MeV means remained -0.773 , -0.841 and -0.960 .

B.3 From likelihood ratios to a calibrated limit

The statistic is the bounded, one-sided $\tilde{q}_\mu$ of Ref. [36], evaluated with the actual profile likelihood. Its denominator uses the physical null fit when the unconstrained fitted signal is negative, and the statistic vanishes when that fit exceeds the tested signal. For each truth,

$$CL_s^{(t)}(\mu) = \frac{P_{\mu,t}(\tilde{q}_\mu \geq \tilde{q}_\mu^{\text{obs}})}{P_{0,t}(\tilde{q}_\mu \geq \tilde{q}_\mu^{\text{obs}})}. \quad (106)$$

Both probabilities are upper tails of the same statistic, including ties. Their ratio implements the modified-frequentist construction of Ref. [38]. Each truth gives an upper endpoint at $CL_s = 0.10$; the larger endpoint is the reported finite-set envelope. This procedure cannot be reproduced by multiplying all significances by one scale factor.

To sample rare tails efficiently, the release generated equal strata from several full-spectrum Poisson proposal means, including strengths near the observed crossing. Every generated spectrum was fitted and weighted with its exact full-spectrum Poisson density divided by the proposal-mixture density. This reuse follows the importance-sampling identity discussed by Berns [43]. A linear approximation helps choose proposals but does not replace the fitted test statistic. An independent 120,000-draw one-bin Poisson test checked the weighting and variance calculation.

The completed selection contains 12,080,640 calibration spectra and 1,368,000 separately seeded validation spectra. Refinement was triggered by censoring and Monte Carlo diagnostics, using fresh independent banks; it did not select methods by observed limits or validation outcomes. The same holdout validation spectra were rescored and counted once. Dense-reference, scalar-versus-batch and source-identity checks retain failed calculations rather than counting them as successful fits.

A numerically resolved endpoint requires both tail effective sample sizes of at least 100, a relative Monte Carlo half-width no larger than 10%, a numerical bracket no wider than 1.5%, and the recorded normalization and monotonicity checks. The approximate 95% Monte Carlo interval is uncertainty in a numerical estimate of a 90% limit. It is neither an expected experimental band nor a simultaneous confidence band.

Scope	Completed	Profile resolved	Fixed resolved	Raw ratio	Calibrated ratio
2015	72/72	70	72	0.422	1.446
2016	142/142	133	141	0.522	1.333
2021 10%	201/201	201	200	0.607	1.199
All three	41/41	41	41	0.595	1.346

Table 12: Completion and declared Monte Carlo precision of the frozen calibration release. The final columns are median fixed/profiled observed-limit ratios on matched completed coordinates. Resolution of the numerical endpoint does not qualify the background family or resolve the inherited 2016 numerical exception.

B.4 Why some observed limits and probabilities change strongly

Most profiled limits change modestly: the median calibrated/asymptotic ratio is approximately 1.08 in 2015, 1.09 in 2016, 1.04 in 2021 and 1.21 in the common-range combination. Localized exceptions are much larger. At 75 MeV the profiled 2016 limit changes from 1.81×10^{-6} to 4.19×10^{-5} in ϵ^2 . At combined 73 MeV it changes from 8.94×10^{-7} to 1.14×10^{-5} . A downward fluctuation had made the raw limits tight; under the stress construction the interpolation already tends to overpredict the background and can hide an injected positive signal. The calibrated test must account for that behavior.

At combined 66 MeV, the stress background gives a mean signed local significance mapping of +8.94 with spread 0.97, while the local-GP control gives -0.11 with spread 1.01. Their similar spreads and different centers show that the effect is predominantly a shifted expected fitted signal. All 500 stress-background-only validation spectra exceed the observed statistic. The envelope local probability is consequently near one. It does not identify an observed particle, and removing the stress shape because it weakens the result would not establish a justified background choice.

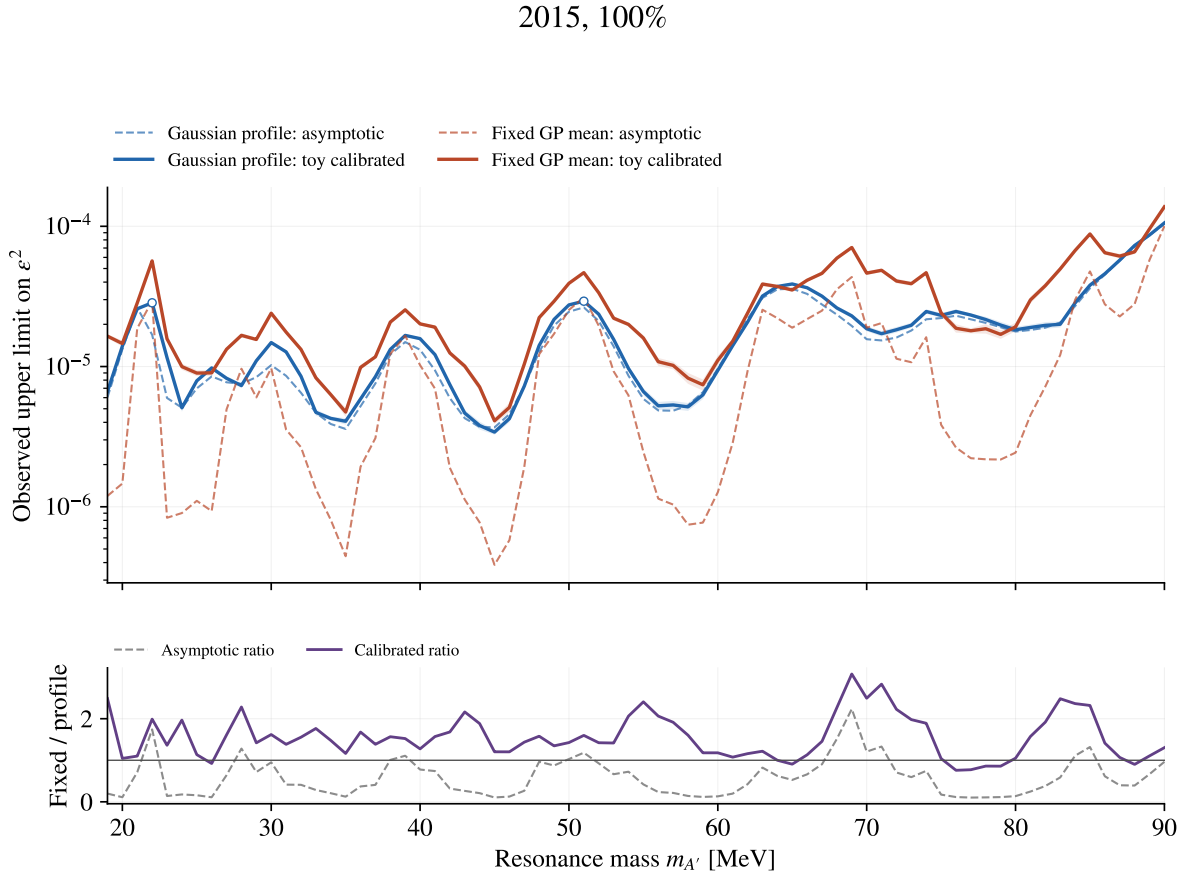


Figure 84: Observed asymptotic and calibrated 90% CL_s limits for 2015 using the two background treatments. Calibration envelopes cover the two stated generating shapes. Shading and open precision markers refer to finite Monte Carlo accuracy; they are not expected-limit bands.

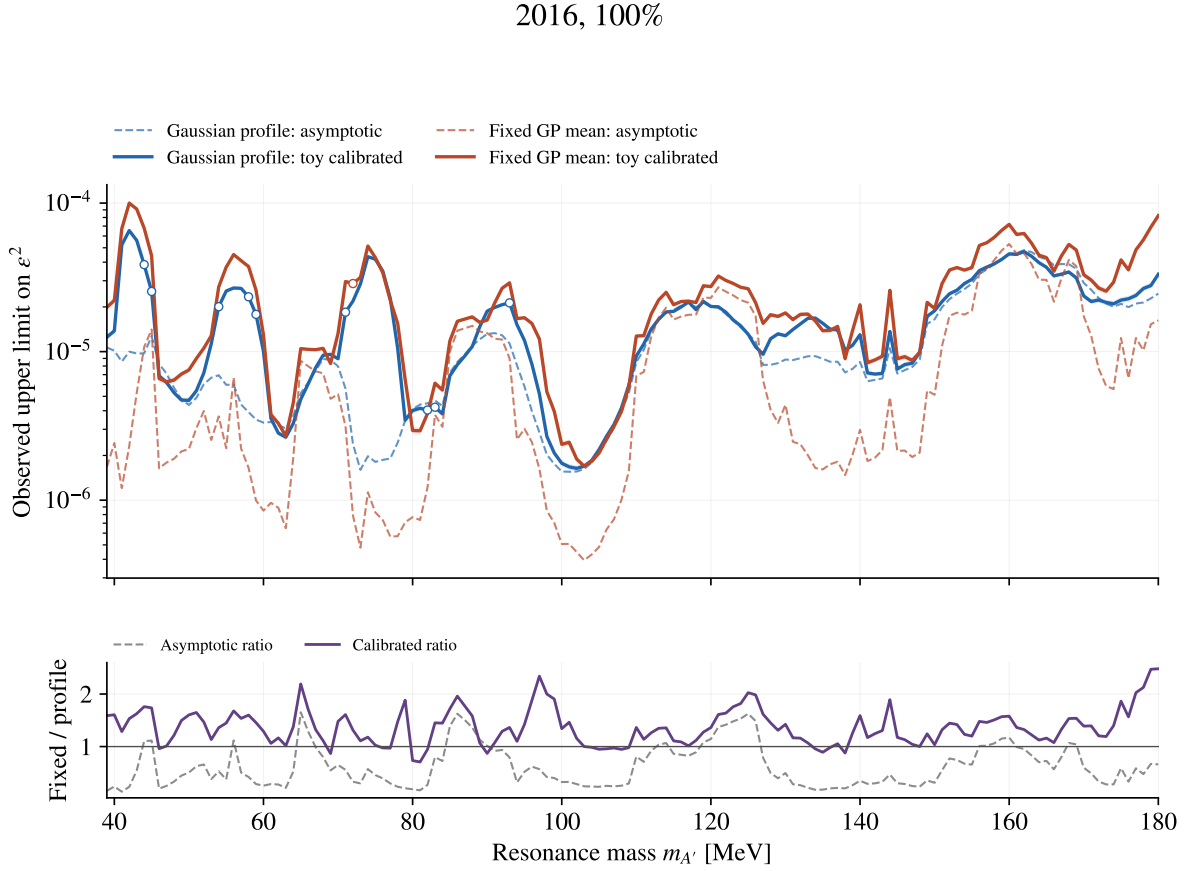


Figure 85: The same comparison for full 2016. The pronounced change is localized and depends on the archived stress construction. A resolved calibration endpoint does not remove the source-convergence qualifications or inherited numerical exception.

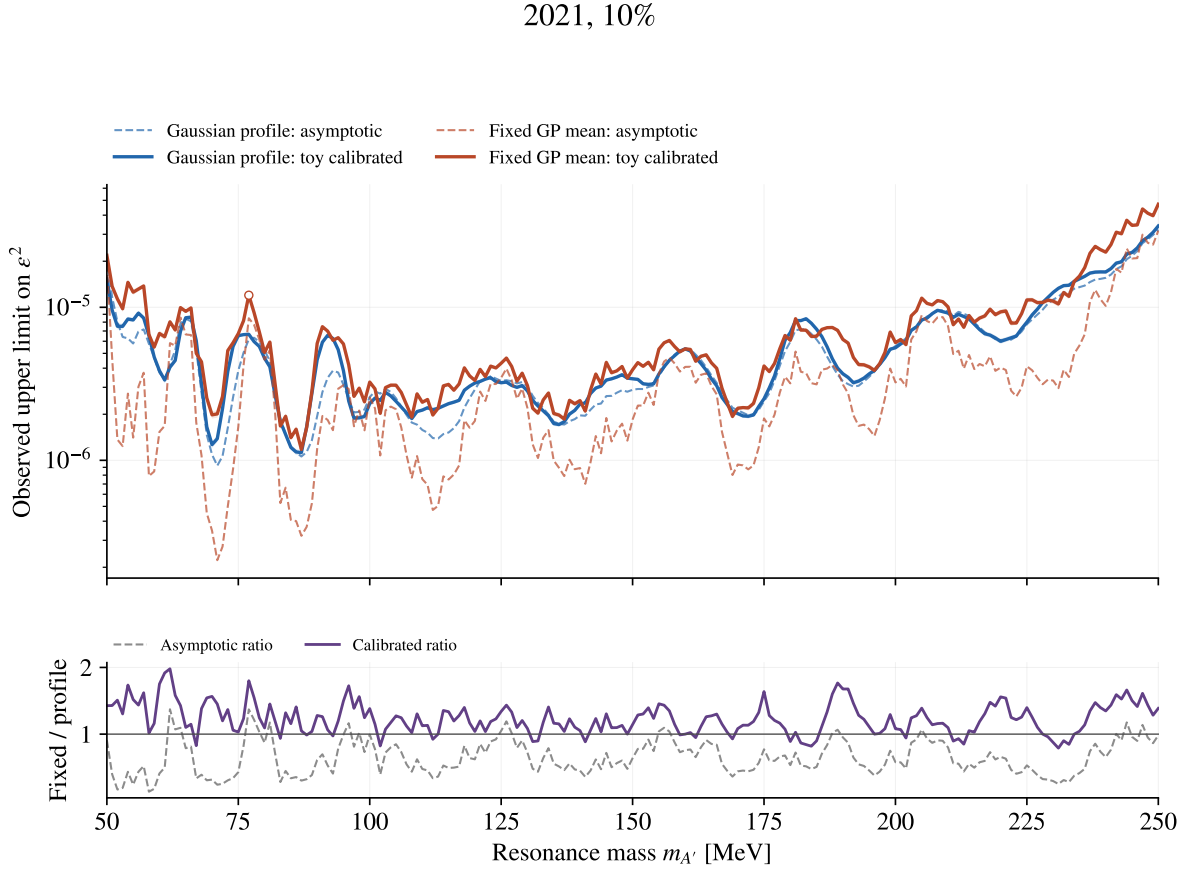


Figure 86: The same comparison for native 2021 10%. Both methods use the same physical sample and background-truth alternatives; the observed comparison does not measure an expected sensitivity gain.

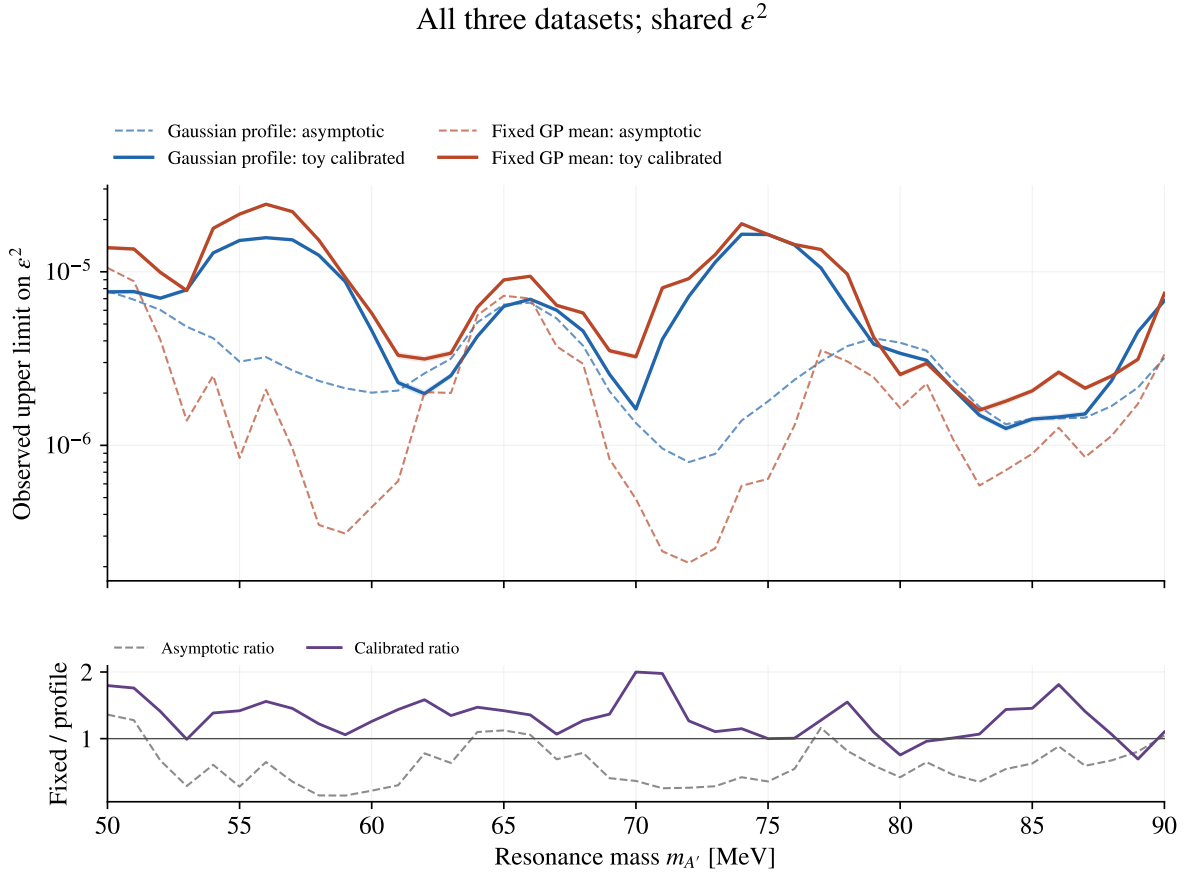


Figure 87: The three-dataset common-range comparison. This calibration covers 41 masses, rather than the later 232-mass union scan. It uses a shared physical coupling and either all local-GP or all archived-stress constituent backgrounds; mixed truth assignments are not included.

Dependence on the generating truth

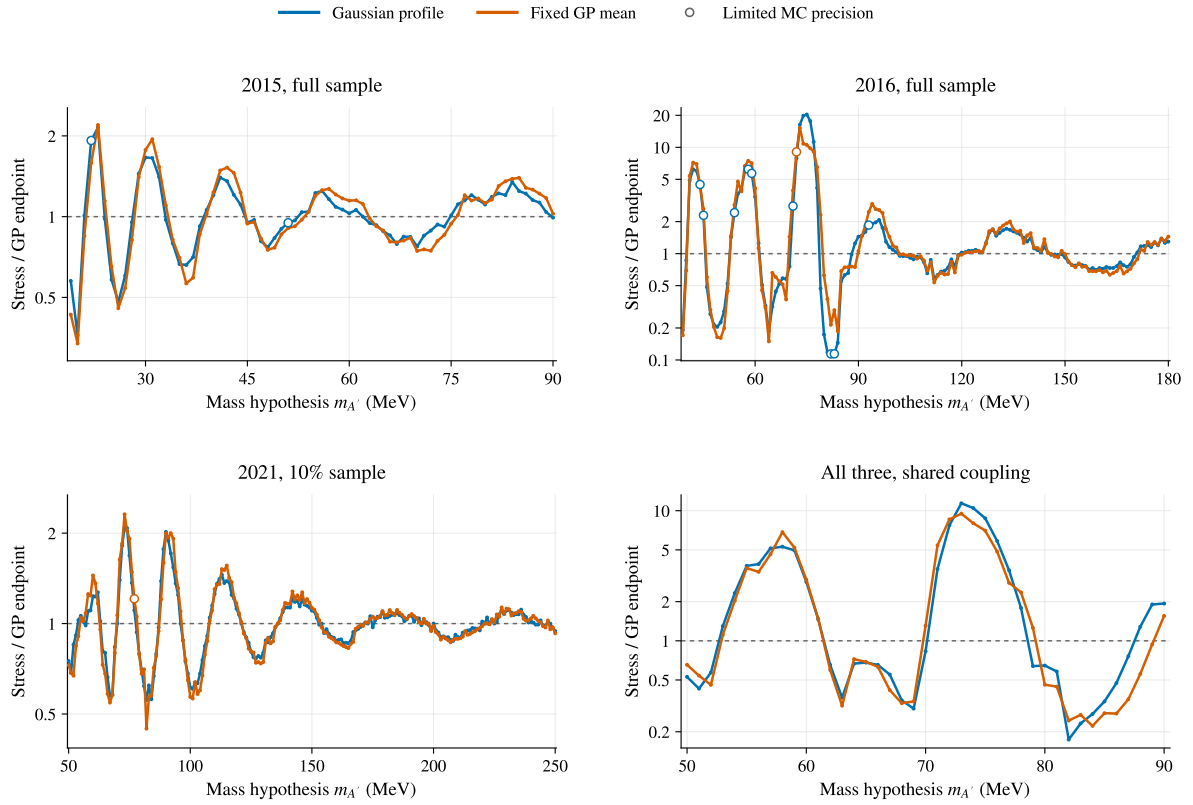


Figure 88: Ratio of the stress-truth to local-GP calibrated endpoint. Values above one indicate that the stress background controls the larger endpoint; values below one indicate that the local-GP control does. Open circles mark limited precision in at least one component, and gaps retain unavailable or censored ratios. This comparison diagnoses the chosen background family rather than certifying either member.

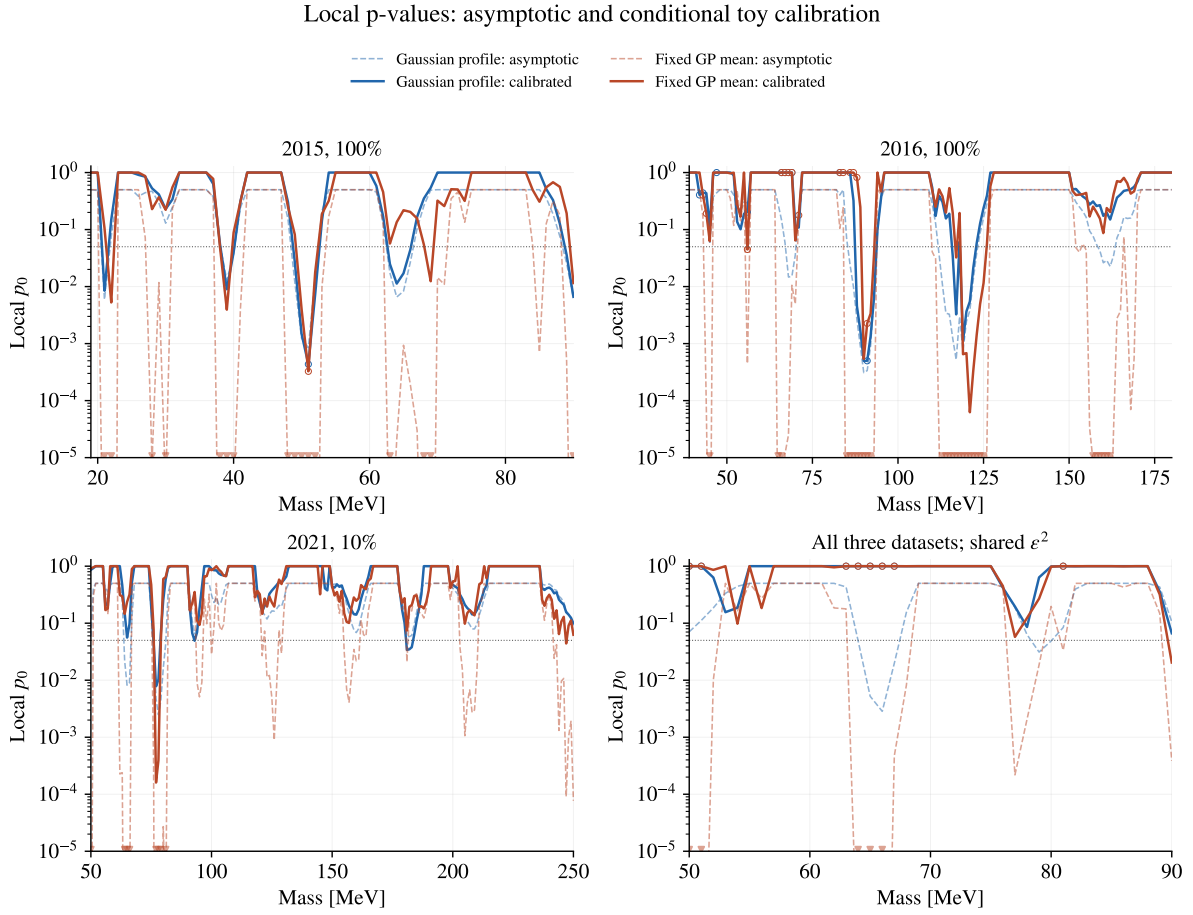


Figure 89: Asymptotic local probabilities and the larger background-only tail from the two calibration truths. The empirical bounded-statistic convention gives one when the discovery statistic is zero; the earlier asymptotic display gives 0.5. Triangles retain asymptotic values below the plotting floor. Open circles flag limited Monte Carlo precision or a noisy estimate displayed at one; numerical ledgers retain unclipped values. No curve in this figure includes a global trials correction.

B.5 Interpolation offsets and a useful diagnostic

An unfluctuated generating spectrum can already reveal why the fitted signal is shifted. Train the background model on the sidebands of that spectrum, obtaining $\bar{b}_t$, then compare its interpolation with the known generating counts b_t inside the gap. Project that residual onto the signal template:

$$\delta A_{\text{lin}} = \frac{w^T V^{-1}(b_t - \bar{b}_t)}{w^T V^{-1} w}, \quad V = \text{diag}(\bar{b}_t) + C. \quad (107)$$

The fixed-background version sets $C = 0$. A positive projection means that the smooth interpolation leaves a residual resembling a positive signal; a negative projection means it tends to absorb one.

Background-only signal extraction

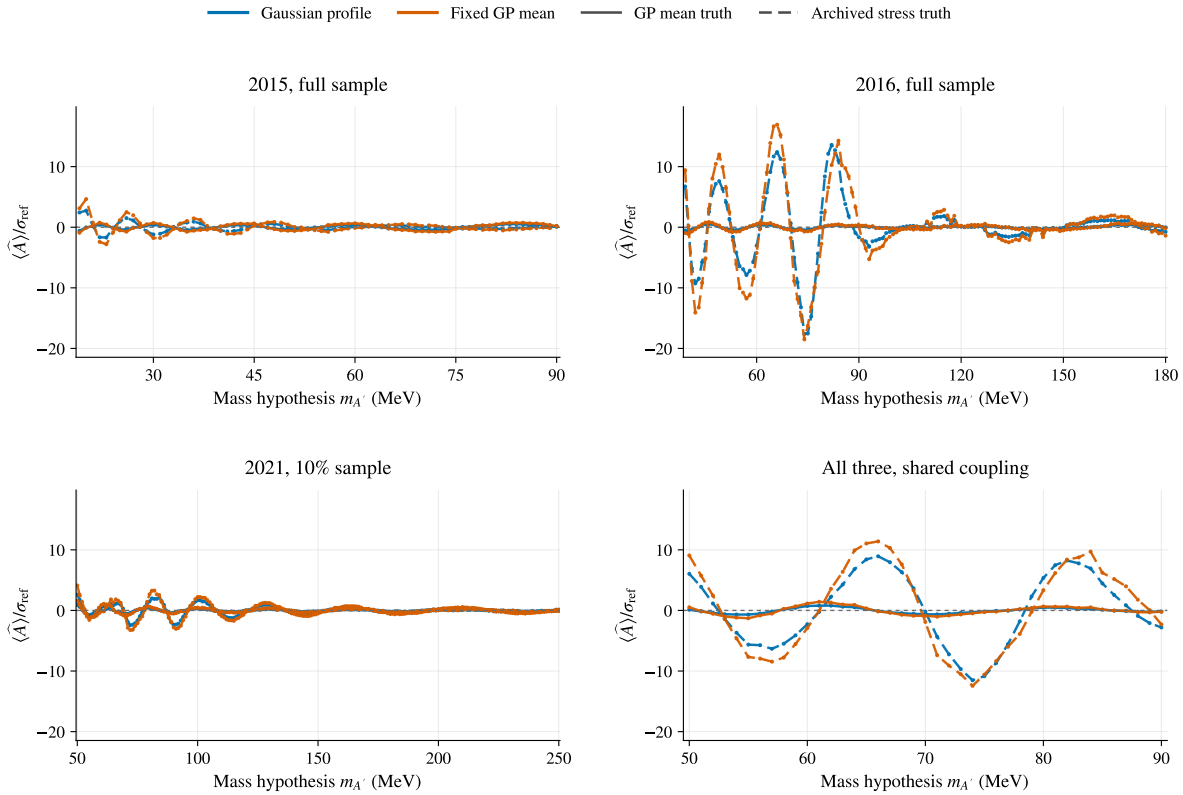


Figure 90: Mean fitted background-only yield in units of one fixed reference error, from 500 independently generated spectra per mass and truth. Solid and dashed curves distinguish the two backgrounds. Pointwise 95% intervals describe Monte Carlo precision on each mean. This normalization differs from a pull using each toy’s random fitted error; the plot does not measure a bias in observed data.

For the profiled 2016 stress ensembles, the median absolute mean offset is 1.157 reference errors, while the median absolute discrepancy between the ensemble mean and Eq. 107 is 0.031. For the common-range combination these quantities are 5.361 and 0.035. Thus a deterministic interpolation residual explains much of the conditional shift without requiring random fluctuations to generate it. This makes the projection useful for locating problematic mass regions, comparing independently

justified background shapes, and designing paired checks of kernel policies before expensive simulations. It is an approximation and a diagnostic, not an automatic correction to data or a substitute for validation of the full likelihood-ratio tails.

B.6 Independent validation in plain language

A positive injection supplies a known answer: the true signal yield. If a claimed 90% upper bound falls below that yield too often, the procedure excludes a signal that was actually present more often than its nominal benchmark permits. The validation plot asks this question before and after calibration on separately seeded spectra. At the true injected yield, the envelope excludes only when both background calibrations give $CL_s < 0.10$. Each point represents one mass, one generating truth and one method. Counts are not pooled across masses.

Independent validation of 90% CL_s exclusion

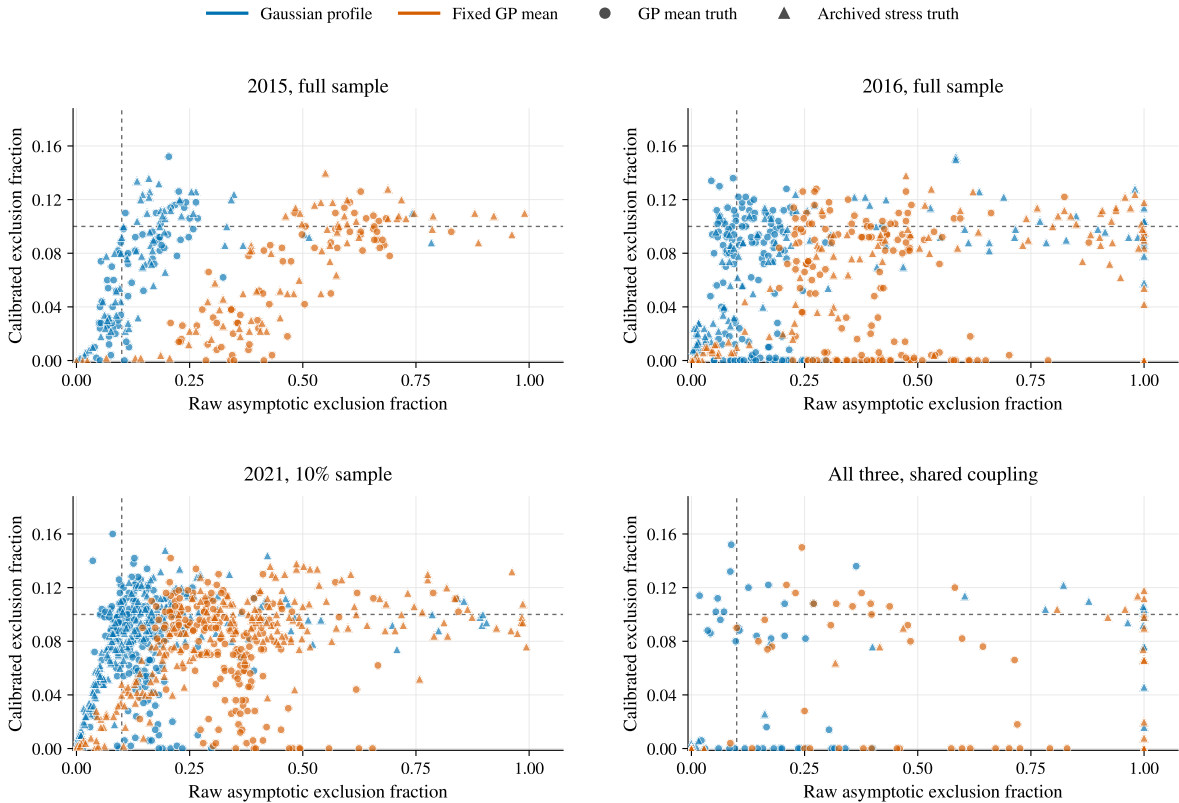


Figure 91: Independent validation with five-reference-error injections. The horizontal coordinate is the fraction of 500 spectra excluding the known injected yield with the raw asymptotic rule; the vertical coordinate uses the calibrated two-truth envelope. A point far to the right has an excessive raw exclusion rate. A point below the horizontal 0.10 guide indicates a conservative calibrated rate in that cell. The guides mark the nominal benchmark, not an expected discovery probability. Finite counts and Monte Carlo uncertainty limit the precision of each conclusion.

Validation family	Cells	Raw adjusted flags	Calibrated adjusted flags
True-yield exclusion	3,648	2,039	0
Background-only local rejection	1,824	952	0

Table 13: One-sided binomial tests for rates above 10% for true-yield exclusion and above 5% for local background-only rejection, with Holm adjustment within each family at 0.05. A flag counts a failed cell-level rate test; it is not an individual excluded toy. The full table includes both background treatments and both truths.

Even the profiled method has raw excess-rate flags: the GP and stress truths give respectively 179/912 and 326/912 flagged exclusion cells, and 50/456 and 140/456 flagged local-rejection cells. These results prevent a claim that large event counts or small historical mean pulls alone establish an adequate asymptotic mapping over the tested family. After calibration, no adjusted flags remain in the complete suite. That result supports the stated conditional procedure, but the tests have limited power. With 500 spectra per cell, the first Holm exclusion step requires at least 81/500 exclusions, or 16.2%, to reject the 10% null in this family. The analogous local-rejection step begins at 48/500, or 9.6%, against a 5% null. No flags therefore does not certify percent-level accuracy at every mass.

The same background disagreement can reduce detection power. At combined 66 MeV a five-reference-error signal generated on the local-GP truth passes the raw local $p_0 < 0.05$ criterion in 499/500 trials but the calibrated criterion in 0/500. Under the stress background the background-only raw test fires in 500/500 trials, compared with 26/500 after calibration. The conservative envelope protects against this false-positive mechanism at a cost for signals on the other truth. An observed-limit ratio alone cannot measure that tradeoff; a fair power comparison needs fixed physical signals and comparable error control.

B.7 The targeted 71 MeV follow-up and remaining qualification

The reverse-injection smooth truth that produced the 71 MeV failure is not included among the two calibration truths. A retrospective check replayed its original 1,500 spectra, 500 at each of zero, two and five reference errors, with dense GP refits and the full conditioned covariance. At the stronger injection the calibrated true-yield exclusion point estimates are 3/500 for the profile and 2/500 for the fixed mean, compared with their raw values of 145/500 and 349/500. This is conservative behavior under that selected truth. However, 682 of the 3,000 method-toy decisions have limited calibration Monte Carlo precision. These reused spectra supply an auxiliary diagnosis, not a new independent validation sample or truth-independent coverage.

The remaining scientific task is to qualify a background family using source fits, sidebands and independent or explicitly held-out predictive controls, and then validate the complete selected analysis policy. A poorly qualified stress model need not become a mandatory physical possibility, but its effect cannot be dismissed solely because it weakens a result. The inherited 2016 numerical exception must be resolved separately. These requirements define the scope of the present unblinding review and remain visible alongside the proposed asymptotic primary results.

C Candidate removal and traditional background fits

Removing the candidate regions changes the local peak–dip pattern, but does not remove the oscillatory response. With both primary regions replaced, the remote observed scan retains 67%, 78% and 92% of its original variation in 2015, 2016 and 2021, respectively. The corresponding reference-spectrum response retains 110%, 93% and 86%. Traditional fits to the original counts give substantially different peak strengths, and even reverse the sign of the 2021 feature at 78 MeV. The evidence supports coupled, background-dependent fitted structure. It does not yet establish a robust particle signal.

The large oscillating “background” in the probability study is primarily the *signal-like response to a reference spectrum*, $a_m = r_m(B)$. It is not itself an event-count spectrum. The experiment below therefore addresses both objects separately: the scan of the observed counts, and the scan of the archived reference counts. Modifying one does not redefine probabilities for the other.

C.1 What was removed and how it was replaced

The two leading positive observed GP peaks per year were fixed from the existing extraction catalogue before examining these interventions: 51 and 21 MeV in 2015; 90 and 117 MeV in 2016; 78 and 65 MeV in 2021. Selection did not use the extreme reference-centered probabilities. Full 2015, full 2016 and only the released 2021 10% sample are used.

Data	Candidate [MeV]	σ_m [MeV]	Replaced bin edges [MeV]	Bins
2015	51	2.622	45.000–57.000	48
2015	21	1.025	18.750–23.250	18
2016	90	3.698	81.750–98.250	66
2016	117	4.989	105.750–128.250	90
2021	78	2.263	72.875–82.875	16
2021	65	2.122	60.375–69.750	15

Table 14: Primary holes contain whole analysis bins centered within $2.25\sigma_m$ of a selected peak. The tabulated edges include the full boundary bins. Native analysis widths are 0.25 MeV for 2015, 0.25 MeV for 2016 and 0.625 MeV for 2021. No display binning enters a fit.

The primary replacement GP is trained outside *both* holes, with the frozen kernel from that year’s leading selected peak. It predicts the joint latent log-intensity (μ, V) in the missing bins. The deterministic replacement is the expected count $\exp(\mu_i + V_{ii}/2)$. Ten additional replacements draw a joint latent function, then Poisson counts. The latent covariance does not include a second copy of Poisson observation noise. The archived reference spectrum receives its own deterministic replacements learned only from its own retained bins.

Each region is replaced separately and both are replaced together. The single-region comparisons reuse the same joint draws. Every complete modified spectrum stays fixed throughout its mass scan; every bin outside the chosen holes is exactly unchanged. The original moving-mask GP is then retrained at every hypothesis with its frozen per-mass kernel, followed by the exact profiled fit. This gives 42 spectra per year and 17,430 complete mass tests across the three grids. These conditional replacements ask how retained data and the fitting procedure respond to a local edit. They are not independent background experiments or additional calibration toys.

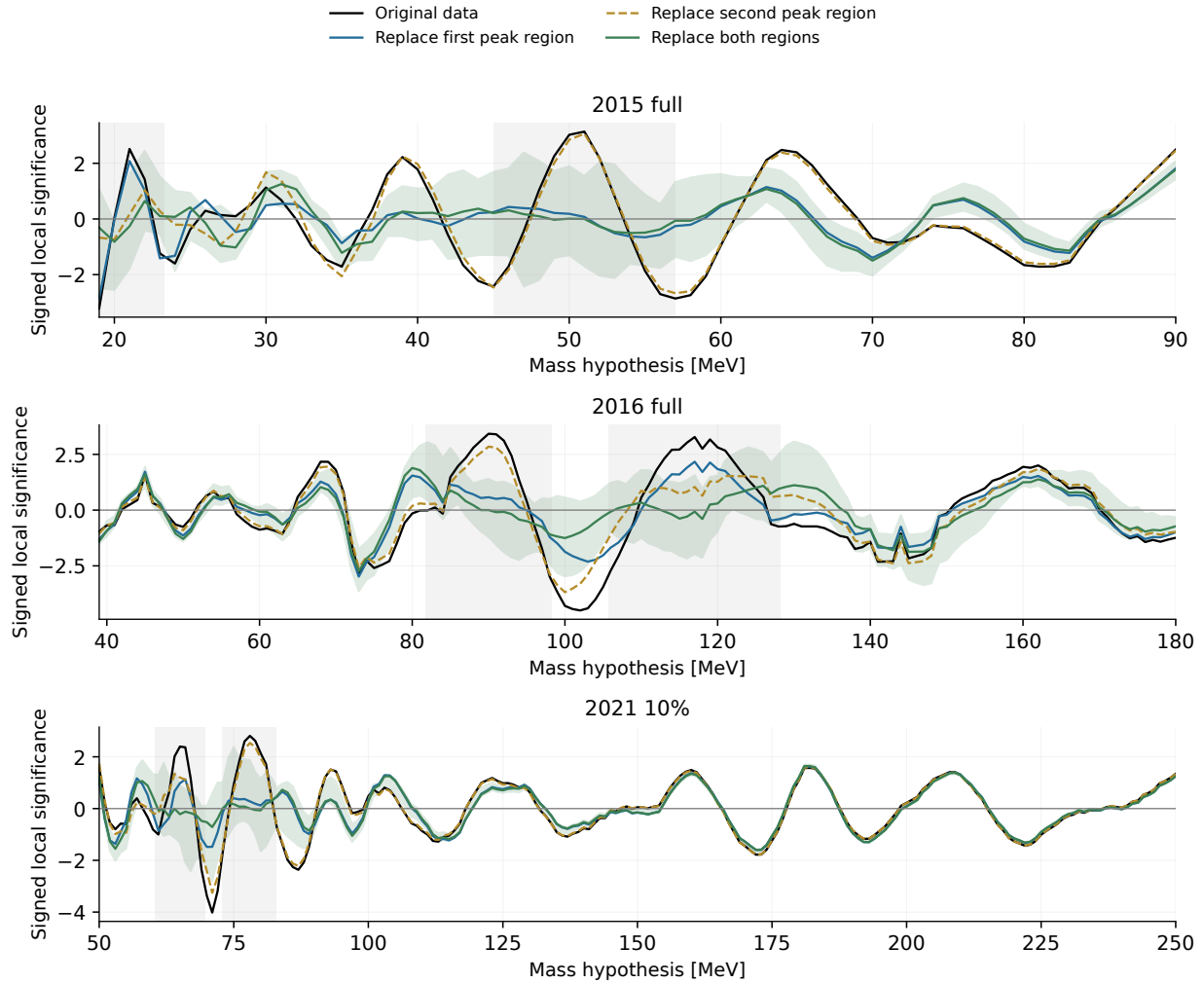


Figure 92: Observed scans before and after replacement. Black: original data. Blue and dashed orange: deterministic replacement of the first or second listed region. Green: both. The light green envelope spans the ten paired, conditional GP-plus-Poisson replacements with both holes filled; it is not a confidence band. Gray vertical regions show the primary holes. Curves join the existing 1 MeV hypotheses without extra smoothing. The ordinate is a signed local significance, not a newly calibrated significance. Removing a peak's own bins predictably weakens that feature; the informative changes are in the surrounding pattern and retained regions.

C.2 What changes locally, and what persists farther away

The neighboring deficits weaken strongly. At 19 MeV in 2015, r changes from -3.21 to -0.31 after both replacements; at 102 MeV in 2016 it changes from -4.50 to -1.00 ; at 71 MeV in 2021 it changes from -4.02 to -0.72 . The last change is already $-4.02 \rightarrow -1.48$ when only the 78 MeV region is removed, and $-4.02 \rightarrow -3.26$ when only 65 MeV is removed. These masses were retained as illustrative deficit checks, not selected to maximize the replacement effect. Their signal-fit windows overlap removed bins, so they are deliberately excluded from the remote metric.

There is also coupling between separated candidate regions. Replacing only the 78 MeV region in 2021 changes the 65 MeV root from $+2.40$ to $+0.98$, although the 65 MeV candidate bins are unchanged. The fitted background uses information shared across hypotheses. Together with the controlled signal injections in Sec. 6.10, this shows that a peak and its surrounding dips need not be independent features. Excision alone cannot tell whether the original cause was a particle, background mismatch, detector structure, or a fluctuation.

For a stricter comparison, a mass is called remote only when its *entire native signal-fit window* avoids both primary holes. The statistic below measures the remaining scan variation about its own mean. It is a descriptive measure of amplitude, not spectral power at a selected frequency or a probability of oscillation.

Data	Masses	GP mean	10-fill range	Polynomial	Wider GP
2015	40	0.673	0.573–0.781	1.227	0.719
2016	74	0.778	0.762–1.096	0.783	0.624
2021	170	0.921	0.908–0.954	1.107	0.861

Table 15: Fraction of original remote variation retained in the *observed* signed local significance: $\text{sd}(r_{\text{filled}})/\text{sd}(r_{\text{original}})$, each standard deviation taken about its own mean. Both candidate regions are replaced. The 10-fill column is the minimum–maximum across ten conditional replicas, not a confidence interval. All columns use the same primary remote mass set, including the wider-hole column.

Data	GP mean	Polynomial	Wider GP	GP mean correlation
2015	1.102	3.626	0.836	0.916
2016	0.928	0.978	0.671	0.998
2021	0.865	1.162	0.382	0.928

Table 16: The same comparison for the response to the archived *reference* spectrum, $a_m = r_m(B)$. Every fill is learned from that reference’s own retained bins. Correlation is with the original reference response on the primary remote mass set. Poor polynomial interpolation can create additional structure; a smaller or larger ratio does not rank physical background models.

All six primary observed/reference comparisons retain at least half the original remote standard deviation and at least two sign changes within contiguous remote intervals. This predeclared routing condition triggered the traditional fits at all six masses. It is not a rejection threshold for a physical null hypothesis. Persistence also does not mean the pattern is unchanged: the 2015 observed deterministic correlation is only 0.56, and across its ten replacements it ranges from 0.07 to 0.82.

C.3 Dependence on the replacement model

The polynomial alternative fits sidebands within $\pm 7\sigma_m$, excluding both primary holes: degree five at 2015/21 MeV and degree three elsewhere. Its expected counts fill the holes. The wider GP alternative excludes $\pm 3\sigma_m$ around both candidates and conditions the same leading-peak kernel on the retained bins. The wider GP replacement reduces the 2021 reference's remote variation to

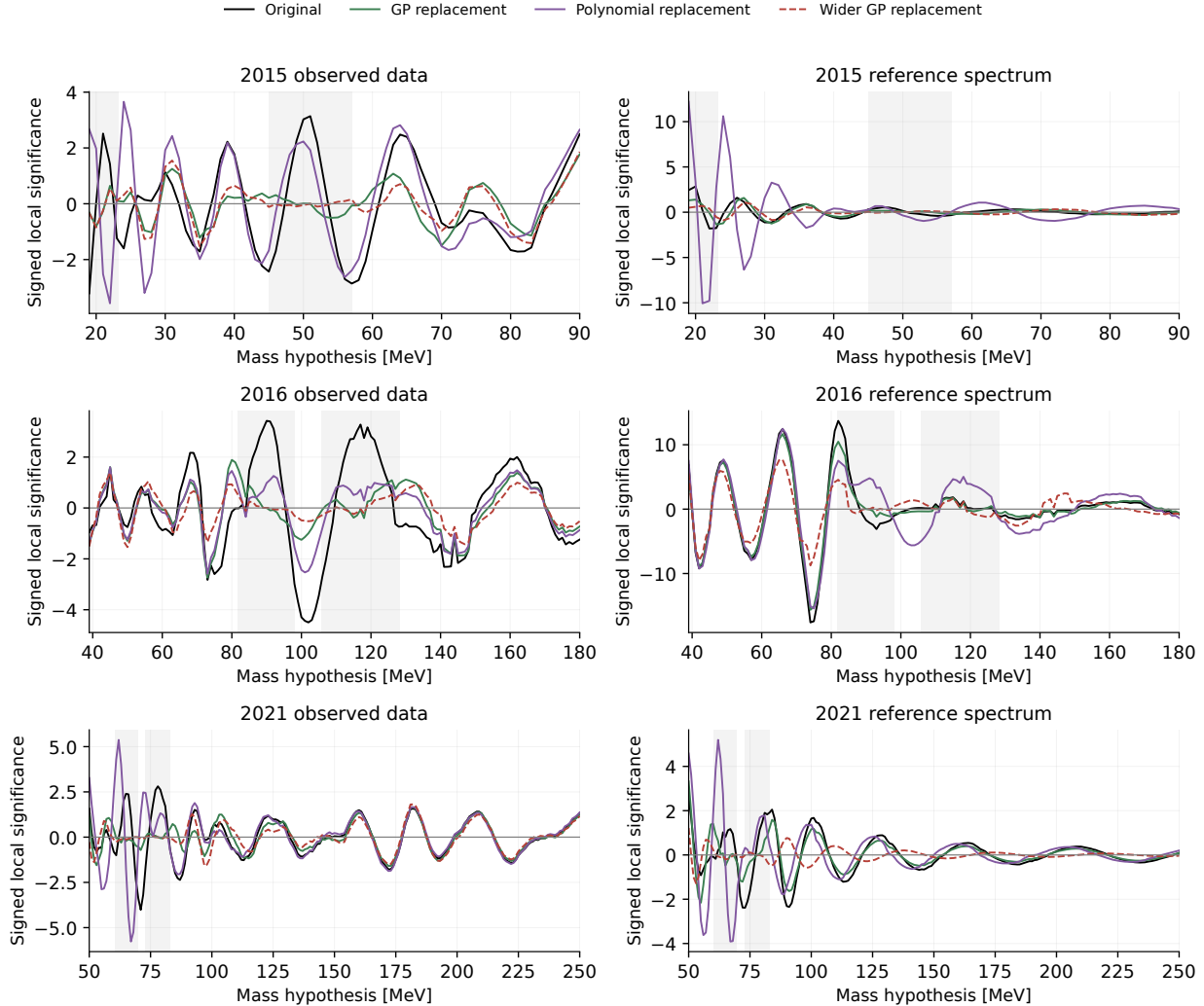


Figure 93: Both-hole replacements in observed counts (left) and the reference spectrum (right). Green: primary GP mean. Purple: exponential-polynomial fill. Dashed red: wider GP holes. The reference ordinate $a_m = r_m(B)$ is the fitted root, not the background event rate. Gray shading marks primary holes; wider holes extend beyond it. All outcomes remain visible, including unsuccessful interpolation. No altered spectrum supplies a new observed-data probability.

38%, while the observed scan retains 86%. These use the same primary remote mass set; some windows touch the wider holes. The 2016 reference still oscillates strongly below the removed regions. Persistence therefore has appreciable model dependence.

The polynomial filler can create new distortions. At the 2015 21 MeV reference hole, its sideband deviance is 267.20/33 degrees of freedom; its filled reference scan retains 3.63 times the original

remote variation. At the 2021 65 MeV hole, the polynomial sideband deviances are 96.12/17 for observed counts and 86.17/17 for the reference. For a deterministic reference, this deviance is a mismatch scale, not a calibrated goodness-of-fit probability. These are poor smooth descriptions of the retained sidebands. Their altered oscillations do not identify a better physical background. The GP fill is also conditional on its kernel and on excluding both candidate regions; it cannot be treated as known signal-free truth.

C.4 Traditional local fits on the original data

The follow-up uses a positive exponential-polynomial background and a bin-integrated Gaussian signal of fixed mass and archived resolution. All polynomial coefficients and the signed signal yield are fitted together with a Poisson likelihood; the null fit profiles the same coefficients at zero signal. Unlike the GP fit, this diagnostic has no Gaussian background constraint learned from the surrounding spectrum. Its different window and nuisance model change both the fitted signal and its uncertainty.

The choices are inspired by published HPS searches [3, 4], with explicit adaptations to the frozen analysis bins. The 2015 baseline uses a degree-five exponential Chebyshev polynomial and total width $14\sigma_m$ at 21 MeV, and degree three with width $13\sigma_m$ at 51 MeV. For 2016 and 2021 it uses a degree-three exponential Legendre polynomial in a total $8\sigma_m$ window. This is a specified cross-check, not an exact reproduction of the published event selection, binning or mass-dependent window schedule.

The actual baseline edges are 14–28.25 and 34–68 MeV for the two 2015 masses, 75.25–104.75 and 97–137 MeV for 2016, and 56.625–73.5 and 69.125–87.25 MeV for 2021. The 21 MeV window is shifted at the frozen support boundary. Four predetermined alternatives change the polynomial degree by ± 1 or the total window width by $\pm 2\sigma_m$, one change at a time. All thirty fits are retained.

Data	Mass [MeV]	r_{GP}	r_{poly}	Nominal p_0	D/dof
2015	51	+3.14	+0.13	0.448	141.7/131
2015	21	+2.52	+5.58	1.21×10^{-8}	92.5/50
2016	90	+3.42	+2.00	0.023	125.3/113
2016	117	+3.28	+1.53	0.064	165.6/155
2021	78	+2.81	-1.35	0.500	29.2/24
2021	65	+2.40	+1.96	0.025	17.4/22

Table 17: Traditional baseline fits to the *original* observed data at the fixed GP-selected masses. The two methods use different background models and windows. Here $p_0 = \bar{\Phi}[\max(0, r)]$ with 0.5 for nonpositive roots. These are uncalibrated local references at masses selected with the same events. In particular, the small 2015/21 MeV value accompanies a poor background description and must not be quoted as a particle significance.

There is no uniform confirmation of the GP peak strengths. At 2015/51 MeV the baseline root falls from +3.14 to +0.13. The two 2016 roots fall from +3.42, +3.28 to +2.00, +1.53. At 2021/78 MeV the sign changes from +2.81 to -1.35; all five traditional variants give a negative root. The 2021/65 MeV baseline is +1.96, with several alternatives near +1. This is substantial background-model dependence, not a comparison of two independent measurements.

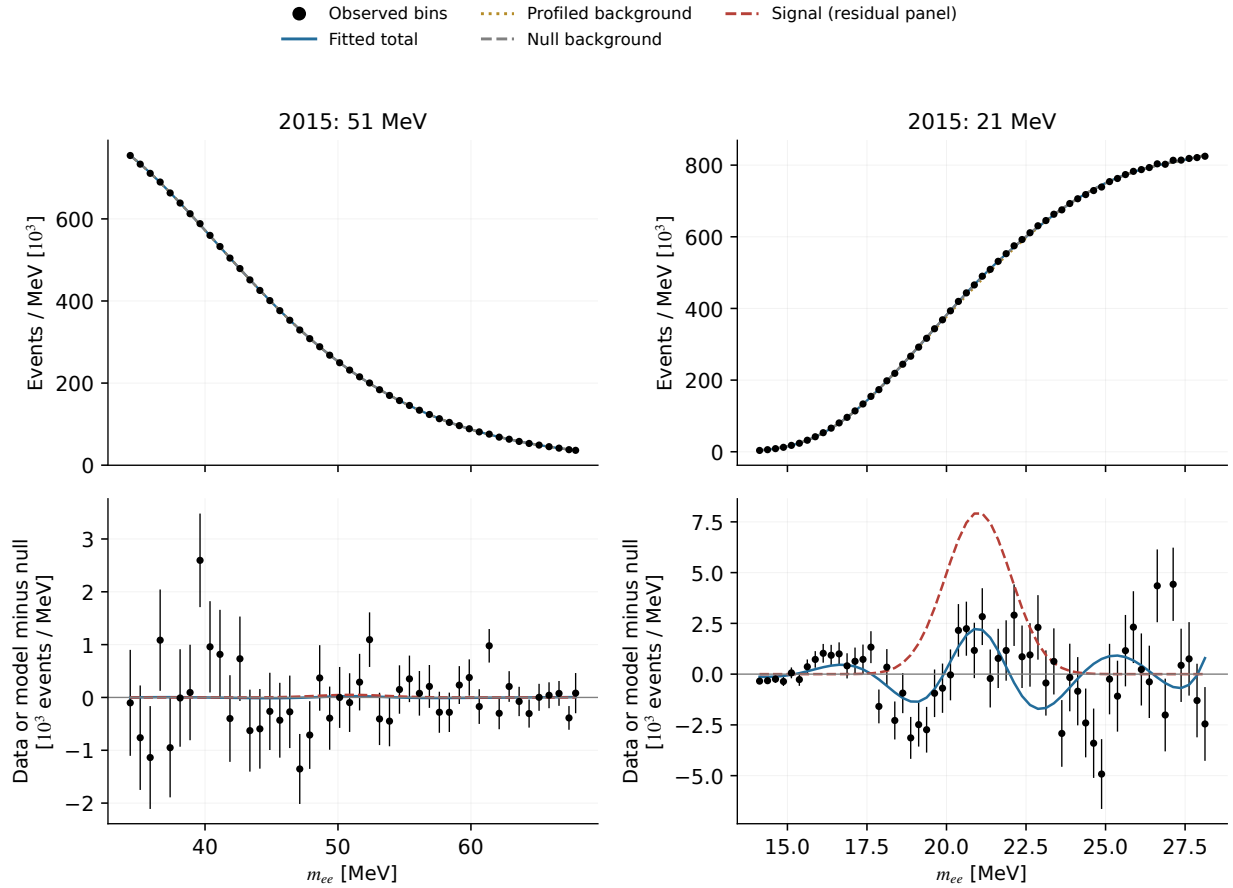


Figure 94: Traditional baseline fits to full 2015 data. At 51 and 21 MeV the nominal local coordinates are 0.131 and 5.580, with $D/\text{dof} = 141.69/131$ and $92.50/50$. Adjacent bins are summed for display: 0.75 MeV at the 51 MeV hypothesis, and 0.25 MeV at 21 MeV. Every fit still uses the original 0.25 MeV bins. Counts and residuals are divided by the actual displayed width, including a partial endpoint group. Bottom: data minus the refitted null background, with counting errors only. The red Gaussian is not the blue total change: the polynomial background also moves. Subtraction induces correlated residuals. The large 21 MeV root accompanies an inadequate overall fit. Table 18 retains all variants, including the badly fitting degree-two model at 51 MeV; neither extreme is a defensible physical significance.

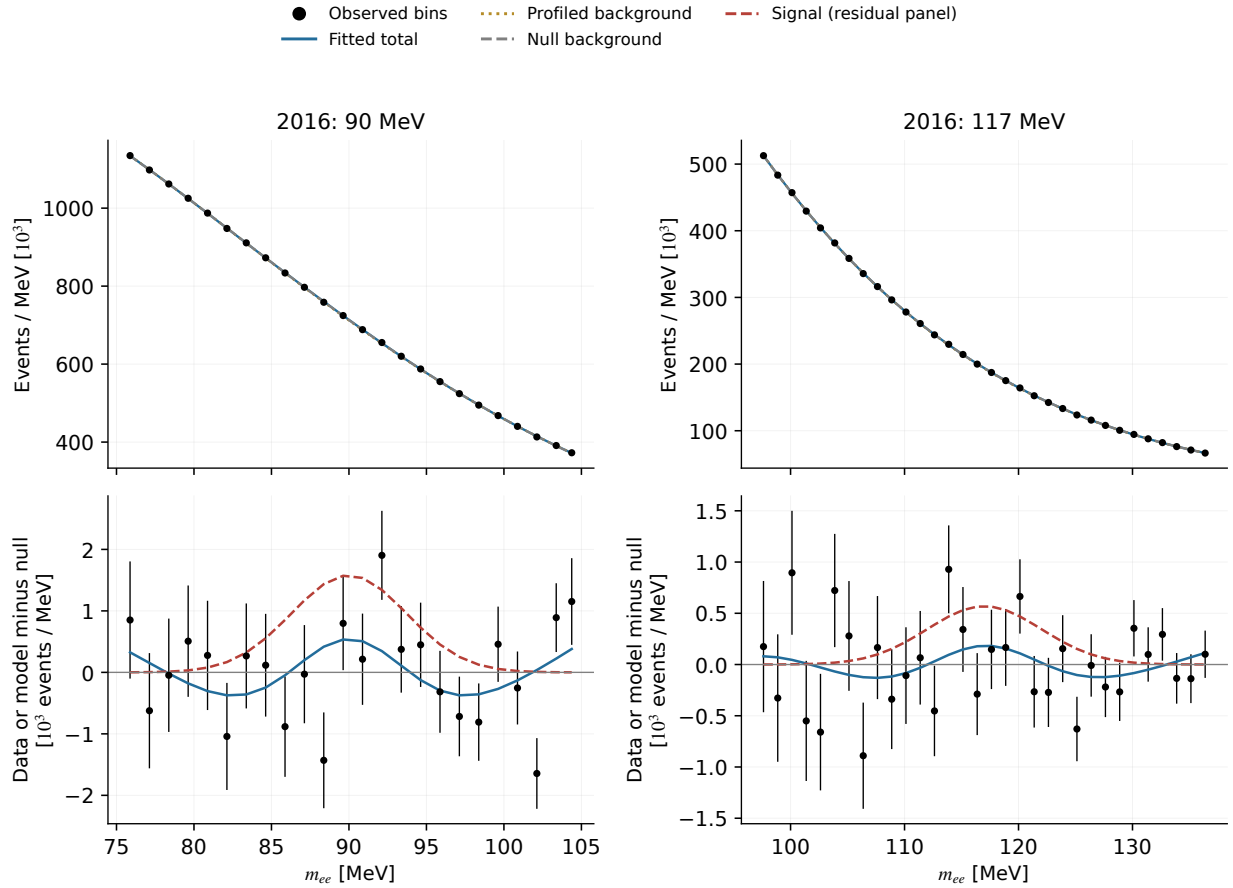


Figure 95: Traditional baseline fits to full 2016 data. At 90 and 117 MeV the nominal local coordinates are 1.998 and 1.525, with $D/\text{dof} = 125.35/113$ and $165.59/155$. The display sums five adjacent native bins to 1.25 MeV; any partial endpoint is retained. Counts, residuals and predictions are divided by each displayed width. Fits and probabilities still use the original 0.25 MeV bins. Both baseline positive roots are weaker than their GP values. Reducing the polynomial degree gives negative yields and larger deviances (Table 18). All alternatives are retained. Numerical convergence does not establish that a background describes the data; nominal local probabilities refer only to that model at the already selected mass.

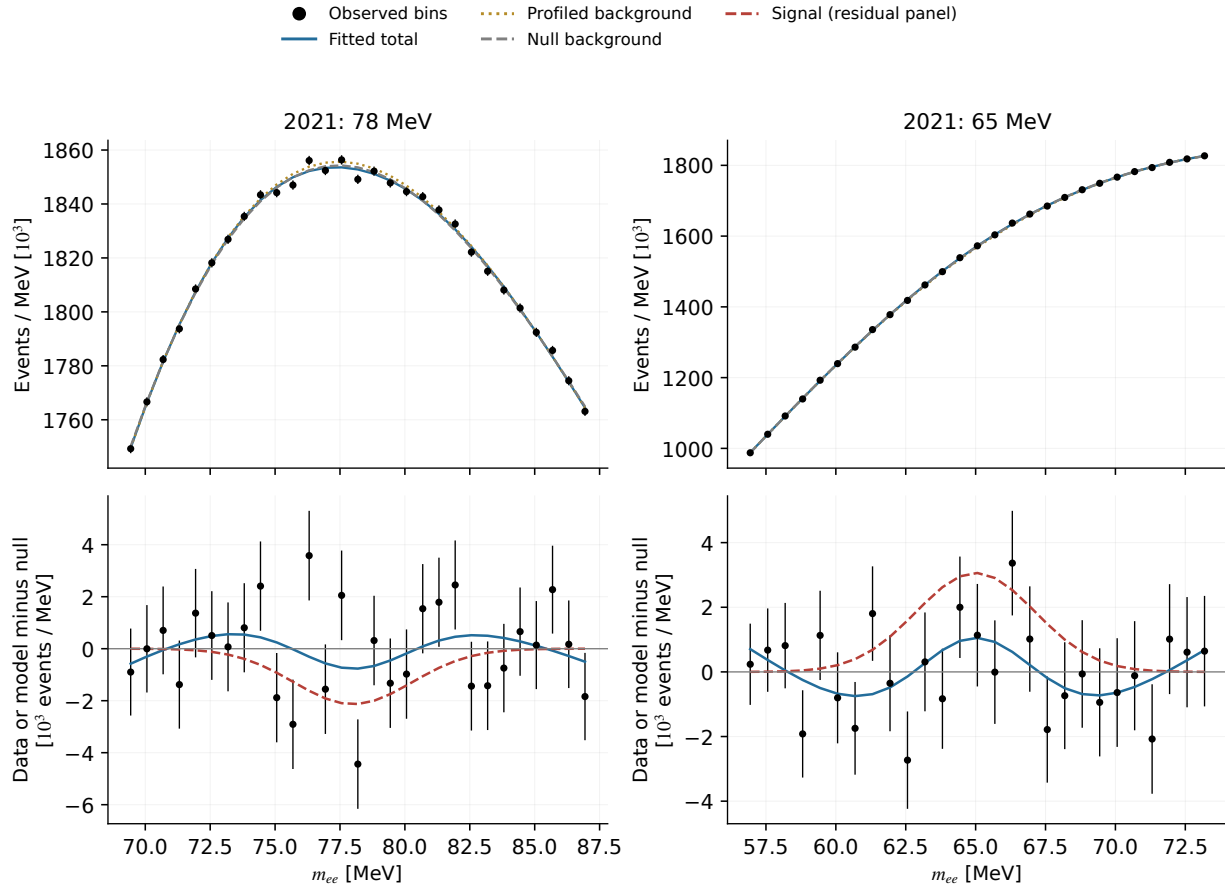


Figure 96: Traditional baseline fits to the released 2021 10% sample. At 78 and 65 MeV the signed local coordinates are -1.350 and 1.960 , with $D/\text{dof} = 29.16/24$ and $17.36/22$. Original 0.625 MeV bins are retained and displayed as counts per MeV. The 78 MeV feature becomes a deficit for every specified conventional fit. The 65 MeV baseline remains a modest positive excess. Its degree-two alternative gives a very large positive root but $D/\text{dof} = 28.10$, showing a severe background-model failure (Table 18). Counting bars and the fitted Gaussian component are visual aids, not independent residual significances. No additional 2021 data were opened.

C.5 How to interpret the strongest-looking conventional outcomes

Data, MeV	Baseline	Degree -1	Degree $+1$	Width -2σ	Width $+2\sigma$
2015, 51	+0.13 (1.08)	-34.63 (22.39)	+1.44 (1.03)	+0.54 (1.10)	-0.93 (1.11)
2015, 21	+5.58 (1.85)	-12.56 (10.41)	+3.46 (0.96)	+5.11 (1.38)	-1.30 (3.91)
2016, 90	+2.00 (1.11)	-2.48 (2.17)	+1.19 (1.12)	+0.80 (1.03)	+2.19 (1.10)
2016, 117	+1.53 (1.07)	-4.03 (1.44)	+1.13 (1.08)	+2.10 (1.04)	+1.84 (0.98)
2021, 78	-1.35 (1.22)	-0.21 (7.95)	-0.44 (1.27)	-1.35 (1.50)	-3.32 (1.19)
2021, 65	+1.96 (0.79)	+8.46 (28.10)	+0.97 (0.83)	+0.86 (0.98)	+0.91 (1.07)

Table 18: All thirty retained traditional fits: each cell gives signed local significance r followed by deviance per degree of freedom in parentheses. Width changes refer to the *total* window width. Large roots accompanied by a large deviance expose inadequate local background functions. No row or variant is selected by its resulting probability. Actual edges, yields, covariance matrices and nominal local probabilities are retained in the numerical archive.

The most striking baseline is 2015/21 MeV: $r = 5.58$ and the formal local reference is 1.21×10^{-8} . But the total fitted spectrum has $D/\text{dof} = 92.50/50$; a nominal chi-square goodness-of-fit reference is about 2.4×10^{-4} . Adding one polynomial degree changes the root to 3.46 and the deviance to 46.95/49. Widening the degree-five window reverses the sign and makes the fit much worse. The large baseline number therefore cannot be defended by quoting an asymptotic tail alone. These controls and the existing low-mass side study must be considered together.

The same failure mode is conspicuous in the 2021/65 MeV degree-two fit: $r = 8.46$ accompanies $D/\text{dof} = 28.10$. A polynomial that cannot follow the continuum can force its residual shape into the Gaussian signal coordinate. The extreme signed deficits in some other low-degree fits have the same interpretive problem. These variants are retained to expose that failure, not to choose a different significance after seeing the outcome.

Goodness of fit is a useful warning here, but not a complete validity test. The masses were selected with the same data, polynomial choices were investigated, and no scan-wide or model-selection correction is included in these nominal local values. Even a satisfactory deviance does not demonstrate unbiased signal extraction or correct coverage. Conversely, weakening under a more flexible background does not by itself prove the absence of a signal. A final analysis needs a background family qualified by independent predictive controls and signal-injection recovery before its tail probabilities can be interpreted physically.

C.6 What this implies for presentation and the next data step

The study shows that removing candidate regions can relax their neighboring deficits, while oscillations elsewhere survive. This is compatible with the fitted echoes demonstrated in Sec. 6.10; it does not identify the observed parent peak or prove that the remaining pattern comes from a resonance. The especially large reference offsets already occur without an explicitly added signal template. Their sensitivity to the continuum construction is itself a reason to qualify the reference before using its very small centered tails as particle evidence.

A useful next test is to freeze the candidate masses, widths, background checks and complete predicted peak-dip response before examining the additional 20% that would take 2021 from 10% to 30%. Analyze that additional portion on its own as well as in the cumulative sample. Real resonances, detector effects and background-model mismatch can all become more visible with more

events, so growth alone is not decisive. The sequential 30% and 100% looks and any model choices must be included in the declared final inference procedure. No such unblinding was performed here.

C.7 Reproducibility and numerical checks

All 17,430 profile tests are complete, with no mass or replacement dropped. An independent reconstruction checks the complete fixed spectra, untouched bins, paired Poisson replacements, remote geometry and 5,160 saved GP likelihoods. It also checks all thirty traditional fits using separate 48-point quadrature and likelihood derivatives. The largest root differences are 5.3×10^{-13} for reconstructed GP fits and 2.9×10^{-11} for conventional fits. All conventional final stationarity, covariance, fixed-multistart and doubled-quadrature checks pass. An initial numerical stall in one variant was resolved without changing the statistical model; the attempt and refinement are archived.

The derivative, scripts, exact modified spectra, fit components, all thirty conventional outcomes, independent HEP review and verification reports are retained under

`study_results/v4p9p16_candidate_removal_20260906/`

The frozen source manifest binds the original numerical products; the v5 manifest separately records the copied figures and revised text. Numerical agreement establishes reproducibility, not physical background validity or calibrated discovery significance. The original combined limits, Figure 52, extraction fits and probability ensembles remain unchanged.

D Joint scan of deficits

A negative auxiliary amplitude describes a signal-shaped deficit, not a physical negative rate. This scan uses the existing fits.

Combined deficit scan: an illustrative check

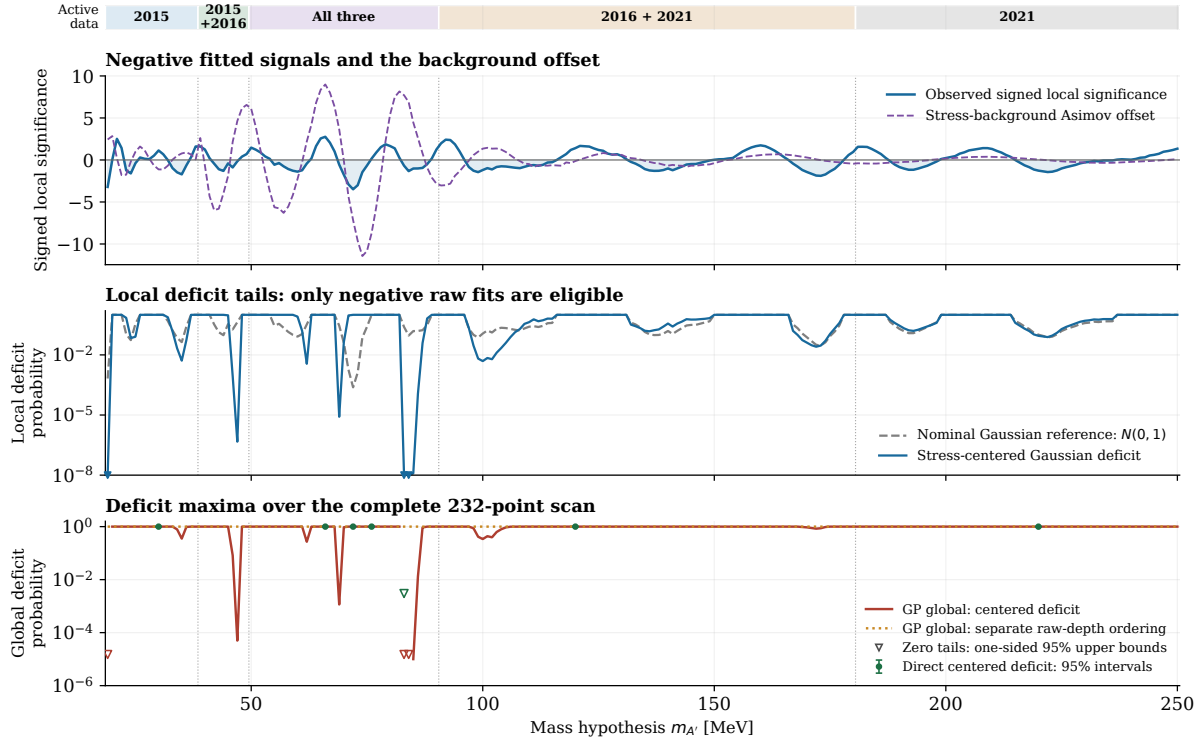


Figure 97: Profiled deficit scan. Top: observed signed local significance and the stress-background offset; shading marks negative fitted contributions. Middle: raw-root Gaussian reference and conditional Gaussian-response tails. Both assign one to nonnegative raw roots. Local triangles mark the 10^{-8} display floor. Bottom: two separate global orderings; triangles mark one-sided 95% zero-count bounds. Direct error bars are central 95% intervals.

For $z = (r - a)/s$, the conditional local rule is $p^- = \Phi(z)$ when $r < 0$, and one otherwise. The principal scan score is $T^- = \max_{m:r_m < 0}(-z_m)$, with $-\infty$ for an empty set. The separate raw-depth statistic is $D^- = \max_m \max(0, -r_m)$.

The deepest raw deficit is at 72 MeV: $r = -3.490$, with uncalibrated local reference 2.41×10^{-4} . The largest stress-centered deficit is at 83 MeV: $r = -0.676$, $a = +7.707$, $s = 0.983$. Its raw reference is 0.2495 but its conditional local value is 7.18×10^{-18} , mainly because of the stress offset.

The latter threshold has 0/200,000 GP and 0/1,000 direct exceedances, giving one-sided 95% global bounds of 1.50×10^{-5} and 0.003. Every simulated raw-depth maximum exceeds the observed deficit. These realizations are shared with the excess scan.

These illustrative tails follow the excess scan and do not adjust for choosing directions, methods or orderings. The unresolved tails establish neither a particle signal nor physical background validity.

E Historical validation studies and their role in v5

This record groups the earlier development studies by the question they answer. Their sources remain at the commit recorded in the v5 provenance. Section 5 retains the full-refit validation and support studies; Section 6 gives the current observed limits.

E.1 Kernel, window and normalization development: June–August 2026

Early window and cross-kernel studies compared local polynomial backgrounds with GP interpolation. Later work widened the fit support beyond the search interval, adopted the moving $\pm 2.25\sigma_m$ exclusion, and retained full-template signal normalization in the local likelihood. These studies explain the current geometry; their old limit illustrations are historical.

The v4.2 follow-up separated display repairs from conditional window replacement and shared-coupling tests. Replacing selected counts creates neither a new background-only observation nor independent signal evidence. Appendix C extends that question with declared whole-bin interventions and comparison fits.

E.2 Source, exposure and threshold tests: August 2026

The v4.5–v4.8 program first used small pilot ensembles, then 100-background full-refit ensembles, to separate the fitted source shape from the imposed exposure. Its matched-reference injections use each background toy’s reference uncertainty, fixed before adding the signal. The displayed results in Section 5 retain the original ensemble definitions and their conditional nature.

Threshold-focused tests showed why passing a smooth-source ensemble does not establish performance near every steep turn-on. The v4.9 qualification and v4.9.5 support continuation examined several lower edges. All eligible edges failed the first, more stringent screen. A later practical screen was fixed before the held-out continuation and selected 36 MeV within the retained tie rule. The surviving under-recovery at 55 MeV is part of the result, not an exception removed from the figure. That sequence explains both the adopted support and the limits of its validation.

E.3 The 2016 support and numerical-state sequence: September 2026

The v4.9.7 study tested lower support edges 28–34 MeV at 44, 49, 54 and 59 MeV using 25 paired backgrounds and matched 0-, 2- and 5-reference-sigma injections. Every edge violated the predeclared gross-bias guard. The rule returned no provisional edge, so the proposed 75-background continuation and new combination did not follow from that study. Its broad-tail generating component carried a failed source-fit flag and was retained only as an explicitly conditional stress input.

Later v4.9.9–v4.9.11 investigations examined predictive controls and reproducibility of the archived 30–210 MeV state. An independent process reproduced the selected state to the required 10^{-6} tolerance at 87 of 142 masses. Subsequent optimizer-off predictions and numerical checks permit a conditional downstream release, documented by the numerical exception retained in v4.9.12. They do not convert the failed support-selection result into a passed physical-background qualification.

Full-refit closure, frozen-GP expected limits, empirical CL_s calibration and coherent global scans test different parts of the procedure. More draws in one cannot replace missing validation of another. This distinction allows reproducible observed curves to coexist with open physics-review questions.

F Historical pseudoexperiment regimes and model decisions

This appendix restores the three generation regimes and full-100 exposure study from the v4.9.7 note, together with the threshold and length-scale studies that explain subsequent decisions. Their original data fractions, statistical conventions and GP policies remain identified. They are historical validation evidence, not additional versions of the current observed curves.

F.1 Three toy-generation regimes

The word “toy” does not specify what is allowed to fluctuate. The three regimes below test different parts of the analysis and cannot be pooled into one calibration sample.

Analytic functional-form pseudoexperiments. A smooth function is fitted to a source spectrum independently of the GP. Its integral in each native bin supplies a background mean, and independent Poisson counts generate the full spectrum. The GP is then retrained on that spectrum’s sidebands before extraction. This tests recovery under a declared alternative shape, including the response of the training step. The exact year-matched and generalized-gamma families are written in Eqs. 98–100. Parameters are chosen using source-fit and normalization checks before extraction; a selected function is not refitted to obtain a more favorable signal pull.

Conditional blind-window GP toys. The observed sidebands are fitted once at a specified mass, giving a background mean and correlated uncertainty inside the excluded window. A background realization and a signal are sampled using that fixed prediction, and only the local extraction is repeated. This isolates the response of the likelihood under its assumed background uncertainty. It omits the variation in sideband fitting, hyperparameter estimation and model selection. It is therefore a conditional extraction test or an expected-limit reference under the stated posterior model, rather than an independent test of GP interpolation.

Full-spectrum GP-mean/refit pseudoexperiments. The propagated count-space GP mean provides an expected background over the complete support. A full spectrum is sampled, the complete Gaussian signal is added, and the sideband-to-window prediction is re-estimated before extraction. Signal tails outside the mask participate in the new training sample, allowing the study to diagnose signal absorption. The generation remains conditional on a GP-derived truth. Whether kernel coordinates are reoptimized or frozen is a separate analysis choice: the historical full-refit studies reoptimize them, while the later calibration and significance-GP studies condition on the reviewed kernels.

Regime	Background source and repeated steps	Principal question
Analytic full spectrum	Independent smooth source fit; Poisson bins; new GP sideband fit	Can the analysis recover signals from this alternative smooth spectrum?
Conditional GP window	Fixed sideband prediction and covariance; repeat local likelihood	Does extraction propagate the uncertainty represented by this fixed local model?
GP-mean full spectrum	GP-derived full-support mean; new spectrum and sideband prediction	How does retraining respond to fluctuations and signal tails under this GP-conditioned truth?

Table 19: Restored classification of the three historical generation regimes. The term “full spectrum” alone does not imply that kernel optimization or all preceding model choices were repeated.

Signal sampling is another independent choice. Poisson-total injection samples each bin with mean $A_{\text{inj}}w_i$; a fixed-total multinomial injection fluctuates only the allocation among bins. Fixed-total background scans are likewise conditional on their chosen event total. The full-100 study below used Poisson signal injection, while its mean strength was set by each background toy’s matched reference fit. Thus neither its reference error nor its physical injected yield is constant across the ensemble. These details distinguish it from the later fixed-physical-yield calibration.

F.2 Historical full-100 2021 source-and-exposure closure

Finding summary. The archived complete ensemble confirmed a local conditional 65 MeV offset in the $1\% \times 10$ lane, with a smaller native-10% response, after a pull-blind optimizer treatment. It does not establish observed-data bias or direct coverage.

This restored v4.9.7 Appendix B.6 records the complete 100-background ensembles before the later threshold replacements. Its 95% intervals and original generating truths remain distinct from the 90% intervals of the updated main-text composite. It was frozen before the reserve toys were inspected: toys 0–9 are the pilot sample and toys 10–99 are an independent confirmation sample for the pilot-selected $1\% \times 10$, 65 MeV zero-signal offset. All 100 background pseudo-histograms are refitted uniformly under the new estimator and restart policy; the v4.5 extraction rows are not pooled with the new rows.

The analysis card remains the declared v4.2 2021 state: a 40–300 MeV GP support, 50–250 MeV search range, $k_{\min} = 1.1$, $k_{\max} = 15$, a $2.25\sigma_m$ blind and training exclusion, two required sidebands, five-bin local rebinning, and 12 randomized optimizer restarts in addition to the nominal start. The four historical ensembles are $1\% \times 10$, native 10%, $1\% \times 100$, and $10\% \times 10$. The common **fGenGammaThresh** family is fitted independently to the native 1% and native 10% spectra. These functions are smooth stress truths, not physical background generators. In particular, the native-10% fit has Pearson $\chi^2/\text{ndf} = 2.781$ and is deliberately not retuned after inspecting closure.

For each scenario s , background toy t , and mass m , an independently optimized matched background-only fit defines $\sigma_{A,stm}^{\text{ref}}$. The injected amplitude and reported pull are

$$A_{stmz}^{\text{inj}} = z \sigma_{A,stm}^{\text{ref}}, \quad p_{stmz} = \frac{\hat{A}_{stmz} - A_{stmz}^{\text{inj}}}{\hat{\sigma}_{A,stmz}}, \quad z \in \{0, 1, 3, 5\}. \quad (108)$$

The injection scale is therefore toy-local, while the denominator is always the post-fit uncertainty of the injected pseudoexperiment. The exact v4.2 conversion $K_{stm} = A(\varepsilon^2 = 1)$ is also recomputed

from each native pseudo-spectrum, including the $\pm 1.64\sigma_m$ fractional-bin density integral, so that

$$\varepsilon_{\text{inj},stmz}^2 = A_{stmz}^{\text{inj}}/K_{stm}, \quad \widehat{\varepsilon}_{stmz}^2 = \widehat{A}_{stmz}/K_{stm}. \quad (109)$$

Negative fitted coordinates are retained. They are estimator fluctuations, not physical negative couplings, and none of these distributions is an upper limit or an expected-limit band.

F.2.1 Predeclared optimizer-repeat and exclusion gate

Each optimizer attempt refits exactly the same cached background and signal counts; the pseudoexperiment seed and optimizer-start seed occupy separate deterministic namespaces. An attempt uses the nominal kernel start plus 12 randomized starts and archives its seed, warnings, log marginal likelihood (LML), length scale, constant, extraction uncertainty, and covariance diagnostics. Three independently seeded, identically configured background-only/zero-signal optimizer attempts use the same data and masks. The highest-LML branch must occur at least twice under the predeclared agreement criteria

$$\frac{|\Delta \text{LML}|}{N_{\text{train}}} \leq 10^{-3}, \quad |\Delta \ln \ell| \leq 0.01, \quad |\Delta \ln C| \leq 0.05, \quad |\Delta \ln \sigma_A| \leq 0.02. \quad (110)$$

If it does not, two additional independent attempts are made. A nonzero-injection fit begins with one attempt and receives two, then at most five, attempts if it has an invalid or nonfinite fit/covariance state, retains the exact optimizer-start signature, lies within 2% of a length-scale bound, or has $\widehat{\sigma}_A/\sigma_A^{\text{ref}} \notin [0.5, 2]$. Branch selection uses only maximum GP LML and reproducibility—never $\widehat{A}$, pull sign or size, $\widehat{\varepsilon}^2$, or agreement with the injected amplitude. Optimizer warnings are archived for audit, but are not by themselves a repeat or exclusion trigger; only the pull- and amplitude-blind numerical diagnostics listed above activate the adaptive attempts.

An irreproducible reference state excludes all four strengths at that scenario-toy-mass key; an irreproducible injected refit excludes only that strength. Statistical extremes that pass this pull-blind technical gate remain accepted; repeat-triggered boundary solutions are retained when replicated. A closure cell was predeclared reportable only if at least 95 of its 100 generated rows survived. This fit-state policy avoids both post-hoc pull clipping and the scientifically incorrect removal of a complete pseudoexperiment because it fluctuates unfavorably.

The production run contains 400 scenario-toy tasks, 8000 predeclared extraction-state keys, 7996 finite raw amplitude fits, and 12,845 archived optimizer-attempt records. Of the 7994 accepted states, 5593 required one attempt, 2384 required three, and 17 required five. Six states are excluded: the $1\% \times 100$ toy-83, 90 MeV, $z = 3$ refit; the $10\% \times 10$ toy-28, 180 MeV, $z = 5$ refit; and all four strengths of the $10\% \times 10$ toy-66, 90 MeV reference state. The first two fail injected-fit reproducibility and the last fails reference-branch reproducibility after five attempts. No entire background-toy index is removed, every one of the 80 closure cells retains 99 or 100 rows, and all pass the predeclared minimum-size gate.

The count-space covariance produced by the frozen lognormal-moment mapping is classified as numerically repairable, not necessarily positive semidefinite, when it is finite and symmetric and satisfies $\lambda_{\min}/\max_i |C_{ii}| \geq -0.01$. The extraction likelihood then follows its frozen numerical path: progressively larger diagonal jitter through $10^{-3} \max_i |C_{ii}|$, followed, when needed, by an eigenvalue-clipped square root. Among the 7994 accepted states, 498 archived raw matrices have the eigenvalue ratio below -10^{-3} and 22 lie below -0.005 ; the latter are confined to 90 MeV in the two high-exposure ensembles. A pull-blind influence check that omits those 22 rows changes any closure-cell mean by at most 0.031 and any width by at most 0.014, but would reduce one

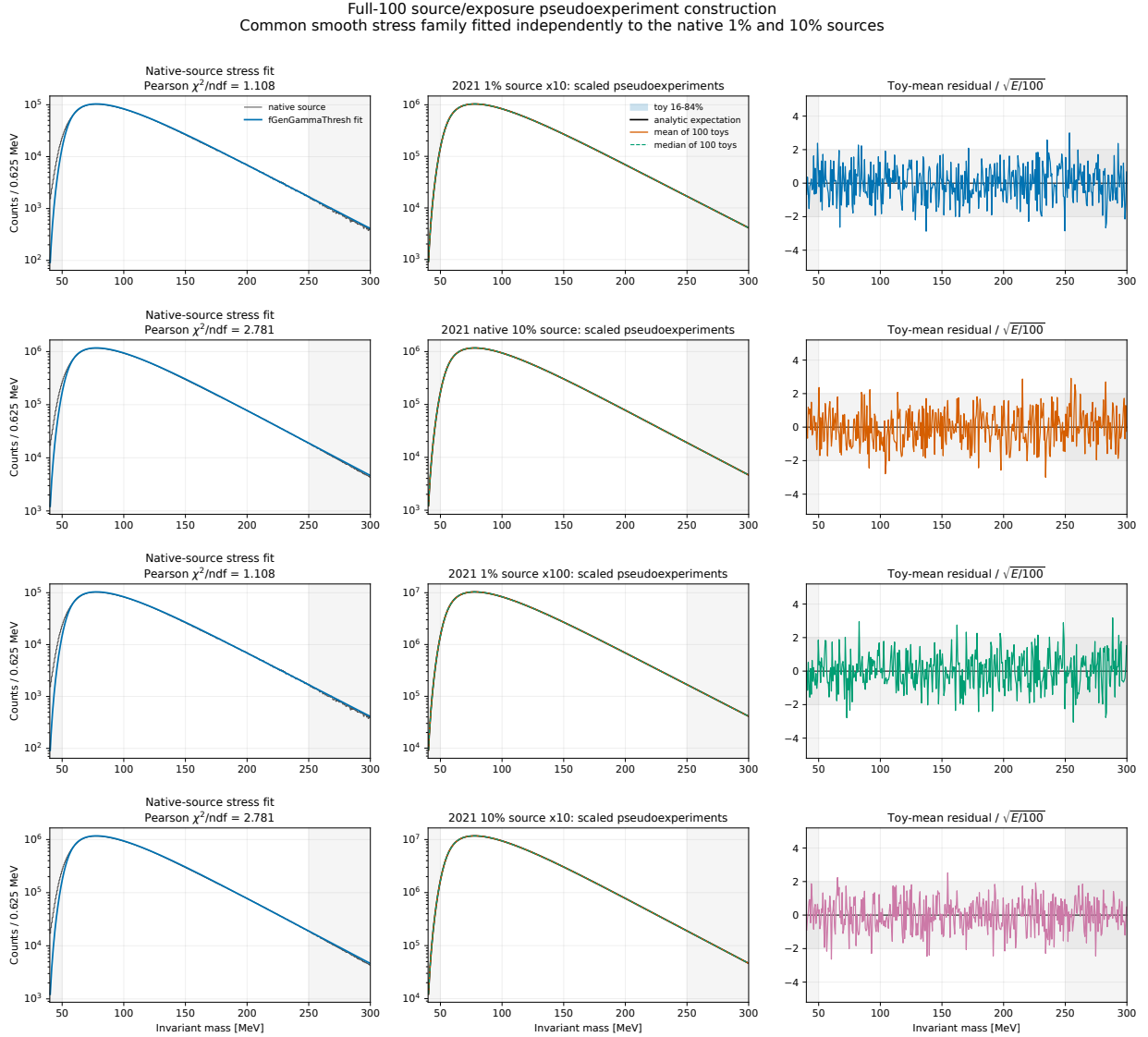


Figure 98: Invariant-mass construction of the full-100 ensembles, following the functional-form display of Fig. 46. Each row shows the native source and independently fitted stress truth (left), the scaled analytic expectation with the mean, median, and central 68% range of 100 background pseudoexperiments (middle), and the toy-mean residual divided by its expected standard error (right). Grey bands are support-only regions outside the 50–250 MeV search. The common analytic family isolates source/selection and normalization differences without claiming that either smooth function is a physical background generator.

cell below the predeclared 95-row reporting threshold. The rows therefore remain in the headline moments. An all-state deterministic reconstruction replays the actual repair path and compares it with the minimally sufficient isotropic diagonal loading. The replay agrees with the archived pull to within 0.0031 and with its uncertainty to within 3.2×10^{-4} relatively; changing to minimal diagonal loading moves any pull by at most 0.0029 and any uncertainty by at most 6.94×10^{-4} relatively; the largest changes in a cell mean and width are 5.0×10^{-5} and 1.1×10^{-4} , respectively. The archived ratios, rather than reconstructed threshold counts, define the 498 and 22 figures above. This tolerance and repair audit is a numerical qualification of the estimator, not evidence that the unrepaired matrices are positive semidefinite.

F.2.2 Spurious-signal bias and independent confirmation

Figure 100 isolates the $z = 0$ tests. The apparent $1\% \times 10$, 65 MeV pilot offset does not disappear with more toys. In the predeclared independent reserve (toys 10–99), the mean pull is 0.768 with sample width 0.944 and 95% Student- t interval [0.570, 0.965]; the two-sided test of zero mean gives $p = 1.64 \times 10^{-11}$. This exceeds the predeclared materiality threshold $|\bar{p}| \geq 0.2$. The full-100 precision estimate is 0.789 with width 0.962, median 0.784, and 95% interval [0.598, 0.980].

The deterministic analytic-mean stress spectrum gives a stable zero-signal pull of 0.837 at the same mass. The agreement with the 100-toy mean and the near-normal reserve quantiles in Fig. 101 identify a local functional-form/GP underprediction, not an optimizer failure or a few statistical outliers. The four injection strengths share each background toy and are highly correlated; their similar offsets are recovery diagnostics, not four independent confirmations.

The remaining zero-signal cells were treated as exploratory with Holm correction over all 20 scenario–mass checks. Two additional material offsets survive that correction: native 10% at 65 MeV has mean 0.716 (analytic-mean value 0.723), and $10\% \times 10$ at 180 MeV has mean 0.338 (analytic-mean value 0.194). These are closure properties conditional on the deliberately imperfect smooth stress truth and the frozen card. They do not measure a bias of the observed data result and do not calibrate frequentist coverage.

F.2.3 Injection recovery, pull widths, and signed coupling coordinates

Figures 102–105 use a four-panel layout for mean pull, pull width, residual fitted significance and recovered amplitude. The pull-mean error bars are 95% Student- t intervals and the width bars are exact normal-theory intervals based on the accepted sample size. The lower panels show residual fitted significance and median amplitude recovery with empirical central 68% spreads. Joining lines connect only the five predeclared mass points.

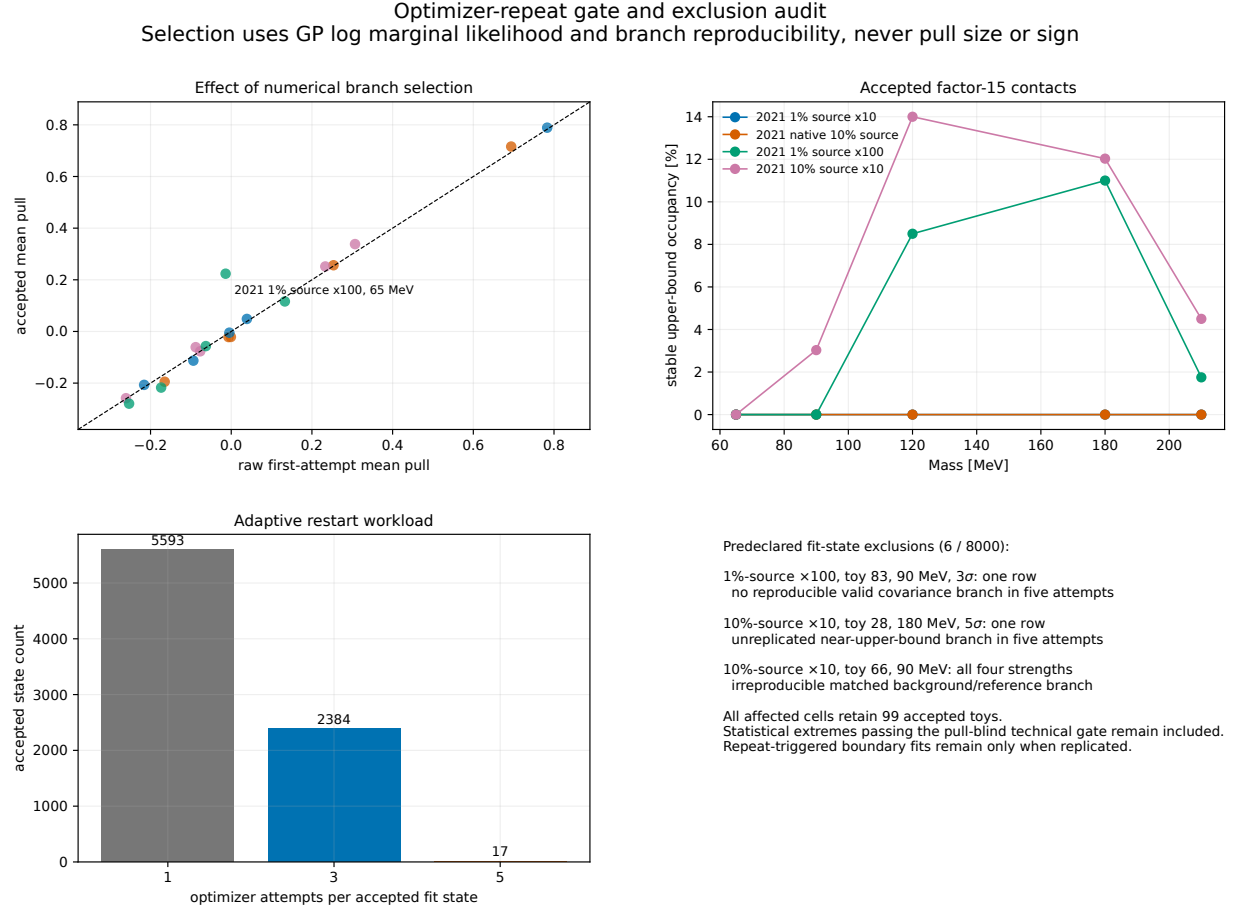


Figure 99: Optimizer-repeat gate and exclusion audit. The upper-left panel compares zero-signal means from the raw first attempts with the accepted LML-selected states; the visibly repaired $1\% \times 100$, 65 MeV point is numerical rather than a pull-based deletion. The upper-right panel reports accepted factor-15 contacts. The lower-left panel gives the adaptive attempt workload, and the lower-right panel lists all six excluded states. Statistical extremes that pass the pull-blind technical gate remain in the headline summaries; repeat-triggered boundary solutions are retained when replicated.

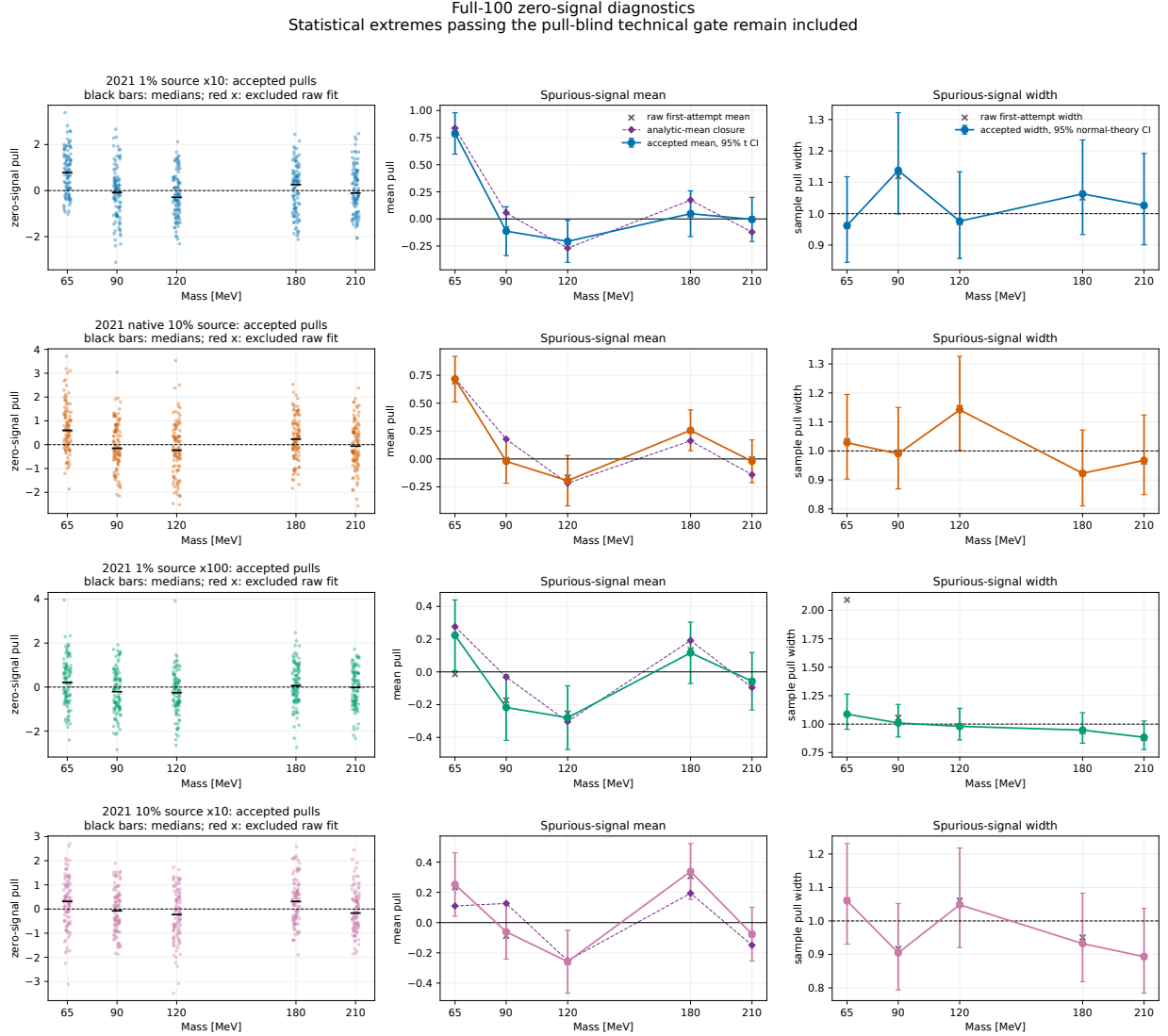


Figure 100: Full-100 spurious-signal study. Each row corresponds to one ensemble. The left panels show every accepted zero-signal pull and its median; the middle panels show raw first-attempt means, accepted means with 95% Student- t intervals, and the deterministic analytic-mean closure; the right panels show accepted sample widths with 95% normal-theory intervals. Red crosses mark excluded raw fit states where present. No accepted pull is clipped. The common mass dependence between analytic-mean and ensemble results diagnoses stress-truth/model closure rather than a statistical outlier.

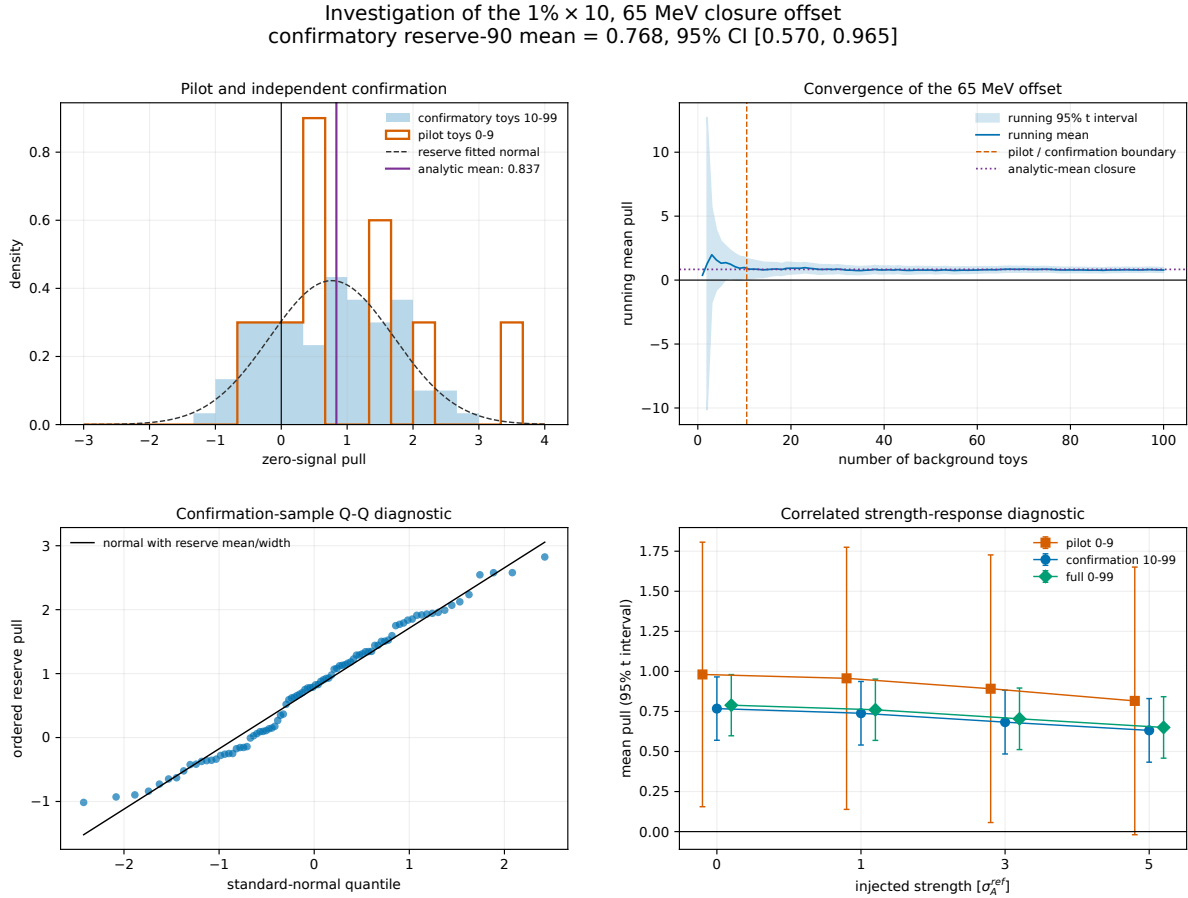


Figure 101: Investigation of the $1\% \times 10$, 65 MeV closure offset. The top-left panel separates the pilot and independent reserve distributions; the top-right tracks the running mean and its 95% Student- t interval, with the pilot/reserve boundary fixed at toy 10. The lower-left normal Q-Q diagnostic shows no isolated tail capable of explaining the mean, and the lower-right panel shows the correlated response across injection strengths. The analytic-mean pull of 0.837 was obtained from the exact stress-truth mean with independent optimizer seeds.

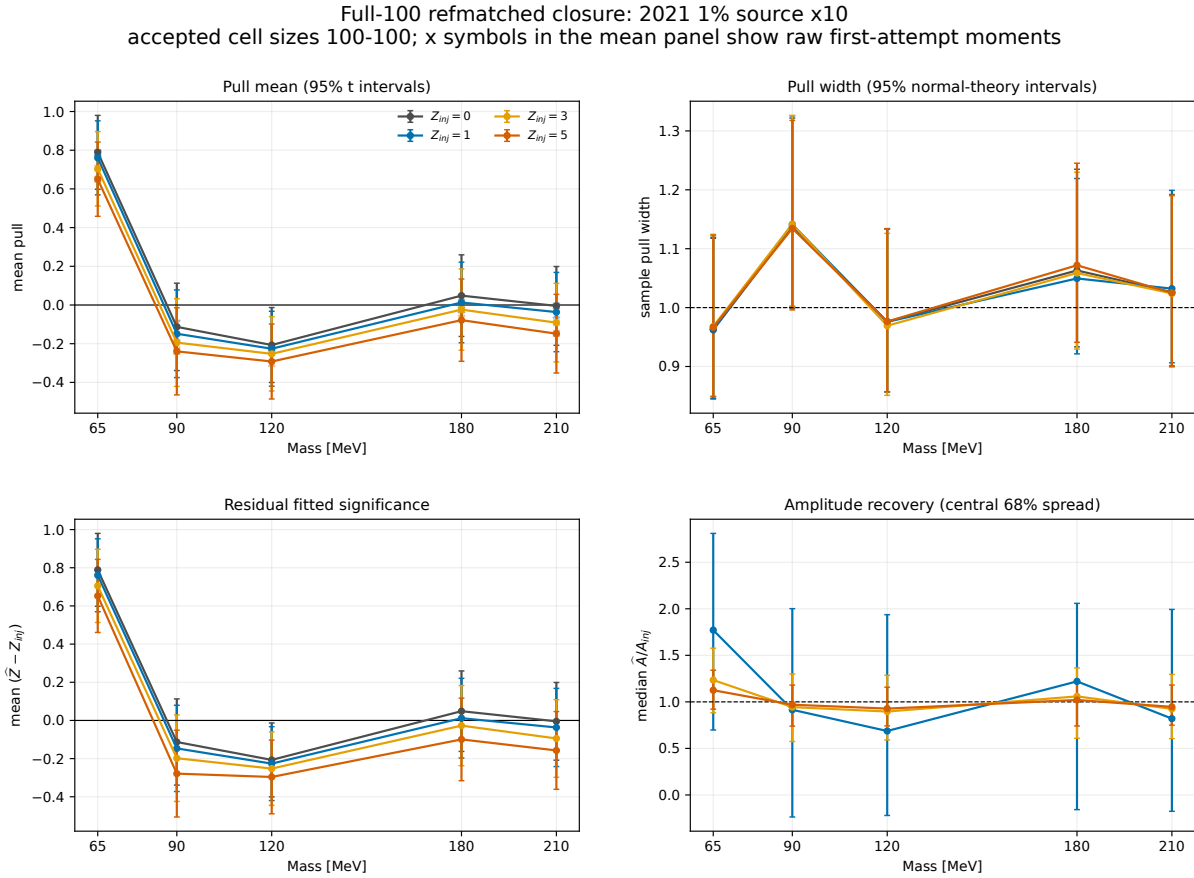


Figure 102: Full-100 refmatched closure for $1\% \times 10$. Every cell retains 100 accepted toys. The persistent positive 65 MeV pull means and the corresponding analytic-mean result are local stress-truth/model offsets; the other masses have near-unit widths and substantially smaller means. Crosses in the mean panel show raw first-attempt moments before the predeclared LML/reproducibility selection.

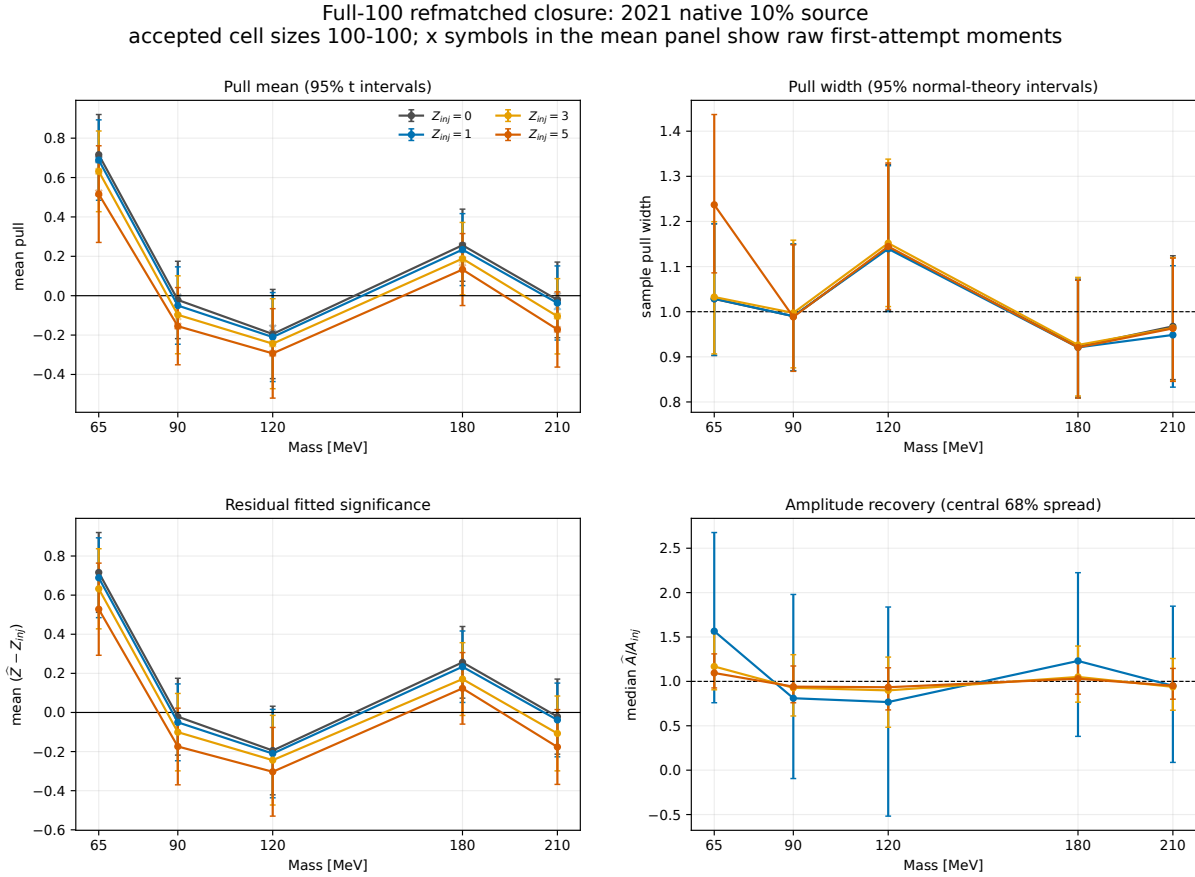


Figure 103: As in Fig. 102, for the native-10% ensemble. All cells retain 100 toys. The largest mean offset is at 65 MeV. The only accepted width outside 0.8–1.2 in the complete study is 1.237 for the 65 MeV, $z = 5$ cell; its normal-theory 95% interval is [1.086, 1.437].

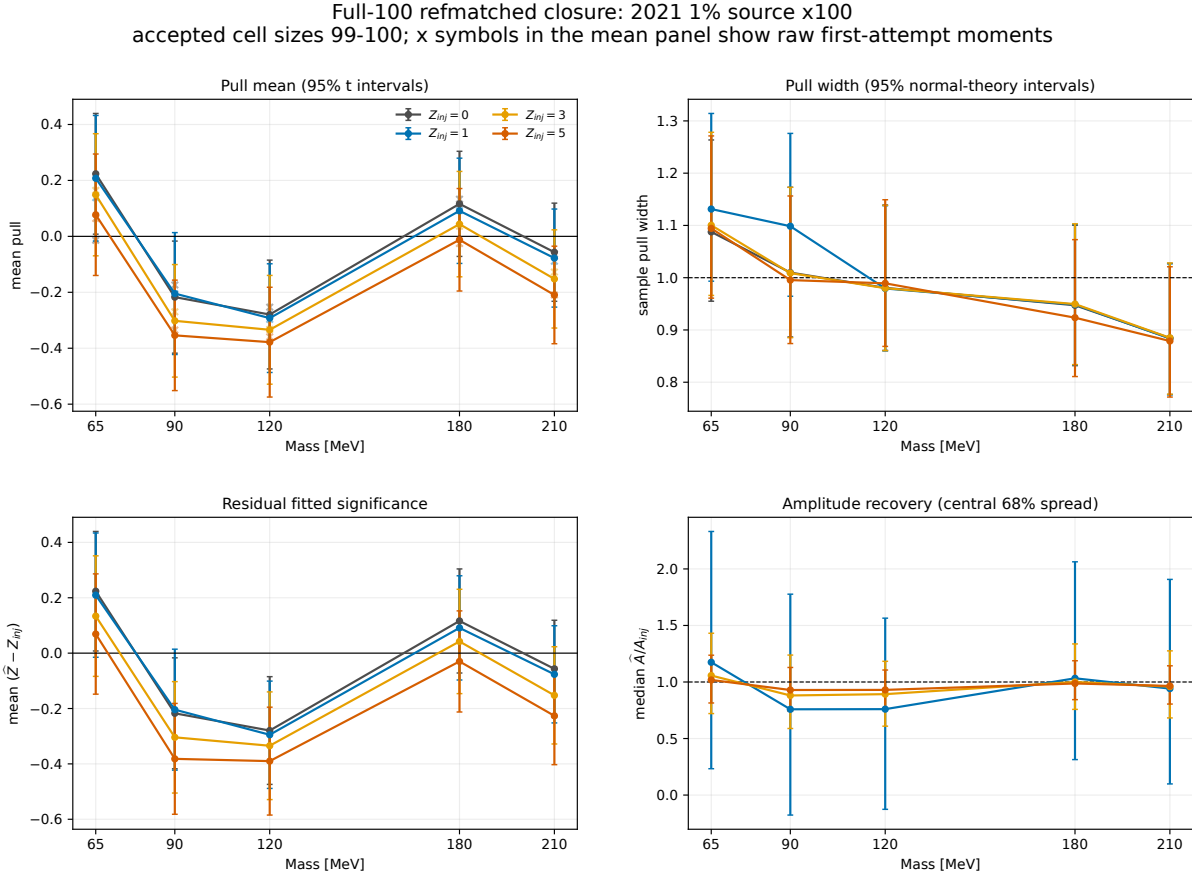


Figure 104: As in Fig. 102, for $1\% \times 100$. The 90 MeV, $z = 3$ cell retains 99 toys after one irreproducible injected refit is excluded; all other cells retain 100. Eighty-five of 1999 accepted fits contact the factor-15 upper length-scale bound. Stable contacts are reported rather than removed or used to change the frozen card.

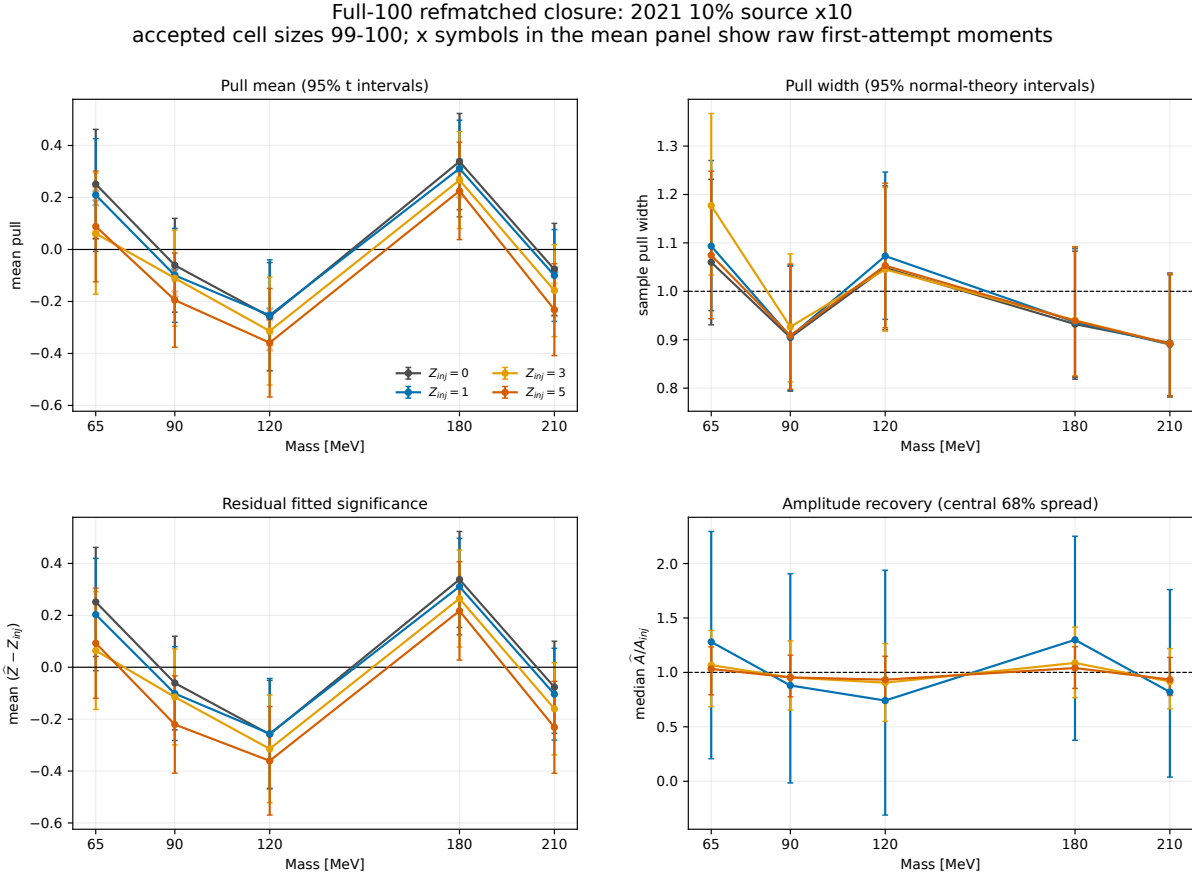


Figure 105: As in Fig. 102, for $10\% \times 10$. The four 90 MeV cells and the 180 MeV, $z = 5$ cell retain 99 toys; all others retain 100. The repeat gate repairs the dramatic 65 and 180 MeV optimizer branches seen in the v4.5 pilot without discarding those background pseudoexperiments. Among 1995 accepted fits, 134 stably contact the factor-15 upper bound.

Table 20: Full-100 accepted closure summary over the five requested masses. The maximum mean entries give the mass, and for nonzero injections also z , at which the maximum absolute mean occurs. Width ranges span five zero-signal or 15 nonzero cells. Boundary contacts are accepted, stable fit states, not exclusions.

Ensemble	minimum n	max $ \bar{p} $, $z = 0$	width range, $z = 0$	max $ \bar{p} $, $z > 0$	width range, $z > 0$
$1\% \times 10$	100	0.789 (65)	0.962–1.138	0.760 (65, 1)	0.964–1.142
native 10%	100	0.716 (65)	0.923–1.142	0.689 (65, 1)	0.921–1.237
$1\% \times 100$	99	0.280 (120)	0.884–1.088	0.378 (120, 5)	0.879–1.131
$10\% \times 10$	99	0.338 (180)	0.893–1.060	0.359 (120, 5)	0.890–1.177

The accepted upper-bound occupancies are 0/2000 for $1\% \times 10$, 0/2000 for native 10%, 85/1999 (4.25%) for $1\% \times 100$, and 134/1995 (6.72%) for $10\% \times 10$. The last two totals correspond to 23 and 37 unique scenario–toy–mass states (14 and 18 unique toy indices), respectively, because strengths sharing one background toy are correlated. Factor 15 is therefore still a frozen candidate-card choice with non-negligible high-exposure boundary contact, not a universally nonbinding bound. The closure ensemble does not by itself authorize a change of card: such a change would require a separately frozen same-input upper-bound study and renewed observed and coverage validation.

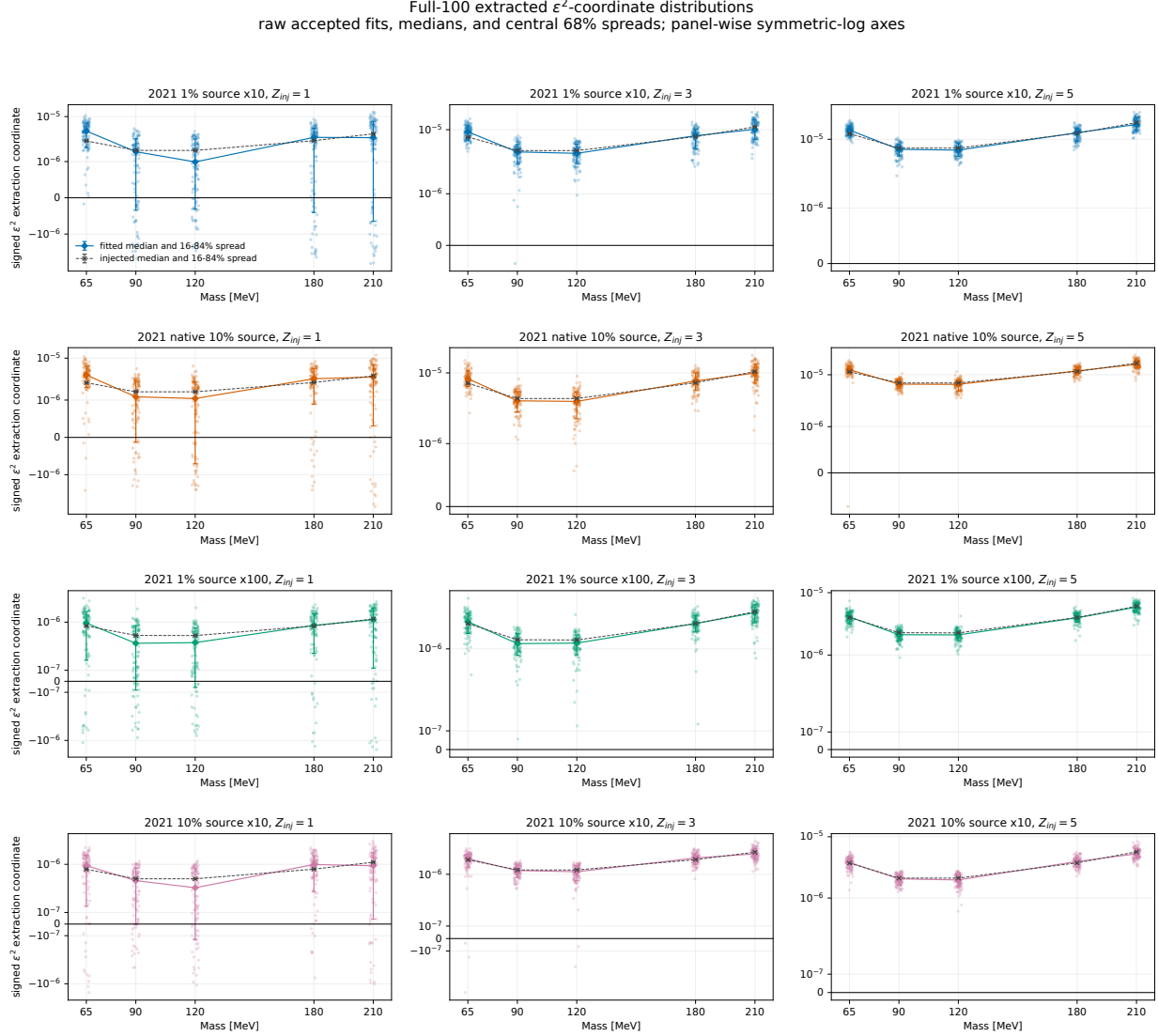


Figure 106: Signed ε^2 extraction-coordinate distributions for the 1-, 3-, and 5- σ_A injections. Points are all accepted fits; solid summaries are medians and empirical 16–84% intervals, while dashed summaries describe the toy-local injected coordinates. Panel-wise symmetric-log axes retain negative fitted values and broad 1- σ fluctuations. Because both K_{stm} and $\sigma_{A,stm}^{\text{ref}}$ are toy-local, the injected coordinates are distributions. These are extraction diagnostics, not limits, physical negative couplings, or expected bands.

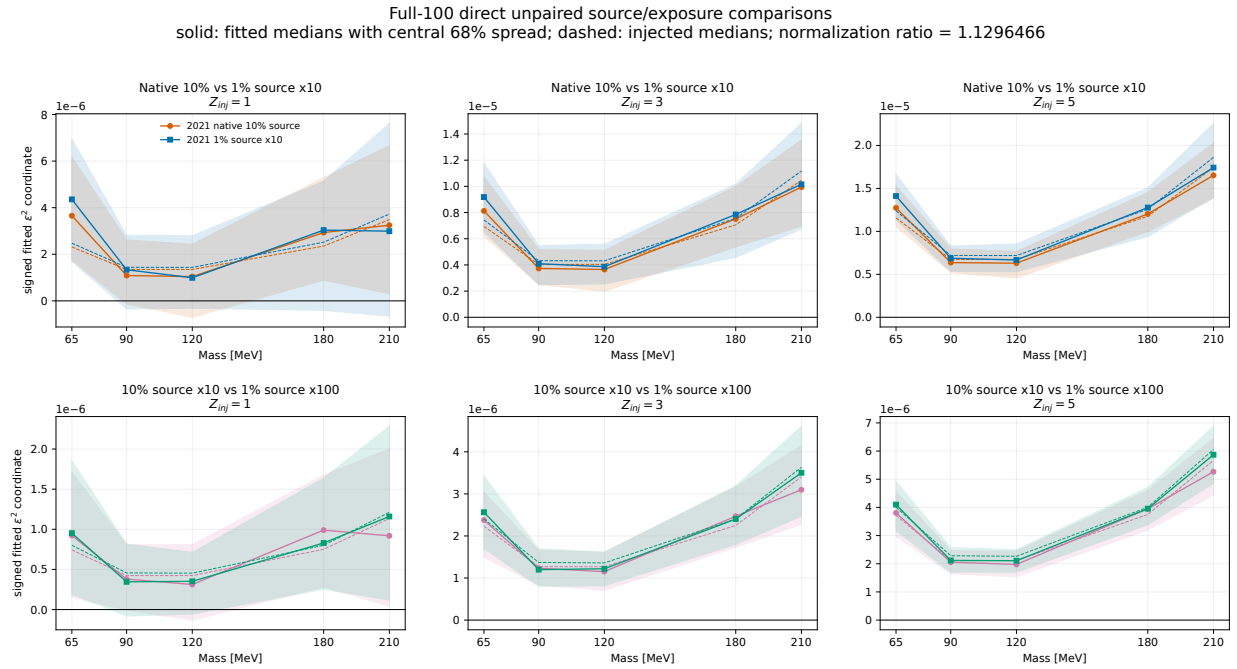


Figure 107: Direct full-100 comparisons between native 10% and 1% $\times$ 10 (top) and between 10% $\times$ 10 and 1% $\times$ 100 (bottom). Solid curves and bands are fitted medians and empirical 16–84% spreads; dashed curves are injected medians. The source functions are fitted independently and the cross-source ensembles are unpaired. Their expected-normalization ratio is 1.1296466 in both comparisons, so differences combine source/selection shape and normalization and are not pure exposure effects or paired tests. The ribbons are descriptive coordinate spreads, not expected-limit bands.

The full ensemble resolves the principal v4.5 ambiguity. The two spectacular pilot branches were numerical and are repaired by reproducible maximum-LML selection, while the $1\% \times 10$, 65 MeV offset is a stable local closure failure of the declared smooth stress truth under the frozen GP analysis. Only irreproducible numerical fit states are excluded; statistical extremes that pass the pull-blind gate remain in every arithmetic moment. The result is a full-100 conditional injection–extraction closure study, but it is not direct CL_s coverage, a scan-wise global-significance calibration, or evidence that the observed 65 MeV feature is signal or background. The accepted v4.2 observed result and its conditional limit bands remain unchanged.

F.3 Why threshold modeling was tested at 65 MeV

The historical full-100 offsets motivated a test of the steep 2021 turn-on. The original generalized-gamma truth gave mean pulls of 0.789 for $1\% \times 10$ and 0.716 for native 10% at 65 MeV. Their deterministic analytic-mean counterparts, 0.837 and 0.723, showed that the effect did not require rare Poisson fluctuations. The first targeted study changed the generating spectrum while retaining the 40–300 MeV GP support and matched-reference extraction.

For each native source, the local intensity was fitted over 35–80 MeV:

$$\lambda_{d,p}(m) = S(m; t_d, w_d) \exp \left[\sum_{k=0}^p c_{dk} T_k \left(\frac{m - 0.0575}{0.0225} \right) \right], \quad (111)$$

where T_k is a Chebyshev polynomial and m is in GeV. Degrees 3, 4 and 5 were tested with twelve deterministic optimizer starts. The fixed source-fit gates required native-bin deviance/ndf at most 1.5, five-bin deviance/ndf at most 2.0, maximum absolute five-bin Pearson residual at most five, and no parameter-bound contact. Degree five was the lowest common passing choice: its native-bin/five-bin deviances were 0.994/0.850 for the 1% source and 1.188/1.589 for native 10%. No signal-extraction result entered that degree choice.

Between 75 and 80 MeV the local form was joined to the archived tail using $q(u) = u^3(10 - 15u + 6u^2)$, with $u = (m - 0.075)/0.005$ clipped to $[0, 1]$. The bin expectation integrates $(1 - q)\lambda_{d,5} + q c_d G_d$ over the transition. A common normalization preserved the full support count before exposure scaling. This also rescaled the tail above 80 MeV by about 1.3%; the construction was not strictly a local replacement. Its tail deviances/ndf were 1.122 and 2.861 for the two sources, respectively.

The new 20-background study reduced the $1\% \times 10$ 65 MeV mean pull to 0.300, with 95% interval $[-0.219, 0.819]$. Native 10% retained 0.641, with interval $[0.082, 1.199]$. Both scenarios had negative 55 MeV offsets, and $1\% \times 10$ had a positive 70 MeV offset. The old and new ensembles are unpaired and use different truths; these results show partial conditional mitigation, not a universal correction or a completed support choice.

Two later 65 MeV replacements entered the principal validation display. The $1\% \times 10$ replacement continued the degree-five turn-on study to 100 backgrounds on 40–300 MeV support and gave mean pull 0.139 with 90% interval $[-0.050, 0.329]$. The native-10% replacement used the explicit degree-six intensity

$$H_{10\%,6}(m) = S(m; t, w) \exp \left[\sum_{k=0}^6 a_k T_k \left(\frac{m - 0.055}{0.025} \right) \right] \quad (112)$$

over 30–80 MeV, joined over 75–85 MeV to an archived `fSigPowExpQ` tail with the same $q(u)$ but $u = (m - 0.075)/0.010$. It analyzed the full 30–300 MeV support. It gave -0.246 with 90% interval $[-0.417, -0.076]$; an independent continuation gave -0.284 . Its 85–300 MeV tail deviance/ndf remained 6.220. These replacements explain the changed markers in the main plots,

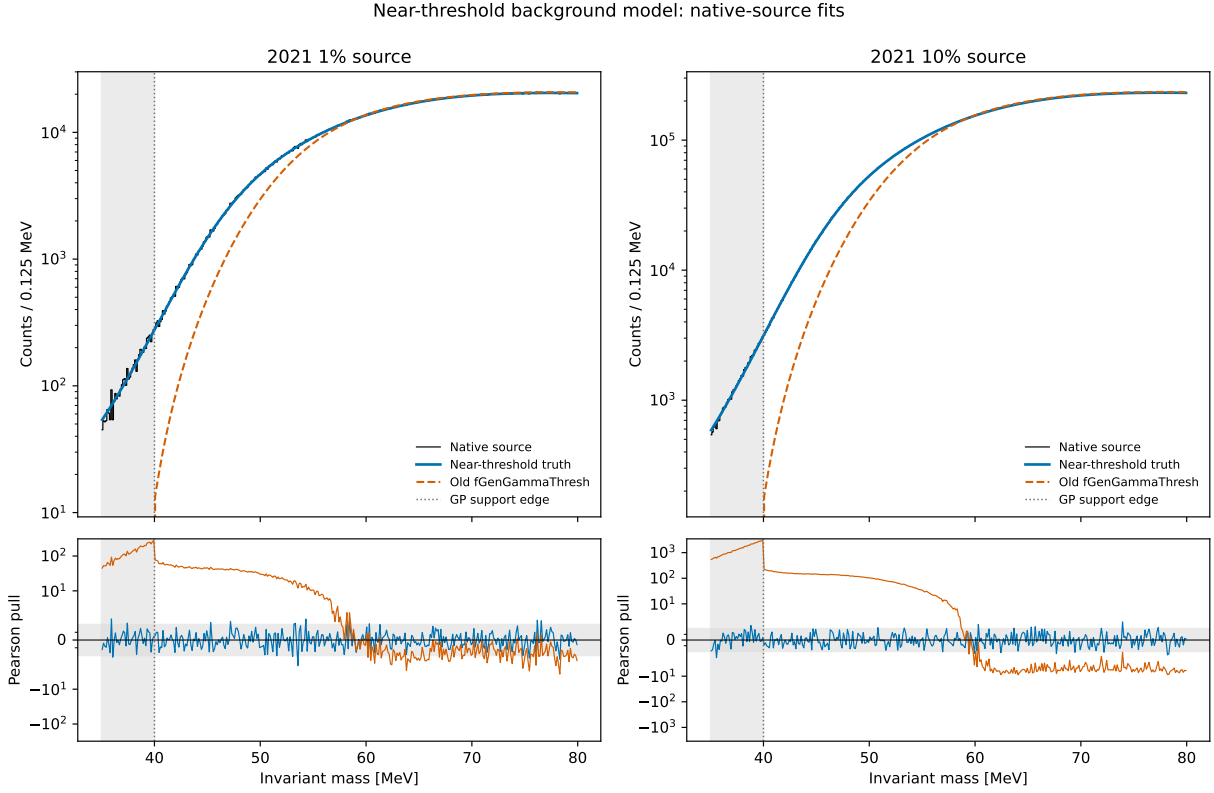


Figure 108: Historical source fits for the dedicated 2021 turn-on study. Native 0.125-MeV source bins are compared with the local degree-five model and the old generalized-gamma expectation. The local fit is shown through 35–80 MeV, but the generated truth begins at 40 MeV and blends to a rescaled tail from 75 to 80 MeV. The lower panels retain the large old-model residuals on a symmetric-log scale. These are source-model comparisons, not signal significances.

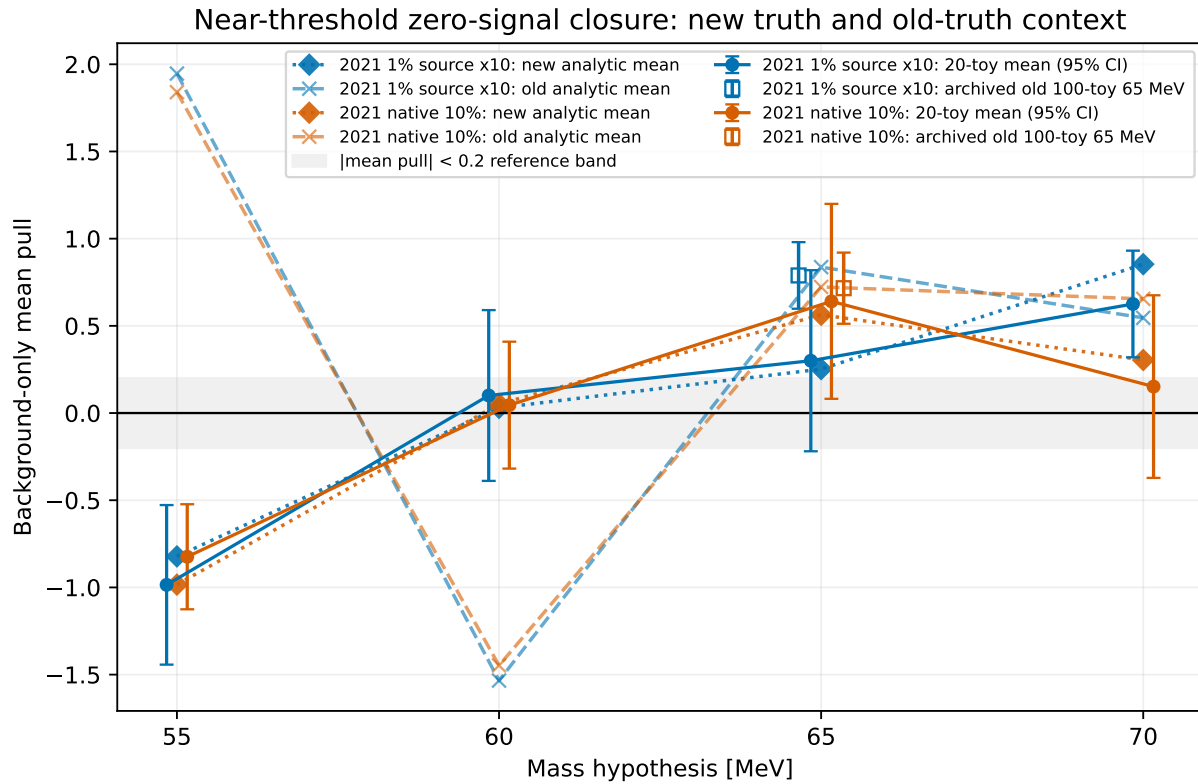


Figure 109: Historical 20-background near-threshold closure. Points carry 95% Student- t intervals; deterministic responses to the old and new analytic means expose the truth-shape dependence. The known 65 MeV response improves in the 1%-source test, while neighboring offsets remain. The later 100-background replacement results in the main text use their own identified truth and support; they are not pooled into these curves.

while preserving the historical full-100 failure here. The later 36 MeV support decision is a further, separately documented study.

F.4 Controlled 2016 upper length-scale range

The observed full-2016 study scanned upper factors $k_{\max} = 8, 10, 12, 15, 20$ at the same 142 hypotheses from 39 to 180 MeV. A fit occupied the upper boundary when $\ell_{\text{opt}}/\ell_{\max} \geq 0.999$. The occupancies were 142/142 at factor eight, 56/142 at ten, and zero at twelve, fifteen and twenty. This tested whether the imposed optimizer range was restricting the fit.

Questionable branches were repeated with the data and analysis card fixed. Fourteen selected repair rows used the largest repeatable GP log marginal likelihood; no limit, significance or kernel coordinate was interpolated. From factor twelve to fifteen the largest observed-yield-limit change was 0.083%, the largest local-significance change was 0.00255, and the log-marginal-likelihood differences lay in $[-4.5, 8.0] \times 10^{-6}$. The fifteen-to-twenty comparison was similarly stable. This supported factor twelve as the first nonbinding plateau setting. The direction of the observed limit and the local p-values were not selection criteria. Because the investigation followed an observed boundary diagnosis, it is historical evidence for the range choice, not a pre-unblinding freeze or a resolution of the later 2016 background-qualification exception.

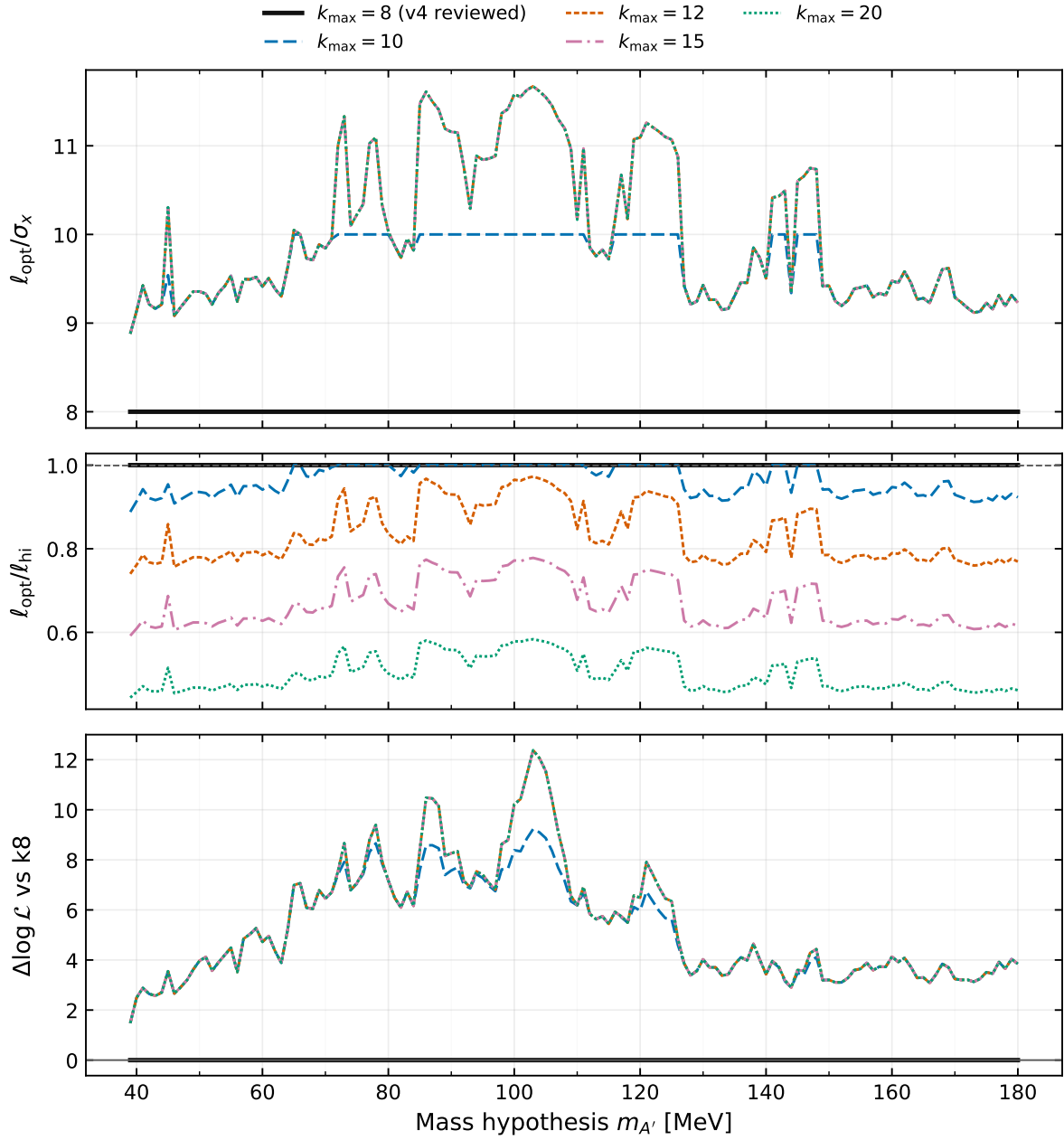


Figure 110: The controlled 2016 upper-length-scale study, restored from v4.9.7 Figure 74. The optimized scales, boundary occupancy and GP log marginal likelihood are compared on identical data and mass hypotheses. Factor twelve is the first nonbinding tested setting and is followed by the factor-fifteen/factor-twenty plateau. This observed-data numerical range diagnosis is not an expected-sensitivity or coverage optimization.

G Historical comparisons with the 2015 and 2016 resonance searches

These comparisons use the already analyzed full 2015 and 2016 engineering-run samples. Here “95%” specifies the confidence level of an upper limit; “full sample” specifies the data exposure. Neither term refers to the unreleased part of the 2021 sample. The curves below are saved historical results, with their original background models and settings. They provide a check of the observed yield-to- ε^2 scale and mass-dependent behavior, while the closure and calibration studies address the separate questions of bias, signal recovery and coverage.

G.1 The original and corrected 2015 polynomial analyses

The published 2015 search fitted a Gaussian resonance with the exponential of a Chebyshev polynomial in a local mass window. Its search covered approximately 19 MeV to 81 MeV, with fifth-order background below about 39 MeV and third order above. The result used a 95% confidence profile-likelihood limit with a power constraint: an observed upper limit stronger than the median background-only limit was replaced by that median. Its radiative-fraction convention was $f_{\text{rad}} = 0.085$ [3, 31].

Appendix J of the 2016 internal note documents a later re-analysis of the same 2015 sample with the corrected 2016 analysis code [28]. The archived account identifies two separate coordinate-consistency errors in the original implementation. The fit window was mapped to $[-0.25, 0.25]$ instead of the intended $[-1, 1]$, changing the interval over which the polynomial basis had been selected. The mass resolution was also not consistently transformed from the physical mass coordinate to the scaled fit coordinate, so the signal width used by the likelihood was inconsistent with that coordinate. After those corrections the recorded model choices were

$$(N, n_\sigma) = \begin{cases} (5, 10), & m_{A'} < 40 \text{ MeV}, \\ (3, 8), & m_{A'} \geq 40 \text{ MeV}, \end{cases} \quad (113)$$

where N is the polynomial order and n_σ is the window-size parameter reported in the internal note. The corrected contour is its Fig. 71; its Fig. 72 compares the original and corrected contours. The corrected re-analysis retained the 2015 detector and radiative inputs and used the later bounded CL_s procedure. It is a distinct reference from the literal published, power-constrained 2015 curve.

This history is also separate from the RooFit binned extended-likelihood issue discussed in the 2015 internal note. That note states that its final results used an independent vanilla ROOT fitter; the coordinate mapping and signal-width corrections above concern the resonance-search implementation described in the later internal note [31, 28].

G.2 The historical 2015 GPR comparison

The original comparison retained in v4.9.7 used a 2015 GPR configuration with a $2.25\sigma_m$ blind and training-exclusion interval, a 1 MeV mass step, a 7% radiative-normalization penalty, and the local event density integrated over the physical $m \pm 1.64\sigma_m$ interval. Its 95% upper limit used the bounded profile-likelihood CL_s construction. The GP was trained outside the excluded interval and its count-space background prediction and correlated uncertainty entered the signal-extraction likelihood. The Gaussian signal was normalized on the histogram support and sliced into the extraction window without renormalization, so its amplitude retained the convention of a total signal yield.

Figure 111 restores that earlier comparison with the corrected 2015 re-analysis. The reference curve was digitized from the solid black contour in Fig. 71 of the 2016 internal note. On the common integer-MeV grid from 20 MeV to 81 MeV, the pointwise ratio

$$R(m) = \frac{\varepsilon_{95,\text{GPR}}^2(m)}{\varepsilon_{95,\text{reference}}^2(m)} \quad (114)$$

has median 0.99 and mean 1.06. The central 68% of ratios lies approximately between 0.73 and 1.42, and all but two grid points lie within a factor of two. This spread is across masses in two observed curves; it is not a confidence interval or an expected-limit band.

A separate comparison campaign on 5 August 2026 recomputed the 95% limits at the reviewed v4 2015 kernel coordinates with upper length-scale factor 8. That campaign obtained median ratios of 1.1094 against the literal published 2015 contour and 0.8437 against the corrected recast. The first comparison used log-linear interpolation of the published digitization on 20 MeV to 81 MeV; the second used exact matched masses on 19 MeV to 81 MeV. These later numbers belong to that later frozen GP state and are not replacements for the 0.99 ratio in the restored figure.

G.3 The published 2016 method and the saved GPR comparison

The published 2016 analysis used a Gaussian signal plus an *exponentiated* Legendre-polynomial background,

$$f(m) = \mu G(m; m_{A'}, \sigma_m) + 10^{L_N(m;t)}, \quad (115)$$

with the polynomial evaluated in the coordinate scaled to the local fit interval. The background is therefore positive; it is not a raw Legendre polynomial added directly to the signal. The selected order was five below 66 MeV and three above, and the full fit-window width ranged from six to ten mass resolutions. Windows that reached the event-range boundaries were shifted to stay within those boundaries. The published upper limit used the 95% asymptotic CL_s construction. The reported systematic treatment included 10 000 variations of the signal width for the 3.4% resolution uncertainty, taking the 84th-percentile upper limit, and a downward 7.4% adjustment of the radiative fraction for its total uncertainty [4].

Figure 112 uses the saved 5 August 2026 GPR comparison. The GPR curve is the historical v4.1 2016 candidate with upper length-scale factor 12, selected through the boundary-release and likelihood-plateau study described in Section F.4. It is not the current v5 observed curve. For this comparison the reviewed kernel amplitude and length scale were held fixed, with no optimizer restarts; only the CL_s test size was changed from $\alpha = 0.10$ to 0.05. The individual 2016 likelihood used fit support 30 MeV to 210 MeV, search 39 MeV to 180 MeV, and $2.25\sigma_m$ extraction and training exclusion. The GPR conversion used its recorded 7% radiative penalty and the uncropped, fractional-bin density integral over $m \pm 1.64\sigma_m$. Thus the confidence levels match, while the background treatment and systematic prescriptions remain different.

The archived reconstruction reproduced the selected 90% signal-yield limits with maximum relative difference 1.378×10^{-10} before the change to 95%. Updating only the density integral shifted the corresponding 2016 coupling conversion by -0.176% to $+0.270\%$; the likelihood yield was unchanged. No pseudoexperiments or expected bands were generated for that comparison. On the 141 matched masses from 39 MeV to 179 MeV, its GPR/published ratio has median 0.5579 and is below one at 117 masses. The minimum and maximum ratios are 0.2049 at 80 MeV and 1.7344 at 160 MeV. The median corresponds to a 44.2% reduction in the observed limit on that comparison grid, not a demonstrated 44.2% sensitivity gain.

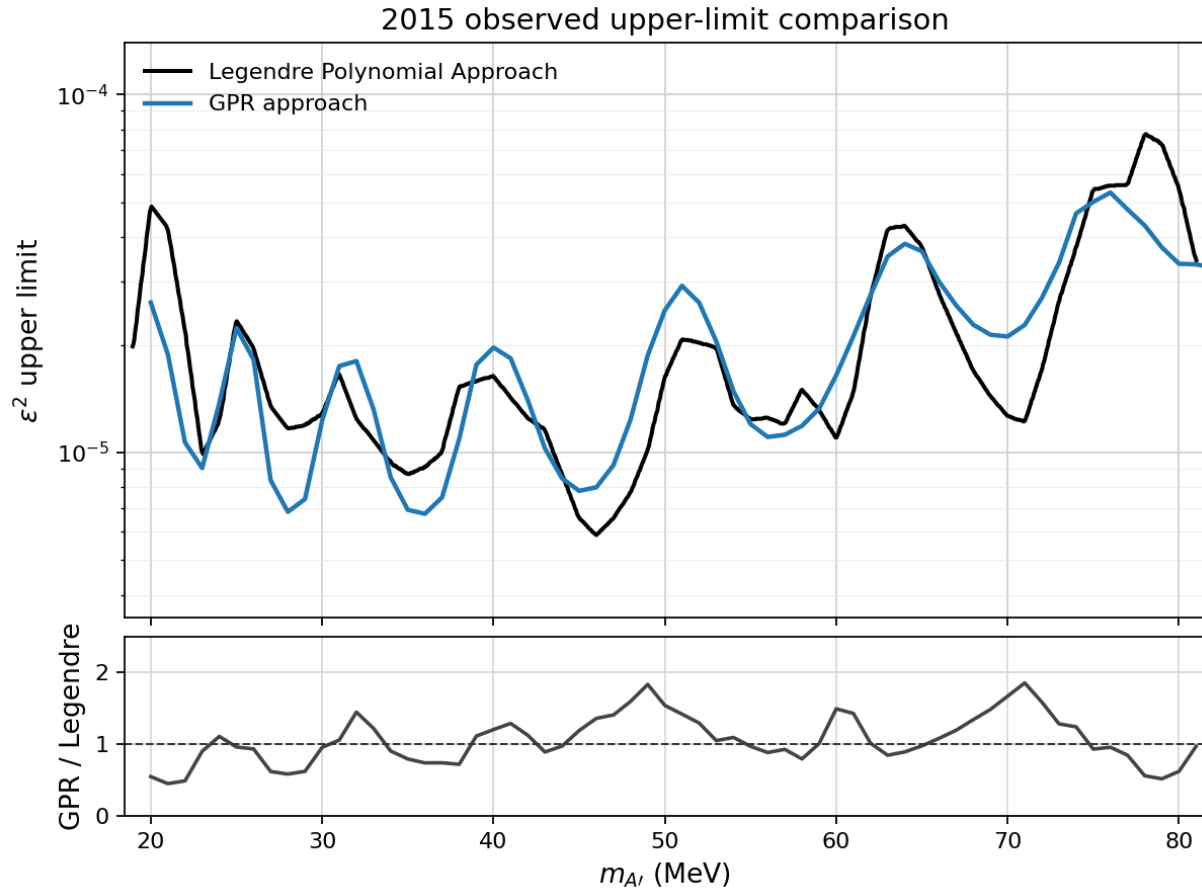


Figure 111: Restored historical 2015 observed 95% confidence-limit comparison from Appendix E of v4.9.7. The black “Legendre Polynomial Approach” curve is the corrected 2015 re-analysis digitized from Fig. 71 of the 2016 internal note; the blue curve is the earlier GPR comparison for the same full 2015 sample. The lower panel is the pointwise ratio of these saved curves, with median 0.99 and mean 1.06 over 20 MeV to 81 MeV. Both curves predate the current v5 settings. Their similar normalization scale does not establish coverage or absence of signal absorption.

Feature	Published 2016 analysis	Archived v4.1 GPR comparison
Data	Full 2016 engineering-run sample	Same full 2016 sample
Background	Local exponentiated Legendre fit; coefficients profiled with the signal	Sideband-trained log-count GP; predicted count-space mean and correlated uncertainty enter the extraction
Window	Mass-dependent local fit, six to ten resolutions in full width	GP support 30–210 MeV; local extraction $\pm 2.25\sigma_m$
Signal	Gaussian with detector resolution and reported width-systematic variations	Gaussian normalized on the full histogram support and sliced into the extraction interval
Upper limit	95% asymptotic CL_s with the published systematic treatment	95% asymptotic bounded-profile CL_s at the saved GP coordinates
Interpretation	Published observed result	Historical observed diagnostic; no new toy calibration or expected bands

Table 21: Methods entering the saved 2016 comparison. Matching confidence level and data exposure does not make the two inference procedures identical.

Shared counts, resolution and radiative inputs explain part of the agreement; different background responses and realized fluctuations can change individual masses. Coverage and signal recovery require the distinct toy tests in Section 5 and Appendix B.

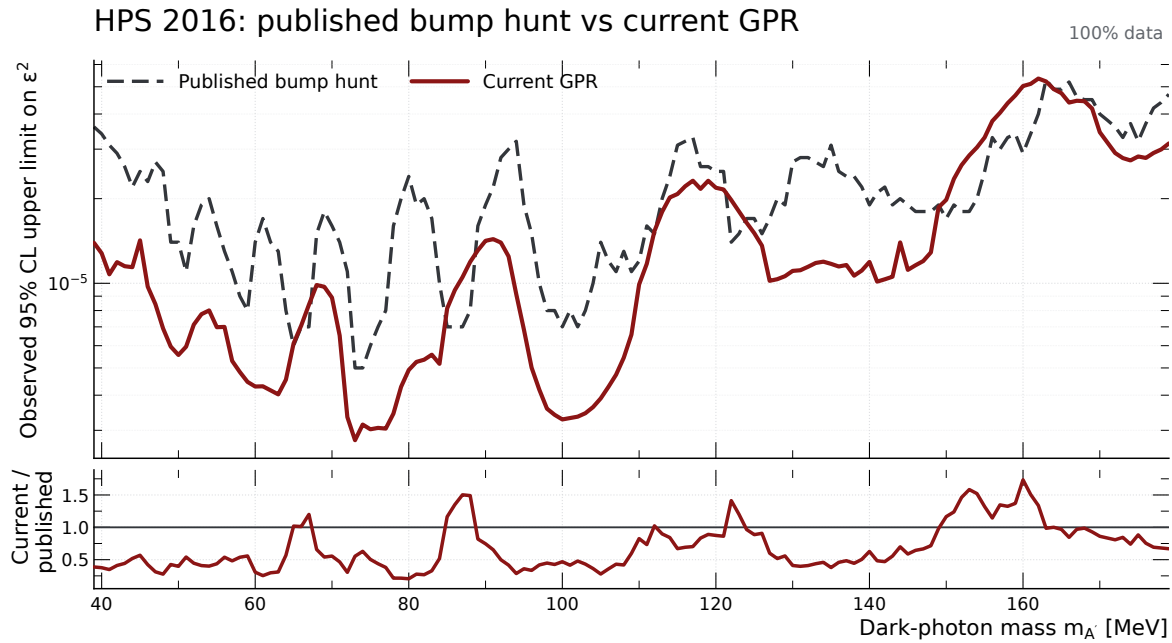


Figure 112: Saved 2016 observed 95% confidence-limit comparison from 5 August 2026, using the full already analyzed 2016 sample. The published reference uses the polynomial CL_s method; the curve labeled “current” in this archived graphic means the then-current v4.1 factor-12 GPR candidate. It does not mean the v5.0.1 configuration. The lower panel gives the pointwise observed GPR/published ratio, whose median is 0.5579 over 141 matched masses. This was an observed, asymptotic comparison with fixed reviewed kernel coordinates, no new toys, and no expected-limit bands.

H Exploratory 2015 search on the 15–20 MeV rising edge

Result. This short-support study does not reveal a persuasive narrow excess in 15–20 MeV. The largest nominal-GP upward fluctuation is at 17.25 MeV. Releasing an active smoothness ceiling gives a stable fit with $r = 1.58$ and local asymptotic $p_0 = 0.057$; the independently fitted polynomial background gives $p_0 = 0.254$ there. These values are model-conditional, selected from a scan, and do not establish or exclude a heavy photon.

H.1 Why a separate support is useful

The released 2015 scan begins at 19 MeV, with GP training support 14–135 MeV. Its input histogram extends below that support. The spectrum contains 81 events in 12–13 MeV, 7,653 in 14–15 MeV and 330,090 in 19–20 MeV. A local GP can isolate this turn-on without fitting the distant falling spectrum. It must still describe the steep background accurately across a missing signal window.

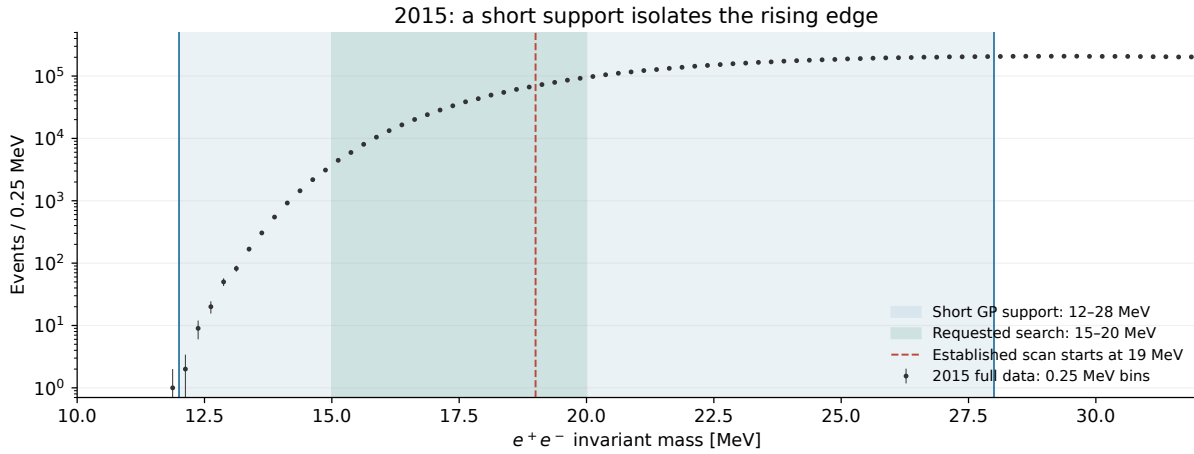


Figure 113: The same released full-2015 histogram, summed from 0.05 to 0.25 MeV bins. The chosen GP support is 12–28 MeV; the requested search is 15–20 MeV. The dashed line marks the lower boundary of the established scan, not the lower boundary of the available counts.

The new scan has 21 hypotheses, separated by 0.25 MeV. A labelled continuation to 22 MeV connects it to the previously selected 21 MeV point; that continuation is excluded from selection of the leading 15–20 MeV feature. The comparison supports 12–26, 12–30 and 12.5–28 MeV were declared before examining the new signal fits. Their p-values do not select the nominal support.

Detector boundary. The inherited Gaussian width is $\sigma_m = -0.09223 + 0.053219m$ MeV for m in MeV, giving about 0.71–0.97 MeV here. Below 19 MeV it is an extrapolated shape assumption. The published 2015 prompt search covered 19–81 MeV [3]. Low-mass signal efficiency, acceptance and the selected signal shape need direct simulation and control checks before a physical coupling result can be quoted.

H.2 Local probabilities and dependence on the background

The GP fits log counts versus log mass outside $m \pm 2.25\sigma_m$. Its prediction and correlated count covariance constrain the background in an exact Poisson window likelihood. Both the signal amplitude and background nuisance coordinates are fitted; the auxiliary amplitude may be negative while every total expectation remains positive. We report

$$r = \text{sign}(\hat{A})\sqrt{2[\text{NLL}(0) - \text{NLL}(\hat{A})]}, \quad p_0 = \bar{\Phi}(\max(r, 0)).$$

Thus the asymptotic convention assigns $p_0 = 0.5$ to negative fits [36]. No GP estimate of a global trials factor is used in this section.

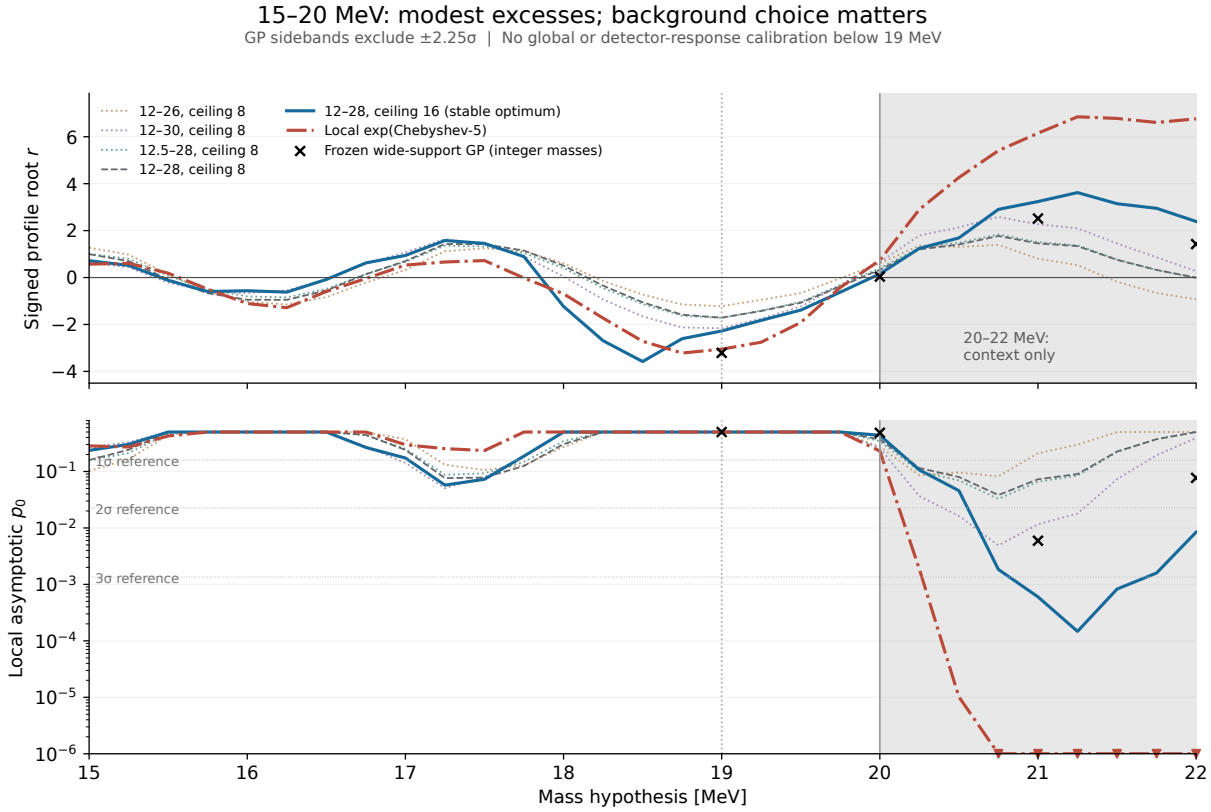


Figure 114: Signed local significance and local asymptotic excess probabilities. All four initial GP supports reach the inherited ceiling of eight resolution units. Raising the nominal-support ceiling to 16 removes every active boundary; ceilings 32 and 64 recover the same optima, with maximum root changes below 0.00054. The local polynomial is a separate background-family check. Gray shading identifies the 20–22 MeV bridge. Triangles mark the 10^{-6} display floor, not resolved global tails.

For the polynomial check, a positive $\exp(\text{Chebyshev}_5)$ background and Gaussian signal are integrated over each 0.25 MeV bin. All background coefficients are profiled over a moving $\pm 7\sigma_m$ support, clipped at 12 MeV. Its form and window scale follow the published low-mass procedure [3]; its suitability below 19 MeV is tested here rather than assumed. In 15–20 MeV its largest positive root is only 0.74. At 21 MeV it gives a much larger root, 6.15, but a signal-plus-background Poisson deviance of 98.7 for 51 nominal degrees of freedom. That poor-fit diagnostic and the GP dependence prevent interpreting the large value as a robust signal.

H.3 Signal extraction at fixed low-mass anchors

The displays use the stable ceiling-16 GP on 12–28 MeV. The 15 and 17 MeV hypotheses were fixed before the scan. Curves for profiled components are confined to the fitted window. Outside it, only data and the sideband GP constraint are shown. The gold uncertainty is the GP constraint before fitting the window, not an uncertainty band for the final fitted background.

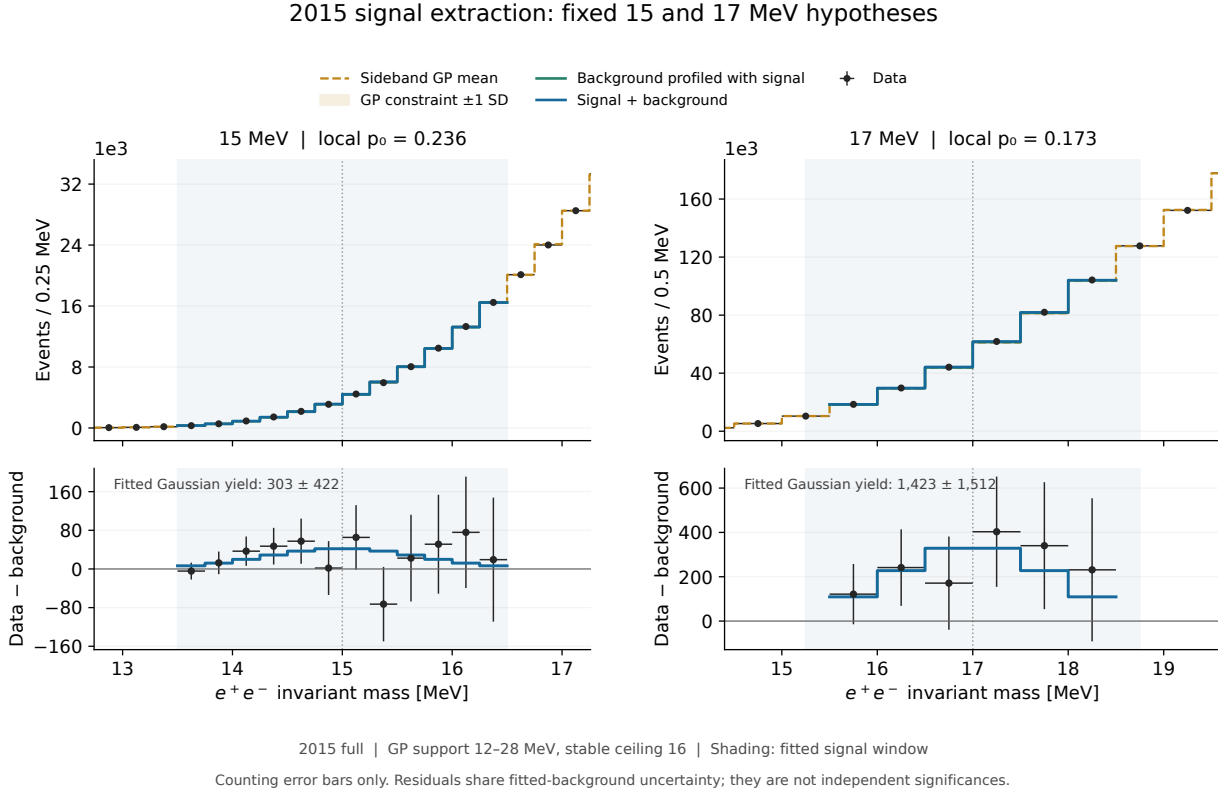


Figure 115: Fixed 15 and 17 MeV hypotheses. Top: data, sideband background constraint, profiled background and signal-plus-background prediction. Bottom: data minus the background profiled with signal, with the fitted Gaussian contribution. The error bars describe counting uncertainty only. The subtraction uses the same data to fit the background and is not an independent pull test.

The likelihood uses 0.25 MeV bins. For these figures, complete groups of bins are chosen nearest to half a mass resolution and anchored at the original zero edge. The resulting display widths are 0.25 MeV at 15 MeV and 0.50 MeV at the other displayed masses. Boundary groups crossing the fitted window are omitted from the profiled display, while all original window bins remain in the likelihood. No grouping or phase is chosen to make a peak sharper.

m [MeV]	r , ceiling 8	r , ceiling 16	p_0 , ceiling 16	p_0 , polynomial	Fitted yield, ceiling 16
15	1.00	0.72	0.236	0.284	303 ± 422
17	0.70	0.94	0.173	0.298	$1,423 \pm 1,512$
17.25	1.43	1.58	0.057	0.254	$2,542 \pm 1,612$
20	0.33	0.16	0.438	0.230	$514 \pm 3,312$

Yield errors are local curvature errors for the assumed Gaussian template; they are not post-selection confidence intervals. No conversion to ϵ^2 is made.

H.4 The leading fluctuation and the upper edge of the requested interval

At 17.25 MeV the stable GP fit gives approximately $2,542 \pm 1,612$ Gaussian-template events. Its maximum is selected after scanning 15–20 MeV, so this amplitude can be upward biased. It does not acquire a stronger interpretation because the scan was restricted to a rising edge. The 20 MeV panel supplies the fixed upper anchor.

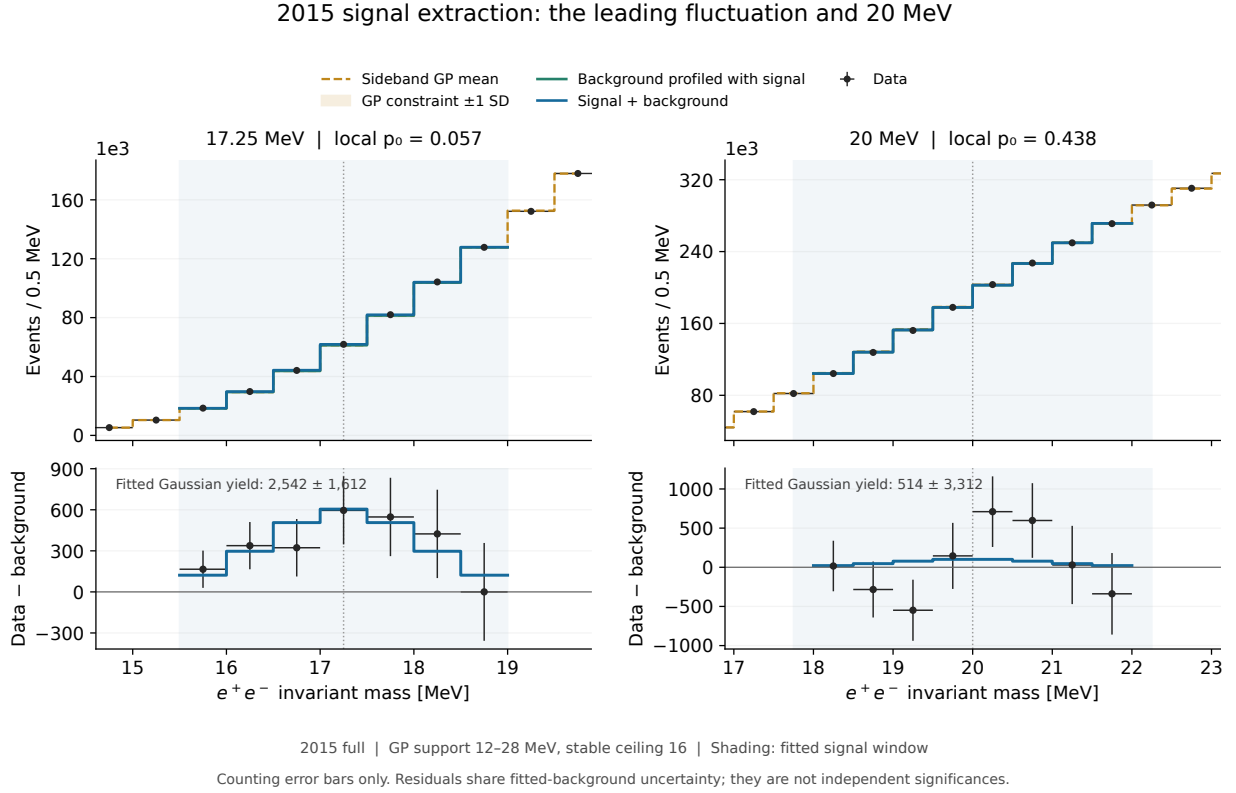


Figure 116: The selected 17.25 MeV upward fluctuation and the fixed 20 MeV hypothesis, with the same conventions and fixed-phase grouping as the preceding displays. Their local asymptotic roots are 1.58 and 0.16. A convincing resonance would need a stable excess across defensible background choices and a validated accepted signal shape.

At 21 MeV, the nominal GP changes from $r = 1.46$ to $r = 3.24$ when its active smoothness restriction is removed; the frozen wide-support fit has $r \simeq 2.52$. This is evidence that the inference is sensitive to the background description. It is not evidence that the low-mass study has uncovered a second established particle feature. Resolving this dependence requires predictive background controls and detector-level checks, not choosing the smallest p-value.

H.5 Small conditional toy checks and what remains to establish

Ten pilot spectra were first generated at each of the four displayed masses, then extended to 100 at each mass for each of ceilings 8 and 16: 800 distinct mass/model toy fits in total. The first ten are included in each 100. Each Poisson spectrum covers the complete 12–28 MeV support, and its GP hyperparameters and background are fitted again. The generating mean is the continuous sideband-conditioned GP prediction at that mass, without an observed-window adjustment. These mass-local backgrounds cannot be joined into one global experiment.

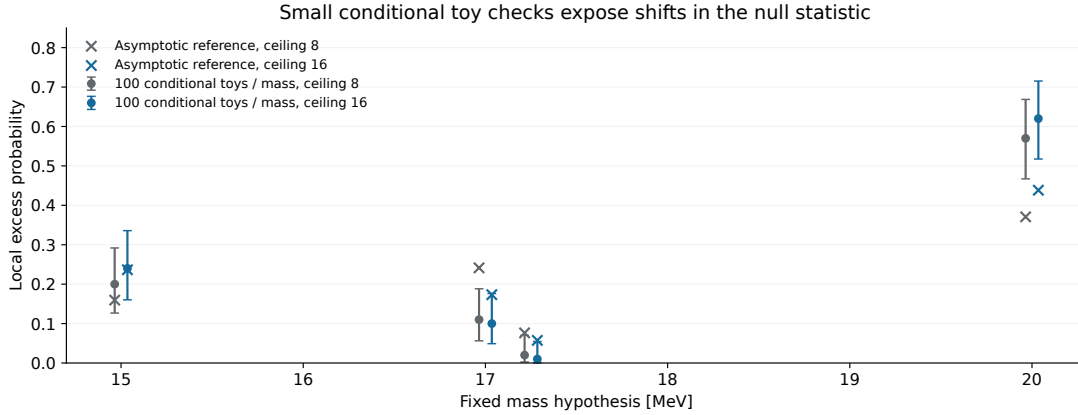


Figure 117: Toy upper-tail fractions with central 95% binomial intervals, compared with the asymptotic local reference. The two ceiling choices have their own conditional generating backgrounds. These small plug-in ensembles assess this fitting procedure under those means; they are not independent background validation or a final p-value calibration.

Ceiling	m [MeV]	Tails/toys	Fraction	95% interval	Mean r	SD r
8	15	20/100	0.20	[0.127, 0.292]	+0.23	0.89
8	17	11/100	0.11	[0.056, 0.188]	-0.48	0.95
8	17.25	2/100	0.02	[0.002, 0.070]	-0.43	0.99
8	20	57/100	0.57	[0.467, 0.669]	+0.52	0.92
16	15	24/100	0.24	[0.160, 0.336]	+0.07	1.04
16	17	10/100	0.10	[0.049, 0.176]	-0.31	0.97
16	17.25	1/100	0.01	[0.00025, 0.054]	-0.38	0.94
16	20	62/100	0.62	[0.517, 0.715]	+0.42	1.01

The shifts of the toy-root means away from zero demonstrate why an asymptotic reference alone is insufficient to qualify this extension. A small toy-tail fraction is conditional on a data-derived mean and has finite sampling uncertainty. It does not remove the choice of mass, support or background family.

Next, validate accepted signal templates and the turn-on; test background prediction and signal recovery under alternative smooth truths; fix choices independently of signal-window residuals; and calibrate a coherent scan for a global probability. A shorter GP support does not remove look-elsewhere accounting. The accompanying study bundle retains scripts, fit components, display maps, 800 toy roots and source hashes.

References

- [1] Bob Holdom. Two U(1)’s and epsilon charge shifts. *Phys. Lett. B*, 166:196–198, 1986.
- [2] Rouven Essig, Philip Schuster, Natalia Toro, and Bogdan Wojtsekhowski. An electron fixed target experiment to search for a new vector boson A' decaying to e^+e^- . *JHEP*, 02:009, 2011.
- [3] P. H. Adrian et al. Search for a dark photon in electroproduced e^+e^- pairs with the heavy photon search experiment at JLab. *Phys. Rev. Lett.*, 121:031802, 2018.
- [4] P. H. Adrian et al. Searching for prompt and long-lived dark photons in electro-produced e^+e^- pairs with the heavy photon search experiment at JLab. *arXiv preprint arXiv:2212.10629*, 2023.
- [5] J. P. Lees et al. Search for a dark photon in e^+e^- collisions at BaBar. *Phys. Rev. Lett.*, 113:201801, 2014.
- [6] D. Babusci et al. Limit on the production of a light vector gauge boson in ϕ meson decays with the KLOE detector. *Phys. Lett. B*, 720:111–115, 2013.
- [7] A. Anastasi et al. Limit on the production of a low-mass vector boson in $e^+e^- \rightarrow U\gamma$, $U \rightarrow e^+e^-$ with the KLOE experiment. *Phys. Lett. B*, 750:633–637, 2015.
- [8] H. Merkel et al. Search at the Mainz Microtron for light massive gauge bosons relevant for the muon $g - 2$ anomaly. *Phys. Rev. Lett.*, 112:221802, 2014.
- [9] S. Abrahamyan et al. Search for a new gauge boson in electron-nucleus fixed-target scattering by the APEX experiment. *Phys. Rev. Lett.*, 107:191804, 2011.
- [10] J. R. Batley et al. Search for the dark photon in π^0 decays. *Phys. Lett. B*, 746:178–185, 2015.
- [11] G. Agakishiev et al. Searching a dark photon with HADES. *Phys. Lett. B*, 731:265–271, 2014.
- [12] A. Adare et al. Search for dark photons from neutral meson decays in $p + p$ and $d + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV. *Phys. Rev. C*, 91:031901, 2015.
- [13] D. Banerjee et al. Search for a hypothetical 16.7 mev gauge boson and dark photons in the NA64 experiment at CERN. *Phys. Rev. Lett.*, 120:231802, 2018.
- [14] Dipanwita Banerjee et al. Search for a new $X(16.7)$ boson and dark gauge bosons in the NA64 experiment at CERN. *Phys. Rev. D*, 104:L111102, 2021.
- [15] Roel Aaij et al. Search for $A' \rightarrow \mu^+\mu^-$ decays. *Phys. Rev. Lett.*, 124:041801, 2020.
- [16] J. D. Bjorken et al. Search for neutral metastable penetrating particles produced in the SLAC beam dump. *Phys. Rev. D*, 38:3375, 1988.
- [17] E. M. Riordan et al. Search for short-lived axions in an electron-beam-dump experiment. *Phys. Rev. Lett.*, 59:755, 1987.
- [18] A. Bross et al. A search for short-lived particles produced in an electron beam dump. *Phys. Rev. Lett.*, 67:2942, 1991.
- [19] M. Davier and H. Nguyen Ngoc. An unambiguous search for a light higgs boson. *Phys. Lett. B*, 229:150–155, 1989.

- [20] A. Konaka et al. Search for neutral particles in electron-beam-dump experiment. *Phys. Rev. Lett.*, 57:659, 1986.
- [21] J. Blümlein and J. Brunner. New exclusion limits on dark gauge forces from proton bremsstrahlung in beam-dump data. *Phys. Lett. B*, 731:320–326, 2014.
- [22] D. G. Krige. A statistical approach to some basic mine valuation problems on the Witwatersrand. *Journal of the Chemical, Metallurgical and Mining Society of South Africa*, 52:119–139, 1951.
- [23] Georges Matheron. Principles of geostatistics. *Economic Geology*, 58(8):1246–1266, 1963.
- [24] Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian Processes for Machine Learning*. Adaptive Computation and Machine Learning. MIT Press, Cambridge, Massachusetts, 2006.
- [25] Meghan Frate, Kyle Cranmer, Saarik Kalia, Alexander Vandenberg-Rodes, and Daniel Whiteson. Modeling smooth backgrounds and generic localized signals with gaussian processes. *arXiv preprint*, 2017.
- [26] A. Gandrakota, A. Lath, A. V. Morozov, and S. Murthy. Model selection and signal extraction using gaussian process regression. *JHEP*, 02:230, 2023.
- [27] Omar Moreno et al. The heavy photon search experiment at Jefferson Lab. *arXiv preprint arXiv:2203.08324*, 2022.
- [28] HPS Collaboration. 2016 bump hunt internal note, 2017. HPS internal note.
- [29] Vitaliy Fadeyev et al. New technology for the heavy photon search silicon vertex tracker. *Nucl. Instrum. Meth. A*, 959:163567, 2020.
- [30] Elizabeth Berzin, Sarah Gaiser, Matthew Gignac, Rory O’Dwyer, and Lewis Wolf. HPS-ANALYSIS-NOTE v2: 2021 displaced vertex search analysis note, 2026. HPS internal analysis note.
- [31] HPS Collaboration. 2015 resonance search internal note, 2016. HPS internal note.
- [32] J. Barr and B. Liu. Gaussian process regression as a sustainable data-driven background estimate method at the (HL)-LHC. *Eur. Phys. J. C*, 85:846, 2025.
- [33] Fabian Pedregosa, Gael Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Edouard Duchesnay. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [34] scikit-learn developers. Gaussian processes: scikit-learn 1.6.1 documentation, 2025. Version 1.6.1; accessed August 4, 2026.
- [35] Kyle Cranmer et al. HistFactory: A tool for creating statistical models for use with RooFit and RooStats. Technical Report CERN-OPEN-2012-016, CERN, 2012. CERN Open Report.
- [36] Glen Cowan, Kyle Cranmer, Eilam Gross, and Ofer Vitells. Asymptotic formulae for likelihood-based tests of new physics. *Eur. Phys. J. C*, 71:1554, 2011.

- [37] Thomas Junk. Confidence level computation for combining searches with small statistics. *Nucl. Instrum. Meth. A*, 434:435–443, 1999.
- [38] Alexander L. Read. Presentation of search results: The CL_s technique. *J. Phys. G*, 28:2693–2704, 2002.
- [39] ATLAS Collaboration and CMS Collaboration. Measurements of the higgs boson production and decay rates and constraints on its couplings from a combined ATLAS and CMS analysis of the LHC pp collision data at $\sqrt{s} = 7$ and 8 tev. *JHEP*, 08:045, 2016.
- [40] Eilam Gross and Ofer Vitells. Trial factors for the look elsewhere effect in high energy physics. *Eur. Phys. J. C*, 70:525–530, 2010.
- [41] Marco Fabbrichesi, Emidio Gabrielli, and Gaia Lanfranchi. *The Dark Photon*. SpringerBriefs in Physics. Springer, 2020.
- [42] V. Ananiev and A. L. Read. Gaussian process-based calculation of look-elsewhere trials factor. *JINST*, 18:P05041, 2023.
- [43] L. Berns. An importance sampling method for feldman-cousins confidence intervals. *Phys. Rev. D*, 109:092002, 2024.